



UBX-F10/M10 SPG7.0x

Standard precision GNSS firmware

Protocol version 52.00

Interface description

Abstract

This document provides a common interface description for the firmware variants listed in the Document information. It lists the protocols and messages, explains the configuration concept, and provides the default configuration values for protocol version 52.00.

Document information

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This document applies to the following firmware versions:

Firmware name	Version and revision identifier	Protocol version
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SPG 7.00	EXT SPGL1L5 7.00 (9b2693)	52.00
SPG 7.01	EXT SPGL1L5 7.01 (61eed)	52.00

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1 General information

1.1 Document overview

This document describes the interface of the SPG 7.0x standard precision GNSS firmware. The interface consists of the following parts:

- [NMEA protocol](#)
- [UBX protocol](#)
- [Configuration interface](#)



Some of the features described here may not be available in the receiver, and some may require specific configurations to be enabled. See the applicable data sheet for availability of the features and the integration manual for instructions for enabling them.



Previous versions of u-blox receiver documentation combined general receiver description and interface specification. In the current documentation the receiver description is included in the integration manual.

See also [Related documents](#).

1.2 Firmware and protocol versions

u-blox receivers execute firmware from internal ROM or load an external image and execute it from internal code-RAM.

- If the product does not have internal code-RAM, the firmware runs from the ROM.
- If the product has internal code-RAM but an external image is not available, the firmware runs from the ROM. Some products have only limited ROM and enter boot mode with no GNSS function if an external image is not available.
- If the external firmware image is stored in a flash memory, it is loaded into the code-RAM before execution.
- In some products, the firmware image can be stored in the host system and loaded into the code-RAM from there.

The location and the version of the currently running firmware can be found in the boot screen and in the [UBX-MON-VER](#) message. If the firmware has been loaded from the flash memory or from the host processor, it is indicated by text "EXT", whereas running the firmware from the internal ROM is indicated by text "ROM".

The u-blox receivers output the boot screen automatically upon receiver start or after hardware reset over the serial interfaces in [UBX-INF-NOTICE](#) or [NMEA-Standard-TXT](#) messages if configured using [CFG-INFMSG](#). The UBX-MON-VER message can be polled using the [UBX polling mechanism](#). An example of the boot screen and the firmware version information in u-center 2 is shown in [Figure 1](#).

Time	Message	UBX-MON-VER
07:37:44	R→ \$GNTXT,01,01,02,u-blox AG - www.u-blox.com*4E..	Field Value
07:37:44	R→ \$GNTXT,01,01,02,HW UBX 10 000A0000*53..	swVersion ROM SPG 5.10 (7b202e)
07:37:44	R→ \$GNTXT,01,01,02,ROM SPG 5.10 (7b202e)*7C..	hwVersion 000A0000
07:37:44	R→ \$GNTXT,01,01,02,FWVER=SPG 5.10*40..	
07:37:44	R→ \$GNTXT,01,01,02,PROTVER=34.10*1E..	group1 ▾
07:37:44	R→ \$GNTXT,01,01,02,CHIPID=000000E9BC858F4254*78..	extension
07:37:44	R→ \$GNTXT,01,01,02,MOD=EVK-M101*20..	FWVER=SPG 5.10
07:37:44	R→ \$GNTXT,01,01,02,GPS;GLO;GAL;BDS*77..	PROTVER=34.10
07:37:44	R→ \$GNTXT,01,01,02,SBAS;QZSS*60..	MOD=EVK-M101
07:37:44	R→ \$GNTXT,01,01,02,ANTSUPERV=*22..	GPS;GLO;GAL;BDS
07:37:44	R→ \$GNTXT,01,01,02,ANTSTATUS=DONTKNOW*2D..	SBAS;QZSS
07:37:44	R→ \$GNTXT,01,01,02,PF=F7FB*4B..	
07:37:44	R→ \$GNTXT,01,01,02,Starting GNSS*5A..	

Figure 1: An example of u-center 2 showing the Text console with the boot screen output on the left and the Message view with the UBX-MON-VER version information on the right

The following information is available (✓) from the boot screen (**B**) and the UBX-MON-VER message (**M**):

B	M	Example	Information
✓		u-blox AG - www.u-blox.com	Start of the boot screen.
✓		HW UBX 10 00000000	Hardware version of the u-blox receiver.
✓	✓	00000000	
✓	✓	ROM SPG 5.10 (000000)	Firmware version and revision identifier.
✓	✓	ROM BASE 0x118B2060	Revision of the underlying boot loader firmware in ROM.
✓	✓	FWVER=SPG 5.10	Product firmware version, where: - SPG = Standard precision GNSS product - SIG = Standard Precision Infrastructure GNSS product - HPG = High precision GNSS product - ADR = Automotive dead reckoning product - TIM = Time sync product - LAP = Lane accurate positioning product - HPS = High precision sensor fusion product - DBS = Dual band standard precision - MDR = Multi-mode dead reckoning product - PMP = L-Band Inmarsat point-to-multipoint receiver - SME = Safe measurement engine - SCAP = Safety critical automotive positioning - QZSS = QZSS L6 centimeter level augmentation service (CLAS) message receiver - DBD = Dual band dead reckoning product - ASP = Automotive standard precision - LDR = ROM bootloader, no GNSS functionality - AHP = Automotive high precision product
✓	✓	PROTVER=34.00	Supported protocol version.
✓		CHIPID=000000D0D69D0F7A54	Unique chip identification number.
✓	✓	MOD=EVK-M101	Module name.
✓	✓	GPS;GLO;GAL;BDS	List of supported major GNSS (see GNSS identifiers).
✓	✓	SBAS;QZSS	List of supported augmentation systems (see GNSS identifiers).
✓	✓	NAVIC	Extended list of supported GNSS (see GNSS identifiers).

B	M	Example	Information
✓		ANTSUPERV=AC SD PDoS SR	Configuration of the antenna supervisor, where: - AC = Active antenna control enabled - SD = Short circuit detection enabled - OD = Open circuit detection enabled - PDoS = Short circuit power down logic enabled - SR = Automatic recovery from short state enabled
✓		PF=FFF79	Product configuration.
✓		BD=E01C	GNSS band configuration.

-  The "FWVER" product firmware version indicates which firmware is currently running. This is referred to as "firmware version" in this and other documents.
-  The version and revision numbers should only be used to identify a known firmware version. They are not necessarily numeric nor are they guaranteed to increase with later firmware versions.
-  All u-blox receivers output the start text, hardware version, and firmware version and revision. Some of the other entries in the boot screen example may be omitted.

The product firmware version and revision relate to the protocol version:

Firmware version	Version and revision identifier	Protocol version
SPG 7.00	EXT SPGL1L5 7.00 (a239ce)	52.00
SPG 7.00	EXT SPGL1L5 7.00 (9b2693)	52.00
SPG 7.01	EXT SPGL1L5 7.01 (61eed)	52.00

1.3 Receiver configuration

u-blox positioning receivers are fully configurable with UBX protocol messages. The configuration used by the receiver during normal operation is called the "current configuration". The current configuration can be changed during normal operation by sending [UBX-CFG-VALSET](#) messages over any I/O port. The receiver changes its current configuration immediately after receiving a configuration message. The receiver always uses the current configuration only.

The current configuration is loaded from permanent configuration hard-coded in the receiver firmware (the defaults) and from non-volatile memory (user configuration) on startup of the receiver. Changes made to the current configuration at run-time will be lost when there is a power cycle, a hardware reset or a (complete) controlled software reset (see [Configuration reset behavior](#)).

See [Configuration interface](#) for a detailed description of the receiver configuration system, the explanation of the configuration concept and its principles and interfaces.

-  The configuration interface has changed from earlier u-blox positioning receivers. There is some backwards compatibility provided in UBX-CFG configuration messages. Users are strongly advised to only use the [Configuration interface](#). See also .
-  See the integration manual for a basic receiver configuration most commonly used.

1.4 Message naming

Message names are written in full with the parts of the name separated by hyphens ("-"). The full message name consists of the protocol name (e.g. *UBX*), the class name (e.g. *NAV*) and the message name (e.g. *PVT*). For example, the receiver software version information message is referred to as *UBX-MON-VER*. Similarly, the *NMEA-Standard-GGA* is the NMEA standard message (sentence) with the global positioning fix data.

References to fields of the message add the field name separated by a dot ("."), e.g. *UBX-MON-VER.swVersion*.

Some messages use a fourth level of naming, called the message version. One example is the *UBX-MGA-GPS* message for GPS assistance data, which exists in versions for ephemerides (*UBX-MGA-GPS-EPH*) and almanacs (*UBX-MGA-GPS-ALM*).

Names of configuration items are of the form *CFG-GROUP-ITEM*. For example, *CFG-NAVSPG-DYNMODEL* refers to the navigation dynamic platform model the receiver uses. Constants add a fourth level to the item name, such as *CFG-NAVSPG-DYNMODEL-AUTOMOT* for the automotive platform model. In the context of describing an item's value, only the last part of the constant name can be used (e.g. "set *CFG-NAVSPG-DYNMODEL* to *PORT* for portable applications").

1.5 GNSS, satellite, and signal identifiers

1.5.1 Overview

Many [UBX protocol](#) messages contain information about specific satellites. Any single satellite can be identified by a `gnssId` field indicating the GNSS the satellite is part of and an `svId` (SV for space vehicle) field indicating the number of the satellite in that system. Usually, the `svId` is the native number associated with the satellite in the specific GNSS. For example, the Galileo SV4 is identified as `gnssId 2`, `svId 4`, while the GPS SV4 is `gnssId 0`, `svId 4`.

Some legacy UBX protocol messages combine both the satellite number and the GNSS identification into a one-byte (type U1) field. See the single `svId` mapping in [Satellite identifiers](#) to identify the corresponding GNSS and satellite.

GLONASS satellites can be tracked before they have been identified. In UBX messages, the unknown satellites are reported with `svId 255`. In NMEA messages, the unknown satellites are null (empty) fields. Product-related documentation and u-center use R? to label unidentified GLONASS satellites.

Signal identifiers are used when different signals from the same GNSS satellite need to be distinguished (e.g. in the [UBX-NAV-SIG](#) message). A separate `sigId` field identifies the signal. These signal identifiers are only valid when combined with a GNSS identifier (`gnssId` field).

The [NMEA protocol](#) (version 4.10 and later) identifies GNSS satellites with a one-digit system ID and a two-digit satellite number. u-blox receivers support this method in their NMEA output when "strict" SV numbering is selected. In most cases this is the default setting, but it can be checked or changed using the [Configuration interface](#) (see also [NMEA GNSS, satellite, and signal numbering](#)).

In order to support some GNSS (e.g. BeiDou, Galileo, QZSS), which are not supported by some or all NMEA protocol versions, an "extended" SV numbering scheme can be enabled. This uses the NMEA-defined numbers where possible but adds other number ranges to support other GNSS. Note however that these non-standard extensions require 3-digit numbers, which may not be supported by some NMEA parsing software. For example, QZSS satellites use numbers in the range 193 to 202.

The NMEA standard defines signal identifiers to distinguish different signals sent by a single GNSS satellite (e.g. L2 CL and CM). u-blox positioning receivers use those identifiers for signal identification, as far as the corresponding standard is supported in a particular product.



Note that the following sections are a generic overview for different u-blox positioning receivers. A particular product may not support all of the described GNSS identifiers, satellite numbers, signal identifiers or combinations thereof.

1.5.2 GNSS identifiers

Table 1 lists each GNSS along with the GNSS identifier ([UBX protocol](#)), the NMEA system identifiers ([NMEA protocol](#)), and abbreviations used in this document:

GNSS	Abbreviations		UBX gnssId	NMEA system ID		
				2.3 - 4.0	4.10	4.11
GPS	GPS	G	0	1	1	1
SBAS	SBAS	S	1	1	1	1
Galileo	GAL	E	2	n/a	3	3
BeiDou	BDS	B	3	n/a	(4) ¹	4
QZSS	QZSS	Q	5	n/a	(1) ¹	5
GLONASS	GLO	R	6	2	2	2
NavIC	NavIC	N	7	n/a	n/a	6

Table 1: GNSS identifiers

See also [NMEA Talker ID](#).

1.5.3 Satellite identifiers

The satellite numbering scheme for the [UBX protocol](#) is provided in Table 2. The satellite numbering scheme for the [NMEA protocol](#) is provided in Table 3.

GNSS	SV Range	gnssId:svId	single svId
GPS	G1-G32	0:1-32	1-32
SBAS	S120-S158	1:120-158	120-158
Galileo	E1-E36	2:1-36	211-246
BeiDou	B1-B5	3:1-5	159-163
	B6-B37	3:6-37	33-64
	B38-B63	3:38-63	n/a
QZSS	Q1-Q10	5:1-10	193-202
GLONASS	R1-R31	6:1-31	65-95
	R?	6:255	255
NavIC	N1-N7	7:1-7	247-253
	N8-N14	7:8-14	n/a

Table 2: UBX protocol satellite numbering scheme

GNSS	SV Range	NMEA 2.3 - 4.0		NMEA 4.10		NMEA 4.11	
		strict	extended	strict	extended	strict	extended
GPS	G1-G32	1-32	1-32	1-32	1-32	1-32	1-32
SBAS	S120-S158	33-64	33-64, 152-158	33-64	33-64, 152-158	33-64	33-64, 152-158
Galileo	E1-E36	n/a	301-336	1-36	1-36	1-36	1-36
BeiDou	B1-B5	n/a	401-405	1-5	1-5	1-5	1-5
	B6-B37	n/a	406-437	6-37	6-37	6-37	6-37

¹ While not defined by NMEA 4.10, in this mode, u-blox receivers use system ID 4 for BeiDou and, if extended satellite numbering is enabled, system ID 1 for QZSS.

GNSS	SV Range	NMEA 2.3 - 4.0		NMEA 4.10		NMEA 4.11	
		strict	extended	strict	extended	strict	extended
	B38-B63	n/a	438-463	38-63	38-63	38-63	38-63
QZSS	Q1-Q10	n/a	193-202	n/a	193-202	1-10	1-10
GLONASS	R1-R32	65-96	65-96	65-96	65-96	65-96	65-96
	R?	null	null	null	null	null	null
NavIC	N1-N7	n/a	n/a	n/a	n/a	1-7	1-7
	N8-N14	n/a	n/a	n/a	n/a	8-14	8-14

Table 3: NMEA protocol satellite numbering scheme

1.5.4 Signal identifiers

A summary of all the signal identification schemes used in the [NMEA protocol](#) and the [UBX protocol](#) is provided in the following table. (Only a subset of the signals is supported by each product.) In the NMEA protocol, system and signal identifiers are in hexadecimal format. An unknown signal identifier is presented as 0 in the NMEA protocol.

Signal	UBX Protocol		NMEA Protocol 4.10		NMEA Protocol 4.11	
	gnssId	sigId	System ID	Signal ID	System ID	Signal ID
GPS L1C/A ²	0	0	1	1	1	1
GPS L2 CL	0	3	1	6	1	6
GPS L2 CM	0	4	1	5	1	5
GPS L5 I	0	6	1	7	1	7
GPS L5 Q	0	7	1	8	1	8
SBAS L1C/A ²	1	0	1	1	1	1
Galileo E1 C ²	2	0	3	7	3	7
Galileo E1 B ²	2	1	3	7	3	7
Galileo E5 aI	2	3	3	1	3	1
Galileo E5 aQ	2	4	3	1	3	1
Galileo E5 bI	2	5	3	2	3	2
Galileo E5 bQ	2	6	3	2	3	2
Galileo E6 B	2	8	3	5	3	5
Galileo E6 C	2	9	3	5	3	5
Galileo E6 A	2	10	3	4	3	4
BeiDou B1I D1 ²	3	0	(4) ³	(1) ⁴	4	1
BeiDou B1I D2 ²	3	1	(4) ³	(1) ⁴	4	1
BeiDou B2I D1	3	2	(4) ³	(3) ⁴	4	B
BeiDou B2I D2	3	3	(4) ³	(3) ⁴	4	B
BeiDou B3I D1	3	4	(4) ³	N/A	4	8
BeiDou B3I D2	3	10	(4) ³	N/A	4	8
BeiDou B1 Cp (pilot)	3	5	(4) ³	N/A	4	3

² This signal belongs to the group of signals reported in the UBX messages that do not have an explicit `sigId` field.

³ While not defined by NMEA 4.10, in this mode, u-blox receivers use system ID 4 for BeiDou and, if extended satellite numbering is enabled, system ID 1 for QZSS.

⁴ BeiDou and QZSS signal ID are not defined in the NMEA protocol version 4.10. Values shown in the table are only valid for u-blox products and, for QZSS signal ID, if extended satellite numbering is enabled.

Signal	UBX Protocol		NMEA Protocol 4.10		NMEA Protocol 4.11	
	gnssId	sigId	System ID	Signal ID	System ID	Signal ID
BeiDou B1 Cd (data)	3	6	(4) ³	N/A	4	3
BeiDou B2 ap (pilot)	3	7	(4) ³	N/A	4	5
BeiDou B2 ad (data)	3	8	(4) ³	N/A	4	5
QZSS L1C/A ²	5	0	(1) ³	(1) ⁴	5	1
QZSS L1S	5	1	(1) ³	(4) ⁴	5	4
QZSS L2 CM	5	4	(1) ³	(5) ⁴	5	5
QZSS L2 CL	5	5	(1) ³	(6) ⁴	5	6
QZSS L5 I	5	8	(1) ³	N/A	5	7
QZSS L5 Q	5	9	(1) ³	N/A	5	8
QZSS L1C/B ²	5	12	(1) ³	N/A	5	N/A
GLONASS L1 OF ²	6	0	2	1	2	1
GLONASS L2 OF	6	2	2	3	2	3
NavIC L5 A ²	7	0	N/A	N/A	6	1

Table 4: Signal identifiers

1.6 Message types

The following message types are defined:

Message type	Description
Input	Messages that are input to the receiver and never output. E.g. UBX-MGA-GPS-EPH .
Output	Messages that are output by the receiver in no particular interval and never input. E.g. UBX-ACK-ACK .
Input/output	Messages that can be output by or input to the receiver. E.g. UBX-MGA-DBD-DATA0 .
Periodic	Messages that are output in regular intervals but cannot be polled. E.g. UBX-NAV-EOE .
Periodic/pollable	Messages that are output in regular intervals and can be polled. E.g. UBX-NAV-PVT .
Command	Messages that are a command to the receiver. Similar to type <i>Input</i> these are input-only. E.g. UBX-CFG-RST .
Get	Output-only configuration or command messages. E.g. UBX-CFG-DAT .
Set	Input-only configuration or command messages. E.g. UBX-CFG-VALDEL .
Get/set	Input/output configuration or command messages. E.g. UBX-CFG-NAVX5 .
Polled	Non-periodic messages that can only be polled. E.g. UBX-MON-VER .
Poll request	Poll request. E.g. UBX-MGA-DBD-POLL .

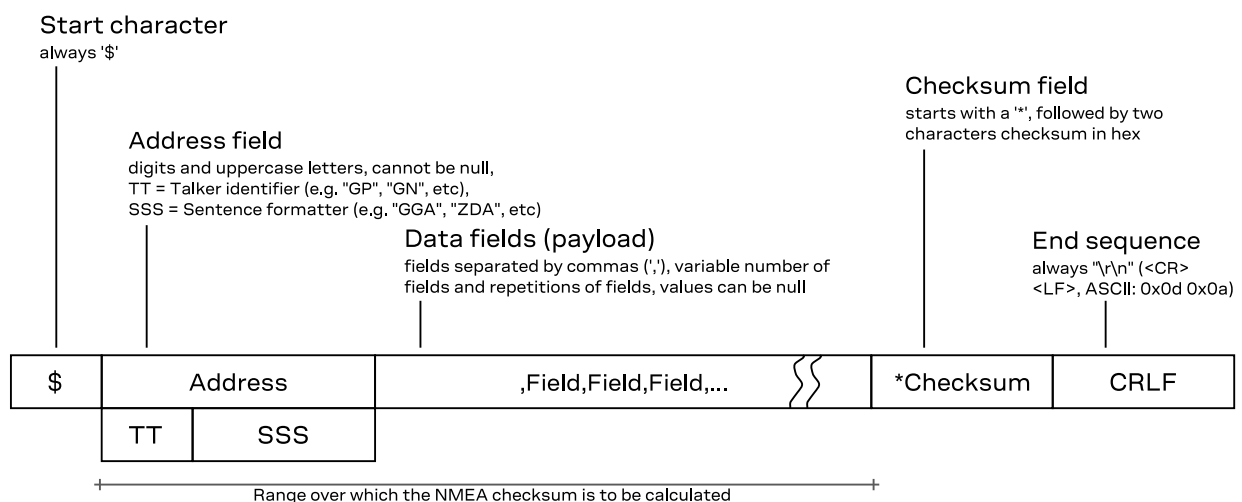
2 NMEA protocol

The following sections give an overview of the NMEA messages used by u-blox positioning receivers.

By default, the NMEA messages sent by u-blox positioning receivers are based on the NMEA 0183 version 4.11 standard. For further information on the NMEA standard, refer to the *NMEA 0183 Standard for Interfacing Marine Electronic Devices*, Version 4.11, November 2018, which is available on <http://www.nmea.org/>.

2.1 NMEA frame structure

The following figure shows the structure of a NMEA protocol message (called "sentences" in the standard).



Example

\$	GP	ZDA	,141644.00,22,03,2002,00,00	*67	\r\n
----	----	-----	-----------------------------	-----	------

2.2 NMEA protocol configuration

The [NMEA protocol](#) on u-blox receivers can be configured for customer applications by using the [Configuration interface](#) (CFG-NMEA-* items).

Several NMEA standard versions are supported. Version 4.11 (not in all products), 4.10, 4.00, 2.3, or 2.1 can be configured. See [Configuration defaults](#) for the default version. See [CFG-NMEA-PROTVER](#) to configure the version. See [NMEA multi-GNSS operation](#) and [NMEA data fields](#) for details on how this affects the output.

The following filtering flags can be used to configure the output of some NMEA message fields:

Filter	Configuration Item	Description
Position filtering	CFG-NMEA-OUT_INVFIX	Enable to permit positions from failed or invalid fixes to be reported (with the "V" status flag to indicate that the data is not valid).
Valid position filtering	CFG-NMEA-OUT_MSKFIX	Enable to permit positions from invalid fixes to be reported (with the "V" status flag to indicate that the data is not valid).
Time filtering	CFG-NMEA-OUT_INVTIME	Enable to permit the receiver's best knowledge of time to be output, even though it might be wrong.

Filter	Configuration Item	Description
Date filtering	CFG-NMEA-OUT_INVDATE	Enable to permit the receiver's best knowledge of date to be output, even though it might be wrong.
GPS-only filtering	CFG-NMEA-OUT_ONLYGPS	Enable to restrict output to only report GPS satellites.
Track filtering	CFG-NMEA-OUT_FROZENCOD	Enable to permit course over ground (COG) to be reported even when it would otherwise be frozen.

The following filtering flags can be used to configure the output of some NMEA message flags:

Mode	Configuration Item	Description
Compatibility mode	CFG-NMEA-COMPAT	Some older NMEA applications expect the NMEA output to be formatted in a specific way, for example, they will only work if the latitude and longitude have exactly four digits behind the decimal point. u-blox receivers offer a compatibility mode to support these legacy applications.
Consideration mode	CFG-NMEA-CONSIDER	u-blox receivers use a sophisticated signal quality detection scheme, in order to produce the best possible position output. This algorithm considers all SV measurements, and may eventually decide to only use a subset thereof, if it improves the overall position accuracy. If consideration mode is enabled, all satellites, which were considered for navigation, are communicated as being used for the position determination. If consideration mode is disabled, only those satellites which after the consideration step remained in the position output are marked as being used.
Limit length mode	CFG-NMEA-LIMIT82	Enabling this mode will limit the NMEA sentence length to a maximum of 82 characters.
High precision mode	CFG-NMEA-HIGHPREC	Enabling this mode increases precision of the position output. Latitude and longitude then have seven digits after the decimal point, and altitude has three digits after the decimal point. Note: The high precision mode cannot be set in conjunction with either compatibility mode or Limit82 mode.

The following extended configuration options are available:

Option	Configuration Item(s)	Description
GNSS to filter	CFG-NMEA-FILT_GPS etc.	Filters satellites based on the GNSS they belong to.
Satellite numbering	CFG-NMEA-SVNUMBERING	This field configures the display of satellites that do not have an NMEA-defined value. Note: this does not apply to satellites with an unknown ID. See also Satellite identifiers .
Main Talker ID	CFG-NMEA-MAINTALKERID	By default the main Talker ID (i.e. the Talker ID used for all messages other than GSV) is determined by the GNSS assignment of the receiver's channels (see configuration items CFG-SIGNAL-*). This field enables the main Talker ID to be overridden. See also NMEA Talker ID .
GSV Talker ID	CFG-NMEA-GSVTALKERID	By default the Talker ID for GSV messages is GNSS-specific (as defined by NMEA). This field enables the GSV Talker ID to be overridden.
BDS Talker ID	CFG-NMEA-BDSTALKERID	By default the Talker ID for BeiDou is "GB". This field enables the BeiDou Talker ID to be overridden.

2.3 NMEA proprietary messages

The NMEA standard allows for proprietary, manufacturer-specific messages to be added. These shall be marked with a manufacturer mnemonic. The mnemonic assigned to u-blox is UBX and is used for all non-standard messages. These proprietary NMEA messages therefore have the address field set to PUBX. The first data field in a PUBX message identifies the message number with two digits.

2.4 NMEA multi-GNSS operation

Many applications that process NMEA messages assume that only a single GNSS is active. However, when multiple GNSS are configured, the NMEA specification requires the output to change in the following ways:

Main Talker ID The main [NMEA Talker ID](#) is "GN" (e.g. instead of "GP" for a GPS-only receiver).

GSV Talker and Signal IDs The [GSV](#) message reports the signal strength of the visible satellites. In multi-GNSS operation, other messages use the main Talker ID "GN" but the Talker ID in the GSV message is specific to the GNSS it is reporting information for.

The GSV messages are grouped by the Talker and Signal IDs. Separate sets of GSV messages are sent for each GNSS and signal. The Signal ID of a satellite may be unknown. Such satellites are presented in their own set with Signal ID 0. Grouping the GSV messages by the Signal ID is supported in protocol versions 27.12 and later.

Multiple GSA and GRS messages Multiple [GSA](#) and [GRS](#) messages are output for each fix, one for each GNSS. This may confuse applications that assume they are output only once per position fix (as is the case for a single GNSS receiver).

GGA Talker IDs The NMEA specification indicates that the GGA message is GPS-specific. However, u-blox receivers support the output of a GGA message for each of the Talker IDs.

BeiDou and Galileo Only NMEA version 4.10 and later have support for these systems.

QZSS Only NMEA version 4.11 and later have support for this system.

Extended satellite numbering In order to support some GNSS (e.g. BeiDou, Galileo, QZSS) that are not supported by some or all NMEA protocol versions, an "extended" SV numbering scheme can be enabled. This uses the NMEA-defined numbers where possible, but adds other number ranges to support other GNSS. Note however that these non-standard extensions require 3-digit numbers, which may not be supported by some NMEA parsing software. For example, QZSS satellites use numbers in the range 193 to 202. See [NMEA protocol configuration](#) and [Satellite identifiers](#).

2.5 NMEA data fields

Various data fields in NMEA messages depend on [NMEA protocol configuration](#) or require a definition for their interpretation.

2.5.1 NMEA Talker ID

One of the ways the NMEA standard differs depending on the GNSS is by using a two-letter message identifier, the "Talker ID". The specific Talker ID used by a u-blox receiver will depend on the product and its configuration. The table below shows the Talker ID that will be used for various GNSS configurations by default.

GNSS	Talker ID	Comments
GPS, SBAS	GP	NMEA 2.3+
GLONASS	GL	NMEA 2.3+
Galileo	GA	NMEA 4.10+
BeiDou	GB	NMEA 4.10+ (official NMEA only since 4.11)
NavIC	GI	NMEA 4.11+
QZSS	GQ	NMEA 4.11+ (GP for NMEA 2.3 - 4.10)

GNSS	Talker ID	Comments
Any combination of GNSS	GN	

2.5.2 NMEA extra fields

The following extra fields are available in NMEA 4.10 and later.

Message	Extra fields
NMEA-Standard-GBS	systemId and signalId
NMEA-Standard-GNS	navStatus
NMEA-Standard-GRS	systemId and signalId
NMEA-Standard-GSA	systemId
NMEA-Standard-GSV	signalId
NMEA-Standard-RMC	navStatus

2.5.3 NMEA latitude and longitude format

According to the NMEA standard, latitude and longitude are output in the format *degrees, minutes and (decimal) fractions of minutes*. To convert to *degrees and fractions of degrees*, or *degrees, minutes, seconds and fractions of seconds*, the *minutes* and *fractional minutes* parts need to be converted. For example:

Format	Latitude	Longitude
Receiver output	\$GNRMC,014230.00,A,4722.80340,N,00831.68218,E,0.000,,120477,,,A,V*14	
(d)ddmm.mmmmm	4722.80340 North	00831.68218 East
Degrees and minutes	47 degrees, 22.80340 minutes	8 degrees, 31.68218 minutes
Degrees	47.38005667 degrees	8.52803633 degrees
Degrees, minutes and seconds	47 degrees, 22 minutes, 48.2040 seconds	8 degrees, 31 minutes, 40.9308 seconds

2.5.4 NMEA GNSS, satellite, and signal numbering

See [GNSS, satellite, and signal identifiers](#) for details on how GNSS, satellites and signals are numbered in the NMEA protocol.

NMEA defines satellite numbering systems for some, but not all GNSS. The exact behavior depends on the configured NMEA protocol version and ("extended" or "strict") mode. See [NMEA protocol configuration](#) for details.

2.5.5 NMEA position fix flags

This section shows how u-blox positioning receivers implement the NMEA protocol and the conditions determining how flags are set.

The following flags are used in NMEA 4.10 and later.

NMEA Message	GLL, RMC	GGA	GLL, VTG	RMC, GNS
Field	status ⁵	quality ⁶	posMode ⁷	posMode ⁷
No position fix (at power-up, after losing satellite lock)	V	0	N	N

⁵ Possible *status* values: V = data invalid, A = data valid

⁶ Possible values for *quality*: 0 = No fix, 1 = autonomous GNSS fix, 2 = differential GNSS fix, 4 = RTK fixed, 5 = RTK float, 6 = estimated/dead reckoning fix

⁷ Possible values for *posMode*: N = No fix, E = estimated/dead reckoning fix, A = autonomous GNSS fix, D = differential GNSS fix, F = RTK float, R = RTK fixed. In NMEA GNS, u-blox uses a non-standard implementation where same single status is reported for all enabled and not filtered out constellations.

NMEA Message	GLL, RMC	GGA	GLL, VTG	RMC, GNS
Field	status ⁵	quality ⁶	posMode ⁷	posMode ⁷
GNSS fix, but user limits exceeded	V	0	N	N
Dead reckoning fix, but user limits exceeded	V	6	E	E
Dead reckoning fix	A	6	E	E
RTK float	A	5	D	F
RTK fixed	A	4	D	R
2D GNSS fix	A	1 / 2	A / D	A / D
3D GNSS fix	A	1 / 2	A / D	A / D
Combined GNSS/dead reckoning fix	A	1 / 2	A / D	A / D

In high precision GNSS (HPG) products it is recommended to select NMEA version 4.10 or above. Earlier versions do not support the float RTK (F) and real time kinematic (R) mode indicator flags in all messages.

The following flags are used in NMEA 2.3 - 4.0.

NMEA Message	GLL, RMC	GGA	GSA	GLL, VTG, RMC, GNS
Field	status ⁸	quality ⁹	navMode ¹⁰	posMode ¹¹
No position fix (at power-up, after losing satellite lock)	V	0	1	N
GNSS fix, but user limits exceeded	V	0	1	N
Dead reckoning fix, but user limits exceeded	V	6	2	E
Dead reckoning fix	A	6	2	E
2D GNSS fix	A	1 / 2	2	A / D
3D GNSS fix	A	1 / 2	3	A / D
Combined GNSS/dead reckoning fix	A	1 / 2	3	A / D

The flags in NMEA 2.1 and earlier are the same as NMEA 2.3 but with the following differences:

- The *posMode* field is not output for GLL, RMC and VTG messages (each message has one field less).
- The GGA *quality* field is set to 1 (instead of 6) for both types of dead reckoning fix.

2.5.6 NMEA output of invalid or unknown data

By default the receiver will not output invalid data. In such cases, it will output empty fields. See [NMEA protocol configuration](#) for options to adjust this behavior.

A valid position fix is reported as follows:

```
$GPGLL,4717.11634,N,00833.91297,E,124923.00,A,A*6E
```

An invalid position fix (but valid time) is reported as follows:

```
$GPGLL,,,,,124924.00,V,N*42
```

⁸ Possible values for *status*: V = data invalid, A = data valid

⁹ Possible values for *quality*: 0 = no fix, 1 = autonomous GNSS fix, 2 = differential GNSS fix, 4 = RTK fixed, 5 = RTK float, 6 = estimated/dead reckoning fix

¹⁰ Possible values for *navMode*: 1 = No fix, 2 = 2D fix, 3 = 3D fix

¹¹ Possible values for *posMode*: N = No fix, E = estimated/dead reckoning fix, A = autonomous GNSS fix, D = differential GNSS fix. In NMEA GNS, u-blox uses a non-standard implementation where same single status is reported for all enabled and not filtered out constellations.

If the time is unknown (e.g. during a cold start):

\$GPGLL,,,,,,V,N*64



Unlike the NMEA standard behavior to invalid data, dead reckoning products always report a position. It is marked as invalid (V) when the user limits are exceeded or valid (A) if the user limits are met.

2.6 NMEA messages overview

Message	Class/ID	Description (Type)
NMEA-Standard – Standard NMEA messages		
NMEA-Standard-DTM	0xf0 0x0a	• Datum reference (Output)
NMEA-Standard-GAQ	0xf0 0x45	• Poll a standard message (Talker ID GA) (Poll request)
NMEA-Standard-GBQ	0xf0 0x44	• Poll a standard message (Talker ID GB) (Poll request)
NMEA-Standard-GBS	0xf0 0x09	• GNSS satellite fault detection (Output)
NMEA-Standard-GGA	0xf0 0x00	• Global positioning system fix data (Output)
NMEA-Standard-GLL	0xf0 0x01	• Latitude and longitude, with time of position fix and status (Output)
NMEA-Standard-GLQ	0xf0 0x43	• Poll a standard message (Talker ID GL) (Poll request)
NMEA-Standard-GNQ	0xf0 0x42	• Poll a standard message (Talker ID GN) (Poll request)
NMEA-Standard-GNS	0xf0 0x0d	• GNSS fix data (Output)
NMEA-Standard-GPQ	0xf0 0x40	• Poll a standard message (Talker ID GP) (Poll request)
NMEA-Standard-GQQ	0xf0 0x47	• Poll a standard message (Talker ID GQ) (Poll request)
NMEA-Standard-GRS	0xf0 0x06	• GNSS range residuals (Output)
NMEA-Standard-GSA	0xf0 0x02	• GNSS DOP and active satellites (Output)
NMEA-Standard-GST	0xf0 0x07	• GNSS pseudorange error statistics (Output)
NMEA-Standard-GSV	0xf0 0x03	• GNSS satellites in view (Output)
NMEA-Standard-RLM	0xf0 0x0b	• Return link message (RLM) (Output)
NMEA-Standard-RMC	0xf0 0x04	• Recommended minimum data (Output)
NMEA-Standard-TXT	0xf0 0x41	• Text transmission (Output)
NMEA-Standard-VLW	0xf0 0x0f	• Dual ground/water distance (Output)
NMEA-Standard-VTG	0xf0 0x05	• Course over ground and ground speed (Output)
NMEA-Standard-ZDA	0xf0 0x08	• Time and date (Output)
NMEA-PUBX – u-blox proprietary NMEA messages		
NMEA-PUBX-CONFIG	0xf1 0x41	• Set protocols and baud rate (Set)
NMEA-PUBX-POSITION	0xf1 0x00	• Poll a PUBX,00 message (Poll request) • Lat/Long position data (Output)
NMEA-PUBX-RATE	0xf1 0x40	• Set NMEA message output rate (Set)
NMEA-PUBX-SVSTATUS	0xf1 0x03	• Poll a PUBX,03 message (Poll request) • Satellite status (Output)
NMEA-PUBX-TIME	0xf1 0x04	• Poll a PUBX,04 message (Poll request) • Time of day and clock information (Output)

2.7 Standard messages

Standard NMEA messages as defined by the NMEA 0183 standard. See [NMEA protocol](#) for details.

2.7.1 DTM

2.7.1.1 Datum reference

Message	NMEA-Standard-DTM				
	Datum reference				
Type	Output				
Comment	This message gives the difference between the current datum and the reference datum. The current datum is set to WGS84 by default. The reference datum cannot be changed and is always set to WGS84.				
Information	Class/ID: 0xF0 0x0a		Number of fields: 11		
Structure	\$xxDTM, datum, subDatum, lat, NS, lon, EW, alt, refDatum*cs\r\n				
Examples	\$GPDTM, W84, , 0.0, N, 0.0, E, 0.0, W84*6F\r\n \$GPDTM, 999, , 0.08, N, 0.07, E, -47.7, W84*1C\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	xxDTM	string	-	\$GPDTM	DTM Message ID (xx = current Talker ID, see NMEA Talker IDs table)
1	datum	string	-	W84	Local datum code: W84 = WGS84, P90 = PZ90, 999 = user-defined
2	subDatum	string	-	-	A null field (or a string describing the currently selected datum for protocol versions less than 14.00)
3	lat	numeric	min	0.08	Offset in Latitude
4	NS	character	-	S	North/South indicator
5	lon	numeric	min	0.07	Offset in Longitude
6	EW	character	-	E	East/West indicator
7	alt	numeric	m	-2.8	Offset in altitude
8	refDatum	string	-	W84	Reference datum code: W84 (WGS 84, fixed field)
9	cs	hexadecimal	-	*67	Checksum
10	CRLF	character	-	-	Carriage return and line feed

2.7.2 GAQ

2.7.2.1 Poll a standard message (Talker ID GA)

Message		NMEA-Standard-GAQ			
		Poll a standard message (Talker ID GA)			
Type	Poll request				
Comment	Polls a standard NMEA message if the current Talker ID is GA.				
Information	Class/ID: 0xf0 0x45		Number of fields: 4		
Structure	\$xxGAQ, msgId*cs\r\n				
Example	\$EIGAQ, RMC*2B\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	xxGAQ	string	-	\$EIGAQ	GAQ Message ID (xx = Talker ID of the device requesting the poll)
1	msgId	string	-	RMC	Message ID of the message to be polled
2	cs	hexadecimal	-	*2B	Checksum

3	CRLF	character	-	-	Carriage return and line feed
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2.7.3 GBQ

2.7.3.1 Poll a standard message (Talker ID GB)

Message	NMEA-Standard-GBQ Poll a standard message (Talker ID GB)				
Type	Poll request				
Comment	Polls a standard NMEA message if the current Talker ID is GB				
Information	Class/ID: 0xf0 0x44		Number of fields: 4		
Structure	\$xxGBQ,msgId*cs\r\n				
Example	\$EIGBQ,RMC*28\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	xxGBQ	string	-	\$EIGBQ	GBQ Message ID (xx = Talker ID of the device requesting the poll)
1	msgId	string	-	RMC	Message ID of the message to be polled
2	cs	hexadecimal	-	*28	Checksum
3	CRLF	character	-	-	Carriage return and line feed

2.7.4 GBS

2.7.4.1 GNSS satellite fault detection

Message		NMEA-Standard-GBS			
		GNSS satellite fault detection			
Type	Output				
Comment	This message outputs the results of the Receiver Autonomous Integrity Monitoring Algorithm (RAIM). - The fields errLat, errLon and errAlt output the standard deviation of the position calculation, using all satellites that pass the RAIM test successfully. - The fields errLat, errLon and errAlt are only output if the RAIM process passed successfully (i.e. no or successful edits happened). These fields are never output if 4 or fewer satellites are used for the navigation calculation (because, in such cases, integrity cannot be determined by the receiver autonomously). - The fields prob, bias and stdev are only output if at least one satellite failed in the RAIM test. If more than one satellites fail the RAIM test, only the information for the worst satellite is output in this message.				
Information	Class/ID: 0xf0 0x09		Number of fields: 13		
Structure	\$xxGBS,time,errLat,errLon,errAlt,svid,prob,bias,stddev,systemId,signalId*cs\r\n				
Examples	\$GPGBS,235503.00,1.6,1.4,3.2,,,,,*40\r\n\$GPGBS,235458.00,1.4,1.3,3.1,03,, -21.4,3.8,1,0*5B\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	xxGBS	string	-	\$GPGBS	GBS Message ID (xx = current Talker ID, see NMEA Talker IDs table)
1	time	hhmmss.ss	-	235503.00	UTC time to which this RAIM sentence belongs. See section UTC representation in the integration manual for details.
2	errLat	numeric	m	1.6	Expected error in latitude
3	errLon	numeric	m	1.4	Expected error in longitude

4	errAlt	numeric	m	3.2	Expected error in altitude
5	svid	numeric	-	03	Satellite ID of most likely failed satellite
6	prob	numeric	-	-	Probability of missed detection: null (not supported, fixed field)
7	bias	numeric	m	-21.4	Estimated bias of most likely failed satellite (a priori residual)
8	stddev	numeric	m	3.8	Standard deviation of estimated bias
9	systemId	hexadecimal	-	1	NMEA-defined GNSS system ID, see Signal Identifiers table (only available in NMEA 4.10 and later)
10	signalId	hexadecimal	-	-	NMEA-defined GNSS signal ID, see Signal Identifiers table (only available in NMEA 4.10 and later)
11	cs	hexadecimal	-	*5B	Checksum
12	CRLF	character	-	-	Carriage return and line feed

2.7.5 GGA

2.7.5.1 Global positioning system fix data

Message	NMEA-Standard-GGA Global positioning system fix data				
Type	Output				
Comment	Time and position, together with GPS fixing-related data (number of satellites in use, and the resulting HDOP, age of differential data if in use, etc.). 📘 The output of this message is dependent on the currently selected datum (default: WGS84). The NMEA specification indicates that the GGA message is GPS-specific. However, when the receiver is configured for multi-GNSS, the GGA message contents will be generated from the multi-GNSS solution. For multi-GNSS use, it is recommended that the NMEA-GNS message is used instead.				
Information	Class/ID: 0xf0 0x00		Number of fields: 17		
Structure	\$xxGGA,time,lat,NS,lon,EW,quality,numSV,HDOP,alt,altUnit,sep,sepUnit,diffAge,diffStation*cs\r\n				
Example	\$GPGGA,092725.00,4717.11399,N,00833.91590,E,1,08,1.01,499.6,M,48.0,M,,*5B\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	xxGGA	string	-	\$GPGGA	GGA Message ID (xx = current Talker ID, see NMEA Talker IDs table)
1	time	hhmmss.ss	-	092725.00	UTC time. See section UTC representation in the integration manual for details.
2	lat	ddmm.mmmmm	-	4717.11399	Latitude (degrees and minutes), see format description
3	NS	character	-	N	North/South indicator
4	lon	dddmm.mmmmm	-	00833.91590	Longitude (degrees and minutes), see format description
5	EW	character	-	E	East/West indicator
6	quality	digit	-	1	Quality indicator for position fix, see position fix flags description
7	numSV	numeric	-	08	Number of satellites used. If compatibility mode is enabled, the range is limited to 12 satellites (see configuration item CFG-NMEA-COMPAT).
8	HDOP	numeric	-	1.01	Horizontal Dilution of Precision
9	alt	numeric	m	499.6	Altitude above mean sea level

10	altUnit	character	-	M	Altitude units: M (meters, fixed field)
11	sep	numeric	m	48.0	Geoid separation: difference between ellipsoid and mean sea level
12	sepUnit	character	-	M	Geoid separation units: M (meters, fixed field)
13	diffAge	numeric	s	-	Age of differential corrections (null when DGPS is not used)
14	diffStation	numeric	-	-	ID of station providing differential corrections (null when DGPS is not used)
15	cs	hexadecimal	-	*5B	Checksum
16	CRLF	character	-	-	Carriage return and line feed

2.7.6 GLL

2.7.6.1 Latitude and longitude, with time of position fix and status

Message	NMEA-Standard-GLL				
	Latitude and longitude, with time of position fix and status				
Type	Output				
Comment	 The output of this message is dependent on the currently selected datum (default: WGS84)				
Information	Class/ID: 0xf0 0x01		Number of fields: 10		
Structure	\$xxGLL, lat , NS , lon , EW , time , status , posMode * cs \r \n				
Example	\$GPGLL, 4717.11364, N, 00833.91565, E, 092321.00, A, A * 60 \r \n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	xxGLL	string	-	\$GPGLL	GLL Message ID (xx = current Talker ID, see NMEA Talker IDs table)
1	lat	ddmm. mmmmmm	-	4717.11364	Latitude (degrees and minutes), see format description
2	NS	character	-	N	North/South indicator
3	lon	dddmm. mmmmmm	-	00833.91565	Longitude (degrees and minutes), see format description
4	EW	character	-	E	East/West indicator
5	time	hhmmss.ss	-	092321.00	UTC time. See section UTC representation in the integration manual for details.
6	status	character	-	A	Data validity status, see position fix flags description
7	posMode	character	-	A	Positioning mode, see position fix flags description (only available in NMEA 2.3 and later)
8	cs	hexadecimal	-	*60	Checksum
9	CRLF	character	-	-	Carriage return and line feed

2.7.7 GLQ

2.7.7.1 Poll a standard message (Talker ID GL)

Message	NMEA-Standard-GLQ	
	Poll a standard message (Talker ID GL)	
Type	Poll request	
Comment	Polls a standard NMEA message if the current Talker ID is GL	
Information	Class/ID: 0xF0 0x43	Number of fields: 4

Structure \$xxGLQ,msgId*cs\r\n

Example \$EIGLQ,RMC*3A\r\n

Payload:

Field	Name	Format	Unit	Example	Description
0	xxGLQ	string	-	\$EIGLQ	GLQ Message ID (xx = Talker ID of the device requesting the poll)
1	msgId	string	-	RMC	Message ID of the message to be polled
2	cs	hexadecimal	-	*3A	Checksum
3	CRLF	character	-	-	Carriage return and line feed

2.7.8 GNQ

2.7.8.1 Poll a standard message (Talker ID GN)

Message	NMEA-Standard-GNQ Poll a standard message (Talker ID GN)				
Type	Poll request				
Comment	Polls a standard NMEA message if the current Talker ID is GN				
Information	Class/ID: 0xf0 0x42		Number of fields: 4		
Structure	\$xxGNQ,msgId*cs\r\n				
Example	\$EIGNQ,RMC*3A\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	xxGNQ	string	-	\$EIGNQ	GNQ Message ID (xx = Talker ID of the device requesting the poll)
1	msgId	string	-	RMC	Message ID of the message to be polled
2	cs	hexadecimal	-	*3A	Checksum
3	CRLF	character	-	-	Carriage return and line feed

2.7.9 GNS

2.7.9.1 GNSS fix data

Message	NMEA-Standard-GNS GNSS fix data				
Type	Output				
Comment	Time and position, together with GNSS fixing-related data (number of satellites in use, and the resulting HDOP, age of differential data if in use, etc.).  The output of this message is dependent on the currently selected datum (default: WGS84)				
Information	Class/ID: 0xf0 0x0d		Number of fields: 16		
Structure	\$xxGNS,time,lat,NS,lon,EW,posMode,numSV,HDOP,alt,sep,diffAge,diffStation,navStatus*c s\r\n				
Examples	\$GNGNS,103600.01,5114.51176,N,00012.29380,W,ANNN,07,1.18,111.5,45.6,,,V*00\r\n\$GNGNS,122310.2,3722.425671,N,12258.856215,W,DAAA,14,0.9,1005.543,6.5,,,V*0E\r\n\$GPGNS,122310.2,,,,,07,,,,,5.2,23,V*02\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	xxGNS	string	-	\$GPGNS	GNS Message ID (xx = current Talker ID, see NMEA Talker IDs table)

1	time	hhmmss.ss	-	091547.00	UTC time. See section UTC representation in the integration manual for details.
2	lat	ddmm. mmmm	-	5114.50897	Latitude (degrees and minutes), see format description
3	NS	character	-	N	North/South indicator
4	lon	dddmm. mmmm	-	00012.28663	Longitude (degrees and minutes), see format description
5	EW	character	-	E	East/West indicator
6	posMode	character	-	AAAA	Positioning mode, see position fix flags description . The first four characters indicate the status for GPS, GLONASS, Galileo and BeiDou. Note that the NMEA GNS message only reports a single status. It indicates the status for all enabled constellations that have not been filtered out. To obtain a more detailed status report, refer to the status provided in the UBX messages.
7	numSV	numeric	-	10	Number of satellites used (range: 0-99)
8	HDOP	numeric	-	0.83	Horizontal Dilution of Precision
9	alt	numeric	m	111.1	Altitude above mean sea level
10	sep	numeric	m	45.6	Geoid separation: difference between ellipsoid and mean sea level
11	diffAge	numeric	s	-	Age of differential corrections (null when DGPS is not used)
12	diffStation	numeric	-	-	ID of station providing differential corrections (null when DGPS is not used)
13	navStatus	character	-	V	Navigational status indicator: V (Equipment is not providing navigational status information, fixed field, only available in NMEA 4.10 and later)
14	cs	hexadecimal	-	*71	Checksum
15	CRLF	character	-	-	Carriage return and line feed

2.7.10 GPQ

2.7.10.1 Poll a standard message (Talker ID GP)

Message	NMEA-Standard-GPQ				
	Poll a standard message (Talker ID GP)				
Type	Poll request				
Comment	Polls a standard NMEA message if the current Talker ID is GP				
Information	Class/ID: 0xf0 0x40		Number of fields: 4		
Structure	\$xxGPQ,msgId*cs\r\n				
Example	\$EIGPQ,RMC*3A\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	xxGPQ	string	-	\$EIGPQ	GPQ Message ID (xx = Talker ID of the device requesting the poll)
1	msgId	string	-	RMC	Message ID of the message to be polled
2	cs	hexadecimal	-	*3A	Checksum

3	CRLF	character	-	-	Carriage return and line feed
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2.7.11 GQQ

2.7.11.1 Poll a standard message (Talker ID GQ)

Message	NMEA-Standard-GQQ Poll a standard message (Talker ID GQ)				
Type	Poll request				
Comment	Polls a standard NMEA message if the current Talker ID is GQ				
Information	Class/ID: 0xf0 0x47		Number of fields: 4		
Structure	\$xxGQQ,msgId*cs\r\n				
Example	\$EIGQQ,RMC*3A\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	xxGQQ	string	-	\$EIGQQ	GQQ Message ID (xx = Talker ID of the device requesting the poll)
1	msgId	string	-	RMC	Message ID of the message to be polled
2	cs	hexadecimal	-	*3A	Checksum
3	CRLF	character	-	-	Carriage return and line feed

2.7.12 GRS

2.7.12.1 GNSS range residuals

Message	NMEA-Standard-GRS GNSS range residuals				
Type	Output				
Comment	If less than 12 SVs are available, the remaining fields are output empty. If more than 12 SVs are used, only the residuals of the first 12 SVs are output, in order to remain consistent with the NMEA standard. In a multi-GNSS system this message will be output multiple times, once for each GNSS. 🔗 This message relates to associated GGA and GSA messages.				
Information	Class/ID: 0xf0 0x06		Number of fields: 19		
Structure	\$xxGRS,time,mode{,residual},systemId,signalId*cs\r\n				
Examples	\$GNGRS,104148.00,1,2.6,2.2,-1.6,-1.1,-1.7,-1.5,5.8,1.7,,,,,1,1*52\r\n \$GNGRS,104148.00,1,,0.0,2.5,0.0,,2.8,,,,,,1,5*52\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	xxGRS	string	-	\$GPGRS	GRS Message ID (xx = current Talker ID, see NMEA Talker IDs table)
1	time	hhmmss.ss	-	082632.00	UTC time of associated position fix. See section UTC representation in the integration manual for details.
2	mode	digit	-	1	Computation method used: 1 = Residuals were recomputed after the GGA position was computed (fixed)
Start of repeated group (12 times)					
3 + n	residual	numeric	m	0.54	Range residuals for SVs used in navigation. The SV order matches the order from the GSA sentence
End of repeated group (12 times)					

15	systemId	hexadecimal -	1	NMEA-defined GNSS system ID, see Signal Identifiers table (only available in NMEA 4.10 and later)
16	signalId	hexadecimal -	-	NMEA-defined GNSS signal ID, see Signal Identifiers table (only available in NMEA 4.10 and later)
17	cs	hexadecimal -	*70	Checksum
18	CRLF	character -	-	Carriage return and line feed

2.7.13 GSA

2.7.13.1 GNSS DOP and active satellites

Message	NMEA-Standard-GSA GNSS DOP and active satellites				
Type	Output				
Comment	The GNSS receiver operating mode, satellites used for navigation, and DOP values. - If less than 12 SVs are used for navigation, the remaining fields are left empty. If more than 12 SVs are used for navigation, only the IDs of the first 12 are output. - The SV numbers (fields 'svid') are in the range of 1 to 32 for GPS satellites, and 33 to 64 for SBAS satellites (33 = SBAS PRN 120, 34 = SBAS PRN 121, and so on) In a multi-GNSS system this message will be output multiple times, once for each GNSS.				
Information	Class/ID: 0xf0 0x02		Number of fields: 21		
Structure	\$xxGSA,opMode,navMode{,svid},PDOP,HDOP,VDOP,systemId*cs\r\n				
Example	\$GPGSA,A,3,23,29,07,08,09,18,26,28,,,,,1.94,1.18,1.54,1*0D\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	xxGSA	string	-	\$GPGSA	GSA Message ID (xx = current Talker ID, see NMEA Talker IDs table)
1	opMode	character	-	A	Operation mode: - M = Manually set to operate in 2D or 3D mode - A = Automatically switching between 2D or 3D mode
2	navMode	digit	-	3	Navigation mode, see position fix flags description
Start of repeated group (12 times)					
3 + n	svid	numeric	-	29	Satellite number
End of repeated group (12 times)					
15	PDOP	numeric	-	1.94	Position dilution of precision
16	HDOP	numeric	-	1.18	Horizontal dilution of precision
17	VDOP	numeric	-	1.54	Vertical dilution of precision
18	systemId	hexadecimal	-	1	NMEA-defined GNSS system ID, see Signal Identifiers table (only available in NMEA 4.10 and later)
19	cs	hexadecimal	-	*0D	Checksum
20	CRLF	character	-	-	Carriage return and line feed

2.7.14 GST

2.7.14.1 GNSS pseudorange error statistics

Message	NMEA-Standard-GST GNSS pseudorange error statistics			
Type	Output			

Comment	This message reports statistical information on the quality of the position solution.				
Information	Class/ID: 0xf0 0x07		Number of fields: 11		
Structure	\$xxGST,time,rangeRms,stdMajor,stdMinor,orient,stdLat,stdLong,stdAlt*cs\r\n				
Example	\$GPGST,082356.00,1.8,,,1.7,1.3,2.2*7E\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	xxGST	string	-	\$GPGST	GST Message ID (xx = current Talker ID, see NMEA Talker IDs table)
1	time	hhmmss.ss	-	082356.00	UTC time of associated position fix. See section UTC representation in the integration manual for details.
2	rangeRms	numeric	m	1.8	RMS value of the standard deviation of the ranges
3	stdMajor	numeric	m	-	Standard deviation of semi-major axis
4	stdMinor	numeric	m	-	Standard deviation of semi-minor axis
5	orient	numeric	deg	-	Orientation of semi-major axis
6	stdLat	numeric	m	1.7	Standard deviation of latitude error
7	stdLong	numeric	m	1.3	Standard deviation of longitude error
8	stdAlt	numeric	m	2.2	Standard deviation of altitude error
9	cs	hexadecimal	-	*7E	Checksum
10	CRLF	character	-	-	Carriage return and line feed

2.7.15 GSV

2.7.15.1 GNSS satellites in view

Message	NMEA-Standard-GSV GNSS satellites in view				
Type	Output				
Comment	The number of satellites in view, together with each SV ID, elevation azimuth, and signal strength (C/No) value. Only four satellite details are transmitted in one message. In a multi-GNSS system, sets of GSV messages will be output multiple times, one set for each GNSS. The messages are grouped by the signal ID and separate messages are output for each signal ID. (supported for protocol versions 27.12 and later)				
Information	Class/ID: 0xf0 0x03		Number of fields: 7 + [1..4]·4		
Structure	\$xxGSV,numMsg,msgNum,numSV{,svid,elv,az,cno},signalId*cs\r\n				
Examples	<pre>\$GPGSV,3,1,09,09,,,17,10,,,40,12,,,49,13,,,35,1*6F\r\n \$GPGSV,3,2,09,15,,,44,17,,,45,19,,,44,24,,,50,1*64\r\n \$GPGSV,3,3,09,25,,,40,1*6E\r\n \$GPGSV,1,1,03,12,,,42,24,,,47,32,,,37,5*66\r\n \$GPGSV,1,1,01,03,05,218,,0*59\r\n \$GAGSV,1,1,00,2*76\r\n</pre>				
Payload:					
Field	Name	Format	Unit	Example	Description
0	xxGSV	string	-	\$GPGSV	GSV Message ID (xx = GSV Talker ID, see NMEA Talker IDs table). Talker ID GN shall not be used.
1	numMsg	digit	-	3	Number of messages, total number of GSV messages being output (range: 1-9)
2	msgNum	digit	-	1	Number of this message (range: 1-numMsg)
3	numSV	numeric	-	10	Number of known satellites in view regarding both the talker ID and the signalId

Start of repeated group (1...4 times)

4 + n·4	svid	numeric	-	23	Satellite ID
5 + n·4	elv	numeric	deg	38	Elevation (<= 90)
6 + n·4	az	numeric	deg	230	Azimuth (range: 0-359)
7 + n·4	cno	numeric	dBHz	44	Signal strength (C/N0, range: 0-99), null when not tracking

End of repeated group (1...4 times)

4 + N·4	signalId	hexadecimal	-	-	NMEA-defined GNSS signal ID, see Signal Identifiers table (only available in NMEA 4.10 and later)
5 + N·4	cs	hexadecimal	-	*7F	Checksum
6 + N·4	CRLF	character	-	-	Carriage return and line feed

2.7.16 RLM

2.7.16.1 Return link message (RLM)

Message	NMEA-Standard-RLM Return link message (RLM)				
Type	Output				
Comment	The RLM sentence is used to transfer a Return link message from a Cospas-Sarsat recognized Return link service provider (RLSP). The RLM sentence supports communications to an emitting beacon once a distress alert has been detected, located and confirmed. The communications may include acknowledgement of the alert to the emitting beacon as well as optional text messages, and may also include remote beacon configuration and testing.				
Information	Class/ID: 0xf0 0x0b		Number of fields: 7		
Structure	\$xxRLM, beacon, time, code, body*cs\r\n				
Examples	\$GARLM, 00000078A9FBAD5, 083559.00, 3, C45B*57\r\n \$GARLM, F7129D41BC6A78C, 034433.02, 3, B63CA732AFD419D2*57\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	xxRLM	string	-	\$GARLM	RLM message ID (xx = current Talker ID, see NMEA Talker IDs table)
1	beacon	hexadecimal	-	00000078A9FBAD5	Beacon ID, identifies beacon intended to receive this message (fixed length 15 hexadecimal character field)
2	time	hhmmss.ss	-	083559.00	Time of reception field to indicate RLM timestamp in UTC. See section UTC representation in the integration manual for details.
3	code	character	-	3	Message code field to identify type of RLM Message Service: 0 = Reserved for future RLM services 1 = Acknowledgement service RLM 2 = Command service RLM 3 = Message service RLM 4-E = Reserved for future RLM services - F = Test service RLM (currently used only by the Galileo program)
4	body	hexadecimal	-	C45B	Message body encapsulates the data parameters provided by the RLSP into hexadecimal format.
5	cs	hexadecimal	-	*57	Checksum

6	CRLF	character	-	-	Carriage return and line feed
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2.7.17 RMC

2.7.17.1 Recommended minimum data

Message	NMEA-Standard-RMC Recommended minimum data				
Type	Output				
Comment	The recommended minimum sentence defined by NMEA for GNSS system data. 🔗 The output of this message is dependent on the currently selected datum (default: WGS84)				
Information	Class/ID: 0xF0 0x04		Number of fields: 16		
Structure	\$xxRMC,time,status,lat,NS,lon,EW,spd,cog,date,mv,mvEW,posMode,navStatus*cs\r\n				
Example	\$GPRMC,083559.00,A,4717.11437,N,00833.91522,E,0.004,77.52,091202,,A,V*57\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	xxRMC	string	-	\$GPRMC	RMC Message ID (xx = current Talker ID, see NMEA Talker IDs table)
1	time	hhmmss.ss	-	083559.00	UTC time. See section UTC representation in the integration manual for details.
2	status	character	-	A	Data validity status, see position fix flags description
3	lat	ddmm.mmmmm	-	4717.11437	Latitude (degrees and minutes), see format description
4	NS	character	-	N	North/South indicator
5	lon	dddmm.mmmmm	-	00833.91522	Longitude (degrees and minutes), see format description
6	EW	character	-	E	East/West indicator
7	spd	numeric	knots	0.004	Speed over ground
8	cog	numeric	deg	77.52	Course over ground
9	date	ddmmyy	-	091202	Date in day, month, year format. See section UTC representation in the integration manual for details.
10	mv	numeric	deg	-	Magnetic variation value
11	mvEW	character	-	-	Magnetic variation E/W indicator
12	posMode	character	-	A	Mode Indicator, see position fix flags description (only available in NMEA 2.3 and later)
13	navStatus	character	-	V	Navigational status indicator: V (Equipment is not providing navigational status information, fixed field, only available in NMEA 4.10 and later)
14	cs	hexadecimal	-	*57	Checksum
15	CRLF	character	-	-	Carriage return and line feed

2.7.18 TXT

2.7.18.1 Text transmission

Message	NMEA-Standard-TXT Text transmission
Type	Output
Comment	

Information	Class/ID: 0xF0 0x41		Number of fields: 7		
Structure	\$xxTXT, numMsg, msgNum, msgType, text*cs\r\n				
Examples	\$GPTXT, 01, 01, 02, u-blox ag - www.u-blox.com*50\r\n				
	\$GPTXT, 01, 01, 02, ANTARIS ATR0620 HW 00000040*67\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	xxTXT	string	-	\$GPTXT	TXT Message ID (xx = current Talker ID, see NMEA Talker IDs table)
1	numMsg	numeric	-	01	Total number of messages in this transmission (range: 1-99)
2	msgNum	numeric	-	01	Message number in this transmission (range: 1-numMsg)
3	msgType	numeric	-	02	Text identifier (u-blox receivers specify the type of the message with this number): • 00 = Error • 01 = Warning • 02 = Notice • 07 = User
4	text	string	-	www.u-blox.com	Any ASCII text
5	cs	hexadecimal	-	*67	Checksum
6	CRLF	character	-	-	Carriage return and line feed

2.7.19 VLW

2.7.19.1 Dual ground/water distance

Message	NMEA-Standard-VLW Dual ground/water distance				
Type	Output				
Comment	The distance traveled, relative to the water and over the ground. This message relates to the odometer feature detailed in the integration manual.				
Information	Class/ID: 0xF0 0x0F		Number of fields: 11		
Structure	\$xxVLW,twd,twdUnit,wd,wdUnit,tgd,tgdUnit,gd,gdUnit*cs\r\n				
Example	\$GPVLW,,N,,N,15.8,N,1.2,N*06\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	xxVLW	string	-	\$GPVLW	VLW Message ID (xx = current Talker ID, see NMEA Talker IDs table)
1	twd	numeric	nmi	-	Total cumulative water distance: null (fixed field)
2	twdUnit	character	-	N	Total cumulative water distance units: N (nautical miles, fixed field)
3	wd	numeric	nmi	-	Water distance since reset: null (fixed field)
4	wdUnit	character	-	N	Water distance since reset units: N (nautical miles, fixed field)
5	tgd	numeric	nmi	15.8	Total cumulative ground distance (only available in NMEA 4.00 and later)
6	tgdUnit	character	-	N	Total cumulative ground distance units: N (nautical miles, fixed field, only available in NMEA 4.00 and later)

7	gd	numeric	nmi	1.2	Ground distance since reset (only available in NMEA 4.00 and later)
8	gdUnit	character	-	N	Ground distance since reset units: N (nautical miles, fixed field, only available in NMEA 4.00 and later)
9	cs	hexadecimal	-	*06	Checksum
10	CRLF	character	-	-	Carriage return and line feed

2.7.20 VTG

2.7.20.1 Course over ground and ground speed

Message	NMEA-Standard-VTG				
	Course over ground and ground speed				
Type	Output				
Comment	Velocity is given as course over ground (COG) and speed over ground (SOG).				
Information	Class/ID: 0xf0 0x05		Number of fields: 12		
Structure	\$xxVTG,cogt,cogtUnit,cogm,cogmUnit,sogn,sognUnit,sogk,sogkUnit,posMode*cs\r\n				
Example	\$GPVTG,77.52,T,,M,0.004,N,0.008,K,A*06\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	xxVTG	string	-	\$GPVTG	VTG Message ID (xx = current Talker ID, see NMEA Talker IDs table)
1	cogt	numeric	degrees	77.52	Course over ground (true)
2	cogtUnit	character	-	T	Course over ground units: T (degrees true, fixed field)
3	cogm	numeric	degrees	-	Course over ground (magnetic)
4	cogmUnit	character	-	M	Course over ground units: M (degrees magnetic, fixed field)
5	sogn	numeric	knots	0.004	Speed over ground
6	sognUnit	character	-	N	Speed over ground units: N (knots, fixed field)
7	sogk	numeric	km/h	0.008	Speed over ground
8	sogkUnit	character	-	K	Speed over ground units: K (kilometers per hour, fixed field)
9	posMode	character	-	A	Mode indicator, see position fix flags description (only available in NMEA 2.3 and later)
10	cs	hexadecimal	-	*06	Checksum
11	CRLF	character	-	-	Carriage return and line feed

2.7.21 ZDA

2.7.21.1 Time and date

Message	NMEA-Standard-ZDA
	Time and date
Type	Output
Comment	UTC, day, month, year and local time zone.
Information	Class/ID: 0xf0 0x08 Number of fields: 9
Structure	\$xxZDA,time,day,month,year,ltzh,ltzn*cs\r\n
Example	\$GPZDA,082710.00,16,09,2002,00,00*64\r\n

Payload:					
Field	Name	Format	Unit	Example	Description
0	xxZDA	string	-	\$GPZDA	ZDA Message ID (xx = current Talker ID, see NMEA Talker IDs table)
1	time	hhmmss.ss	-	082710.00	UTC Time. See section UTC representation in the integration manual for details.
2	day	dd	day	16	UTC day (range: 1-31)
3	month	mm	month	09	UTC month (range: 1-12)
4	year	yyyy	year	2002	UTC year
5	ltzh	xx	-	00	Local time zone hours (fixed field, always 00)
6	ltzn	zz	-	00	Local time zone minutes (fixed field, always 00)
7	cs	hexadecimal	-	*64	Checksum
8	CRLF	character	-	-	Carriage return and line feed

2.8 PUBX messages

Proprietary NMEA messages for u-blox positioning receivers. See also [NMEA proprietary messages](#).

2.8.1 CONFIG (PUBX,41)

2.8.1.1 Set protocols and baud rate

Message	NMEA-PUBX-CONFIG Set protocols and baud rate				
Type	Set				
Comment					
Information	Class/ID: 0xf1 0x41		Number of fields: 9		
Structure	\$PUBX,41,portId,inProto,outProto,baudrate,autobauding*cs\r\n				
Example	\$PUBX,41,1,0007,0003,19200,0*25\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	PUBX	string	-	\$PUBX	Message ID, UBX protocol header, proprietary sentence
1	msgId	numeric	-	41	Proprietary message identifier
2	portId	numeric	-	1	ID of communication port. See section Communication ports in the integration manual for details.
3	inProto	hexadecimal	-	0007	Input protocol mask. Bitmask, specifying which protocols(s) are allowed for input. See section Communication ports in the integration manual for details.
4	outProto	hexadecimal	-	0003	Output protocol mask. Bitmask, specifying which protocols(s) are allowed for input. See section Communication ports in the integration manual for details.
5	baudrate	numeric	bits/s	19200	Baud rate
6	autobauding	numeric	-	-	Autobauding: 1=enable, 0=disable (not supported on u-blox 5, set to 0)
7	cs	hexadecimal	-	*25	Checksum

8	CRLF	character	-	-	Carriage return and line feed
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2.8.2 POSITION (PUBX,00)

2.8.2.1 Poll a PUBX,00 message

Message	NMEA-PUBX-POSITION Poll a PUBX,00 message				
Type	Poll request				
Comment	A PUBX,00 message is polled by sending the PUBX,00 message without any data fields.				
Information	Class/ID: 0xf1 0x00		Number of fields: 4		
Structure	\$PUBX,00*33\r\n				
Example	\$PUBX,00*33\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	PUBX	string	-	\$PUBX	Message ID, UBX protocol header, proprietary sentence
1	msgId	numeric	-	00	Set to 00 to poll a PUBX,00 message
2	cs	hexadecimal	-	*33	Checksum
3	CRLF	character	-	-	Carriage return and line feed

2.8.2.2 Lat/Long position data

Message	NMEA-PUBX-POSITION				
	Lat/Long position data				
Type	Output				
Comment	This message contains position solution data. The datum selection may be changed using the config item CFG-NAVSPG-USE_USRDAT . 🔗 The output of this message is dependent on the currently selected datum (default: WGS84).				
Information	Class/ID: 0xf1 0x00		Number of fields: 23		
Structure	\$PUBX,00,time,lat,NS,long,EW,altRef,navStat,hAcc,vAcc,SOG,COG,vVel,diffAge,HDOP,VDOP,TDOP,numSvs,reserved,DR*cs\r\n				
Example	\$PUBX,00,081350.00,4717.113210,N,00833.915187,E,546.589,G3,2.1,2.0,0.007,77.52,0.007,0.92,1.19,0.77,9,0,0*5F\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	PUBX	string	-	\$PUBX	Message ID, UBX protocol header, proprietary sentence
1	msgId	numeric	-	00	Proprietary message identifier: 00
2	time	hhmmss.ss	-	081350.00	UTC time. See section UTC representation in the integration manual for details.
3	lat	ddmm.mmmmm	-	4717.113210	Latitude (degrees and minutes), see format description
4	NS	character	-	N	North/South Indicator
5	long	dddmm.mmmmm	-	00833.915187	Longitude (degrees and minutes), see format description
6	EW	character	-	E	East/West indicator
7	altRef	numeric	m	546.589	Altitude above user datum ellipsoid

8	navStat	string	-	G3	Navigation Status: - NF = No Fix - DR = Dead reckoning only solution - G2 = Stand alone 2D solution - G3 = Stand alone 3D solution - D2 = Differential 2D solution - D3 = Differential 3D solution - RK = Combined GPS + dead reckoning solution - TT = Time only solution
9	hAcc	numeric	m	2.1	Horizontal accuracy estimate
10	vAcc	numeric	m	2.0	Vertical accuracy estimate
11	SOG	numeric	km/h	0.007	Speed over ground
12	COG	numeric	deg	77.52	Course over ground
13	vVel	numeric	m/s	0.007	Vertical velocity (positive downwards)
14	diffAge	numeric	s	-	Age of differential corrections (blank when DGPS is not used)
15	HDOP	numeric	-	0.92	HDOP, Horizontal Dilution of Precision
16	VDOP	numeric	-	1.19	VDOP, Vertical Dilution of Precision
17	TDOP	numeric	-	0.77	TDOP, Time Dilution of Precision
18	numSvs	numeric	-	9	Number of satellites used in the navigation solution
19	reserved	numeric	-	-	Reserved, always set to 0
20	DR	numeric	-	-	DR only used
21	cs	hexadecimal	-	*5B	Checksum
22	CRLF	character	-	-	Carriage return and line feed

2.8.3 RATE (PUBX,40)

2.8.3.1 Set NMEA message output rate

Message	NMEA-PUBX-RATE Set NMEA message output rate				
Type	Set				
Comment	Set/Get message rate configuration (s) to/from the receiver. Send rate is relative to the event a message is registered on. For example, if the rate of a navigation message is set to 2, the message is sent every second navigation solution.				
Information	Class/ID: 0xf1 0x40		Number of fields: 11		
Structure	\$PUBX, 40, msgId, rddc, rus1, rus2, rusb, rspl, reserved*cs\r\n				
Example	\$PUBX, 40, GLL, 1, 0, 0, 0, 0, 0*5D\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	PUBX	string	-	\$PUBX	Message ID, UBX protocol header, proprietary sentence
1	ID	numeric	-	40	Proprietary message identifier
2	msgId	string	-	GLL	NMEA message identifier
3	rddc	numeric	cycles	1	output rate on DDC 0 disables that message from being output on this port 1 means that this message is output every epoch

4	rus1	numeric	cycles	1	output rate on USART 1 0 disables that message from being output on this port 1 means that this message is output every epoch
5	rus2	numeric	cycles	1	output rate on USART 2 0 disables that message from being output on this port 1 means that this message is output every epoch
6	rusb	numeric	cycles	1	output rate on USB 0 disables that message from being output on this port 1 means that this message is output every epoch
7	rspl	numeric	cycles	1	output rate on SPI 0 disables that message from being output on this port 1 means that this message is output every epoch
8	reserved	numeric	-	-	Reserved: always fill with 0
9	cs	hexadecimal	-	*5D	Checksum
10	CRLF	character	-	-	Carriage return and line feed

2.8.4 SVSTATUS (PUBX,03)

2.8.4.1 Poll a PUBX,03 message

Message	NMEA-PUBX-SVSTATUS Poll a PUBX,03 message				
Type	Poll request				
Comment	A PUBX,03 message is polled by sending the PUBX,03 message without any data fields.				
Information	Class/ID: 0xf1 0x03		Number of fields: 4		
Structure	\$PUBX,03*30\r\n				
Example	\$PUBX,03*30\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	PUBX	string	-	\$PUBX	Message ID, UBX protocol header, proprietary sentence
1	msgId	numeric	-	03	Set to 03 to poll a PUBX,03 message
2	cs	hexadecimal	-	*30	Checksum
3	CRLF	character	-	-	Carriage return and line feed

2.8.4.2 Satellite status

Message	NMEA-PUBX-SVSTATUS				
	Satellite status				
Type	Output				
Comment	The PUBX,03 message contains satellite status information.				
Information	Class/ID: 0xf1 0x03		Number of fields: 5 + n-6		
Structure	\$PUBX,03,n{,sv,s,az,el,cno,lck}*cs\r\n				
Example	\$PUBX,03,11,23,-,,45,010,29,-,,46,013,07,-,,42,015,08,U,067,31,42,025,10,U,195,33,46,026,18,U,326,08,39,026,17,-,,32,015,26,U,306,66,48,025,27,U,073,10,36,026,28,U,089,61,46,024,15,-,,39,014*0D\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description

0	PUBX	string	-	\$PUBX	Message ID, UBX protocol header, proprietary sentence
1	msgId	numeric	-	03	Proprietary message identifier: 03
2	n	numeric	-	11	Number of GNSS satellites tracked
<i>Start of repeated group (n times)</i>					
3 + n·6	sv	numeric	-	23	Satellite ID according to UBX svId mapping (see Satellite Numbering)
4 + n·6	s	character	-	-	Satellite status: - = Not used - U = Used in solution - e = Ephemeris available, but not used for navigation
5 + n·6	az	numeric	deg	-	Satellite azimuth (range: 0-359)
6 + n·6	el	numeric	deg	-	Satellite elevation (<= 90)
7 + n·6	cno	numeric	dBHz	45	Signal strength (C/N0, range 0-99), blank when not tracking
8 + n·6	lck	numeric	s	010	Satellite carrier lock time (range: 0-64) 0 = code lock only 64 = lock for 64 seconds or more
<i>End of repeated group (n times)</i>					
3 + n·6	cs	hexadecimal	-	*0D	Checksum
4 + n·6	CRLF	character	-	-	Carriage return and line feed

2.8.5 TIME (PUBX,04)

2.8.5.1 Poll a PUBX,04 message

Message		NMEA-PUBX-TIME			
		Poll a PUBX,04 message			
Type	Poll request				
Comment	A PUBX,04 message is polled by sending the PUBX,04 message without any data fields.				
Information	Class/ID: 0xf1 0x04		Number of fields: 4		
Structure	\$PUBX,04*37\r\n				
Example	\$PUBX,04*37\r\n				
Payload:					
Field	Name	Format	Unit	Example	Description
0	PUBX	string	-	\$PUBX	Message ID, UBX protocol header, proprietary sentence
1	msgId	numeric	-	04	Set to 04 to poll a PUBX,04 message
2	cs	hexadecimal	-	*37	Checksum
3	CRLF	character	-	-	Carriage return and line feed

2.8.5.2 Time of day and clock information

Message	NMEA-PUBX-TIME	
	Time of day and clock information	
Type	Output	
Comment		
Information	Class/ID: 0xf1 0x04	Number of fields: 12
Structure	\$PUBX,04,time,date,utcTow,utcWk,leapSec,clkBias,clkDrift,tpGran*cs\r\n	

Example \$PUBX,04,073731.00,091202,113851.00,1196,15D,1930035,-2660.664,43*3C\r\n

Payload:

Field	Name	Format	Unit	Example	Description
0	PUBX	string	-	\$PUBX	Message ID, UBX protocol header, proprietary sentence
1	msgId	numeric	-	04	Proprietary message identifier: 04
2	time	hhmmss.ss	-	073731.00	UTC time. See section UTC representation in the integration manual for details.
3	date	ddmmyy	-	091202	UTC date, day, month, year. See section UTC representation in the integration manual for details.
4	utcTow	numeric	s	113851.00	UTC time of week
5	utcWk	numeric	-	1196	UTC week number, continues beyond 1023
6	leapSec	numeric/ text	s	15D	Leap seconds (not supported for protocol versions less than 13.01) The number is marked with a <i>D</i> if the value is the firmware default value. If the value is not marked it has been received from a satellite.
7	clkBias	numeric	ns	1930035	Receiver clock bias
8	clkDrift	numeric	ns/s	-2660.664	Receiver clock drift
9	tpGran	numeric	ns	43	Time pulse granularity, the quantization error of the TIMEPULSE pin
10	cs	hexadecimal	-	*3C	Checksum
11	CRLF	character	-	-	Carriage return and line feed

3 UBX protocol

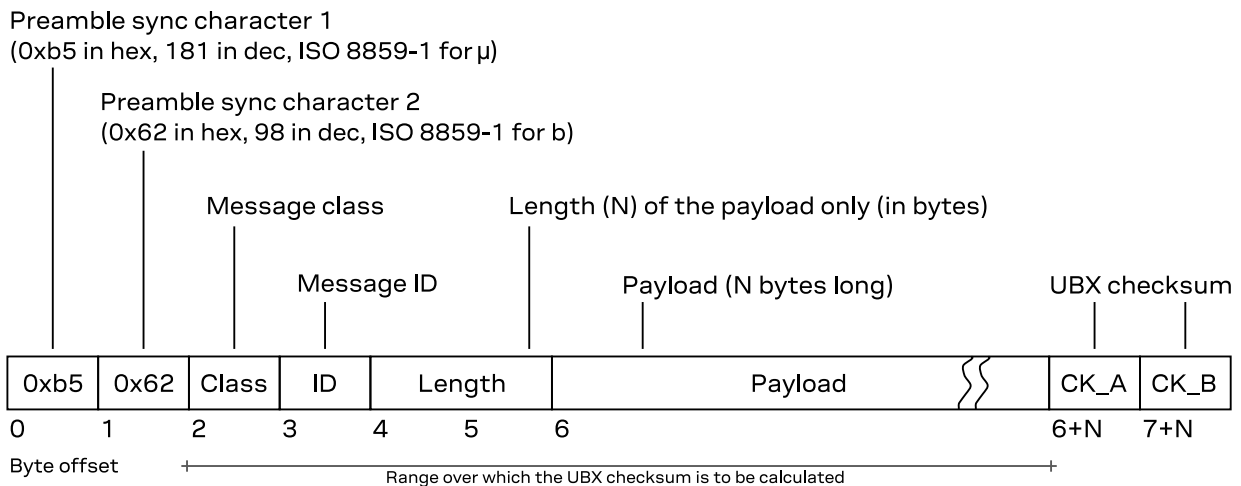
3.1 UBX protocol key features

u-blox receivers support a u-blox-proprietary protocol to communicate with a host computer. This protocol has the following key features:

- Compact – uses 8-bit binary data
- Checksum protected – uses a low-overhead checksum algorithm
- Modular – uses a two-stage message identifier (Class and Message ID)

3.2 UBX frame structure

The structure of a basic UBX frame is shown in the following diagram.



- Every *frame* starts with a 2-byte *preamble* consisting of two synchronization characters: 0xb5 and 0x62.
- A 1-byte *message class* field follows. A class is a group of messages that are related to each other.
- A 1-byte *message ID* field defines the message that is to follow.
- A 2-byte *length* field follows. The length is defined as being that of the payload only. It does not include the preamble, message class, message ID, length, or [UBX checksum](#) fields. The number format of the length field is an unsigned little-endian 16-bit integer (a "U2" in [UBX data types](#)).
- The *payload* field contains a variable number (= *length*) of bytes.
- The two 1-byte *CK_A* and *CK_B* fields hold a 16-bit checksum whose calculation is defined in [UBX checksum](#) section. This concludes the frame.

3.3 UBX payload definition rules

This section contains the rules and guidelines for UBX message payloads. See also [UBX message example](#).

3.3.1 UBX structure packing

Values are placed in such an order that structure packing is not a problem. This means that two-byte values shall start on offsets that are a multiple of two; four-byte values shall start at a multiple of four; and so on.

3.3.2 UBX reserved elements

Some messages contain reserved fields or bits to allow for future expansion. The contents of these elements should be ignored in output messages and must be set to zero in input messages. Where a message is output and subsequently returned to the receiver as an input message, reserved elements can either be explicitly set to zero or left with whatever value they were output with.

For fields in a bitfield the same rules apply. Note that bits not described are automatically reserved and are not explicitly stated (see [UBX message example](#)).

3.3.3 UBX undefined values

The description of some fields provide specific meanings for specific values. For example, the field `gnssId` appears in many UBX messages and uses 0 to indicate GPS, 1 for SBAS and so on (see [GNSS identifiers](#) for details); however it is usually stored in a byte with far more possible values than the handful currently defined. All such undefined values are reserved for future expansion and therefore should not be used.

3.3.4 UBX conditional values

Some UBX messages use validity flag fields to indicate whether the values of some value fields are valid. For example, the [UBX-NAV-PVT](#) message has the `validDate` and `validTime` fields that indicate whether the date (`year`, `month` and `day` fields), and, respectively, the time (`hour`, `min` and `sec` fields) are valid. This means that these value fields will only contain meaningful data if the corresponding flag field is set (has the value 1).

3.3.5 UBX data types

The following data types (number formats) are defined.

Name	Type	Size (Bytes)	Range	Resolution
U1	unsigned 8-bit integer	1	$0 \dots 2^8 - 1$	1
I1	signed 8-bit integer, two's complement	1	$-2^7 \dots 2^7 - 1$	1
X1	8-bit bitfield	1	n/a	n/a
U2	unsigned little-endian 16-bit integer	2	$0 \dots 2^{16} - 1$	1
I2	signed little-endian 16-bit integer, two's complement	2	$-2^{15} \dots 2^{15} - 1$	1
X2	16-bit little-endian bitfield	2	n/a	n/a
U4	unsigned little-endian 32-bit integer	4	$0 \dots 2^{32} - 1$	1
I4	signed little-endian 32-bit integer, two's complement	4	$-2^{31} \dots 2^{31} - 1$	1
X4	32-bit little-endian bitfield	4	n/a	n/a

Name	Type	Size (Bytes)	Range	Resolution
R4	IEEE 754 single (32-bit) precision	4	$-2^{127} \dots 2^{127}$	$\sim \text{value} \cdot 2^{-24}$
R8	IEEE 754 double (64-bit) precision	8	$-2^{1023} \dots 2^{1023}$	$\sim \text{value} \cdot 2^{-53}$
CH	ASCII / ISO 8859-1 char (8-bit)	1	n/a	n/a
U _n	unsigned bitfield value of <i>n</i> bits width	var.	variable	variable
I _n	signed (two's complement) bitfield value of <i>n</i> bits width	var.	variable	variable
S _n	signed bitfield value of <i>n</i> bits width, in sign (most significant bit) and magnitude (remaining bits) notation	var.	variable	variable

3.3.6 UBX fields scale and unit

Fields in UBX messages can have a unit defined. Whenever possible, SI units and symbols are used (e.g. "m" for meters, "s" for seconds). For civil (UTC) time representation units of years (y), months (month), days (d), hours (h), minutes (min) and seconds (s) are used.

Fields in UBX messages can have a scale factor defined. Unity (factor 1) is assumed if no scale is specified. For integer type fields this is often combined with a unit. When a scale is combined with a unit, the scale represents the smallest storage unit. For example, if meters (m) are expressed (stored) in centimeters the scale would be 0.01 (or 1e-2). This is equivalent of specifying a unit of centimeters (cm) and no scale.

The description of some integer values (e.g. U2, I4 or I8) indicates a fixed-point format (e.g. [UU.FF], [IIII.FFF] or [IIIIII.FFFFFFFF]). The fixed-point value can be retrieved from the integer value by first casting it to appropriate type (e.g. as a floating-point number) and then scaling it with the indicated scaling factor.

3.3.7 UBX repeated fields

There are two types of repetitions in UBX messages. The first type specifies that a single field is repeated a constant number of times. This repetition is defined in the type of the field. For example, the [UBX message example](#) can specify a field `data` of type `U1[5]`. In this case the `data` field should be interpreted as an array of five U1 values.

The second type of repetition in messages is referred to as *repeated groups*, which groups one or more fields into a block of payload data. There are several types of repetition:

- The number of repetitions of *variable-by-field group* is indicated by another, earlier field in the same message. The number of repetitions can be zero or more, depending on the value of the referenced field.
- A *constant group* has a constant number of repetitions.
- An *optional group* is repeated zero or one times, depending on the available payload data. That is, the fields are present in the message only if the payload of the message is large enough to cover the whole group of fields.
- The number of repetitions of a *variable-by-size group* is given by the available payload size. The group will repeat until there is not enough payload data left to cover the whole group of fields another time.

Note that only some combinations of repeated groups of fields are possible in a single message. See also [UBX payload decoding](#).

3.3.8 UBX payload decoding

UBX message payloads are designed so that the data (fields) can be extracted by a single pass through the payload from start to end. Fixed-size messages are the trivial case where the offset of all fields is unambiguously defined. Variable-size messages have variable number of repetitions of one or multiple groups of fields. For groups where the number of repetitions is given by the value of another field, that field can always be found at a fixed offset in the message payload before the respective group of fields. Groups whose number of repetitions depend on the payload size can only be the last group of fields in a message and only one such group may exist in a message. See also [UBX repeated fields](#).

3.4 UBX checksum

The checksum is calculated over the message, starting and including the class field up until, but excluding, the checksum fields (see the figure [UBX frame structure](#)).

The checksum algorithm used is the 8-bit Fletcher algorithm, which is used in the TCP standard [RFC 1145](#)). This algorithm works as follows:

- `Buffer[N]` is an array of bytes that contains the data over which the checksum is to be calculated.
- The two `CK_A` and `CK_B` values are 8-bit unsigned integers, only! If implementing with larger-sized integer values, make sure to mask both `CK_A` and `CK_B` with the value `0xff` after both operations in the loop.
- After the loop, the two `U1` values contain the checksum, transmitted after the message payload, which concludes the frame.

```

1 CK_A = 0, CK_B = 0
2 For (I = 0; I < N; I++)
3 {
4     CK_A = CK_A + Buffer[I]
5     CK_B = CK_B + CK_A
6 }
```

3.5 UBX message flow

There are certain features associated with the messages being sent back and forth:

3.5.1 UBX acknowledgement

When messages from the class CFG are sent to the receiver, the receiver will send an "acknowledge" ([UBX-ACK-ACK](#)) or a "not acknowledge" ([UBX-ACK-NAK](#)) message back to the sender, depending on whether or not the message was processed correctly.

Some messages from other classes also use the same acknowledgement mechanism.

3.5.2 UBX polling mechanism

The UBX protocol is designed so that messages can be polled by sending the message required to the receiver but without a payload (or with just a single parameter that identifies the poll request). The receiver then responds with the same message with the payload populated.

3.6 GNSS, satellite, and signal numbering

See [GNSS, satellite, and signal identifiers](#) for details on how GNSS, satellites and signals are numbered in the UBX protocol.

3.7 UBX message example

This is an example of the definition of UBX messages as shown in the following sections.

Message	UBX-DEMO-EXAMPLE				
①	Example demo message				
Type ②	Periodic/poll				
Comment ③	This is a comment that describes the use of the demo example message. There can be references to other sections in the documentation (such as: UBX protocol). Note that there can be important remarks here.				
Message ④	<i>Header</i>	<i>Class ID</i>	<i>Length (bytes)</i>	<i>Payload</i>	<i>Checksum</i>
Structure	0xb5 0x62 0x01 0x07	16 + numRepeat*4		see below	CK_A CK_B
Payload description: ⑤					
<i>Byte offset</i>	<i>Type</i>	<i>Name</i>	<i>Scale</i>	<i>Unit</i>	<i>Description</i>
0	U4	aField	-	-	a field that contains an unsigned integer with no particular scale or unit
4	I4	anotherField	1e-2	m	a field that contains a length in meters (m) with a scale of 1e-2 (= 0.01), i.e. a length in centimeters
8	X2	bitfield ⑥	-	-	this field contains flags or values smaller than one byte, whose definition follows below (bits not described are reserved)
bit 0	U:1	aFieldValid	-	-	the first bit in bitfield indicates whether the aField is valid or not (see UBX conditional values)
bit 1	U:1	someFlag	-	-	the second bit is a flag (1 = true, 0 = false)
bits 5...2	U:4	aBitFieldValue	-	-	a 4-bits value (range: 0...15)
10	U1[5] ⑦	reserved0	-	-	a reserved field, whose value shall be ignored (in output messages) or set to 0 (in input messages)
15	U1	numRepeat	-	-	number of repetitions in the group of fields below
Start of repeated group (numRepeat times) ⑧					
16 + n*4	I2	someValue	-	-	a signed value in a repeated group of fields
18 + n*4	U2	anotherValue	-	-	another value in a repeated group of fields
End of repeated group (numRepeat times)					

① The first line shows the message name (see [Message naming](#)). The second line shows a short description of the message.

② The message type (see [Message types](#)).

③ This section contains comments that describe the message. Often links to other related sections in the documentation or other related messages are found here.

④ The message structure gives the parameters for the [UBX frame structure](#), notably the message class and message ID values and the payload length. For many messages the payload length is a fixed number (of bytes). Messages that contain repeated blocks of information (fields) have a variable payload (see [UBX repeated fields](#)).

⑤ The message payload definition is given as a list of fields and their parameters. Each field starts at a specified offset (in bytes) in the payload (see also [UBX structure packing](#)), is of a specific type (see [UBX data types](#)), has a unique name (within the message), and a description. Optionally, fields can have a scale and/or a unit (see [UBX fields scale and unit](#)).

⑥ Bitfields ("X" types) are broken down into smaller parts. Each part can be one or more bits wide. Values that are two or more bits wide can be unsigned or one of two signed value representation (see [UBX data types](#)). Note that the ten unused bits 15...6 are not explicitly stated as [UBX reserved elements](#).

⑦ Fields can be arrays of values of the same type (see [UBX repeated fields](#)).

⑧ Groups of fields can be repeated in the payload. The number of repetitions can be given by another field in the message (this example), a constant number, zero or one times (known as "optional group"), or derived from the remaining payload size (labeled as "repeated N times"). See also [UBX repeated fields](#) and [UBX payload decoding](#).

3.8 UBX messages overview

Message	Class/ID	Description (Type)
UBX-ACK – Acknowledgement and negative acknowledgement messages		
UBX-ACK-ACK	0x05 0x01	• Message acknowledged (Output)
UBX-ACK-NAK	0x05 0x00	• Message not acknowledged (Output)
UBX-CFG – Configuration and command messages		
UBX-CFG-ANT	0x06 0x13	• Antenna control settings (Get/set)
UBX-CFG-RST	0x06 0x04	• Reset receiver / Clear backup data structures (Command)
UBX-CFG-VALDEL	0x06 0x8c	• Delete configuration item values (Set) • Delete configuration item values (with transaction) (Set)
UBX-CFG-VALGET	0x06 0x8b	• Get configuration items (Poll request)
UBX-CFG-VALSET	0x06 0x8a	• Set configuration item values (Set)
UBX-INF – Information messages		
UBX-INF-DEBUG	0x04 0x04	• ASCII output with debug contents (Output)
UBX-INF-ERROR	0x04 0x00	• ASCII output with error contents (Output)
UBX-INF-NOTICE	0x04 0x02	• ASCII output with informational contents (Output)
UBX-INF-TEST	0x04 0x03	• ASCII output with test contents (Output)
UBX-INF-WARNING	0x04 0x01	• ASCII output with warning contents (Output)
UBX-LOG – Logging messages		
UBX-LOG-CREATE	0x21 0x07	• Create log file (Command)
UBX-LOG-ERASE	0x21 0x03	• Erase logged data (Command)
UBX-LOG-FINDTIME	0x21 0x0e	• Find index of a log entry based on a given time (Input) • Response to FINDTIME request (Output)
UBX-LOG-INFO	0x21 0x08	• Poll for log information (Poll request) • Log information (Output)
UBX-LOG-RETRIEVE	0x21 0x09	• Request log data (Command)
UBX-LOG-RETRIEVEBATCH	0x21 0x10	• Request batch data (Command)

Message	Class/ID	Description (Type)
UBX-LOG-RETRIEVEPOS	0x21 0x0b	• Position fix log entry (Output)
UBX-LOG-RETRIEVEPOSEXTRA	0x21 0x0f	• Odometer log entry (Output)
UBX-LOG-STRING	0x21 0x04	• Store arbitrary string in on-board flash (Command)
UBX-MGA – GNSS assistance (A-GNSS) messages		
UBX-MGA-ACK	0x13 0x60	• Multiple GNSS acknowledge message (Output)
UBX-MGA-ANO	0x13 0x20	• Multiple GNSS AssistNow Predictive Orbits assistance (Input)
UBX-MGA-BDS	0x13 0x03	• BeiDou ephemeris assistance for satellites svld 1..37 (Input) • BeiDou almanac assistance (Input) • BeiDou health assistance (Input) • BeiDou UTC assistance (Input) • BeiDou ionosphere assistance (Input) • BeiDou ephemeris assistance (Input)
UBX-MGA-DBD	0x13 0x80	• Poll the navigation database (Poll request) • Navigation database dump entry (Input/output)
UBX-MGA-FLASH	0x13 0x21	• Transfer MGA-ANO data block to flash (Input) • Finish flashing MGA-ANO data (Input) • Acknowledge last FLASH-DATA or -STOP (Output)
UBX-MGA-GAL	0x13 0x02	• Galileo ephemeris assistance (Input) • Galileo almanac assistance (Input) • Galileo GPS time offset assistance (Input) • Galileo UTC assistance (Input)
UBX-MGA-GLO	0x13 0x06	• GLONASS ephemeris assistance (Input) • GLONASS almanac assistance (Input) • GLONASS auxiliary time offset assistance (Input)
UBX-MGA-GPS	0x13 0x00	• GPS ephemeris assistance (Input) • GPS almanac assistance (Input) • GPS health assistance (Input) • GPS UTC assistance (Input) • GPS ionosphere assistance (Input)
UBX-MGA-INI	0x13 0x40	• Initial position assistance XYZ (Input) • Initial position assistance LLH (Input) • Initial time assistance UTC (Input) • Initial time assistance GNSS (Input) • Initial clock drift assistance (Input) • Initial frequency assistance (Input) • Earth orientation parameters assistance (Input)
UBX-MGA-QZSS	0x13 0x05	• QZSS ephemeris assistance (Input) • QZSS almanac assistance (Input) • QZSS health assistance (Input)
UBX-MON – Monitoring messages		
UBX-MON-BATCH	0x0a 0x32	• Data batching buffer status (Polled)
UBX-MON-COMMS	0x0a 0x36	• Communication port information (Periodic/polled)
UBX-MON-GNSS	0x0a 0x28	• Information message major GNSS selection (Polled) • Information message with supported GNSS and signal plans (Polled)
UBX-MON-HW3	0x0a 0x37	• I/O pin status (Periodic/polled)
UBX-MON-IO	0x0a 0x02	• I/O system status (Periodic/polled)
UBX-MON-MSGPP	0x0a 0x06	• Message parse and process status (Periodic/polled)
UBX-MON-PATCH	0x0a 0x27	• Poll request for installed patches (Poll request) • Installed patches (Polled)

Message	Class/ID	Description (Type)
UBX-MON-POST	0x0a 0x3b	• Power on self test (POST) information (Polled)
UBX-MON-RF	0x0a 0x38	• RF information (Periodic/polled)
UBX-MON-RXBUF	0x0a 0x07	• Receiver buffer status (Periodic/polled)
UBX-MON-RXR	0x0a 0x21	• Receiver status information (Output)
UBX-MON-SPAN	0x0a 0x31	• Signal characteristics (Periodic/polled)
UBX-MON-SYS	0x0a 0x39	• Current system performance information (Periodic/polled)
UBX-MON-TXBUF	0x0a 0x08	• Transmitter buffer status (Periodic/polled)
UBX-MON-VER	0x0a 0x04	• Poll receiver and software version (Poll request) • Receiver and software version (Polled)
UBX-NAV – Navigation solution messages		
UBX-NAV-AOPSTATUS	0x01 0x60	• AssistNow Autonomous status (Periodic/polled)
UBX-NAV-CLOCK	0x01 0x22	• Clock solution (Periodic/polled)
UBX-NAV-COV	0x01 0x36	• Covariance matrices (Periodic/polled)
UBX-NAV-DOP	0x01 0x04	• Dilution of precision (Periodic/polled)
UBX-NAV-EOE	0x01 0x61	• End of epoch (Periodic)
UBX-NAV-GEOFENCE	0x01 0x39	• Geofencing status (Periodic/polled)
UBX-NAV-ODO	0x01 0x09	• Odometer solution (Periodic/polled)
UBX-NAV-ORB	0x01 0x34	• GNSS orbit database info (Periodic/polled)
UBX-NAV-POSECEF	0x01 0x01	• Position solution in ECEF (Periodic/polled)
UBX-NAV-POSLLH	0x01 0x02	• Geodetic position solution (Periodic/polled)
UBX-NAV-PVT	0x01 0x07	• Navigation position velocity time solution (Periodic/polled)
UBX-NAV-RESETODO	0x01 0x10	• Reset odometer (Command)
UBX-NAV-SAT	0x01 0x35	• Satellite information (Periodic/polled)
UBX-NAV-SBAS	0x01 0x32	• SBAS status data (Periodic/polled)
UBX-NAV-SIG	0x01 0x43	• Signal information (Periodic/polled)
UBX-NAV-SLAS	0x01 0x42	• QZSS L1S SLAS status data (Periodic/polled)
UBX-NAV-STATUS	0x01 0x03	• Receiver navigation status (Periodic/polled)
UBX-NAV-TIMEBDS	0x01 0x24	• BeiDou time solution (Periodic/polled)
UBX-NAV-TIMEGAL	0x01 0x25	• Galileo time solution (Periodic/polled)
UBX-NAV-TIMEGLO	0x01 0x23	• GLONASS time solution (Periodic/polled)
UBX-NAV-TIMEGPS	0x01 0x20	• GPS time solution (Periodic/polled)
UBX-NAV-TIMELS	0x01 0x26	• Leap second event information (Periodic/polled)
UBX-NAV-TIMENAVIC	0x01 0x63	• NavIC time solution (Periodic/polled)
UBX-NAV-TIMEQZSS	0x01 0x27	• QZSS time solution (Periodic/polled)
UBX-NAV-TIMEUTC	0x01 0x21	• UTC time solution (Periodic/polled)
UBX-NAV-VELECEF	0x01 0x11	• Velocity solution in ECEF (Periodic/polled)
UBX-NAV-VELNED	0x01 0x12	• Velocity solution in NED frame (Periodic/polled)
UBX-RXM – Receiver manager messages		
UBX-RXM-MEASX	0x02 0x14	• Satellite measurements for RRLP (Periodic/polled)
UBX-RXM-PMREQ	0x02 0x41	• Power management request (Command)
UBX-RXM-RAWX	0x02 0x15	• Multi-GNSS raw measurements (Periodic/polled)
UBX-RXM-RLM	0x02 0x59	• Galileo SAR short-RLM report (Output) • Galileo SAR long-RLM report (Output)
UBX-RXM-SFRBX	0x02 0x13	• Broadcast navigation data subframe (Output)

Message	Class/ID	Description (Type)
UBX-SEC – Security messages		
UBX-SEC-SIG	0x27 0x09	• Signal security information (Periodic/poll)
UBX-SEC-SIGLOG	0x27 0x10	• Signal security log (Periodic/poll)
UBX-SEC-UNIQID	0x27 0x03	• Unique chip ID (Output)
UBX-TIM – Timing messages		
UBX-TIM-TM2	0x0d 0x03	• Time mark data (Periodic/poll)
UBX-TIM-TP	0x0d 0x01	• Time pulse time data (Periodic/poll)
UBX-TIM-VRFY	0x0d 0x06	• Sourced time verification (Periodic/poll)
UBX-UPD – Firmware update messages		
UBX-UPD-SOS	0x09 0x14	• Poll backup restore status (Poll request) • Create backup in flash (Command) • Clear backup in flash (Command) • Backup creation acknowledge (Output) • System restored from backup (Output)

3.9 UBX-ACK (0x05)

The messages in the UBX-ACK class are used to indicate acknowledgement or rejection (i.e. negative acknowledgement) of input messages, such as UBX-CFG messages.

3.9.1 UBX-ACK-ACK (0x05 0x01)

3.9.1.1 Message acknowledged

Message	UBX-ACK-ACK					
	Message acknowledged					
Type	Output					
Comment	Output upon processing of an input message. A UBX-ACK-ACK is sent as soon as possible but at least within one second.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x05	0x01	2	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U1	clsID	-	-	Class ID of the Acknowledged Message	
1	U1	msgID	-	-	Message ID of the Acknowledged Message	

3.9.2 UBX-ACK-NAK (0x05 0x00)

3.9.2.1 Message not acknowledged

Message	UBX-ACK-NAK					
	Message not acknowledged					
Type	Output					
Comment	Output upon processing of an input message. A UBX-ACK-NAK is sent as soon as possible but at least within one second.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x05	0x00	2	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	

0	U1	clsID	-	-	Class ID of the Not-Acknowledged Message
1	U1	msgID	-	-	Message ID of the Not-Acknowledged Message

3.10 UBX-CFG (0x06)

The messages in the UBX-CFG class are used to configure the receiver and poll current configuration values as well as for sending commands to the receiver. Unless stated otherwise, any message in this class sent to the receiver is either acknowledged (by a [UBX-ACK-ACK](#) message) if processed successfully or rejected (with a [UBX-ACK-NAK](#) message) if processed unsuccessfully.

3.10.1 UBX-CFG-ANT (0x06 0x13)

3.10.1.1 Antenna control settings

Message	UBX-CFG-ANT						
	Antenna control settings						
Type	Get/set						
Comment	This message allows the user to configure the antenna supervisor. The antenna supervisor can be used to detect the status of an active antenna and control it. It can be used to turn off the supply to the antenna in the event of a short circuit (for example) or to manage power consumption in power save mode. Refer to antenna supervisor configuration in the integration manual for more information regarding the behavior of the antenna supervisor. Refer to UBX-MON-RF for a description of the fields in the message used to obtain the status of the antenna. Note that not all pins can be used for antenna supervisor operation, the default pins are recommended. Consult the integration manual if you need to use the other pins.						
Message structure	Header	Class	ID	Length (Bytes)		Payload	Checksum
	0xb5 0x62	0x06	0x13	4		see below	CK_A CK_B
Payload description:							
Byte offset	Type	Name		Scale	Unit	Description	
0	X2	flags		-	-	Antenna flag mask	
bit 0	U:1	svcs		-	-	Enable antenna supply voltage control signal	
bit 1	U:1	scd		-	-	Enable short circuit detection	
bit 2	U:1	ocd		-	-	Enable open circuit detection	
bit 3	U:1	pdwnOnSCD		-	-	Power down antenna supply if short circuit is detected. (only in combination with bit 1)	
bit 4	U:1	recovery		-	-	Enable automatic recovery from short state	
2	X2	pins		-	-	Antenna pin configuration	
bits 4...0	U:5	pinSwitch		-	-	PIO-pin used for switching antenna supply	
bits 9...5	U:5	pinSCD		-	-	PIO-pin used for detecting a short in the antenna supply	
bits 14...10	U:5	pinOCD		-	-	PIO-pin used for detecting open/not connected antenna	
bit 15	U:1	reconfig		-	-	if set to one, and this command is sent to the receiver, the receiver will reconfigure the pins as specified.	

3.10.2 UBX-CFG-RST (0x06 0x04)

3.10.2.1 Reset receiver / Clear backup data structures

Message	UBX-CFG-RST					
	Reset receiver / Clear backup data structures					
Type	Command					
Comment	Do not expect this message to be acknowledged by the receiver. - Newer FW version will not acknowledge this message at all. - Older FW version will acknowledge this message but the acknowledge may not be sent completely before the receiver is reset.					
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i>	<i>Checksum</i>
	0xb5 0x62	0x06	0x04	4	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	X2	navBbrMask	-	-	BBR sections to clear. The following special sets apply: 0x0000 Hot start 0x0001 Warm start 0xFFFF Cold start	
bit 0	U ₁	eph	-	-	Ephemeris	
bit 1	U ₁	alm	-	-	Almanac	
bit 2	U ₁	health	-	-	Health	
bit 3	U ₁	klob	-	-	Klobuchar parameters	
bit 4	U ₁	pos	-	-	Position	
bit 5	U ₁	clkd	-	-	Clock drift	
bit 6	U ₁	osc	-	-	Oscillator parameter	
bit 7	U ₁	utc	-	-	UTC correction + GPS leap seconds parameters	
bit 8	U ₁	rtc	-	-	RTC	
bit 11	U ₁	sfdr	-	-	SFDR Parameters (only available on the ADR/UDR/HPS product variant) and weak signal compensation estimates	
bit 12	U ₁	vmon	-	-	SFDR Vehicle Monitoring Parameter (only available on the ADR/UDR/HPS product variant)	
bit 13	U ₁	tct	-	-	TCT Parameters (only available on the ADR/UDR/HPS product variant)	
bit 15	U ₁	aop	-	-	Autonomous orbit parameters	
2	U1	resetMode	-	-	Reset Type 0x00 = Hardware reset (watchdog) immediately 0x01 = Controlled software reset 0x02 = Controlled software reset (GNSS only) 0x04 = Hardware reset (watchdog) after shutdown 0x08 = Controlled GNSS stop 0x09 = Controlled GNSS start	
3	U1	reserved0	-	-	Reserved	

3.10.3 UBX-CFG-VALDEL (0x06 0x8c)

3.10.3.1 Delete configuration item values

Message	UBX-CFG-VALDEL Delete configuration item values				
Type	Set				
Comment	Overview: - This message can be used to delete saved configuration to effectively revert the item values to defaults. - This message can delete saved configuration from the flash configuration layer and the BBR configuration layer. The changes will not be effective until these layers are loaded into the RAM layer. - This message is limited to containing a maximum of 64 keys up for deletion; i.e. N is a maximum of 64. - This message can be used multiple times and every time the result will be applied immediately. To send this message multiple times with the result being applied at the end, see version 1 of UBX-CFG-VALDEL that supports transactions. - This message does not check if the resulting configuration is valid. - See Receiver configuration for details. This message returns a UBX-ACK-NAK and no configuration is applied: - if any key is unknown to the receiver FW - if the layer's bitfield does not specify a layer to delete a value from. Notes: - If a key is sent multiple times within the same message, the value is effectively deleted only once. - Attempting to delete items that have not been set before, or that have already been deleted, is considered a valid request. - The provided keys can be complete key values (group and item specifiers) or wild-card specifications. A complete key value constitutes a deletion request for one key-value pair. A key value with a valid group specifier and 0xffff in the item part of the key value (bits 0-15) constitutes a deletion request for all items in the specified group. A key with a value of 0xffff in the group part of the key value (bits 16-27) is a deletion request for all items known to the receiver in all groups.				
Message structure	Header	Class	ID	Length (Bytes)	Payload
	0xb5 0x62	0x06	0x8c	4 + [0..n]·4	see below
Checksum					
					CK_A CK_B
Payload description:					
Byte offset	Type	Name	Scale	Unit	Description
0	U1	version	-	-	Message version (0x00 for this version)
1	X1	layers	-	-	The layers where the configuration should be deleted from
bit 1	U ₁	bbr	-	-	Delete configuration from the BBR layer
bit 2	U ₁	flash	-	-	Delete configuration from the Flash layer
2	U1[2]	reserved0	-	-	Reserved
Start of repeated group (N times)					
4 + n·4	U4	keys	-	-	Configuration key IDs of the configuration items to be deleted
End of repeated group (N times)					

3.10.3.2 Delete configuration item values (with transaction)

Message	UBX-CFG-VALDEL Delete configuration item values (with transaction)				
Type	Set				
Comment	Overview: - This message can be used to delete saved configuration to effectively revert them to defaults. - This message can delete saved configuration from the flash configuration layer and the BBR configuration layer. The changes will not be effective until these layers are loaded into the RAM layer. - This message is limited to containing a maximum of 64 keys up for deletion; i.e. N is a maximum of 64. - This message can be used multiple times with the result being managed within a transaction. - This message does not check if the resulting configuration is valid.				

- See [Receiver configuration](#) for details.
- See version 0 of [UBX-CFG-VALDEL](#) for simplified version of this message.

This message returns a UBX-ACK-NAK, cancels any started transaction, and no configuration is applied:

- if any key within a transaction is unknown to the receiver FW
- if an invalid transaction state transition is requested
- if the layer's bitfield changes within a transaction
- if the layer's bitfield does not specify a layer to delete a value from.

Notes:

- Any request for another UBX-CFG- message type (including UBX-CFG-VALSET and UBX-CFG-VALGET) will cancel any started transaction, and no configuration is applied.
- This message can be sent with no keys to delete for the purposes of managing the transaction state transition.
- If a key is sent multiple times within the same message or within the same transaction, the value is effectively deleted only once.
- Attempting to delete items that have not been set before, or that have already been deleted, is considered a valid request.
- The provided keys can be complete key values (group and item specifiers) or wild-card specifications. A complete key value constitutes a deletion request for one key-value pair. A key value with a valid group specifier and 0xffff in the item part of the key value (bits 0-15) constitutes a deletion request for all items in the specified group. A key with a value of 0xffff in the group part of the key value (bits 16-27) is a deletion request for all items known to the receiver in all groups.

Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x06	0x8c	4 + [0..n]·4	see below	CK_A CK_B

Payload description:

Byte offset	Type	Name	Scale	Unit	Description
0	U1	version	-	-	Message version (0x01 for this version)
1	X1	layers	-	-	The layers where the configuration should be deleted from
bit 1	U ₁	bbr	-	-	Delete configuration from the BBR layer
bit 2	U ₁	flash	-	-	Delete configuration from the Flash layer
2	X1	transaction	-	-	Transaction action to be applied:
bits 1...0	U ₂	action	-	-	Transaction action to be applied: • 0 = Transactionless UBX-CFG-VALDEL: In the next UBX-CFG-VALDEL, it can be either 0 or 1. If a transaction has not yet been started, the incoming configuration is applied. If a transaction has already been started, cancels any started transaction and the incoming configuration is applied. • 1 = (Re)Start deletion transaction: In the next UBX-CFG-VALDEL, it can be either 0, 1, 2 or 3. If a transaction has not yet been started, a transaction will be started. If a transaction has already been started, restarts the transaction, effectively removing all previous non-applied UBX-CFG-VALDEL messages. • 2 = Deletion transaction ongoing: In the next UBX-CFG-VALDEL, it can be either 0, 1, 2 or 3. • 3 = Apply and end a deletion transaction: In the next UBX-CFG-VALDEL, it can be either 0 or 1.
3	U1	reserved0	-	-	Reserved

Start of repeated group (N times)

4 + n·4	U4	keys	-	-	Configuration key IDs of the configuration items to be deleted
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End of repeated group (N times)

3.10.4 UBX-CFG-VALGET (0x06 0x8b)

3.10.4.1 Get configuration items

Message	UBX-CFG-VALGET					
	Get configuration items					
Type	Poll request					
Comment	Overview: - This message is used to get configuration values by providing a list of configuration key IDs, which identify the configuration items to retrieve. - This message can specify the configuration layer where the values of the specified configuration items are retrieved from. - This message is limited to containing a maximum of 64 key IDs. - See Receiver configuration for details. This message returns a UBX-ACK-NAK: - if any key is unknown to the receiver FW - if the layer field specifies an invalid layer to get the value from - if the keys array specifies more than 64 key IDs. Notes: - If a value is requested multiple times within the same poll request, then the reply will contain it multiple times. - The provided keys can be complete key values (group and item specifiers) or wild-card specifications. A complete key value will constitute a request for one key-value pair. A key value that has a valid group specifier and 0xffff in the item part of the key value (bits 0-15) constitutes a request for all items in the specified group. A key with a value of 0xffff in the group part of the key value (bits 16-27) is a request for all items known to the receiver in all groups. - The response message is limited to containing a maximum of 64 key-value pairs. If there are wild-card specifications then there may be more than 64 possible responses. In order to handle this, the 'position' field can specify that the response message should skip this number of key-value pairs before it starts constructing the message. This allows a large set of values to be retrieved 64 at a time. If the response contains less than 64 key-value pairs then all values have been reported, otherwise there may be more to read. - It is not possible to retrieve configuration values for the same configuration item from multiple configuration layers. Separate poll requests must be made for each desired layer.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x06	0x8b	4 + [0..n]·4	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U1	version		-	-	Message version (0x00 for this version)
1	U1	layer		-	-	The layer from which the configuration items should be retrieved: 0 - RAM layer 1 - BBR layer 2 - Flash layer 7 - Default layer
2	U2	position		-	-	Skip this many key values before constructing output message
Start of repeated group (N times)						
4 + n·4	U4	keys		-	-	Configuration key IDs of the configuration items to be retrieved

End of repeated group (N times)

3.10.5 UBX-CFG-VALSET (0x06 0x8a)

3.10.5.1 Set configuration item values

Message	UBX-CFG-VALSET Set configuration item values					
Type	Set					
Comment	Overview: - This message is used to set a configuration by providing configuration data (a list of key and value pairs), which identify the configuration items to change, and their new values. - This message is limited to containing a maximum of 64 key-value pairs. - This message can be used multiple times and every time the result will be applied immediately. To send this message multiple times with the result being applied at the end, see version 1 of UBX-CFG-VALSET that supports transactions. - See Receiver configuration for details. This message returns a UBX-ACK-NAK and no configuration is applied: - if any key is unknown to the receiver FW - if the layer's bitfield does not specify a layer to save a value to - if the requested configuration is not valid. The validity of a configuration is checked only if the message requests to apply the configuration to the RAM configuration layer. Notes: - If a key is sent multiple times within the same message, then the value eventually being applied is the last sent.					
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i>	<i>Checksum</i>
	0xb5 0x62	0x06	0x8a	4 + [0..n]	see below	CK_A CK_B
<i>Payload description:</i>						
<i>Byte offset</i>	<i>Type</i>	<i>Name</i>		<i>Scale</i>	<i>Unit</i>	<i>Description</i>
0	U1	version		-	-	Message version (0x00 for this version)
1	X1	layers		-	-	The layers where the configuration should be applied
bit 0	U ₁	ram		-	-	Update configuration in the RAM layer
bit 1	U ₁	bbr		-	-	Update configuration in the BBR layer
bit 2	U ₁	flash		-	-	Update configuration in the Flash layer
2	U1[2]	reserved0		-	-	Reserved
<i>Start of repeated group (N times)</i>						
4 + n	U1	cfgData		-	-	Configuration data (key and value pairs)
<i>End of repeated group (N times)</i>						

3.11 UBX-INF (0x04)

Messages in the UBX-INF class are used to output strings from the firmware or application code. All messages have an associated type to indicate the nature or priority of the message.

3.11.1 UBX-INF-DEBUG (0x04 0x04)

3.11.1.1 ASCII output with debug contents

Message	UBX-INF-DEBUG ASCII output with debug contents					
Type	Output					
Comment	This message has a variable length payload, representing an ASCII string.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x04	0x04	[0..n]	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
Start of repeated group (N times)						
0 + n	CH	str		-	-	ASCII Character
End of repeated group (N times)						

3.11.2 UBX-INF-ERROR (0x04 0x00)

3.11.2.1 ASCII output with error contents

Message	UBX-INF-ERROR ASCII output with error contents					
Type	Output					
Comment	This message has a variable length payload, representing an ASCII string.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x04	0x00	[0..n]	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
Start of repeated group (N times)						
0 + n	CH	str		-	-	ASCII Character
End of repeated group (N times)						

3.11.3 UBX-INF-NOTICE (0x04 0x02)

3.11.3.1 ASCII output with informational contents

Message	UBX-INF-NOTICE ASCII output with informational contents					
Type	Output					
Comment	This message has a variable length payload, representing an ASCII string.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x04	0x02	[0..n]	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
Start of repeated group (N times)						
0 + n	CH	str		-	-	ASCII Character
End of repeated group (N times)						

3.11.4 UBX-INF-TEST (0x04 0x03)

3.11.4.1 ASCII output with test contents

Message	UBX-INF-TEST ASCII output with test contents					
Type	Output					
Comment	This message has a variable length payload, representing an ASCII string.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x04	0x03	[0..n]	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
Start of repeated group (N times)						
0 + n	CH	str		-	-	ASCII Character
End of repeated group (N times)						

3.11.5 UBX-INF-WARNING (0x04 0x01)

3.11.5.1 ASCII output with warning contents

Message	UBX-INF-WARNING ASCII output with warning contents					
Type	Output					
Comment	This message has a variable length payload, representing an ASCII string.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x04	0x01	[0..n]	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
Start of repeated group (N times)						
0 + n	CH	str		-	-	ASCII Character
End of repeated group (N times)						

3.12 UBX-LOG (0x21)

The messages in the UBX-LOG class are used to configure and report status information of the logging and data batching features.

3.12.1 UBX-LOG-CREATE (0x21 0x07)

3.12.1.1 Create log file

Message	UBX-LOG-CREATE Create log file					
Type	Command					
Comment	This message is used to create an initial logging file and activate the logging subsystem. UBX-ACK-ACK or UBX-ACK-NAK are returned to indicate success or failure.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x21	0x07	8	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U1	version		-	-	Message version (0x00 for this version)

1	X1	logCfg	-	-	Config flags
	bit 0 U:1	circular	-	-	Log is circular (new entries overwrite old ones in a full log) if this bit set
2	U1	reserved0	-	-	Reserved
3	U1	logSize	-	-	Indicates the size of the log: 0 (maximum safe size) = Ensures that logging will not be interrupted and enough space will be left available for all other uses of the filestore 1 (minimum size) = 2 (user-defined) = See 'userDefinedSize' below
4	U4	userDefinedSize	-	bytes	Sets the maximum amount of space in the filestore that can be used by the logging task. This field is only applicable if logSize is set to user-defined.

3.12.2 UBX-LOG-ERASE (0x21 0x03)

3.12.2.1 Erase logged data

Message	UBX-LOG-ERASE Erase logged data					
Type	Command					
Comment	This message deactivates the logging system and erases all logged data. UBX-ACK-ACK or UBX-ACK-NAK are returned to indicate success or failure.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x21	0x03	0	see below	CK_A CK_B
Payload	This message has no payload.					

3.12.3 UBX-LOG-FINDTIME (0x21 0x0e)

3.12.3.1 Find index of a log entry based on a given time

Message	UBX-LOG-FINDTIME Find index of a log entry based on a given time					
Type	Input					
Comment	This message can be used for a time-based search of a log. It can find the index of the first log entry with time equal to the given time, otherwise the index of the most recent entry with time less than the given time. This index can then be used with the UBX-LOG-RETRIEVE message to provide time-based retrieval of log entries. Searching a log is effective for a given time later than the base date (January 1st, 2004). Searching a log for a given time earlier than the base date will result in an 'entry not found' response. (Searching a log for a given time earlier than the base date will result in a UBX-ACK-NAK message for protocol versions less than 18.00). Searching a log for a given time greater than the last recorded entry's time will return the index of the last recorded entry. (If the logging has stopped due to lack of file space, such a search will result in a UBX-ACK-NAK message for protocol versions less than 18.00).					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x21	0x0e	10	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U1	version	-	-	Message version (0x00 for this version)	
1	U1	type	-	-	Message type, 0 for request	
2	U2	year	-	-	Year (1-65635) of UTC time	

4	U1	month	-	-	Month (1-12) of UTC time
5	U1	day	-	-	Day (1-31) of UTC time
6	U1	hour	-	-	Hour (0-23) of UTC time
7	U1	minute	-	-	Minute (0-59) of UTC time
8	U1	second	-	-	Second (0-60) of UTC time
9	U1	reserved0	-	-	Reserved

3.12.3.2 Response to FINDTIME request

Message	UBX-LOG-FINDTIME					
	Response to FINDTIME request					
Type	Output					
Comment						
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x21	0x0e	8	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U1	version		-	-	Message version (0x01 for this version)
1	U1	type		-	-	Message type, 1 for response
2	U1[2]	reserved0		-	-	Reserved
4	U4	entryNumber		-	-	Index of the first log entry with time = given time, otherwise index of the most recent entry with time < given time. If 0xFFFFFFFF, no log entry found with time <= given time. The indexing of log entries is zero-based.

3.12.4 UBX-LOG-INFO (0x21 0x08)

3.12.4.1 Poll for log information

Message	UBX-LOG-INFO					
	Poll for log information					
Type	Poll request					
Comment	Upon sending of this message, the receiver returns UBX-LOG-INFO as defined below.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x21	0x08	0	see below	CK_A CK_B
Payload	This message has no payload.					

3.12.4.2 Log information

Message	UBX-LOG-INFO Log information
Type	Output
Comment	This message is used to report information about the logging subsystem. Note: • The reported maximum log size will be smaller than that originally specified in LOG-CREATE due to logging and filestore implementation overheads. • Log entries are compressed in a variable length fashion, so it may be difficult to predict log space usage with any precision.

- There may be times when the receiver does not have an accurate time (e.g. if the week number is not yet known), in which case some entries will not have a timestamp. This may result in the oldest/newest entry time values not taking account of these entries.

Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x21	0x08	48	see below	CK_A CK_B

Payload description:

Byte offset	Type	Name	Scale	Unit	Description
0	U1	version	-	-	Message version (0x01 for this version)
1	U1[3]	reserved0	-	-	Reserved
4	U4	filestore Capacity	-	bytes	The capacity of the filestore
8	U1[8]	reserved1	-	-	Reserved
16	U4	currentMaxLog Size	-	bytes	The maximum size the current log is allowed to grow to
20	U4	currentLogSize	-	bytes	Approximate amount of space in log currently occupied
24	U4	entryCount	-	-	Number of entries in the log. Note: for circular logs this value will decrease when a group of entries is deleted to make space for new ones.
28	U2	oldestYear	-	-	Oldest entry UTC year (1-65635) or zero if there are no entries with known time
30	U1	oldestMonth	-	-	Oldest month (1-12)
31	U1	oldestDay	-	-	Oldest day (1-31)
32	U1	oldestHour	-	-	Oldest hour (0-23)
33	U1	oldestMinute	-	-	Oldest minute (0-59)
34	U1	oldestSecond	-	-	Oldest second (0-60)
35	U1	reserved2	-	-	Reserved
36	U2	newestYear	-	-	Newest year (1-65635) or zero if there are no entries with known time
38	U1	newestMonth	-	-	Newest month (1-12)
39	U1	newestDay	-	-	Newest day (1-31)
40	U1	newestHour	-	-	Newest hour (0-23)
41	U1	newestMinute	-	-	Newest minute (0-59)
42	U1	newestSecond	-	-	Newest second (0-60)
43	U1	reserved3	-	-	Reserved
44	X1	status	-	-	Log status flags
bit 3	U ₁	recording	-	-	Log entry recording is currently turned on
bit 4	U ₁	inactive	-	-	Logging system not active - no log present
bit 5	U ₁	circular	-	-	The current log is circular
45	U1[3]	reserved4	-	-	Reserved

3.12.5 UBX-LOG-RETRIEVE (0x21 0x09)

3.12.5.1 Request log data

Message	UBX-LOG-RETRIEVE					
	Request log data					
Type	Command					
Comment						
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x21	0x09	12	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U4	startNumber		-	-	Index of first log entry to be transferred. If it is larger than the index of the last available log entry, then the first log entry to be transferred is the last available log entry. The indexing of log entries is zero-based.
4	U4	entryCount		-	-	Number of log entries to transfer in total including the first entry to be transferred. If it is larger than the log entries available starting from the first entry to be transferred, then only the available log entries are transferred followed by a UBX-ACK-NAK . The maximum is 256.
8	U1	version		-	-	Message version (0x00 for this version)
9	U1[3]	reserved0		-	-	Reserved

3.12.6 UBX-LOG-RETRIEVEBATCH (0x21 0x10)

3.12.6.1 Request batch data

Message	UBX-LOG-RETRIEVEBATCH					
	Request batch data					
Type	Command					
Comment	This message is used to request batched data. Batch entries are returned in chronological order, using one UBX-LOG-BATCH per navigation epoch. The speed of transfer can be maximized by using a high data rate. See Data batching section in the integration manual for more information.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x21	0x10	4	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U1	version		-	-	Message version (0x00 for this version)
1	X1	flags		-	-	Flags
bit 0	U:1	sendMonFirst		-	-	Send UBX-MON-BATCH message before sending the UBX-LOG-BATCH message(s).
2	U1[2]	reserved0		-	-	Reserved

3.12.7 UBX-LOG-RETRIEVEPOS (0x21 0x0b)

3.12.7.1 Position fix log entry

Message	UBX-LOG-RETRIEVEPOS Position fix log entry
Type	Output

Comment	This message is used to report a position fix log entry					
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i>	<i>Checksum</i>
	0xb5 0x62	0x21	0x0b	40	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U4	entryIndex	-	-	The index of this log entry	
4	I4	lon	1e-7	deg	Longitude	
8	I4	lat	1e-7	deg	Latitude	
12	I4	hMSL	-	mm	Height above mean sea level	
16	U4	hAcc	-	mm	Horizontal accuracy estimate	
20	U4	gSpeed	-	mm/s	Ground speed (2-D)	
24	U4	heading	1e-5	deg	Heading	
28	U1	version	-	-	Message version (0x00 for this version)	
29	U1	fixType	-	-	Fix type: • 0x01 = Dead Reckoning only • 0x02 = 2D-Fix • 0x03 = 3D-Fix • 0x04 = GNSS + Dead Reckoning combined	
30	U2	year	-	-	Year (1-65635) of UTC time	
32	U1	month	-	-	Month (1-12) of UTC time	
33	U1	day	-	-	Day (1-31) of UTC time	
34	U1	hour	-	-	Hour (0-23) of UTC time	
35	U1	minute	-	-	Minute (0-59) of UTC time	
36	U1	second	-	-	Second (0-60) of UTC time	
37	U1	reserved0	-	-	Reserved	
38	U1	numSV	-	-	Number of satellites used in the position fix	
39	U1	reserved1	-	-	Reserved	

3.12.8 UBX-LOG-RETRIEVEPOSEXTRA (0x21 0x0f)

3.12.8.1 Odometer log entry

Message	UBX-LOG-RETRIEVEPOSEXTRA Odometer log entry					
Type	Output					
Comment	This message is used to report an odometer log entry					
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i>	<i>Checksum</i>
	0xb5 0x62	0x21	0x0f	32	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U4	entryIndex	-	-	The index of this log entry	
4	U1	version	-	-	Message version (0x00 for this version)	
5	U1	reserved0	-	-	Reserved	
6	U2	year	-	-	Year (1-65635) of UTC time. Will be zero if time not known	
8	U1	month	-	-	Month (1-12) of UTC time	

9	U1	day	-	-	Day (1-31) of UTC time
10	U1	hour	-	-	Hour (0-23) of UTC time
11	U1	minute	-	-	Minute (0-59) of UTC time
12	U1	second	-	-	Second (0-60) of UTC time
13	U1[3]	reserved1	-	-	Reserved
16	U4	distance	-	-	Odometer distance traveled since the last time the odometer was reset by a UBX-NAV-RESETO
20	U1[12]	reserved2	-	-	Reserved

3.12.9 UBX-LOG-STRING (0x21 0x04)

3.12.9.1 Store arbitrary string in on-board flash

Message	UBX-LOG-STRING Store arbitrary string in on-board flash					
Type	Command					
Comment	This message can be used to store an arbitrary byte string in the on-board flash memory. The maximum length that can be stored is 256 bytes.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x21	0x04	[0..n]	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
Start of repeated group (N times)						
0 + n	U1	bytes		-	-	The string of bytes to be logged (maximum 256)
End of repeated group (N times)						

3.13 UBX-MGA (0x13)

The messages in the UBX-MGA class are used for sending GNSS assistance (A-GNSS, aiding) information to the receiver as well as backing up the navigation database from the receiver to a host.

3.13.1 UBX-MGA-ACK (0x13 0x60)

3.13.1.1 Multiple GNSS acknowledge message

Message	UBX-MGA-ACK-DATA0 Multiple GNSS acknowledge message					
Type	Output					
Comment	This message is sent by a u-blox receiver to acknowledge the receipt of an assistance message. Acknowledgments are enabled by setting the CFG-NAVSPG-ACKAIDING item. See section Flow control in the integration manual for details.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x13	0x60	8	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U1	type	-	-	Type of acknowledgment: 0 = The message was not used by the receiver (see infoCode field for an indication of why) 1 = The message was accepted for use by the receiver (the infoCode field will be 0)	

1	U1	version	-	-	Message version (0x00 for this version)
2	U1	infoCode	-	-	Provides greater information on what the receiver chose to do with the message contents: 0 = The receiver accepted the data 1 = The receiver does not know the time so it cannot use the data (To resolve this a UBX-MGA-INI-TIME_UTC message should be supplied first) 2 = The message version is not supported by the receiver 3 = The message size does not match the message version 4 = The message data could not be stored to the database 5 = The receiver is not ready to use the message data 6 = The message type is unknown
3	U1	msgId	-	-	UBX message ID of the acknowledged message
4	U1[4]	msgPayload Start	-	-	The first 4 bytes of the acknowledged message's payload

3.13.2 UBX-MGA-ANO (0x13 0x20)

3.13.2.1 Multiple GNSS AssistNow Predictive Orbits assistance

Message	UBX-MGA-ANO Multiple GNSS AssistNow Predictive Orbits assistance					
Type	Input					
Comment	This message is created by the AssistNow Predictive Orbits service to deliver AssistNow Predictive Orbits assistance to the receiver. See AssistNow Predictive Orbits section in the integration manual for details.					
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i>	<i>Checksum</i>
	0xb5 0x62	0x13	0x20	76	see below	CK_A CK_B
Payload description:						
<i>Byte offset</i>	<i>Type</i>	<i>Name</i>	<i>Scale</i>	<i>Unit</i>	<i>Description</i>	
0	U1	type	-	-	Message type (0x00 for this type)	
1	U1	version	-	-	Message version (0x00 for this version)	
2	U1	svId	-	-	Satellite identifier (see Satellite Numbering)	
3	U1	gnssId	-	-	GNSS identifier (see Satellite Numbering)	
4	U1	year	-	-	years since the year 2000	
5	U1	month	-	-	month (1..12)	
6	U1	day	-	-	day (1..31)	
7	U1	reserved0	-	-	Reserved	
8	U1[64]	data	-	-	assistance data	
72	U1[4]	reserved1	-	-	Reserved	

3.13.3 UBX-MGA-BDS (0x13 0x03)

3.13.3.1 BeiDou ephemeris assistance for satellites svId 1..37

Message	UBX-MGA-BDS-EPH BeiDou ephemeris assistance for satellites svId 1..37					
Type	Input					
Comment	This message is deprecated in this protocol version. Use UBX-MGA-BDS-EPHD instead. This message allows the delivery of BeiDou D1/D2 ephemeris assistance to a receiver. See section AssistNow Live Orbits in the integration manual for details.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x13	0x03	88	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U1	type	-	-	Message type (0x01 for this type)	
1	U1	version	-	-	Message version (0x00 for this version)	
2	U1	svId	-	-	BeiDou satellite identifier (see Satellite Numbering)	
3	U1	reserved0	-	-	Reserved	
4	U1	SatH1	-	-	Autonomous satellite Health flag	
5	U1	IODC	-	-	Issue of Data, Clock	
6	I2	a2	2 ⁻⁶⁶	s/s ²	Time polynomial coefficient 2	
8	I4	a1	2 ⁻⁵⁰	s/s	Time polynomial coefficient 1	
12	I4	a0	2 ⁻³³	s	Time polynomial coefficient 0	
16	U4	toc	2 ³	s	Clock data reference time	
20	I2	TGD1	0.1	ns	Equipment Group Delay Differential	
22	U1	URAI	-	-	User Range Accuracy Index	
23	U1	IODE	-	-	Issue of Data, Ephemeris	
24	U4	toe	2 ³	s	Ephemeris reference time	
28	U4	sqrta	2 ⁻¹⁹	m ^{0.5}	Square root of semi-major axis	
32	U4	e	2 ⁻³³	-	Eccentricity	
36	I4	omega	2 ⁻³¹	semi-circles	Argument of perigee	
40	I2	Deltan	2 ⁻⁴³	semi-circles/s	Mean motion difference from computed value	
42	I2	IDOT	2 ⁻⁴³	semi-circles/s	Rate of inclination angle	
44	I4	M0	2 ⁻³¹	semi-circles	Mean anomaly at reference time	
48	I4	Omega0	2 ⁻³¹	semi-circles	Longitude of ascending node of orbital of plane computed according to reference time	
52	I4	OmegaDot	2 ⁻⁴³	semi-circles/s	Rate of right ascension	
56	I4	i0	2 ⁻³¹	semi-circles	Inclination angle at reference time	
60	I4	Cuc	2 ⁻³¹	radians	Amplitude of cosine harmonic correction term to the argument of latitude	
64	I4	Cus	2 ⁻³¹	radians	Amplitude of sine harmonic correction term to the argument of latitude	
68	I4	Crc	2 ⁻⁶	m	Amplitude of cosine harmonic correction term to the orbit radius	

72	I4	Crs	2 ⁻⁶	m	Amplitude of sine harmonic correction term to the orbit radius
76	I4	Cic	2 ⁻³¹	radians	Amplitude of cosine harmonic correction term to the angle of inclination
80	I4	Cis	2 ⁻³¹	radians	Amplitude of sine harmonic correction term to the angle of inclination
84	U1[4]	reserved1	-	-	Reserved

3.13.3.2 BeiDou almanac assistance

Message	UBX-MGA-BDS-ALM BeiDou almanac assistance				
Type	Input				
Comment	This message allows the delivery of BeiDou almanac assistance to a receiver. See section AssistNow Live Orbits in the integration manual for details.				
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i> <i>Checksum</i>
	0xb5 0x62	0x13	0x03	40	see below CK_A CK_B
Payload description:					
<i>Byte offset</i>	<i>Type</i>	<i>Name</i>	<i>Scale</i>	<i>Unit</i>	<i>Description</i>
0	U1	type	-	-	Message type (0x02 for this version)
1	U1	version	-	-	Message version (0x00 for this version)
2	U1	svId	-	-	BeiDou satellite identifier (see Satellite Numbering)
3	U1	reserved0	-	-	Reserved
4	U1	Wna	-	week	Almanac Week Number
5	U1	toa	2 ⁻¹²	s	Almanac reference time
6	I2	deltaI	2 ⁻¹⁹	semi-circles	Almanac correction of orbit reference inclination at reference time
8	U4	sqrtA	2 ⁻¹¹	m ^{0.5}	Almanac square root of semi-major axis
12	U4	e	2 ⁻²¹	-	Almanac eccentricity
16	I4	omega	2 ⁻²³	semi-circles	Almanac argument of perigee
20	I4	M0	2 ⁻²³	semi-circles	Almanac mean anomaly at reference time
24	I4	Omega0	2 ⁻²³	semi-circles	Almanac longitude of ascending node of orbit plane at computed according to reference time
28	I4	omegaDot	2 ⁻³⁸	semi-circles/s	Almanac rate of right ascension
32	I2	a0	2 ⁻²⁰	s	Almanac satellite clock bias
34	I2	a1	2 ⁻³⁸	s/s	Almanac satellite clock rate
36	U1[4]	reserved1	-	-	Reserved

3.13.3.3 BeiDou health assistance

Message	UBX-MGA-BDS-HEALTH BeiDou health assistance				
Type	Input				
Comment	This message is deprecated in this protocol version. Use UBX-MGA-BDS-HEALTHD instead. This message allows the delivery of BeiDou health assistance from D1/D2 ephemeris to a receiver. See section AssistNow Live Orbits in the integration manual for details.				

This message allows the delivery of health assistance data for all satellites with svld 1 to 30.

Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x13	0x03	68	see below	CK_A CK_B

Payload description:

Byte offset	Type	Name	Scale	Unit	Description
0	U1	type	-	-	Message type (0x04 for this type)
1	U1	version	-	-	Message version (0x00 for this version)
2	U1[2]	reserved0	-	-	Reserved
4	U2[30]	healthCode	-	-	Each two-byte value represents a BeiDou SV (1-30). The 9 LSBs of each byte contain the 9 bit health code from subframe 5 pages 7,8 of the D1 message, and from subframe 5 pages 35,36 of the D2 message.
64	U1[4]	reserved1	-	-	Reserved

3.13.3.4 BeiDou UTC assistance

Message	UBX-MGA-BDS-UTC BeiDou UTC assistance
---------	--

Type	Input
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Comment	This message allows the delivery of BeiDou UTC assistance to a receiver. See section AssistNow Live Orbits in the integration manual for details.
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Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x13	0x03	20	see below	CK_A CK_B

Payload description:

Byte offset	Type	Name	Scale	Unit	Description
0	U1	type	-	-	Message type (0x05 for this type)
1	U1	version	-	-	Message version (0x00 for this version)
2	U1[2]	reserved0	-	-	Reserved
4	I4	a0UTC	2 ⁻³⁰	s	BDT clock bias relative to UTC
8	I4	a1UTC	2 ⁻⁵⁰	s/s	BDT clock rate relative to UTC
12	I1	dtLS	-	s	Delta time due to leap seconds before the new leap second effective
13	U1	reserved1	-	-	Reserved
14	U1	wnRec	-	week	BeiDou week number of reception of this UTC parameter set (8-bit truncated)
15	U1	wnLSF	-	week	Week number of the new leap second
16	U1	dN	-	day	Day number of the new leap second
17	I1	dtLSF	-	s	Delta time due to leap seconds after the new leap second effective
18	U1[2]	reserved2	-	-	Reserved

3.13.3.5 BeiDou ionosphere assistance

Message	UBX-MGA-BDS-IONO BeiDou ionosphere assistance
---------	--

Type	Input
------	-------

Comment	This message allows the delivery of BeiDou ionospheric assistance to a receiver. See section AssistNow Live Orbits in the integration manual for details.
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Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x13	0x03	16	see below	CK_A CK_B

Payload description:

Byte offset	Type	Name	Scale	Unit	Description
0	U1	type	-	-	Message type (0x06 for this type)
1	U1	version	-	-	Message version (0x00 for this version)
2	U1[2]	reserved0	-	-	Reserved
4	I1	alpha0	2 ⁻³⁰	s	Ionospheric parameter alpha0
5	I1	alpha1	2 ⁻²⁷	s/pi	Ionospheric parameter alpha1
6	I1	alpha2	2 ⁻²⁴	s/pi ²	Ionospheric parameter alpha2
7	I1	alpha3	2 ⁻²⁴	s/pi ³	Ionospheric parameter alpha3
8	I1	beta0	2 ⁻¹¹	s	Ionospheric parameter beta0
9	I1	beta1	2 ⁻¹⁴	s/pi	Ionospheric parameter beta1
10	I1	beta2	2 ⁻¹⁶	s/pi ²	Ionospheric parameter beta2
11	I1	beta3	2 ⁻¹⁶	s/pi ³	Ionospheric parameter beta3
12	U1[4]	reserved1	-	-	Reserved

3.13.3.6 BeiDou health assistance

Message	UBX-MGA-BDS-HEALTHD BeiDou health assistance
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Type	Input
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Comment This message allows the delivery of BeiDou health assistance for any PRNs 1-63 in a message. See section AssistNow Live Orbits in the Integration manual for details. Each entry in healthSvData[] pack has a satellite identifier and health code in a U2 bitmask. See the healthSvData field definition for bit layout details.

Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x13	0x03	4 + numEntries·2	see below	CK_A CK_B

Payload description:

Byte offset	Type	Name	Scale	Unit	Description
0	U1	type	-	-	Message type (0x09 for this type)
1	U1	version	-	-	Message version (0x00 for this version)
2	U1	numEntries	-	-	Number of valid entries in healthSvData[] 1-63
3	U1	reserved0	-	-	Reserved

Start of repeated group (numEntries times)

4 + n·2	X2	healthSvData	-	-	Satellite health entries
bits 8...0	U:9	healthCode	-	-	9-bit health code

For satellites 1..19, the 9 LSB contain the 9-bit health code from subframe 5, page 7 of the D1 message, and from subframe 5 page 35 of the D2 message.

For satellites 20..30, the 9 LSB contain the 9-bit health code from subframe 5, page 8 of the D1 message, and from subframe 5 page 36 of the D2 message.

For satellites 31..63, the 9 LSB contain the 9-bit health code from subframe 5, page 24 of the D1 message, and from subframe 5 page 116 of the D2 message.

For satellite identifiers greater than 63, this is reserved.

					(mask 0x01ff, shift 0)
bits 14...9	U ₆	svId	-	-	Satellite identifier (see Satellite Numbering)
					(mask 0x7e00, shift 9)

End of repeated group (*numEntries* times)

3.13.3.7 BeiDou ephemeris assistance

Message	UBX-MGA-BDS-EPHD BeiDou ephemeris assistance					
Type	Input					
Comment	This message allows the delivery of BeiDou D1/D2 ephemeris assistance to a receiver. See section AssistNow Live Orbits in the integration manual for details.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x13	0x03	88	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U1	type	-	-	Message type (0x0B for this type)	
1	U1	version	-	-	Message version (0x00 for this version)	
2	U1	svId	-	-	BeiDou satellite identifier (see Satellite Numbering)	
3	U1	reserved0	-	-	Reserved	
4	U1	SatH1	-	-	Autonomous satellite Health flag	
5	U1	IODC	-	-	Issue of Data, Clock	
6	I2	a2	2 ⁻⁶⁶	s/s ²	Time polynomial coefficient 2	
8	I4	a1	2 ⁻⁵⁰	s/s	Time polynomial coefficient 1	
12	I4	a0	2 ⁻³³	s	Time polynomial coefficient 0	
16	U4	toc	2 ³	s	Clock data reference time	
20	I2	TGD1	0.1	ns	Equipment Group Delay Differential	
22	U1	URAI	-	-	User Range Accuracy Index	
23	U1	IODE	-	-	Issue of Data, Ephemeris	
24	U4	toe	2 ³	s	Ephemeris reference time	
28	U4	sqrta	2 ⁻¹⁹	m ^{0.5}	Square root of semi-major axis	
32	U4	e	2 ⁻³³	-	Eccentricity	
36	I4	omega	2 ⁻³¹	semi-circles	Argument of perigee	
40	I2	Deltan	2 ⁻⁴³	semi-circles/s	Mean motion difference from computed value	
42	I2	IDOT	2 ⁻⁴³	semi-circles/s	Rate of inclination angle	
44	I4	M0	2 ⁻³¹	semi-circles	Mean anomaly at reference time	
48	I4	Omega0	2 ⁻³¹	semi-circles	Longitude of ascending node of orbital of plane computed according to reference time	
52	I4	OmegaDot	2 ⁻⁴³	semi-circles/s	Rate of right ascension	
56	I4	i0	2 ⁻³¹	semi-circles	Inclination angle at reference time	

60	I4	Cuc	2 ⁻³¹	radians	Amplitude of cosine harmonic correction term to the argument of latitude
64	I4	Cus	2 ⁻³¹	radians	Amplitude of sine harmonic correction term to the argument of latitude
68	I4	Crc	2 ⁻⁶	m	Amplitude of cosine harmonic correction term to the orbit radius
72	I4	Crs	2 ⁻⁶	m	Amplitude of sine harmonic correction term to the orbit radius
76	I4	Cic	2 ⁻³¹	radians	Amplitude of cosine harmonic correction term to the angle of inclination
80	I4	Cis	2 ⁻³¹	radians	Amplitude of sine harmonic correction term to the angle of inclination
84	U1[4]	reserved1	-	-	Reserved

3.13.4 UBX-MGA-DBD (0x13 0x80)

3.13.4.1 Poll the navigation database

Message	UBX-MGA-DBD Poll the navigation database				
Type	Poll request				
Comment	Poll the whole navigation data base. The receiver will send all available data from its internal database. The receiver will indicate the finish of the transmission with a UBX-MGA-ACK . The msgPayloadStart field of the UBX-MGA-ACK message will contain a U4 representing the number of UBX-MGA-DBD-DATA* messages sent.				
Message structure	Header	Class	ID	Length (Bytes)	Payload
	0xb5 0x62	0x13	0x80	0	see below
Payload	This message has no payload.				

3.13.4.2 Navigation database dump entry

Message	UBX-MGA-DBD Navigation database dump entry				
Type	Input/output				
Comment	Navigation database entry. The data fields are firmware-specific. Transmission of this type of message will be acknowledged by UBX-MGA-ACK messages, if acknowledgment has been enabled. See section AssistNow Live Orbits in the integration manual for details. ↗ UBX-MGA-DBD messages are only intended to be sent back to the same receiver that generated them.				
Message structure	Header	Class	ID	Length (Bytes)	Payload
	0xb5 0x62	0x13	0x80	12 + [0..n]	see below
Payload description:					
Byte offset	Type	Name	Scale	Unit	Description
0	U1[12]	reserved0	-	-	Reserved
Start of repeated group (N times)					
12 + n	U1	data	-	-	firmware-specific data
End of repeated group (N times)					

3.13.5 UBX-MGA-FLASH (0x13 0x21)

3.13.5.1 Transfer MGA-ANO data block to flash

Message	UBX-MGA-FLASH-DATA Transfer MGA-ANO data block to flash				
Type	Input				
Comment	This message is used to transfer a block of MGA-ANO data from host to the receiver. Upon reception of this message, the receiver will write the payload data to its internal non-volatile memory (flash). Also, on reception of the first MGA-FLASH-DATA message, the receiver will erase the flash allocated to storing any existing MGA-ANO data. The payload can be up to 512 bytes. Payloads larger than this would exceed the receiver's internal buffering capabilities. The receiver will ACK/NACK this message using the message alternatives given below. The host shall wait for an acknowledge message before sending the next data block. See Flash-based AssistNow Predictive Orbits for details.				
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i> <i>Checksum</i>
	0xb5 0x62	0x13	0x21	6 + size	see below CK_A CK_B
Payload description:					
<i>Byte offset</i>	<i>Type</i>	<i>Name</i>	<i>Scale</i>	<i>Unit</i>	<i>Description</i>
0	U1	type	-	-	Message type (0x01 for this type)
1	U1	version	-	-	Message version (0x00 for this version)
2	U2	sequence	-	-	Message sequence number, starting at 0 and incrementing by 1 for each MGA-FLASH-DATA message sent.
4	U2	size	-	-	Payload size in bytes.
Start of repeated group (size times)					
6 + n	U1	data	-	-	Payload data.
End of repeated group (size times)					

3.13.5.2 Finish flashing MGA-ANO data

Message	UBX-MGA-FLASH-STOP Finish flashing MGA-ANO data				
Type	Input				
Comment	This message is used to tell the receiver that there are no more MGA-FLASH type 1 messages coming, and that it can do any final internal operations needed to commit the data to flash as a background activity. A UBX-MGA-ACK message will be sent at the end of this process. Note that there may be a delay of several seconds before the UBX-MGA-ACK for this message is sent because of the time taken for this processing. See Flash-based AssistNow Predictive Orbits for details.				
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i> <i>Checksum</i>
	0xb5 0x62	0x13	0x21	2	see below CK_A CK_B
Payload description:					
<i>Byte offset</i>	<i>Type</i>	<i>Name</i>	<i>Scale</i>	<i>Unit</i>	<i>Description</i>
0	U1	type	-	-	Message type (0x02 for this type)
1	U1	version	-	-	Message version (0x00 for this version)

3.13.5.3 Acknowledge last FLASH-DATA or -STOP

Message	UBX-MGA-FLASH-ACK Acknowledge last FLASH-DATA or -STOP				
Type	Output				
Comment	This message reports an ACK/NACK to the host for the last MGA-FLASH type 1 or type 2 message message received. See Flash-based AssistNow Predictive Orbits for details.				

Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x13	0x21	6	see below	CK_A CK_B

Payload description:

Byte offset	Type	Name	Scale	Unit	Description
0	U1	type	-	-	Message type (0x03 for this type)
1	U1	version	-	-	Message version (0x00 for this version)
2	U1	ack	-	-	Acknowledgment type. 0 - ACK: Message received and written to flash. 1 - NACK: Problem with last message, re-transmission required (this only happens while acknowledging a UBX-MGA_FLASH_DATA message). 2 - NACK: problem with last message, give up.
3	U1	reserved0	-	-	Reserved
4	U2	sequence	-	-	If acknowledging a UBX-MGA-FLASH-DATA message this is the Message sequence number being ack'ed. If acknowledging a UBX-MGA-FLASH-STOP message it will be set to 0xffff.

3.13.6 UBX-MGA-GAL (0x13 0x02)

3.13.6.1 Galileo ephemeris assistance

Message	UBX-MGA-GAL-EPH Galileo ephemeris assistance					
Type	Input					
Comment	This message allows the delivery of Galileo ephemeris assistance to a receiver. See section AssistNow Live Orbits in the integration manual for details.					

Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x13	0x02	76	see below	CK_A CK_B

Payload description:

Byte offset	Type	Name	Scale	Unit	Description
0	U1	type	-	-	Message type (0x01 for this type)
1	U1	version	-	-	Message version (0x00 for this version)
2	U1	svId	-	-	Galileo Satellite identifier (see Satellite Numbering)
3	U1	reserved0	-	-	Reserved
4	U2	iodNav	-	-	Ephemeris and clock correction Issue of Data
6	I2	deltaN	2 ⁻⁴³	semi-circles/s	Mean motion difference from computed value
8	I4	m0	2 ⁻³¹	semi-circles	Mean anomaly at reference time
12	U4	e	2 ⁻³³	-	Eccentricity
16	U4	sqrtA	2 ⁻¹⁹	m ^{0.5}	Square root of the semi-major axis
20	I4	omega0	2 ⁻³¹	semi-circles	Longitude of ascending node of orbital plane at weekly epoch
24	I4	i0	2 ⁻³¹	semi-circles	Inclination angle at reference time
28	I4	omega	2 ⁻³¹	semi-circles	Argument of perigee
32	I4	omegaDot	2 ⁻⁴³	semi-circles/s	Rate of change of right ascension

36	I2	iDot	2 ⁻⁴³	semi-circles/s	Rate of change of inclination angle
38	I2	cuc	2 ⁻²⁹	radians	Amplitude of the cosine harmonic correction term to the argument of latitude
40	I2	cus	2 ⁻²⁹	radians	Amplitude of the sine harmonic correction term to the argument of latitude
42	I2	crc	2 ⁻⁵	radians	Amplitude of the cosine harmonic correction term to the orbit radius
44	I2	crs	2 ⁻⁵	radians	Amplitude of the sine harmonic correction term to the orbit radius
46	I2	cic	2 ⁻²⁹	radians	Amplitude of the cosine harmonic correction term to the angle of inclination
48	I2	cis	2 ⁻²⁹	radians	Amplitude of the sine harmonic correction term to the angle of inclination
50	U2	toe	60	s	Ephemeris reference time
52	I4	af0	2 ⁻³⁴	s	SV clock bias correction coefficient
56	I4	af1	2 ⁻⁴⁶	s/s	SV clock drift correction coefficient
60	I1	af2	2 ⁻⁵⁹	s/s squared	SV clock drift rate correction coefficient
61	U1	sisIndexE1E5b	-	-	Signal-In-Space Accuracy index for dual frequency E1-E5b
62	U2	toc	60	s	Clock correction data reference Time of Week
64	I2	bqdE1E5b	2 ⁻³²	s	E1-E5b Broadcast Group Delay
66	U1[2]	reserved1	-	-	Reserved
68	U1	healthE1B	-	-	E1-B Signal Health Status
69	U1	dataValidityE1B	-	-	E1-B Data Validity Status
70	U1	healthE5b	-	-	E5b Signal Health Status
71	U1	dataValidityE5b	-	-	E5b Data Validity Status
72	U1[4]	reserved2	-	-	Reserved

3.13.6.2 Galileo almanac assistance

Message	UBX-MGA-GAL-ALM Galileo almanac assistance				
Type	Input				
Comment	This message allows the delivery of Galileo almanac assistance to a receiver. See section AssistNow Live Orbits in the integration manual for details.				
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i> <i>Checksum</i>
	0xb5 0x62	0x13	0x02	32	<i>see below</i> CK_A CK_B
Payload description:					
Byte offset	Type	Name	Scale	Unit	Description
0	U1	type	-	-	Message type (0x02 for this type)
1	U1	version	-	-	Message version (0x00 for this version)
2	U1	svId	-	-	Galileo Satellite identifier (see Satellite Numbering)
3	U1	reserved0	-	-	Reserved
4	U1	ioda	-	-	Almanac Issue of Data

5	U1	almWNa	-	week	Almanac reference week number
6	U2	toa	600	s	Almanac reference time
8	I2	deltaSqrtA	2 ⁻⁹	m ^{0.5}	Difference with respect to the square root of the nominal semi-major axis (29 600 km)
10	U2	e	2 ⁻¹⁶	-	Eccentricity
12	I2	deltaI	2 ⁻¹⁴	semi-circles	Inclination at reference time relative to i0 = 56 degree
14	I2	omega0	2 ⁻¹⁵	semi-circles	Longitude of ascending node of orbital plane at weekly epoch
16	I2	omegaDot	2 ⁻³³	semi-circles/s	Rate of change of right ascension
18	I2	omega	2 ⁻¹⁵	semi-circles	Argument of perigee
20	I2	m0	2 ⁻¹⁵	semi-circles	Satellite mean anomaly at reference time
22	I2	af0	2 ⁻¹⁹	s	Satellite clock correction bias 'truncated'
24	I2	af1	2 ⁻³⁸	s/s	Satellite clock correction linear 'truncated'
26	U1	healthE1B	-	-	Satellite E1-B signal health status
27	U1	healthE5b	-	-	Satellite E5b signal health status
28	U1[4]	reserved1	-	-	Reserved

3.13.6.3 Galileo GPS time offset assistance

Message	UBX-MGA-GAL-TIMEOFFSET Galileo GPS time offset assistance					
Type	Input					
Comment	This message allows the delivery of Galileo time to GPS time offset. See section AssistNow Live Orbits in the integration manual for details.					
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i>	<i>Checksum</i>
	0xb5 0x62	0x13	0x02	12	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U1	type	-	-	Message type (0x03 for this type)	
1	U1	version	-	-	Message version (0x00 for this version)	
2	U1[2]	reserved0	-	-	Reserved	
4	I2	a0G	2 ⁻³⁵	s	Constant term of the polynomial describing the offset	
6	I2	a1G	2 ⁻⁵¹	s/s	Rate of change of the offset	
8	U1	t0G	3600	s	Reference time for GGTO data	
9	U1	wn0G	-	weeks	Week Number of GGTO reference	
10	U1[2]	reserved1	-	-	Reserved	

3.13.6.4 Galileo UTC assistance

Message	UBX-MGA-GAL-UTC Galileo UTC assistance					
Type	Input					
Comment	This message allows the delivery of Galileo UTC assistance to a receiver. See section AssistNow Live Orbits in the integration manual for details.					

Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x13	0x02	20	see below	CK_A CK_B
<i>Payload description:</i>						
Byte offset	Type	Name	Scale	Unit	Description	
0	U1	type	-	-	Message type (0x05 for this type)	
1	U1	version	-	-	Message version (0x00 for this version)	
2	U1[2]	reserved0	-	-	Reserved	
4	I4	a0	2 [^] -30	s	First parameter of UTC polynomial	
8	I4	a1	2 [^] -50	s/s	Second parameter of UTC polynomial	
12	I1	dtLS	-	s	Delta time due to current leap seconds	
13	U1	tot	3600	s	UTC parameters reference time of week (Galileo time)	
14	U1	wnt	-	weeks	UTC parameters reference week number (the 8-bit WNT field)	
15	U1	wnLSF	-	weeks	Week number at the end of which the future leap second becomes effective (the 8-bit WNLSF field)	
16	U1	dN	-	days	Day number at the end of which the future leap second becomes effective	
17	I1	dTLSF	-	s	Delta time due to future leap seconds	
18	U1[2]	reserved1	-	-	Reserved	

3.13.7 UBX-MGA-GLO (0x13 0x06)

3.13.7.1 GLONASS ephemeris assistance

Message	UBX-MGA-GLO-EPH GLONASS ephemeris assistance					
Type	Input					
Comment	This message allows the delivery of GLONASS ephemeris assistance to a receiver. See section AssistNow Live Orbits in the integration manual for details.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x13	0x06	48	see below	CK_A CK_B
<i>Payload description:</i>						
Byte offset	Type	Name	Scale	Unit	Description	
0	U1	type	-	-	Message type (0x01 for this type)	
1	U1	version	-	-	Message version (0x00 for this version)	
2	U1	svId	-	-	GLONASS Satellite identifier (see Satellite Numbering)	
3	U1	reserved0	-	-	Reserved	
4	U1	FT	-	-	User range accuracy	
5	U1	B	-	-	Health flag from string 2	
6	U1	M	-	-	Type of GLONASS satellite (1 indicates GLONASS-M)	
7	I1	H	-	-	Carrier frequency number of navigation RF signal, Range=(-7 .. 6), -128 for unknown	
8	I4	x	2 [^] -11	km	X component of the SV position in PZ-90.02 coordinate System	
12	I4	y	2 [^] -11	km	Y component of the SV position in PZ-90.02 coordinate System	

16	I4	z	2 ⁻¹¹	km	Z component of the SV position in PZ-90.02 coordinate System
20	I4	dx	2 ⁻²⁰	km/s	X component of the SV velocity in PZ-90.02 coordinate System
24	I4	dy	2 ⁻²⁰	km/s	Y component of the SV velocity in PZ-90.02 coordinate System
28	I4	dz	2 ⁻²⁰	km/s	Z component of the SV velocity in PZ-90.02 coordinate System
32	I1	ddx	2 ⁻³⁰	km/s ²	X component of the SV acceleration in PZ-90.02 coordinate System
33	I1	ddy	2 ⁻³⁰	km/s ²	Y component of the SV acceleration in PZ-90.02 coordinate System
34	I1	ddz	2 ⁻³⁰	km/s ²	Z component of the SV acceleration in PZ-90.02 coordinate System
35	U1	tb	15	minutes	Index of a time interval within current day according to UTC(SU)
36	I2	gamma	2 ⁻⁴⁰	-	Relative carrier frequency deviation
38	U1	E	-	days	Ephemeris data age indicator
39	I1	deltaTau	2 ⁻³⁰	s	Time difference between L2 and L1 band
40	I4	tau	2 ⁻³⁰	s	SV clock bias
44	U1[4]	reserved1	-	-	Reserved

3.13.7.2 GLONASS almanac assistance

Message	UBX-MGA-GLO-ALM GLONASS almanac assistance					
Type	Input					
Comment	This message allows the delivery of GLONASS almanac assistance to a receiver. See section AssistNow Live Orbits in the integration manual for details.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x13	0x06	36	<i>see below</i>	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U1	type	-	-	Message type (0x02 for this type)	
1	U1	version	-	-	Message version (0x00 for this version)	
2	U1	svId	-	-	GLONASS Satellite identifier (see Satellite Numbering)	
3	U1	reserved0	-	-	Reserved	
4	U2	N	-	days	Reference calendar day number of almanac within the four-year period (from string 5)	
6	U1	M	-	-	Type of GLONASS satellite (1 indicates GLONASS-M)	
7	U1	C	-	-	Unhealthy flag at instant of almanac upload (1 indicates operability of satellite)	
8	I2	tau	2 ⁻¹⁸	s	Coarse time correction to GLONASS time	
10	U2	epsilon	2 ⁻²⁰	-	Eccentricity	
12	I4	lambda	2 ⁻²⁰	semi-circles	Longitude of the first (within the N-day) ascending node of satellite orbit in PZ-90 coordinate system	

16	I4	deltaI	2 ⁻²⁰	semi-circles	Correction to the mean value of inclination
20	U4	tLambda	2 ⁻⁵	s	Time of the first ascending node passage
24	I4	deltaT	2 ⁻⁹	s/orbital-period	Correction to the mean value of Draconian period
28	I1	deltaDT	2 ⁻¹⁴	s/orbital-period ²	Rate of change of Draconian period
29	I1	H	-	-	Carrier frequency number of navigation RF signal, Range=(-7 .. 6)
30	I2	omega	2 ⁻¹⁵	semi-circles	Argument of perigee
32	U1[4]	reserved1	-	-	Reserved

3.13.7.3 GLONASS auxiliary time offset assistance

Message	UBX-MGA-GLO-TIMEOFFSET GLONASS auxiliary time offset assistance					
Type	Input					
Comment	This message allows the delivery of auxiliary GLONASS assistance (including the GLONASS time offsets to other GNSS systems) to a receiver. See section AssistNow Live Orbits in the integration manual for details.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x13	0x06	20	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U1	type	-	-	Message type (0x03 for this type)	
1	U1	version	-	-	Message version (0x00 for this version)	
2	U2	N	-	days	Reference calendar day number within the four-year period of almanac (from string 5)	
4	I4	tauC	2 ⁻²⁷	s	Time scale correction to UTC(SU) time	
8	I4	tauGps	2 ⁻³¹	s	Correction to GPS time relative to GLONASS time	
12	I2	B1	2 ⁻¹⁰	s	Coefficient to determine delta UT1	
14	I2	B2	2 ⁻¹⁶	s/msd	Rate of change of delta UT1	
16	U1[4]	reserved0	-	-	Reserved	

3.13.8 UBX-MGA-GPS (0x13 0x00)

3.13.8.1 GPS ephemeris assistance

Message	UBX-MGA-GPS-EPH GPS ephemeris assistance					
Type	Input					
Comment	This message allows the delivery of GPS ephemeris assistance to a receiver. See section AssistNow Live Orbits in the integration manual for details.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x13	0x00	68	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U1	type	-	-	Message type (0x01 for this type)	

1	U1	version	-	-	Message version (0x00 for this version)
2	U1	svId	-	-	GPS Satellite identifier (see Satellite Numbering)
3	U1	reserved0	-	-	Reserved
4	U1	fitInterval	-	-	Fit interval flag
5	U1	uraIndex	-	-	URA index
6	U1	svHealth	-	-	SV health
7	I1	tgdc	2 ⁻³¹	s	Group delay differential
8	U2	iodc	-	-	IODC
10	U2	toc	2 ⁴	s	Clock data reference time
12	U1	reserved1	-	-	Reserved
13	I1	af2	2 ⁻⁵⁵	s/s squared	Time polynomial coefficient 2
14	I2	af1	2 ⁻⁴³	s/s	Time polynomial coefficient 1
16	I4	af0	2 ⁻³¹	s	Time polynomial coefficient 0
20	I2	crs	2 ⁻⁵	m	Crs
22	I2	deltaN	2 ⁻⁴³	semi-circles/s	Mean motion difference from computed value
24	I4	m0	2 ⁻³¹	semi-circles	Mean anomaly at reference time
28	I2	cuc	2 ⁻²⁹	radians	Amplitude of cosine harmonic correction term to argument of latitude
30	I2	cus	2 ⁻²⁹	radians	Amplitude of sine harmonic correction term to argument of latitude
32	U4	e	2 ⁻³³	-	Eccentricity
36	U4	sqrtA	2 ⁻¹⁹	m ^{0.5}	Square root of the semi-major axis
40	U2	toe	2 ⁴	s	Reference time of ephemeris
42	I2	cic	2 ⁻²⁹	radians	Amplitude of cos harmonic correction term to angle of inclination
44	I4	omega0	2 ⁻³¹	semi-circles	Longitude of ascending node of orbit plane at weekly epoch
48	I2	cis	2 ⁻²⁹	radians	Amplitude of sine harmonic correction term to angle of inclination
50	I2	crc	2 ⁻⁵	m	Amplitude of cosine harmonic correction term to orbit radius
52	I4	i0	2 ⁻³¹	semi-circles	Inclination angle at reference time
56	I4	omega	2 ⁻³¹	semi-circles	Argument of perigee
60	I4	omegaDot	2 ⁻⁴³	semi-circles/s	Rate of right ascension
64	I2	idot	2 ⁻⁴³	semi-circles/s	Rate of inclination angle
66	U1[2]	reserved2	-	-	Reserved

3.13.8.2 GPS almanac assistance

Message	UBX-MGA-GPS-ALM GPS almanac assistance
Type	Input

Comment	This message allows the delivery of GPS almanac assistance to a receiver. See section AssistNow Live Orbits in the integration manual for details.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x13	0x00	36	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U1	type	-	-	Message type (0x02 for this type)	
1	U1	version	-	-	Message version (0x00 for this version)	
2	U1	svId	-	-	GPS Satellite identifier (see Satellite Numbering)	
3	U1	svHealth	-	-	SV health information	
4	U2	e	2 ⁻²¹	-	Eccentricity	
6	U1	almWNa	-	week	Reference week number of almanac (the 8-bit WNa field)	
7	U1	toa	2 ¹²	s	Reference time of almanac	
8	I2	deltaI	2 ⁻¹⁹	semi-circles	Delta inclination angle at reference time	
10	I2	omegaDot	2 ⁻³⁸	semi-circles/s	Rate of right ascension	
12	U4	sqrtA	2 ⁻¹¹	m ^{0.5}	Square root of the semi-major axis	
16	I4	omega0	2 ⁻²³	semi-circles	Longitude of ascending node of orbit plane	
20	I4	omega	2 ⁻²³	semi-circles	Argument of perigee	
24	I4	m0	2 ⁻²³	semi-circles	Mean anomaly at reference time	
28	I2	af0	2 ⁻²⁰	s	Time polynomial coefficient 0 (8 MSBs)	
30	I2	af1	2 ⁻³⁸	s/s	Time polynomial coefficient 1	
32	U1[4]	reserved0	-	-	Reserved	

3.13.8.3 GPS health assistance

Message	UBX-MGA-GPS-HEALTH GPS health assistance					
Type	Input					
Comment	This message allows the delivery of GPS health assistance to a receiver. See section AssistNow Live Orbits in the integration manual for details.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x13	0x00	40	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U1	type	-	-	Message type (0x04 for this type)	
1	U1	version	-	-	Message version (0x00 for this version)	
2	U1[2]	reserved0	-	-	Reserved	
4	U1[32]	healthCode	-	-	Each byte represents a GPS SV (1-32). The 6 LSBs of each byte contains the 6 bit health code from subframes 4/5 page 25.	
36	U1[4]	reserved1	-	-	Reserved	

3.13.8.4 GPS UTC assistance

Message	UBX-MGA-GPS-UTC GPS UTC assistance					
Type	Input					
Comment	This message allows the delivery of GPS UTC assistance to a receiver. See section AssistNow Live Orbits in the integration manual for details.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x13	0x00	20	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U1	type		-	-	Message type (0x05 for this type)
1	U1	version		-	-	Message version (0x00 for this version)
2	U1[2]	reserved0		-	-	Reserved
4	I4	utcA0		2^-30	s	First parameter of UTC polynomial
8	I4	utcA1		2^-50	s/s	Second parameter of UTC polynomial
12	I1	utcDtLS		-	s	Delta time due to current leap seconds
13	U1	utcTot		2^12	s	UTC parameters reference time of week (GPS time)
14	U1	utcWNt		-	weeks	UTC parameters reference week number (the 8-bit WNt field)
15	U1	utcWNlsf		-	weeks	Week number at the end of which the future leap second becomes effective (the 8-bit WNLSF field)
16	U1	utcDn		-	days	Day number at the end of which the future leap second becomes effective
17	I1	utcDtLSF		-	s	Delta time due to future leap seconds
18	U1[2]	reserved1		-	-	Reserved

3.13.8.5 GPS ionosphere assistance

Message	UBX-MGA-GPS-IONO						
	GPS ionosphere assistance						
Type	Input						
Comment	This message allows the delivery of GPS ionospheric assistance to a receiver. See section AssistNow Live Orbits in the integration manual for details.						
Message structure	Header	Class	ID	Length (Bytes)		Payload	Checksum
	0xb5 0x62	0x13	0x00	16		see below	CK_A CK_B
Payload description:							
Byte offset	Type	Name		Scale	Unit	Description	
0	U1	type		-	-	Message type (0x06 for this type)	
1	U1	version		-	-	Message version (0x00 for this version)	
2	U1[2]	reserved0		-	-	Reserved	
4	I1	ionoAlpha0		2^-30	s	Ionospheric parameter alpha0 [s]	
5	I1	ionoAlpha1		2^-27	s/semi-circle	Ionospheric parameter alpha1 [s/semi-circle]	
6	I1	ionoAlpha2		2^-24	s/(semi-circle^2)	Ionospheric parameter alpha2 [s/semi-circle^2]	
7	I1	ionoAlpha3		2^-24	s/(semi-circle^3)	Ionospheric parameter alpha3 [s/semi-circle^3]	

8	I1	ionoBeta0	2 ¹¹	s	Ionospheric parameter beta0 [s]
9	I1	ionoBeta1	2 ¹⁴	s/semi-circle	Ionospheric parameter beta1 [s/semi-circle]
10	I1	ionoBeta2	2 ¹⁶	s/(semi-circle ²)	Ionospheric parameter beta2 [s/semi-circle ²]
11	I1	ionoBeta3	2 ¹⁶	s/(semi-circle ³)	Ionospheric parameter beta3 [s/semi-circle ³]
12	U1[4]	reserved1	-	-	Reserved

3.13.9 UBX-MGA-INI (0x13 0x40)

3.13.9.1 Initial position assistance XYZ

Message	UBX-MGA-INI-POS_XYZ Initial position assistance XYZ					
Type	Input					
Comment	This message allows the delivery of initial position assistance to a receiver in cartesian ECEF coordinates. This message is equivalent to the UBX-MGA-INI-POS_LLH message, except for the coordinate system. See section AssistNow Live Orbits in the integration manual for details. ↗ Supplying position assistance that is inaccurate by more than the specified position accuracy, may lead to substantially degraded receiver performance.					
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i>	<i>Checksum</i>
	0xb5 0x62	0x13	0x40	20	see below	CK_A CK_B
Payload description:						
<i>Byte offset</i>	<i>Type</i>	<i>Name</i>	<i>Scale</i>	<i>Unit</i>	<i>Description</i>	
0	U1	type	-	-	Message type (0x00 for this type)	
1	U1	version	-	-	Message version (0x00 for this version)	
2	U1[2]	reserved0	-	-	Reserved	
4	I4	ecefX	-	cm	WGS84 ECEF X coordinate	
8	I4	ecefY	-	cm	WGS84 ECEF Y coordinate	
12	I4	ecefZ	-	cm	WGS84 ECEF Z coordinate	
16	U4	posAcc	-	cm	Position accuracy (stddev)	

3.13.9.2 Initial position assistance LLH

Message	UBX-MGA-INI-POS_LLH Initial position assistance LLH					
Type	Input					
Comment	This message allows the delivery of initial position assistance to a receiver in WGS84 lat/long/alt coordinates. This message is equivalent to the UBX-MGA-INI-POS_XYZ message, except for the coordinate system. See section AssistNow Live Orbits in the integration manual for details. ↗ Supplying position assistance that is inaccurate by more than the specified position accuracy, may lead to substantially degraded receiver performance.					
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i>	<i>Checksum</i>
	0xb5 0x62	0x13	0x40	20	see below	CK_A CK_B
Payload description:						
<i>Byte offset</i>	<i>Type</i>	<i>Name</i>	<i>Scale</i>	<i>Unit</i>	<i>Description</i>	
0	U1	type	-	-	Message type (0x01 for this type)	
1	U1	version	-	-	Message version (0x00 for this version)	

2	U1[2]	reserved0	-	-	Reserved
4	I4	lat	1e-7	deg	WGS84 Latitude
8	I4	lon	1e-7	deg	WGS84 Longitude
12	I4	alt	-	cm	WGS84 Altitude
16	U4	posAcc	-	cm	Position accuracy (stddev)

3.13.9.3 Initial time assistance UTC

Message **UBX-MGA-INITIME-UTC**
Initial time assistance UTC

Type Input

Comment This message allows the delivery of UTC time assistance to a receiver. This message is equivalent to the [UBX-MGA-INITIME-GNSS](#) message, except for the time base.

See section AssistNow Live Orbits in the integration manual for details.

☞ Supplying time assistance that is inaccurate by more than the specified time accuracy, may lead to substantially degraded receiver performance.

Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x13	0x40	24	see below	CK_A CK_B

Payload description:

Byte offset	Type	Name	Scale	Unit	Description
0	U1	type	-	-	Message type (0x10 for this type)
1	U1	version	-	-	Message version (0x00 for this version)
2	X1	ref	-	-	Reference to be used to set time
bits 3...0	U:4	source	-	-	0 = none, i.e. on receipt of message (will be inaccurate!) 1 = relative to pulse sent to EXTINT0 2 = relative to pulse sent to EXTINT1 3-15 = reserved
bit 4	U:1	fall	-	-	use falling edge of EXTINT pulse (default rising) - only if source is EXTINT
bit 5	U:1	last	-	-	use last EXTINT pulse (default next pulse) - only if source is EXTINT
3	I1	leapSecs	-	s	Number of leap seconds since 1980 (or 0x80 = -128 if unknown)
4	U2	year	-	-	Year
6	U1	month	-	-	Month, starting at 1
7	U1	day	-	-	Day, starting at 1
8	U1	hour	-	-	Hour, from 0 to 23
9	U1	minute	-	-	Minute, from 0 to 59
10	U1	second	-	s	Seconds, from 0 to 59
11	X1	bitfield0	-	-	bitfield:
bit 0	U:1	trustedSource	-	-	Time is provided from a trusted source. Potentially usable for replay attack detection 0: Unknown 1: Time source can be trusted for spoofing detection
12	U4	ns	-	ns	Nanoseconds, from 0 to 999,999,999

16	U2	tAccS	-	s	Seconds part of time accuracy
18	U1[2]	reserved0	-	-	Reserved
20	U4	tAccNs	-	ns	Nanoseconds part of time accuracy, from 0 to 999,999,999

3.13.9.4 Initial time assistance GNSS

Message **UBX-MGA-INI-TIME_GNSS**
Initial time assistance GNSS

Type Input

Comment This message allows the delivery of time assistance to a receiver in a chosen GNSS timebase. This message is equivalent to the [UBX-MGA-INI-TIME_UTC](#) message, except for the time base.

See section AssistNow Live Orbits in the integration manual for details.

☞ Supplying time assistance that is inaccurate by more than the specified time accuracy, may lead to substantially degraded receiver performance.

Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x13	0x40	24	see below	CK_A CK_B

Payload description:

Byte offset	Type	Name	Scale	Unit	Description
0	U1	type	-	-	Message type (0x11 for this type)
1	U1	version	-	-	Message version (0x00 for this version)
2	X1	ref	-	-	Reference to be used to set time
bits 3...0	U:4	source	-	-	0 = none, i.e. on receipt of message (will be inaccurate!) 1 = relative to pulse sent to EXTINT0 2 = relative to pulse sent to EXTINT1 3-15 = reserved
bit 4	U:1	fall	-	-	use falling edge of EXTINT pulse (default rising) - only if source is EXTINT
bit 5	U:1	last	-	-	use last EXTINT pulse (default next pulse) - only if source is EXTINT
3	U1	gnssId	-	-	Source of time information. Currently supported: 0 = GPS time 2 = Galileo time 3 = BeiDou time 6 = GLONASS time 7 = NavIC time
4	X1	bitfield0	-	-	bitfield:
bit 0	U:1	trustedSource	-	-	Time is provided from a trusted source. Potentially usable for replay attack detection 0: Unknown 1: Time source can be trusted for spoofing detection
5	U1	reserved0	-	-	Reserved
6	U2	week	-	-	GNSS week number
8	U4	tow	-	s	GNSS time of week
12	U4	ns	-	ns	GNSS time of week, nanosecond part from 0 to 999,999,999

16	U2	tAccS	-	s	Seconds part of time accuracy
18	U1[2]	reserved1	-	-	Reserved
20	U4	tAccNs	-	ns	Nanoseconds part of time accuracy, from 0 to 999,999,999

3.13.9.5 Initial clock drift assistance

Message	UBX-MGA-INI-CLKD Initial clock drift assistance				
Type	Input				
Comment	This message allows the delivery of clock drift assistance to a receiver. See section AssistNow Live Orbits in the integration manual for details. ☞ Supplying clock drift assistance that is inaccurate by more than the specified accuracy, may lead to substantially degraded receiver performance.				
Message structure	Header	Class	ID	Length (Bytes)	Payload
	0xb5 0x62	0x13	0x40	12	see below
Checksum					
CK_A CK_B					
Payload description:					
Byte offset	Type	Name	Scale	Unit	Description
0	U1	type	-	-	Message type (0x20 for this type)
1	U1	version	-	-	Message version (0x00 for this version)
2	U1[2]	reserved0	-	-	Reserved
4	I4	clkD	-	ns/s	Clock drift
8	U4	clkDAcc	-	ns/s	Clock drift accuracy

3.13.9.6 Initial frequency assistance

Message	UBX-MGA-INI-FREQ Initial frequency assistance				
Type	Input				
Comment	This message allows the delivery of external frequency assistance to a receiver. See section AssistNow Live Orbits in the integration manual for details. ☞ Supplying external frequency assistance that is inaccurate by more than the specified accuracy, may lead to substantially degraded receiver performance.				
Message structure	Header	Class	ID	Length (Bytes)	Payload
	0xb5 0x62	0x13	0x40	12	see below
Checksum					
CK_A CK_B					
Payload description:					
Byte offset	Type	Name	Scale	Unit	Description
0	U1	type	-	-	Message type (0x21 for this type)
1	U1	version	-	-	Message version (0x00 for this version)
2	U1	reserved0	-	-	Reserved
3	X1	flags	-	-	Frequency reference
bits 3...0	U:4	source	-	-	0 = frequency available on EXTINT0 1 = frequency available on EXTINT1 2-15 = reserved
bit 4	U:1	fall	-	-	use falling edge of EXTINT pulse (default rising)
4	I4	freq	1e-2	Hz	Frequency
8	U4	freqAcc	-	ppb	Frequency accuracy

3.13.9.7 Earth orientation parameters assistance

Message	UBX-MGA-INIT-EOP						
Earth orientation parameters assistance							
Type	Input						
Comment	This message allows the delivery of new earth orientation parameters (EOP) to a receiver to improve AssistNow Autonomous operation.						
Message structure	Header	Class	ID	Length (Bytes)		Payload	Checksum
	0xb5 0x62	0x13	0x40	72		see below	CK_A CK_B
Payload description:							
Byte offset	Type	Name		Scale	Unit	Description	
0	U1	type		-	-	Message type (0x30 for this type)	
1	U1	version		-	-	Message version (0x00 for this version)	
2	U1[2]	reserved0		-	-	Reserved	
4	U2	d2kRef		-	d	reference time (days since 1.1.2000 12.00h UTC)	
6	U2	d2kMax		-	d	expiration time (days since 1.1.2000 12.00h UTC)	
8	I4	xpP0		2^-30	arcsec	x_p t^0 polynomial term (offset)	
12	I4	xpP1		2^-30	arcsec/d	x_p t^1 polynomial term (drift)	
16	I4	ypP0		2^-30	arcsec	y_p t^0 polynomial term (offset)	
20	I4	ypP1		2^-30	arcsec/d	y_p t^1 polynomial term (drift)	
24	I4	dUT1		2^-25	s	dUT1 t^0 polynomial term (offset)	
28	I4	ddUT1		2^-30	s/d	dUT1 t^1 polynomial term (drift)	
32	U1[40]	reserved1		-	-	Reserved	

3.13.10 UBX-MGA-QZSS (0x13 0x05)

3.13.10.1 QZSS ephemeris assistance

Message	UBX-MGA-QZSS-EPH						
	QZSS ephemeris assistance						
Type	Input						
Comment	This message allows the delivery of QZSS ephemeris assistance to a receiver. See section AssistNow Live Orbits in the integration manual for details.						
Message structure	Header	Class	ID	Length (Bytes)		Payload	Checksum
	0xb5 0x62	0x13	0x05	68		see below	CK_A CK_B
Payload description:							
Byte offset	Type	Name		Scale	Unit	Description	
0	U1	type		-	-	Message type (0x01 for this type)	
1	U1	version		-	-	Message version (0x00 for this version)	
2	U1	svId		-	-	QZSS Satellite identifier (see Satellite Numbering)	
3	U1	reserved0		-	-	Reserved	
4	U1	fitInterval		-	-	Fit interval flag	
5	U1	uraIndex		-	-	URA index	
6	U1	svHealth		-	-	SV health	
7	I1	tgd		2^-31	s	Group delay differential	
8	U2	iodc		-	-	IODC	

10	U2	toc	2 ⁴	s	Clock data reference time
12	U1	reserved1	-	-	Reserved
13	I1	af2	2 ⁻⁵⁵	s/s squared	Time polynomial coefficient 2
14	I2	af1	2 ⁻⁴³	s/s	Time polynomial coefficient 1
16	I4	af0	2 ⁻³¹	s	Time polynomial coefficient 0
20	I2	crs	2 ⁻⁵	m	Crs
22	I2	deltaN	2 ⁻⁴³	semi-circles/s	Mean motion difference from computed value
24	I4	m0	2 ⁻³¹	semi-circles	Mean anomaly at reference time
28	I2	cuc	2 ⁻²⁹	radians	Amp of cosine harmonic corr term to arg of lat
30	I2	cus	2 ⁻²⁹	radians	Amp of sine harmonic corr term to arg of lat
32	U4	e	2 ⁻³³	-	eccentricity
36	U4	sqrtA	2 ⁻¹⁹	m ^{0.5}	Square root of the semi-major axis A
40	U2	toe	2 ⁴	s	Reference time of ephemeris
42	I2	cic	2 ⁻²⁹	radians	Amp of cos harmonic corr term to angle of inclination
44	I4	omega0	2 ⁻³¹	semi-circles	Long of asc node of orbit plane at weekly epoch
48	I2	cis	2 ⁻²⁹	radians	Amp of sine harmonic corr term to angle of inclination
50	I2	crc	2 ⁻⁵	m	Amp of cosine harmonic corr term to orbit radius
52	I4	i0	2 ⁻³¹	semi-circles	Inclination angle at reference time
56	I4	omega	2 ⁻³¹	semi-circles	Argument of perigee
60	I4	omegaDot	2 ⁻⁴³	semi-circles/s	Rate of right ascension
64	I2	idot	2 ⁻⁴³	semi-circles/s	Rate of inclination angle
66	U1[2]	reserved2	-	-	Reserved

3.13.10.2 QZSS almanac assistance

Message	UBX-MGA-QZSS-ALM QZSS almanac assistance				
Type	Input				
Comment	This message allows the delivery of QZSS almanac assistance to a receiver. See section AssistNow Live Orbits in the integration manual for details.				
Message structure	Header	Class	ID	Length (Bytes)	Payload
	0xb5 0x62	0x13	0x05	36	see below
Payload description:					
Byte offset	Type	Name	Scale	Unit	Description
0	U1	type	-	-	Message type (0x02 for this type)
1	U1	version	-	-	Message version (0x00 for this version)
2	U1	svId	-	-	QZSS Satellite identifier (see Satellite Numbering)
3	U1	svHealth	-	-	Almanac SV health information
4	U2	e	2 ⁻²¹	-	Almanac eccentricity

6	U1	almWNa	-	week	Reference week number of almanac (the 8-bit WNa field)
7	U1	toa	2 ¹²	s	Reference time of almanac
8	I2	deltaI	2 ¹⁹	semi-circles	Delta inclination angle at reference time
10	I2	omegaDot	2 ³⁸	semi-circles/s	Almanac rate of right ascension
12	U4	sqrtA	2 ¹¹	m ^{0.5}	Almanac square root of the semi-major axis A
16	I4	omega0	2 ²³	semi-circles	Almanac long of asc node of orbit plane at weekly
20	I4	omega	2 ²³	semi-circles	Almanac argument of perigee
24	I4	m0	2 ²³	semi-circles	Almanac mean anomaly at reference time
28	I2	af0	2 ²⁰	s	Almanac time polynomial coefficient 0 (8 MSBs)
30	I2	af1	2 ³⁸	s/s	Almanac time polynomial coefficient 1
32	U1[4]	reserved0	-	-	Reserved

3.13.10.3 QZSS health assistance

Message	UBX-MGA-QZSS-HEALTH QZSS health assistance					
Type	Input					
Comment	This message allows the delivery of QZSS health assistance to a receiver. See section AssistNow Live Orbits in the integration manual for details.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x13	0x05	12	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U1	type	-	-	Message type (0x04 for this type)	
1	U1	version	-	-	Message version (0x00 for this version)	
2	U1[2]	reserved0	-	-	Reserved	
4	U1[5]	healthCode	-	-	Each byte represents a QZSS SV (1-5). The 6 LSBs of each byte contains the 6 bit health code from subframes 4/5, data ID = 3, SV ID = 51	
9	U1[3]	reserved1	-	-	Reserved	

3.14 UBX-MON (0x0a)

The messages in the UBX-MON class are used to report the receiver status, such as hardware status or I/O subsystem statistics.

3.14.1 UBX-MON-BATCH (0x0a 0x32)

3.14.1.1 Data batching buffer status

Message	UBX-MON-BATCH Data batching buffer status
Type	Polled
Comment	This message contains status information about the batching buffer.

It can be polled and it can also be sent by the receiver as a response to a [UBX-LOG-RETRIEVEBATCH](#) message before the UBX-LOG-BATCH messages.

See Data batching section in the integration manual for more information.

Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x0a	0x32	12	see below	CK_A CK_B

Payload description:

Byte offset	Type	Name	Scale	Unit	Description
0	U1	version	-	-	Message version (0x00 for this version)
1	U1[3]	reserved0	-	-	Reserved
4	U2	fillLevel	-	-	Current buffer fill level, i.e. number of epochs currently stored
6	U2	dropsAll	-	-	Number of dropped epochs since startup Note: changing the batching configuration will reset this counter.
8	U2	dropsSinceMon	-	-	Number of dropped epochs since last MON-BATCH message
10	U2	nextMsgCnt	-	-	The next retrieved UBX-LOG-BATCH will have this msgCnt value.

3.14.2 UBX-MON-COMMS (0x0a 0x36)

3.14.2.1 Communication port information

Message	UBX-MON-COMMS Communication port information					
Type	Periodic/polled					
Comment	Consolidated communications information for all ports. The size of the message is determined by the number of ports that are in use on the receiver. A port is only included if communication, either send or receive, has been initiated on that port.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x0a	0x36	8 + nPorts·40	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U1	version	-	-	Message version (0x00 for this version)	
1	U1	nPorts	-	-	Number of ports included	
2	X1	txErrors	-	-	TX error bitmask	
bit 0	U:1	mem	-	-	Memory Allocation error	
bit 1	U:1	alloc	-	-	Allocation error (TX buffer full)	
bits 4...2	U:3	outputPort	-	-	Output port: Reports the port from which this message was output from. 0 = N/A 1 = I2C 2 = UART1 3 = UART2 4 = USB 5 = SPI	
3	U1	reserved0	-	-	Reserved	

4	U1[4]	protIds	-		The identifiers of the protocols reported in the msgs array. 0: UBX, 1: NMEA, 2: RTCM2, 5: RTCM3, 6: SPARTN, 0xFF: No protocol reported.
<i>Start of repeated group (nPorts times)</i>					
8 + n·40	U2	portId	-	-	Unique identifier for the port. See section Communications ports in the integration manual for details.
10 + n·40	U2	txPending	-	bytes	Number of bytes pending in transmitter buffer
12 + n·40	U4	txBytes	-	bytes	Number of bytes ever sent
16 + n·40	U1	txUsage	-	%	Maximum usage transmitter buffer during the last sysmon period
17 + n·40	U1	txPeakUsage	-	%	Maximum usage transmitter buffer
18 + n·40	U2	rxPending	-	bytes	Number of bytes in receiver buffer
20 + n·40	U4	rxBytes	-	bytes	Number of bytes ever received
24 + n·40	U1	rxUsage	-	%	Maximum usage receiver buffer during the last sysmon period
25 + n·40	U1	rxPeakUsage	-	%	Maximum usage receiver buffer
26 + n·40	U2	overrunErrs	-	-	Number of 100 ms timeslots with overrun errors
28 + n·40	U2[4]	msgs	-	msg	Number of successfully parsed messages for each protocol. The reported protocols are identified through the protIds field.
36 + n·40	U1[8]	reserved1	-	-	Reserved
44 + n·40	U4	skipped	-	bytes	Number of skipped bytes
<i>End of repeated group (nPorts times)</i>					

3.14.3 UBX-MON-GNSS (0x0a 0x28)

3.14.3.1 Information message major GNSS selection

Message UBX-MON-GNSS						
Information message major GNSS selection						
Type	Polled					
Comment	This message reports major GNSS selection. It does this by means of bit masks in U1 fields. Each bit in a bit mask corresponds to one major GNSS. Augmentation systems are not reported.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x0a	0x28	8	see below	CK_A CK_B
<i>Payload description:</i>						
Byte offset	Type	Name	Scale	Unit	Description	
0	U1	version	-	-	Message version (0x00 for this version)	
1	X1	supported	-	-	A bit mask showing the major GNSS that can be supported by this receiver	
bit 0	U:1	GPSSup	-	-	GPS is supported	
bit 1	U:1	GlonassSup	-	-	GLONASS is supported	
bit 2	U:1	BeidouSup	-	-	BeiDou is supported	
bit 3	U:1	GalileoSup	-	-	Galileo is supported	

2	X1	defaultGnss	-	-	A bit mask showing the default major GNSS selection. If the default major GNSS selection is currently configured in the OTP memory for this receiver, it takes precedence over the default major GNSS selection configured in the executing firmware of this receiver.
bit 0	U ₁	GPSTDef	-	-	GPS is default-enabled
bit 1	U ₁	GlonassDef	-	-	GLONASS is default-enabled
bit 2	U ₁	BeidouDef	-	-	BeiDou is default-enabled
bit 3	U ₁	GalileoDef	-	-	Galileo is default-enabled
3	X1	enabled	-	-	A bit mask showing the current major GNSS selection enabled for this receiver
bit 0	U ₁	GPSEna	-	-	GPS is enabled
bit 1	U ₁	GlonassEna	-	-	GLONASS is enabled
bit 2	U ₁	BeidouEna	-	-	BeiDou is enabled
bit 3	U ₁	GalileoEna	-	-	Galileo is enabled
4	U1	simultaneous	-	-	Maximum number of concurrent major GNSS that can be supported by this receiver
5	U1[3]	reserved0	-	-	Reserved

3.14.3.2 Information message with supported GNSS and signal plans

Message	UBX-MON-GNSS					
Information message with supported GNSS and signal plans						
Type	Polled					
Comment						
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x0a	0x28	4 + numPlans·28	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U1	version	-	-	Message version (0x01 for this version)	
1	U1	numPlans	-	-	Number of signal plans available	
2	X2	activePlanInfo	-	-	Information about currently active signal plan	
bits 7...0	U:8	planCfgId	-	-	Configuration value of the currently active plan	
bits 10...8	U:3	gnssCfgLevel	-	-	Configuration level of currently active plan/ constellation/signal combination 0 - N/A 1 - Standard (currently active constellation/signal combination is supported) 2 - Advanced (an advanced constellation/signal combination is selected; performance may be sub- optimal)	
Start of repeated group (numPlans times)						
4 + n·28	U1	id	-	-	Plan identifier	
5 + n·28	CH[5]	name	-	-	Plan name	

10 + n·28	X2	gpsSup	-	-	GPS supported signals A non-zero value indicates GPS is supported A zero value indicates GPS is not supported
bit 0	U ₁	gps_L1CA	-	-	GPS L1C/A
bit 1	U ₁	gps_L1C	-	-	GPS L1C
bit 2	U ₁	gps_L2C	-	-	GPS L2C
bit 3	U ₁	gps_L5	-	-	GPS L5
12 + n·28	X2	galSup	-	-	GAL supported signals A non-zero value indicates Galileo is supported A zero value indicates Galileo is not supported
bit 0	U ₁	gal_E1	-	-	Galileo E1
bit 1	U ₁	gal_E5A	-	-	Galileo E5a
bit 2	U ₁	gal_E5B	-	-	Galileo E5b
bit 3	U ₁	gal_E6	-	-	Galileo E6
14 + n·28	X2	bdsSup	-	-	BDS supported signals A non-zero value indicates BeiDou is supported A zero value indicates BeiDou is not supported
bit 0	U ₁	bds_B1I	-	-	BeiDou B1I
bit 1	U ₁	bds_B1C	-	-	BeiDou B1C
bit 2	U ₁	bds_B2I	-	-	BeiDou B2I
bit 3	U ₁	bds_B2A	-	-	BeiDou B2a
bit 4	U ₁	bds_B3I	-	-	BeiDou B3I
16 + n·28	X2	gloSup	-	-	GLONASS supported signals A non-zero value indicates GLONASS is supported A zero value indicates GLONASS is not supported
bit 0	U ₁	glo_L1OF	-	-	GLONASS L1OF
bit 1	U ₁	glo_L2OF	-	-	GLONASS L2OF
18 + n·28	X2	sbasSup	-	-	SBAS supported signals A non-zero value indicates SBAS is supported A zero value indicates SBAS is not supported
bit 0	U ₁	sbas_L1CA	-	-	SBAS L1C/A
20 + n·28	X2	qzssSup	-	-	QZSS supported signals A non-zero value indicates QZSS is supported A zero value indicates QZSS is not supported
bit 0	U ₁	qzss_L1CA	-	-	QZSS L1C/A
bit 1	U ₁	qzss_L1CB	-	-	QZSS L1C/B
bit 2	U ₁	qzss_L1C	-	-	QZSS L1C
bit 3	U ₁	qzss_L1S	-	-	QZSS L1S
bit 4	U ₁	qzss_L2C	-	-	QZSS L2C

bit 5	U:1	qzss_L5	-	-	QZSS L5
22 + n·28	X2	navicSup	-	-	NavIC supported signals A non-zero value indicates NavIC is supported A zero value indicates NavIC is not supported
bit 0	U:1	navic_L5	-	-	NavIC L5
24 + n·28	X2	lbandSup	-	-	Lband supported A non-zero value indicates Lband is supported A zero value indicates Lband is not supported
bit 0	U:1	lband_L1	-	-	supported range: 1539-1559 MHz
26 + n·28	U1[6]	reserved0	-	-	Reserved
<i>End of repeated group (numPlans times)</i>					

3.14.4 UBX-MON-HW3 (0x0a 0x37)

3.14.4.1 I/O pin status

Message	UBX-MON-HW3 I/O pin status				
Type	Periodic/pollled				
Comment	This message contains information specific to each HW I/O pin, for example whether the pin is set as Input or Output. For the antenna supervisor status and other RF status information, see the UBX-MON-RF message.				
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i> <i>Checksum</i>
	0xb5 0x62	0x0a	0x37	22 + nPins·6	see below CK_A CK_B
<i>Payload description:</i>					
<i>Byte offset</i>	<i>Type</i>	<i>Name</i>	<i>Scale</i>	<i>Unit</i>	<i>Description</i>
0	U1	version	-	-	Message version (0x00 for this version)
1	U1	nPins	-	-	The number of I/O pins included
2	X1	flags	-	-	Flags
bit 0	U:1	rtcCalib	-	-	RTC is calibrated
bit 1	U:1	safeBoot	-	-	Safeboot mode (0 = inactive, 1 = active)
bit 2	U:1	xtalAbsent	-	-	RTC xtal has been determined to be absent
3	CH[10]	hwVersion	-	-	Zero-terminated hardware version string (same as that returned in the UBX-MON-VER message)
13	U1[9]	reserved0	-	-	Reserved
<i>Start of repeated group (nPins times)</i>					
22 + n·6	U1	reserved1	-	-	Reserved
23 + n·6	U1	pinId	-	-	Identifier for the pin, including both external and internal pins
24 + n·6	X2	pinMask	-	-	Pin mask
bit 0	U:1	periphPIO	-	-	Pin is set to peripheral or PIO? 0=Peripheral 1=PIO
bits 3...1	U:3	pinBank	-	-	Bank the pin belongs to, where 0=A 1=B 2=C 3=D 4=E 5=F 6=G 7=H
bit 4	U:1	direction	-	-	Pin direction? 0=Input 1=Output

bit 5	U:1	value	-	-	Pin value? 0=Low 1=High
bit 6	U:1	vpManager	-	-	Used by virtual pin manager? 0=No 1=Yes
bit 7	U:1	pioIrq	-	-	Interrupt enabled? 0=No 1=Yes
bit 8	U:1	pioPullHigh	-	-	Using pull high resistor? 0=No 1=Yes
bit 9	U:1	pioPullLow	-	-	Using pull low resistor 0=No 1=Yes
26 + n·6	U1	VP	-	-	Virtual pin mapping
27 + n·6	U1	reserved2	-	-	Reserved
<i>End of repeated group (nPins times)</i>					

3.14.5 UBX-MON-IO (0x0a 0x02)

3.14.5.1 I/O system status

Message	UBX-MON-IO				
	I/O system status				
Type	Periodic/pollled				
Comment	The size of the message is determined by the number of ports 'N' the receiver supports, i.e. on u-blox 5 the number of ports is 6.				
Message structure	Header	Class	ID	Length (Bytes)	Payload
	0xb5 0x62	0x0a	0x02	[0..n]·20	see below
<i>Payload description:</i>					
Byte offset	Type	Name	Scale	Unit	Description
<i>Start of repeated group (N times)</i>					
0 + n·20	U4	rxBytes	-	bytes	Number of bytes ever received
4 + n·20	U4	txBytes	-	bytes	Number of bytes ever sent
8 + n·20	U2	parityErrs	-	-	Number of 100 ms timeslots with parity errors
10 + n·20	U2	framingErrs	-	-	Number of 100 ms timeslots with framing errors
12 + n·20	U2	overrunErrs	-	-	Number of 100 ms timeslots with overrun errors
14 + n·20	U2	breakCond	-	-	Number of 100 ms timeslots with break conditions
16 + n·20	U1[4]	reserved0	-	-	Reserved
<i>End of repeated group (N times)</i>					

3.14.6 UBX-MON-MSGPP (0x0a 0x06)

3.14.6.1 Message parse and process status

Message	UBX-MON-MSGPP				
	Message parse and process status				
Type	Periodic/pollled				
Comment					
Message structure	Header	Class	ID	Length (Bytes)	Payload
	0xb5 0x62	0x0a	0x06	120	see below
<i>Payload description:</i>					
Byte offset	Type	Name	Scale	Unit	Description

0	U2[8]	msg1	-	msgs	Number of successfully parsed messages for each protocol on port0
16	U2[8]	msg2	-	msgs	Number of successfully parsed messages for each protocol on port1
32	U2[8]	msg3	-	msgs	Number of successfully parsed messages for each protocol on port2
48	U2[8]	msg4	-	msgs	Number of successfully parsed messages for each protocol on port3
64	U2[8]	msg5	-	msgs	Number of successfully parsed messages for each protocol on port4
80	U2[8]	msg6	-	msgs	Number of successfully parsed messages for each protocol on port5
96	U4[6]	skipped	-	bytes	Number skipped bytes for each port

3.14.7 UBX-MON-PATCH (0x0a 0x27)

3.14.7.1 Poll request for installed patches

Message	UBX-MON-PATCH Poll request for installed patches					
Type	Poll request					
Comment						
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x0a	0x27	0	see below	CK_A CK_B
Payload	This message has no payload.					

3.14.7.2 Installed patches

Message	UBX-MON-PATCH						
	Installed patches						
Type	Polled						
Comment	This message reports information about patches installed and currently enabled on the receiver. It does not report on patches installed and then disabled. An enabled patch is considered active when the receiver executes from the code space where the patch resides on. For example, a ROM patch is reported active only when the system runs from ROM.						
Message structure	Header	Class	ID	Length (Bytes)		Payload	Checksum
	0xb5 0x62	0x0a	0x27	4 + nEntries·16		see below	CK_A CK_B
Payload description:							
Byte offset	Type	Name		Scale	Unit	Description	
0	U2	version		-	-	Message version (0x0001 for this version)	
2	U2	nEntries		-	-	Total number of reported patches	
Start of repeated group (nEntries times)							
4 + n·16	X4	patchInfo		-	-	Status information about the reported patch	
	bit 0	U ₁	activated	-	-	1: the patch is active, 0: otherwise	
	bits 2...1	U ₂	location	-	-	Indicates where the patch is stored. 0: OTP, 1: ROM, 2: BBR, 3: file system	
8 + n·16	U4	comparator Number		-	-	The number of the comparator	
12 + n·16	U4	patchAddress		-	-	The address that is targeted by the patch	

16 + n·16	U4	patchData	-	-	The data that is inserted at the patchAddress
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End of repeated group (*nEntries* times)

3.14.8 UBX-MON-POST (0x0a 0x3b)

3.14.8.1 Power on self test (POST) information

Message	UBX-MON-POST Power on self test (POST) information					
Type	Polled					
Comment	Information on power on self					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x0a	0x3b	12	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U1	version	-	-	Message version (0x00 for this version)	
1	X1	flags	-	-	Flags	
bit 0	U:1	safeBoot	-	-	Safeboot mode (0 = inactive, 1 = active)	
2	U1[2]	reserved0	-	-	Reserved	
4	U4	postStatus	-	-	POST status word	
8	U1[4]	reserved1	-	-	Reserved	

3.14.9 UBX-MON-RF (0x0a 0x38)

3.14.9.1 RF information

Message	UBX-MON-RF					
	RF information					
Type	Periodic/pollled					
Comment	Information for each RF block. There are as many RF blocks reported as bands supported by this receiver.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x0a	0x38	4 + nBlocks·24	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U1	version		-	-	Message version (0x00 for this version)
1	U1	nBlocks		-	-	The number of RF blocks included
2	X1	recInf		-	-	Receiver information
bits 1...0	U:2	msgSource		-	-	0: Single antenna 1: Antenna 1 (RF_IN_1) of dual antenna heading module 2: Antenna 2 (RF_IN_2) of dual antenna heading module 3: Reserved
3	U1	reserved0		-	-	Reserved
Start of repeated group (nBlocks times)						

4 + n·24	U1	blockId	-	-	RF block ID, this is a unique value assigned to every RF block. The band which the RF block represents is subject to product configuration.
5 + n·24	X1	flags	-	-	Flags
bits 1...0	U:2	jammingState	-	-	Output from jamming/interference monitor (0 = unknown or feature disabled or flag unavailable, 1 = ok - no significant jamming, 2 = warning - interference visible but fix OK, 3 = critical - interference visible and no fix). This flag is deprecated in protocol versions that support UBX-SEC-SIG (version 0x02) and always reported as 0; instead jammingState in UBX-SEC-SIG should be monitored.
6 + n·24	U1	antStatus	-	-	Status of the antenna supervisor state machine (0x00=INIT, 0x01=DONTKNOW, 0x02=OK, 0x03=SHORT, 0x04=OPEN)
7 + n·24	U1	antPower	-	-	Current power status of antenna (0x00=OFF, 0x01=ON, 0x02=DONTKNOW)
8 + n·24	U4	postStatus	-	-	POST status word
12 + n·24	U1[4]	reserved1	-	-	Reserved
16 + n·24	U2	noisePerMS	-	-	Noise level as measured by the GPS core
18 + n·24	U2	agcCnt	-	-	AGC Monitor, as percentage of maximum gain, range 0 to 8191 (100%)
20 + n·24	U1	cwSuppression	-	-	CW interference suppression level, scaled (0=no CW jamming, 255 = strong CW jamming)
21 + n·24	I1	ofsI	-	-	Imbalance of I-part of complex signal, scaled (-128 = max. negative imbalance, 127 = max. positive imbalance)
22 + n·24	U1	magI	-	-	Magnitude of I-part of complex signal, scaled (0 = no signal, 255 = max.magnitude)
23 + n·24	I1	ofsQ	-	-	Imbalance of Q-part of complex signal, scaled (-128 = max. negative imbalance, 127 = max. positive imbalance)
24 + n·24	U1	magQ	-	-	Magnitude of Q-part of complex signal, scaled (0 = no signal, 255 = max.magnitude)
25 + n·24	U1	rfBlockGnss Band	-	-	GNSS band associated with the reported RF block. 0 - unknown band or unsupported field 1 - L1 band 2 - L2 band 3 - L3 band 4 - L5 band
26 + n·24	U1[2]	reserved2	-	-	Reserved
End of repeated group (nBlocks times)					

3.14.10 UBX-MON-RXBUF (0x0a 0x07)

3.14.10.1 Receiver buffer status

Message	UBX-MON-RXBUF Receiver buffer status					
Type	Periodic/pollled					
Comment						
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x0a	0x07	24	see below	CK_A CK_B

Payload description:

Byte offset	Type	Name	Scale	Unit	Description
0	U2[6]	pending	-	bytes	Number of bytes pending in receiver buffer for each target
12	U1[6]	usage	-	%	Maximum usage receiver buffer during the last sysmon period for each target
18	U1[6]	peakUsage	-	%	Maximum usage receiver buffer for each target

3.14.11 UBX-MON-RXR (0x0a 0x21)

3.14.11.1 Receiver status information

Message	UBX-MON-RXR					
	Receiver status information					
Type	Output					
Comment	The receiver ready message is sent when the receiver changes from or to backup mode.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x0a	0x21	1	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	X1	flags	-	-	Receiver status flags	
bit 0	U:1	awake	-	-	not in backup mode	

3.14.12 UBX-MON-SPAN (0x0a 0x31)

3.14.12.1 Signal characteristics

Message	UBX-MON-SPAN					
	Signal characteristics					
Type	Periodic/poll					
Comment	This message is to be used as a basic spectrum analyzer, where it displays one spectrum for each of the receiver's existing RF paths. The spectrum is conveyed with the following parameters: The frequency span in Hz, the frequency bin resolution in Hz, the center frequency in Hz, and 256 bins with amplitude data. Additionally, in order to give further insight on the signal captured by the receiver, the current gain of the internal programmable gain amplifier (PGA) is provided. This message gives information for comparative analysis rather than absolute and precise spectrum overview. Users should not expect highly accurate spectrum amplitude. Note that the PGA gain is not included in the spectrum data but is available as a separate field. Neither the spectrum, nor the PGA gain considers the internal fixed LNA gain or an external third-party LNA. The center frequency at each bin, assuming a zero-based bin count, can be computed as $f(i) = center + span * (i - 127) / 256$					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x0a	0x31	4 + numRfBlocks*272	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U1	version	-	-	Message version (0x00 for this version)	
1	U1	numRfBlocks	-	-	Number of RF blocks included	
2	X1	recInf	-	-	Receiver information	
bits 1...0	U:2	msgSource	-	-	0: Single antenna	

- 1: Antenna 1 (RF_IN_1) of dual antenna heading module
2: Antenna 2 (RF_IN_2) of dual antenna heading module
3: Reserved

3	U1	reserved0	-	-	Reserved
<i>Start of repeated group (numRfBlocks times)</i>					
4 + n·272	U1[256]	spectrum	2 ⁻²	dB	Spectrum data (number of points = span/res) [Uuu.fff dB]
260 + n·272	U4	span	-	Hz	Spectrum span
264 + n·272	U4	res	-	Hz	Resolution of the spectrum
268 + n·272	U4	center	-	Hz	Center of spectrum span
272 + n·272	U1	pga	-	dB	Programmable gain amplifier
273 + n·272	U1[3]	reserved1	-	-	Reserved
<i>End of repeated group (numRfBlocks times)</i>					

3.14.13 UBX-MON-SYS (0x0a 0x39)

3.14.13.1 Current system performance information

Message	UBX-MON-SYS				
	Current system performance information				
Type	Periodic/pollled				
Comment	This message contains operationally relevant system information for monitoring purposes. cpuLoadMax value is only valid, if 1 second output frequency is set. Detailed information about ioUsage/ioUsageMax are available in UBX-MON-COMMS message. tempValue has an accuracy of +/- 2 deg.				
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i>
	0xb5 0x62	0x0a	0x39	24	see below
<i>Payload description:</i>					
<i>Byte offset</i>	<i>Type</i>	<i>Name</i>	<i>Scale</i>	<i>Unit</i>	<i>Description</i>
0	U1	msgVer	-	-	Message Version (0x01)
1	U1	bootType	-	-	Boot type system 0-Unknown 1-Cold Start 2-Watchdog 3-Hardware reset 4-Hardware backup 5-Software standby 6-Software reset 7-VIO fail 8-VDD_X fail 9-VDD_RF fail 10-V_CORE_HIGH fail 11-System reset
2	U1	cpuLoad	-	-	Highest actual load of realtime tasks of all CPUs in %
3	U1	cpuLoadMax	-	-	Maximal CPU load value in % seen since last restart

4	U1	memUsage	-	-	Highest actual dynamic memory usage of all CPUs in %
5	U1	memUsageMax	-	-	Maximal dynamic memory usage in % seen since last restart
6	U1	ioUsage	-	-	Highest actual IO bandwidth usage of all rx/tx interfaces in %
7	U1	ioUsageMax	-	-	Maximal bandwidth usage of all rx/tx interfaces in % seen since last restart
8	U4	runTime	-	sec	Time since the last receiver startup
12	U2	noticeCount	-	-	Number of notices occurred since last restart
14	U2	warnCount	-	-	Number of warnings occurred since last restart
16	U2	errorCount	-	-	Number of errors occurred since last restart
18	I1	tempValue	-	-	Temperature value [C]
19	U1[5]	reserved0	-	-	Reserved

3.14.14 UBX-MON-TXBUF (0x0a 0x08)

3.14.14.1 Transmitter buffer status

Message		UBX-MON-TXBUF				
		Transmitter buffer status				
Type		Periodic/pollled				
Comment						
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x0a	0x08	28	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U2[6]	pending	-	bytes	Number of bytes pending in transmitter buffer for each target	
12	U1[6]	usage	-	%	Maximum usage transmitter buffer during the last sysmon period for each target	
18	U1[6]	peakUsage	-	%	Maximum usage transmitter buffer for each target	
24	U1	tUsage	-	%	Maximum usage of transmitter buffer during the last sysmon period for all targets	
25	U1	tPeakusage	-	%	Maximum usage of transmitter buffer for all targets	
26	X1	errors	-	-	Error bitmask	
bits 5...0	U:6	limit	-	-	Buffer limit of corresponding target reached	
	bit 6 U:1	mem	-	-	Memory Allocation error	
	bit 7 U:1	alloc	-	-	Allocation error (TX buffer full)	
27	U1	reserved0	-	-	Reserved	

3.14.15 UBX-MON-VER (0x0a 0x04)

3.14.15.1 Poll receiver and software version

Message	UBX-MON-VER Poll receiver and software version					
Type	Poll request					
Comment						
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x0a	0x04	0	see below	CK_A CK_B
Payload	This message has no payload.					

3.14.15.2 Receiver and software version

Message	UBX-MON-VER					
	Receiver and software version					
Type	Polled					
Comment						
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x0a	0x04	40 + [0..n]·30	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	CH[30]	swVersion		-	-	Nul-terminated software version string.
30	CH[10]	hwVersion		-	-	Nul-terminated hardware version string
Start of repeated group (N times)						
40 + n·30	CH[30]	extension		-	-	Extended software information strings. A series of nul-terminated strings. Each extension field is 30 characters long and contains varying software information. Not all extension fields may appear. Examples of reported information: the software version string of the underlying ROM (when the receiver's firmware is running from flash), the firmware version, the supported protocol version , the module identifier, the flash information structure (FIS) file information, the supported major GNSS, the supported augmentation systems. See Firmware and protocol versions for details.
End of repeated group (N times)						

3.15 UBX-NAV (0x01)

The messages in the UBX-NAV class are used to output navigation results and data, such as position, altitude and velocity in a number of formats, and status flags and accuracy estimate figures, or satellite and signal information. The messages are generated with the configured navigation rate.

3.15.1 UBX-NAV-AOPSTATUS (0x01 0x60)

3.15.1.1 AssistNow Autonomous status

Message	UBX-NAV-AOPSTATUS AssistNow Autonomous status					
Type	Periodic/pollled					

Comment	This message provides information on the status of the <i>AssistNow Autonomous</i> subsystem on the receiver. For example, a host application can determine the optimal time to shut down the receiver by monitoring the <code>status</code> field for a steady 0. See the integration manual for details on this feature.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x01	0x60	16	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U4	iTOW		-	ms	GPS time of week of the navigation epoch. See the description of iTOW for details.
4	U1	aopCfg		-	-	AssistNow Autonomous configuration
	bit 0 U ₁	useAOP		-	-	AOP enabled flag
5	U1	status		-	-	AssistNow Autonomous subsystem is idle (0) or running (not 0)
6	U1[10]	reserved0		-	-	Reserved

3.15.2 UBX-NAV-CLOCK (0x01 0x22)

3.15.2.1 Clock solution

Message	UBX-NAV-CLOCK					
	Clock solution					
Type	Periodic/pollled					
Comment						
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x01	0x22	20	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U4	iTOW		-	ms	GPS time of week of the navigation epoch. See section Navigation epochs in the integration manual for details. See section iTOW timestamps in the integration manual for details.
4	I4	clkB		-	ns	Clock bias
8	I4	clkD		-	ns/s	Clock drift
12	U4	tAcc		-	ns	Time accuracy estimate
16	U4	fAcc		-	ps/s	Frequency accuracy estimate

3.15.3 UBX-NAV-COV (0x01 0x36)

3.15.3.1 Covariance matrices

Message	UBX-NAV-COV					
	Covariance matrices					
Type	Periodic/pollled					
Comment	This message outputs the covariance matrices for the position and velocity solutions in the topocentric coordinate system defined as the local-level North (N), East (E), Down (D) frame. As the covariance matrices are symmetric, only the upper triangular part is output.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x01	0x36	64	see below	CK_A CK_B

Payload description:

Byte offset	Type	Name	Scale	Unit	Description
0	U4	iTOW	-	ms	GPS time of week of the navigation epoch. See section iTOW timestamps in the integration manual for details.
4	U1	version	-	-	Message version (0x00 for this version)
5	U1	posCovValid	-	-	Position covariance matrix validity flag
6	U1	velCovValid	-	-	Velocity covariance matrix validity flag
7	U1[9]	reserved0	-	-	Reserved
16	R4	posCovNN	-	m ²	Position covariance matrix value p _{NN}
20	R4	posCovNE	-	m ²	Position covariance matrix value p _{NE}
24	R4	posCovND	-	m ²	Position covariance matrix value p _{ND}
28	R4	posCovEE	-	m ²	Position covariance matrix value p _{EE}
32	R4	posCovED	-	m ²	Position covariance matrix value p _{ED}
36	R4	posCovDD	-	m ²	Position covariance matrix value p _{DD}
40	R4	velCovNN	-	m ² /s ²	Velocity covariance matrix value v _{NN}
44	R4	velCovNE	-	m ² /s ²	Velocity covariance matrix value v _{NE}
48	R4	velCovND	-	m ² /s ²	Velocity covariance matrix value v _{ND}
52	R4	velCovEE	-	m ² /s ²	Velocity covariance matrix value v _{EE}
56	R4	velCovED	-	m ² /s ²	Velocity covariance matrix value v _{ED}
60	R4	velCovDD	-	m ² /s ²	Velocity covariance matrix value v _{DD}

3.15.4 UBX-NAV-DOP (0x01 0x04)

3.15.4.1 Dilution of precision

Message	UBX-NAV-DOP Dilution of precision					
Type	Periodic/pollled					
Comment	- DOP values are dimensionless. - All DOP values are scaled by a factor of 100. If the unit transmits a value of e.g. 156, the DOP value is 1.56.					
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i>	<i>Checksum</i>
	0xb5 0x62	0x01	0x04	18	see below	CK_A CK_B

Payload description:

Byte offset	Type	Name	Scale	Unit	Description
0	U4	iTOW	-	ms	GPS time of week of the navigation epoch. See section iTOW timestamps in the integration manual for details.
4	U2	gDOP	0.01	-	Geometric DOP
6	U2	pDOP	0.01	-	Position DOP
8	U2	tDOP	0.01	-	Time DOP
10	U2	vDOP	0.01	-	Vertical DOP
12	U2	hDOP	0.01	-	Horizontal DOP
14	U2	nDOP	0.01	-	Northing DOP

16	U2	eDOP	0.01	-	Easting DOP
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3.15.5 UBX-NAV-EOE (0x01 0x61)

3.15.5.1 End of epoch

Message	UBX-NAV-EOE					
	End of epoch					
Type	Periodic					
Comment						
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x01	0x61	4	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U4	iTOW		-	ms	GPS time of week of the navigation epoch. See section iTOW timestamps in the integration manual for details.

3.15.6 UBX-NAV-GEOFENCE (0x01 0x39)

3.15.6.1 Geofencing status

Message	UBX-NAV-GEOFENCE					
	Geofencing status					
Type	Periodic/poll					
Comment	This message outputs the evaluated states of all configured geofences for the current epoch's position. See section Geofencing in the integration manual for feature details.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x01	0x39	8 + numFences·2	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U4	iTOW		-	ms	GPS time of week of the navigation epoch. See section iTOW timestamps in the integration manual for details.
4	U1	version		-	-	Message version (0x00 for this version)
5	U1	status		-	-	Geofencing status 0 - Geofencing not available or not reliable 1 - Geofencing active
6	U1	numFences		-	-	Number of geofences
7	U1	combState		-	-	Combined (logical OR) state of all geofences 0 - Unknown 1 - Inside 2 - Outside
Start of repeated group (numFences times)						
8 + n·2	U1	state		-	-	Geofence state 0 - Unknown 1 - Inside 2 - Outside
9 + n·2	U1	id		-	-	Geofence ID: 0 = not available, 1 to 4 = FENCE1/2/3/4, 5 to 9 = POLYGON1/2/3/4/5

End of repeated group (*numFences* times)

3.15.7 UBX-NAV-ODO (0x01 0x09)

3.15.7.1 Odometer solution

Message	UBX-NAV-ODO Odometer solution					
Type	Periodic/pollled					
Comment	This message outputs the traveled distance since last reset (see UBX-NAV-RESETODO) together with an associated estimated accuracy and the total cumulated ground distance (can only be reset by a cold start of the receiver).					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x01	0x09	20	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U1	version		-	-	Message version (0x00 for this version)
1	U1[3]	reserved0		-	-	Reserved
4	U4	iTOW		-	ms	GPS time of week of the navigation epoch. See section iTOW timestamps in the integration manual for details.
8	U4	distance		-	m	Ground distance since last reset
12	U4	totalDistance		-	m	Total cumulative ground distance
16	U4	distanceStd		-	m	Ground distance accuracy (1-sigma)

3.15.8 UBX-NAV-ORB (0x01 0x34)

3.15.8.1 GNSS orbit database info

Message	UBX-NAV-ORB GNSS orbit database info						
Type	Periodic/pollled						
Comment	Status of the GNSS orbit database knowledge.						
Message structure	Header	Class	ID	Length (Bytes)		Payload	Checksum
	0xb5 0x62	0x01	0x34	8 + numSv-6		see below	CK_A CK_B
Payload description:							
Byte offset	Type	Name		Scale	Unit	Description	
0	U4	iTOW		-	ms	GPS time of week of the navigation epoch. See section iTOW timestamps in the integration manual for details.	
4	U1	version		-	-	Message version (0x01 for this version)	
5	U1	numSv		-	-	Number of SVs in the database	
6	U1[2]	reserved0		-	-	Reserved	
Start of repeated group (numSv times)							
8 + n·6	U1	gnssId		-	-	GNSS ID	
9 + n·6	U1	svId		-	-	Satellite ID	
10 + n·6	X1	svFlag		-	-	Information Flags	

bits 1...0	U:2	health	-	-	SV health: 0 = unknown 1 = healthy 2 = not healthy
bits 3...2	U:2	visibility	-	-	SV health: 0 = unknown 1 = below horizon 2 = above horizon 3 = above elevation mask
11 + n·6	X1	eph	-	-	Ephemeris data In products supporting L5 signals, the receiver may store multiple ephemeris data sets per satellite. ephUsability and ephSource fields show information on one of the data sets. It is not possible to choose which data set's status is shown.
bits 4...0	U:5	ephUsability	-	-	How long the receiver will be able to use the stored ephemeris data from now on: 31 = The usability period is unknown 30 = The usability period is more than 450 minutes 30 > n > 0 = The usability period is between (n-1)*15 and n*15 minutes 0 = Ephemeris can no longer be used
bits 7...5	U:3	ephSource	-	-	0 = not available 1 = GNSS transmission 2 = external aiding 3-7 = other
12 + n·6	X1	alm	-	-	Almanac data
bits 4...0	U:5	almUsability	-	-	How long the receiver will be able to use the stored almanac data from now on: 31 = The usability period is unknown 30 = The usability period is more than 30 days 30 > n > 0 = The usability period is between n-1 and n days 0 = Almanac can no longer be used
bits 7...5	U:3	almSource	-	-	0 = not available 1 = GNSS transmission 2 = external aiding 3-7 = other
13 + n·6	X1	otherOrb	-	-	Other orbit data available
bits 4...0	U:5	anoAop Usability	-	-	How long the receiver will be able to use the orbit data from now on: 31 = The usability period is unknown 30 = The usability period is more than 30 days 30 > n > 0 = The usability period is between n-1 and n days

bits 7...5	U ₃	type	-	-	0 = Data can no longer be used
					Type of orbit data: 0 = No orbit data available 1 = AssistNow Predictive Orbits data 2 = AssistNow Autonomous data 3-7 = Other orbit data

End of repeated group (*numSv* times)

3.15.9 UBX-NAV-POSECEF (0x01 0x01)

3.15.9.1 Position solution in ECEF

Message	UBX-NAV-POSECEF Position solution in ECEF					
Type	Periodic/pollled					
Comment	See important comments concerning validity of position given in section Navigation output filters in the integration manual.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x01	0x01	20	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U4	iTOW		-	ms	GPS time of week of the navigation epoch. See section iTOW timestamps in the integration manual for details.
4	I4	ecefX		-	cm	ECEF X coordinate
8	I4	ecefY		-	cm	ECEF Y coordinate
12	I4	ecefZ		-	cm	ECEF Z coordinate
16	U4	pAcc		-	cm	Position Accuracy Estimate

3.15.10 UBX-NAV-POSLLH (0x01 0x02)

3.15.10.1 Geodetic position solution

Message	UBX-NAV-POSLLH Geodetic position solution					
Type	Periodic/pollled					
Comment	See important comments concerning validity of position given in section Navigation output filters in the integration manual. This message outputs the Geodetic position in the currently selected ellipsoid. The default is the WGS84 Ellipsoid, but can be changed with the message CFG-NAVSPG-USE_USRDAT .					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x01	0x02	28	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U4	iTOW		-	ms	GPS time of week of the navigation epoch. See section iTOW timestamps in the integration manual for details.
4	I4	lon		1e-7	deg	Longitude

8	I4	lat	1e-7	deg	Latitude
12	I4	height	-	mm	Height above ellipsoid
16	I4	hMSL	-	mm	Height above mean sea level
20	U4	hAcc	-	mm	Horizontal accuracy estimate
24	U4	vAcc	-	mm	Vertical accuracy estimate

3.15.11 UBX-NAV-PVT (0x01 0x07)

3.15.11.1 Navigation position velocity time solution

Message	UBX-NAV-PVT				
	Navigation position velocity time solution				
Type	Periodic/poll				
Comment	This message combines position, velocity and time solution, including accuracy figures. Note that during a leap second there may be more or less than 60 seconds in a minute. See description of leap seconds in the integration manual for details.				
Message structure	Header	Class	ID	Length (Bytes)	Payload
	0xb5 0x62	0x01	0x07	92	see below
Payload description:					
Byte offset	Type	Name	Scale	Unit	Description
0	U4	iTOW	-	ms	GPS time of week of the navigation epoch. See section iTOW timestamps in the integration manual for details.
4	U2	year	-	y	Year (UTC)
6	U1	month	-	month	Month, range 1..12 (UTC)
7	U1	day	-	d	Day of month, range 1..31 (UTC)
8	U1	hour	-	h	Hour of day, range 0..23 (UTC)
9	U1	min	-	min	Minute of hour, range 0..59 (UTC)
10	U1	sec	-	s	Seconds of minute, range 0..60 (UTC)
11	X1	valid	-	-	Validity flags
bit 0	U:1	validDate	-	-	1 = valid UTC Date (see section Time validity in the integration manual for details)
bit 1	U:1	validTime	-	-	1 = valid UTC time of day (see section Time validity in the integration manual for details)
bit 2	U:1	fullyResolved	-	-	1 = UTC time of day has been fully resolved (no seconds uncertainty). Cannot be used to check if time is completely solved.
bit 3	U:1	validMag	-	-	1 = valid magnetic declination
12	U4	tAcc	-	ns	Time accuracy estimate (UTC)
16	I4	nano	-	ns	Fraction of second, range -1e9 .. 1e9 (UTC)
20	U1	fixType	-	-	GNSSfix Type: 0 = no fix 1 = dead reckoning only 2 = 2D-fix 3 = 3D-fix 4 = GNSS + dead reckoning combined 5 = time only fix

21	X1	flags	-	-	Fix status flags
bit 0	U:1	gnssFixOK	-	-	1 = valid fix (i.e within DOP & accuracy masks)
bit 1	U:1	diffSoln	-	-	1 = differential corrections were applied
bits 4...2	U:3	psmState	-	-	Power save mode state (see Power management section in the integration manual for details). 0 = PSM is not active 1 = Enabled (an intermediate state before Acquisition state) 2 = Acquisition 3 = Tracking 4 = Power Optimized Tracking 5 = Inactive
bit 5	U:1	headVehValid	-	-	1 = heading of vehicle is valid, only set if the receiver is in sensor fusion mode
bits 7...6	U:2	carrSoln	-	-	Carrier phase range solution status: 0 = no carrier phase range solution 1 = carrier phase range solution with floating ambiguities 2 = carrier phase range solution with fixed ambiguities (not supported for protocol versions less than 20.00)
22	X1	flags2	-	-	Additional flags
bit 5	U:1	confirmedAvai	-	-	1 = information about UTC Date and Time of Day validity confirmation is available (see section Time validity in the integration manual for details) This flag is only supported in Protocol Versions 19.00, 19.10, 20.10, 20.20, 20.30, 22.00, 23.00, 23.01, 27 and 28 .
bit 6	U:1	confirmedDate	-	-	1 = UTC Date validity could be confirmed (see section Time validity in the integration manual for details)
bit 7	U:1	confirmedTime	-	-	1 = UTC Time of Day could be confirmed (see section Time validity in the integration manual for details)
23	U1	numSV	-	-	Number of satellites used in Nav Solution
24	I4	lon	1e-7	deg	Longitude
28	I4	lat	1e-7	deg	Latitude
32	I4	height	-	mm	Height above ellipsoid
36	I4	hMSL	-	mm	Height above mean sea level
40	U4	hAcc	-	mm	Horizontal accuracy estimate
44	U4	vAcc	-	mm	Vertical accuracy estimate
48	I4	velN	-	mm/s	NED north velocity
52	I4	velE	-	mm/s	NED east velocity
56	I4	velD	-	mm/s	NED down velocity
60	I4	gSpeed	-	mm/s	Ground Speed (2-D)
64	I4	headMot	1e-5	deg	Heading of motion (2-D)
68	U4	sAcc	-	mm/s	Speed accuracy estimate

72	U4	headAcc	1e-5	deg	Heading accuracy estimate (both motion and vehicle)
76	U2	pDOP	0.01	-	Position DOP
78	X2	flags3	-	-	Additional flags
bit 0	U:1	invalidLlh	-	-	1 = Invalid lon, lat, height and hMSL (applicable to heading products only)
bits 4...1	U:4	lastCorrection Age	-	-	Age of the most recently received differential correction: 0 = Not available 1 = Age between 0 and 1 second 2 = Age between 1 (inclusive) and 2 seconds 3 = Age between 2 (inclusive) and 5 seconds 4 = Age between 5 (inclusive) and 10 seconds 5 = Age between 10 (inclusive) and 15 seconds 6 = Age between 15 (inclusive) and 20 seconds 7 = Age between 20 (inclusive) and 30 seconds 8 = Age between 30 (inclusive) and 45 seconds 9 = Age between 45 (inclusive) and 60 seconds 10 = Age between 60 (inclusive) and 90 seconds 11 = Age between 90 (inclusive) and 120 seconds >=12 = Age greater or equal than 120 seconds
bit 13	U:1	authTime	-	-	Flag that indicates if the output time has been validated against an external trusted time source 0 = Time is not authenticated 1 = Time is authenticated
bit 14	U:1	nmaFixStatus	-	-	Flag assigned to a fix that has been computed mixing satellites with data authenticated through Navigation Message Authentication (NMA) methods and satellites using unauthenticated data. The fix is flagged as Verified when internal cross-checks validates the unauthenticated signals against the authenticated ones. Note that Not Verified status does not imply directly spoofing attacks, to identify spoofing alerts refer to UBX-SEC-SIG . 0 = Not Verified: The mixed solution does not agree with the NMA authenticated data or the comparison could not be performed, e.g., not enough authenticated SVs to extrapolate the result or cryptographic data not decoded yet 1 = Verified: The mixed solution agrees with the NMA authenticated data Currently, the only existing NMA method is Galileo Open Service Navigation Message Authentication (OSNMA) protocol.
80	U1[4]	reserved0	-	-	Reserved
84	I4	headVeh	1e-5	deg	Heading of vehicle (2-D), this is only valid when headVehValid is set, otherwise the output is set to the heading of motion
88	I2	magDec	1e-2	deg	Magnetic declination. Only supported in ADR 4.10 and later.

90	U2	magAcc	1e-2	deg	Magnetic declination accuracy. Only supported in ADR 4.10 and later.
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3.15.12 UBX-NAV-RESETODO (0x01 0x10)

3.15.12.1 Reset odometer

Message	UBX-NAV-RESETODO Reset odometer					
Type	Command					
Comment	This message resets the traveled distance computed by the odometer (see UBX-NAV-ODO). UBX-ACK-ACK or UBX-ACK-NAK are returned to indicate success or failure.					
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i>	<i>Checksum</i>
	0xb5 0x62	0x01	0x10	0	see below	CK_A CK_B
Payload	This message has no payload.					

3.15.13 UBX-NAV-SAT (0x01 0x35)

3.15.13.1 Satellite information

Message	UBX-NAV-SAT					
	Satellite information					
Type	Periodic/pollled					
Comment	This message displays information about SVs that are either known to be visible or currently tracked by the receiver. All signal related information corresponds to the subset of signals specified in Signal Identifiers . For resolving potential ambiguities on signal related information, refer to UBX-NAV-SIG .					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x01	0x35	8 + numSvs·12	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U4	iTOW		-	ms	GPS time of week of the navigation epoch. See section iTOW timestamps in the integration manual for details.
4	U1	version		-	-	Message version (0x01 for this version)
5	U1	numSvs		-	-	Number of satellites
6	U1[2]	reserved0		-	-	Reserved
Start of repeated group (numSvs times)						
8 + n·12	U1	gnssId		-	-	GNSS identifier (see Satellite Numbering) for assignment
9 + n·12	U1	svId		-	-	Satellite identifier (see Satellite Numbering) for assignment
10 + n·12	U1	cno		-	dBHz	Carrier to noise ratio (signal strength)
11 + n·12	I1	elev		-	deg	Elevation (range: +/-90), unknown if out of range
12 + n·12	I2	azim		-	deg	Azimuth (range 0-360), unknown if elevation is out of range
14 + n·12	I2	prRes		0.1	m	Pseudorange residual
16 + n·12	X4	flags		-	-	Bitmask

bits 2...0	U:3	qualityInd	-	-	Signal quality indicator: 0 = no signal 1 = searching signal 2 = signal acquired 3 = signal detected but unusable 4 = code locked and time synchronized 5, 6, 7 = code and carrier locked and time synchronized
bit 3	U:1	svUsed	-	-	1 = Signal in the subset specified in Signal Identifiers is currently being used for navigation
bits 5...4	U:2	health	-	-	Signal health flag: 0 = unknown 1 = healthy 2 = unhealthy
bit 6	U:1	diffCorr	-	-	1 = differential correction data is available for this SV
bit 7	U:1	smoothed	-	-	1 = carrier smoothed pseudorange used
bits 10...8	U:3	orbitSource	-	-	Orbit source: 0 = no orbit information is available for this SV 1 = ephemeris is used 2 = almanac is used 3 = AssistNow Predictive Orbits orbit is used 4 = AssistNow Autonomous orbit is used 5, 6, 7 = other orbit information is used
bit 11	U:1	ephAvail	-	-	1 = ephemeris is available for this SV
bit 12	U:1	almAvail	-	-	1 = almanac is available for this SV
bit 13	U:1	anoAvail	-	-	1 = AssistNow Predictive Orbits data is available for this SV
bit 14	U:1	aopAvail	-	-	1 = AssistNow Autonomous data is available for this SV
bit 16	U:1	sbasCorrUsed	-	-	1 = SBAS corrections have been used for a signal in the subset specified in Signal Identifiers
bit 17	U:1	rtcmCorrUsed	-	-	1 = RTCM corrections have been used for a signal in the subset specified in Signal Identifiers
bit 18	U:1	slasCorrUsed	-	-	1 = QZSS SLAS corrections have been used for a signal in the subset specified in Signal Identifiers
bit 19	U:1	spartnCorrUsed	-	-	1 = SPARTN corrections have been used for a signal in the subset specified in Signal Identifiers
bit 20	U:1	prCorrUsed	-	-	1 = Pseudorange corrections have been used for a signal in the subset specified in Signal Identifiers
bit 21	U:1	crCorrUsed	-	-	1 = Carrier range corrections have been used for a signal in the subset specified in Signal Identifiers
bit 22	U:1	doCorrUsed	-	-	1 = Range rate (Doppler) corrections have been used for a signal in the subset specified in Signal Identifiers
bit 23	U:1	clasCorrUsed	-	-	1 = CLAS corrections have been used for a signal in the subset specified in Signal Identifiers

bit 24	U ₁	lppCorrUsed	-	-	1 = LPP corrections have been used for a signal in the subset specified in Signal Identifiers
bit 25	U ₁	hasCorrUsed	-	-	1 = HAS corrections have been used for a signal in the subset specified in Signal Identifiers

End of repeated group (*numSvs* times)

3.15.14 UBX-NAV-SBAS (0x01 0x32)

3.15.14.1 SBAS status data

Message	UBX-NAV-SBAS					
	SBAS status data					
Type	Periodic/pollled					
Comment	This message outputs the status of the SBAS sub system					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x01	0x32	12 + cnt*12	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U4	iTOW		-	ms	GPS time of week of the navigation epoch. See the description of iTOW for details.
4	U1	geo		-	-	PRN Number of the GEO where correction and integrity data is used from
5	U1	mode		-	-	SBAS Mode 0 Disabled 1 Enabled integrity 3 Enabled test mode
6	I1	sys		-	-	SBAS System (WAAS/EGNOS/...) -1 Unknown 0 WAAS 1 EGNOS 2 MSAS 3 GAGAN 16 GPS
7	X1	service		-	-	SBAS Services available
	bit 0 U;1	Ranging		-	-	GEO may be used as ranging source
	bit 1 U;1	Corrections		-	-	GEO is providing correction data
	bit 2 U;1	Integrity		-	-	GEO is providing integrity
	bit 3 U;1	Testmode		-	-	GEO is in test mode
	bit 4 U;1	Bad		-	-	Problem with signal or broadcast data indicated
8	U1	cnt		-	-	Number of SV data following
9	X1	statusFlags		-	-	SBAS status flags
	bits 1...0 U;2	integrityUsed		-	-	SBAS integrity used 0 = Unknown 1 = Integrity information is not available or SBAS integrity is not enabled 2 = Receiver uses only GPS satellites for which integrity information is available

10	U1[2]	reserved0	-	-	Reserved
<i>Start of repeated group (cnt times)</i>					
12 + n·12	U1	svid	-	-	SV ID
13 + n·12	U1	reserved1	-	-	Reserved
14 + n·12	U1	udre	-	-	Monitoring status
15 + n·12	U1	svSys	-	-	System (WAAS/EGNOS/...) same as SYS
16 + n·12	U1	svService	-	-	Services available same as SERVICE
17 + n·12	U1	reserved2	-	-	Reserved
18 + n·12	I2	prc	-	cm	Pseudo Range correction in [cm]
20 + n·12	U1[2]	reserved3	-	-	Reserved
22 + n·12	I2	ic	-	cm	Ionosphere correction in [cm]
<i>End of repeated group (cnt times)</i>					

3.15.15 UBX-NAV-SIG (0x01 0x43)

3.15.15.1 Signal information

Message	UBX-NAV-SIG					
	Signal information					
Type	Periodic/poll					
Comment	This message displays information about signals currently tracked or searched by the receiver.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x01	0x43	8 + numSigs·16	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U4	iTOW		-	ms	GPS time of week of the navigation epoch. See section iTOW timestamps in the integration manual for details.
4	U1	version		-	-	Message version (0x00 for this version)
5	U1	numSigs		-	-	Number of signals
6	U1[2]	reserved0		-	-	Reserved
Start of repeated group (numSigs times)						
8 + n·16	U1	gnssId		-	-	GNSS identifier (see Satellite Numbering) for assignment
9 + n·16	U1	svId		-	-	Satellite identifier (see Satellite Numbering) for assignment
10 + n·16	U1	sigId		-	-	New style signal identifier (see Signal Identifiers)
11 + n·16	U1	freqId		-	-	Only used for GLONASS: This is the frequency slot + 7 (range from 0 to 13)
12 + n·16	I2	prRes		0.1	m	Pseudorange residual
14 + n·16	U1	cno		-	dBHz	Carrier-to-noise density ratio (signal strength)

15 + n·16	U1	qualityInd	-	-	Signal quality indicator: 0 = no signal 1 = searching signal 2 = signal acquired 3 = signal detected but unusable 4 = code locked and time synchronized 5, 6, 7 = code and carrier locked and time synchronized
16 + n·16	U1	corrSource	-	-	Correction source: 0 = no corrections 1 = SBAS corrections 2 = BeiDou corrections 3 = RTCM2 corrections 4 = RTCM3 OSR corrections 5 = RTCM3 SSR corrections 6 = QZSS SLAS corrections 7 = SPARTN corrections 9 = CLAS corrections 10 = LPP OSR corrections 11 = LPP SSR corrections 12 = GAL HAS corrections
17 + n·16	U1	ionoModel	-	-	Ionospheric model used: 0 = No model 1 = Klobuchar model transmitted by GPS 2 = SBAS model 3 = Klobuchar model transmitted by BeiDou 4 = NTCM model transmitted by Galileo 7 = Locally estimated iono model 8 = Iono delay derived from dual-frequency observations 9 = Iono delay derived from multi-frequency observations 11 = Default delay values
18 + n·16	X2	sigFlags	-	-	Signal related flags
bits 1...0	U:2	health	-	-	Signal health flag: 0 = unknown 1 = healthy 2 = unhealthy
bit 2	U:1	prSmoothed	-	-	1 = Pseudorange has been smoothed
bit 3	U:1	prUsed	-	-	1 = Pseudorange has been used for this signal
bit 4	U:1	crUsed	-	-	1 = Carrier range has been used for this signal
bit 5	U:1	doUsed	-	-	1 = Range rate (Doppler) has been used for this signal
bit 6	U:1	prCorrUsed	-	-	1 = Pseudorange corrections have been used for this signal
bit 7	U:1	crCorrUsed	-	-	1 = Carrier range corrections have been used for this signal
bit 8	U:1	doCorrUsed	-	-	1 = Range rate (Doppler) corrections have been used for this signal
bit 9	U:1	authStatus	-	-	Authentication status of the navigation data used to compute the satellite's position in current navigation epoch. If the authentication fails, the navigation data

is not used so the authentication status in this message can take only two values:

- 0 = Unknown
- 1 = Authenticated

Note that currently the only data authentication function is provided by Galileo Open Service Navigation Message Authentication (OSNMA) protocol for E1 I/NAV message.

20 + n·16	U1[4]	reserved1	-	-	Reserved
<i>End of repeated group (numSigs times)</i>					

3.15.16 UBX-NAV-SLAS (0x01 0x42)

3.15.16.1 QZSS L1S SLAS status data

Message	UBX-NAV-SLAS QZSS L1S SLAS status data					
Type	Periodic/poll					
Comment	This message outputs the status of the QZSS L1S SLAS sub system					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x01	0x42	20 + cnt·8	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U4	iTOW		-	ms	GPS time of week of the navigation epoch. See the description of iTOW for details.
4	U1	version		-	-	Message version (0x00 for this version)
5	U1[3]	reserved0		-	-	Reserved
8	I4	gmsLon		1e-3	deg	Longitude of the used ground monitoring station
12	I4	gmsLat		1e-3	deg	Latitude of the used ground monitoring station
16	U1	gmsCode		-	-	Code of the used ground monitoring station according to the QZSS SLAS Interface Specification, available from qzss.go.jp/en/
17	U1	qzssSvId		-	-	Satellite identifier of the QZS/GEO whose correction data is used (see Satellite Numbering)
18	X1	serviceFlags		-	-	Flags regarding SLAS service
	bit 0 U:1	gmsAvailable		-	-	1 = Ground monitoring station available
	bit 1 U:1	qzssSv Available		-	-	1 = Correction providing QZSS SV available
	bit 2 U:1	testMode		-	-	1 = Currently used QZSS SV in test mode
19	U1	cnt		-	-	Number of pseudorange corrections following
Start of repeated group (cnt times)						
20 + n·8	U1	gnssId		-	-	GNSS identifier (see Satellite Numbering)
21 + n·8	U1	svId		-	-	Satellite identifier (see Satellite Numbering)
22 + n·8	U1	reserved1		-	-	Reserved
23 + n·8	U1[3]	reserved2		-	-	Reserved
26 + n·8	I2	prc		-	cm	Pseudorange correction

End of repeated group (*cnt* times)

3.15.17 UBX-NAV-STATUS (0x01 0x03)

3.15.17.1 Receiver navigation status

Message	UBX-NAV-STATUS					
	Receiver navigation status					
Type	Periodic/poll					
Comment	See important comments concerning the validity of the position given in section Navigation output filters in the Integration manual.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x01	0x03	16	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U4	iTOW		-	ms	GPS time of week of the navigation epoch. For details, see section iTOW timestamps in the integration manual.
4	U1	gpsFix		-	-	GPSfix Type, this value does not qualify a fix as valid and within the limits. See note on flag gpsFixOk below. 0x00 = no fix 0x01 = dead reckoning only 0x02 = 2D-fix 0x03 = 3D-fix 0x04 = GPS + dead reckoning combined 0x05 = Time only fix 0x06..0xff = reserved
5	X1	flags		-	-	Navigation Status Flags
	bit 0	U:1	gpsFixOk		-	1 = position and velocity valid and within DOP and ACC Masks.
	bit 1	U:1	diffSoln		-	1 = differential corrections were applied
	bit 2	U:1	wknSet		-	1 = Week Number valid (for details, see section Time validity in the Integration manual)
	bit 3	U:1	towSet		-	1 = Time of Week valid (for details, see section Time validity in the integration manual)
6	X1	fixStat		-	-	Fix Status Information
	bit 0	U:1	diffCorr		-	1 = differential corrections available
	bit 1	U:1	carrSolnValid		-	1 = valid carrSoln
	bit 2	U:1	indoorDet		-	1 = indoor environment detected (not supported for protocol versions less than 52.00)
	bits 4...3	U:2	selectiveL5 State		-	Selective L5 state (not supported for protocol versions less than 52.00) 0 = DISABLED 1 = L5_OFF 2 = L5_ON
	bits 7...6	U:2	mapMatching		-	Map matching status: 00: none 01: valid but not used, i.e. map matching data was received, but was too old

					10: valid and used, map matching data was applied 11: valid and used, map matching data was applied. In case of sensor unavailability map matching data enables dead reckoning. This requires map matched latitude/longitude or heading data.	
7	X1	flags2	-	-	Further information about navigation output	
	bits 1...0	U:2	psmState	-	-	Power save mode state (not supported for protocol versions less than 13.01) 0 = ACQUISITION [or when psm disabled] 1 = TRACKING 2 = POWER OPTIMIZED TRACKING 3 = INACTIVE
	bits 4...3	U:2	spoofDetState	-	-	Spoofing detection state (not supported for protocol versions less than 18.00) 0: Unknown or deactivated: Monitor is not enabled, monitor is uninitialized, or the antenna is disconnected. 1: No spoofing indicated: All spoofing detectors are active and report normal behavior. 2: Spoofing indicated or suspected (see also UBX-SEC-SIG.spfState): Spoofing detectors report suspicion of a spoofing attack. 3: Spoofing affirmed or detected (see also UBX-SEC-SIG.spfState): Spoofing detectors confirm the presence of a spoofing attack.
	bits 7...6	U:2	carrSoln	-	-	Carrier phase range solution status: 0 = no carrier phase range solution 1 = carrier phase range solution with floating ambiguities 2 = carrier phase range solution with fixed ambiguities
8	U4	ttff	-	ms	Time to first fix (millisecond time tag)	
12	U4	msss	-	ms	Time since the last receiver startup or GNSS restart	

3.15.18 UBX-NAV-TIMEBDS (0x01 0x24)

3.15.18.1 BeiDou time solution

Message	UBX-NAV-TIMEBDS BeiDou time solution					
Type	Periodic/pollled					
Comment	This message reports the precise BDS time of the most recent navigation solution including validity flags and an accuracy estimate.					
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i>	<i>Checksum</i>
	0xb5 0x62	0x01	0x24	20	see below	CK_A CK_B

Payload description:

Byte offset	Type	Name	Scale	Unit	Description
0	U4	iTOW	-	ms	GPS time of week of the navigation epoch. See section iTOW timestamps in the integration manual for details.
4	U4	SOW	-	s	BDS time of week (rounded to seconds)
8	I4	fSOW	-	ns	Fractional part of SOW (range: +/-500000000). The precise BDS time of week in seconds is: $SOW + fSOW * 1e-9$
12	I2	week	-	-	BDS week number of the navigation epoch
14	I1	leapS	-	s	BDS leap seconds (BDS-UTC)
15	X1	valid	-	-	Validity Flags
bit 0	U:1	sowValid	-	-	1 = Valid SOW and fSOW (see section Time validity in the integration manual for details)
bit 1	U:1	weekValid	-	-	1 = Valid week (see section Time validity in the integration manual for details)
bit 2	U:1	leapSValid	-	-	1 = Valid leap second
16	U4	tAcc	-	ns	Time Accuracy Estimate

3.15.19 UBX-NAV-TIMEGAL (0x01 0x25)

3.15.19.1 Galileo time solution

Message	UBX-NAV-TIMEGAL Galileo time solution					
Type	Periodic/pollled					
Comment	This message reports the precise Galileo time of the most recent navigation solution including validity flags and an accuracy estimate.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x01	0x25	20	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U4	iTOW		-	ms	GPS time of week of the navigation epoch. See section iTOW timestamps in the integration manual for details.
4	U4	galTow		-	s	Galileo time of week (rounded to seconds)
8	I4	fGalTow		-	ns	Fractional part of the Galileo time of week (range: +/-500000000). The precise Galileo time of week in seconds is: $galTow + fGalTow * 1e-9$
12	I2	galWno		-	-	Galileo week number
14	I1	leapS		-	s	Galileo leap seconds (Galileo-UTC)
15	X1	valid		-	-	Validity Flags
bit 0	U:1	galTowValid		-	-	1 = Valid galTow and fGalTow (see section Time validity in the integration manual for details)
bit 1	U:1	galWnoValid		-	-	1 = Valid galWno (see section Time validity in the integration manual for details)
bit 2	U:1	leapSValid		-	-	1 = Valid leapS

16 U4 tAcc - ns Time Accuracy Estimate

3.15.20 UBX-NAV-TIMEGLO (0x01 0x23)

3.15.20.1 GLONASS time solution

Message	UBX-NAV-TIMEGLO GLONASS time solution						
Type	Periodic/pollled						
Comment	This message reports the precise GLO time of the most recent navigation solution including validity flags and an accuracy estimate.						
Message structure	Header	Class	ID	Length (Bytes)		Payload	Checksum
	0xb5 0x62	0x01	0x23	20		see below	CK_A CK_B
Payload description:							
Byte offset	Type	Name		Scale	Unit	Description	
0	U4	iTOW		-	ms	GPS time of week of the navigation epoch. See section iTOW timestamps in the integration manual for details.	
4	U4	TOD		-	s	GLONASS time of day (rounded to integer seconds)	
8	I4	fTOD		-	ns	Fractional part of TOD (range: +/-500000000). The precise GLONASS time of day in seconds is: TOD + fTOD * 1e-9	
12	U2	Nt		-	days	Current date (range: 1-1461), starting at 1 from the 1st Jan of the year indicated by N4 and ending at 1461 at the 31st Dec of the third year after that indicated by N4	
14	U1	N4		-	-	Four-year interval number starting from 1996 (1=1996, 2=2000, 3=2004...)	
15	X1	valid		-	-	Validity flags	
	bit 0 U;1	todValid		-	-	1 = Valid TOD and fTOD (see section Time validity in the integration manual for details)	
	bit 1 U;1	dateValid		-	-	1 = Valid N4 and Nt (see section Time validity in the integration manual for details)	
16	U4	tAcc		-	ns	Time Accuracy Estimate	

3.15.21 UBX-NAV-TIMEGPS (0x01 0x20)

3.15.21.1 GPS time solution

Message	UBX-NAV-TIMEGPS						
	GPS time solution						
Type	Periodic/pollled						
Comment	This message reports the precise GPS time of the most recent navigation solution including validity flags and an accuracy estimate.						
Message structure	Header	Class	ID	Length (Bytes)		Payload	Checksum
	0xb5 0x62	0x01	0x20	16		see below	CK_A CK_B
Payload description:							
Byte offset	Type	Name		Scale	Unit	Description	

0	U4	iTOW	-	ms	GPS time of week of the navigation epoch. See section iTOW timestamps in the integration manual for details.
4	I4	fTOW	-	ns	Fractional part of iTOW (range: +/-500000). The precise GPS time of week in seconds is: $(iTOW * 1e-3) + (fTOW * 1e-9)$
8	I2	week	-	-	GPS week number of the navigation epoch
10	I1	leapS	-	s	GPS leap seconds (GPS-UTC)
11	X1	valid	-	-	Validity Flags
	bit 0 U ₁	towValid	-	-	1 = Valid GPS time of week (iTOW & fTOW, (see section Time validity in the integration manual for details)
	bit 1 U ₁	weekValid	-	-	1 = Valid GPS week number (see section Time validity in the integration manual for details)
	bit 2 U ₁	leapSValid	-	-	1 = Valid GPS leap seconds
12	U4	tAcc	-	ns	Time Accuracy Estimate

3.15.22 UBX-NAV-TIMELS (0x01 0x26)

3.15.22.1 Leap second event information

Message	UBX-NAV-TIMELS Leap second event information					
Type	Periodic/pollled					
Comment	Information about the upcoming leap second event if one is scheduled. Note: Many sources of leap second information provide the week number of a leap second event as an 8-bit unsigned number. For the upcoming leap second events, this can be resolved and displayed in this message. However, for the previous leap second events decoded from these sources, there is an inherent ambiguity of 256 weeks. Therefore, when the time since the previous event is more than 256 weeks, the dateOfLsGpsWn and timeToLsEvent parameters may provide incorrect information.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x01	0x26	24	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U4	iTOW	-	ms	GPS time of week of the navigation epoch. See section iTOW timestamps in the integration manual for details.	
4	U1	version	-	-	Message version (0x00 for this version)	
5	U1[3]	reserved0	-	-	Reserved	

8	U1	srcOfCurrLs	-	-	Information source for the current number of leap seconds. 0 = Default (hardcoded in the firmware, can be outdated) 1 = Derived from time difference between GPS and GLONASS time 2 = GPS 3 = SBAS 4 = BeiDou 5 = Galileo 6 = Aided data 7 = Configured 8 = NavIC 255 = Unknown	
9	I1	currLs	-	s	Current number of leap seconds since start of GPS time (Jan 6, 1980). It reflects how much GPS time is ahead of UTC time. Galileo number of leap seconds is the same as GPS. BeiDou number of leap seconds is 14 less than GPS. GLONASS follows UTC time, so no leap seconds.	
10	U1	srcOfLsChange	-	-	Information source for the future leap second event. 0 = No source 2 = GPS 3 = SBAS 4 = BeiDou 5 = Galileo 6 = GLONASS 7 = NavIC	
11	I1	lsChange	-	s	Future leap second change if one is scheduled. +1 = positive leap second, -1 = negative leap second, 0 = no future leap second event scheduled or no information available. If the value is 0, then the amount of leap seconds did not change and the event should be ignored.	
12	I4	timeToLsEvent	-	s	Number of seconds until the next leap second event, or from the last leap second event if no future event scheduled. If > 0 event is in the future, = 0 event is now, < 0 event is in the past. Valid only if validTimeToLsEvent = 1.	
16	U2	dateOfLsGps Wn	-	-	GPS week number (WN) of the next leap second event or the last one if no future event scheduled. Valid only if validTimeToLsEvent = 1.	
18	U2	dateOfLsGps Dn	-	-	GPS day of week number (DN) for the next leap second event or the last one if no future event scheduled. Valid only if validTimeToLsEvent = 1. (GPS and Galileo DN: from 1 = Sun to 7 = Sat. BeiDou DN: from 0 = Sun to 6 = Sat.)	
20	U1[3]	reserved1	-	-	Reserved	
23	X1	valid	-	-	Validity flags	
	bit 0	U;1	validCurrLs	-	-	1 = Valid current number of leap seconds value.
	bit 1	U;1	validTimeToLs Event	-	-	1 = Valid time to next leap second event or from the last leap second event if no future event scheduled.

3.15.23 UBX-NAV-TIMENAVIC (0x01 0x63)

3.15.23.1 NavIC time solution

Message	UBX-NAV-TIMENAVIC						
	NavIC time solution						
Type	Periodic/polled						
Comment	This message reports the precise NavIC time of the most recent navigation solution including validity flags and an accuracy estimate.						
Message structure	Header	Class	ID	Length (Bytes)		Payload	Checksum
	0xb5 0x62	0x01	0x63	20		see below	CK_A CK_B
Payload description:							
Byte offset	Type	Name		Scale	Unit	Description	
0	U4	iTOW		-	ms	GPS time of week of the navigation epoch. See section iTOW timestamps in the integration manual for details.	
4	U4	NavICTow		-	s	NavIC time of week (rounded to seconds)	
8	I4	fNavICTow		-	ns	Fractional part of the NavIC time of week (range: +/-500000000). The precise NavIC time of week in seconds is: $\text{NavICTow} + \text{fNavICTow} * 1\text{e-}9$	
12	I2	NavICWno		-	-	NavIC week number	
14	I1	leapS		-	s	NavIC leap seconds (NavIC-UTC)	
15	X1	valid		-	-	Validity Flags	
	bit 0	U:1	NavICTow Valid	-	-	1 = Valid NavICTow and fNavICTow (see section Time validity in the integration manual for details)	
	bit 1	U:1	NavICWno Valid	-	-	1 = Valid NavICWno (see section Time validity in the integration manual for details)	
	bit 2	U:1	leapSValid	-	-	1 = Valid leapS	
16	U4	tAcc		-	ns	Time Accuracy Estimate	

3.15.24 UBX-NAV-TIMEQZSS (0x01 0x27)

3.15.24.1 QZSS time solution

Message	UBX-NAV-TIMEQZSS						
	QZSS time solution						
Type	Periodic/polled						
Comment	This message reports the precise QZSS time of the most recent navigation solution including validity flags and an accuracy estimate. See the Clocks and time section in the integration manual for details.						
Message structure	Header	Class	ID	Length (Bytes)		Payload	Checksum
	0xb5 0x62	0x01	0x27	20		see below	CK_A CK_B
Payload description:							
Byte offset	Type	Name		Scale	Unit	Description	
0	U4	iTOW		-	ms	GPS time of week of the navigation epoch.	
4	U4	qzssTow		-	s	QZSS time of week (rounded to seconds)	

8	I4	fQzssTow	-	ns	Fractional part of QZSS time of week (range: +/-500000000). The precise QZSS time of week in seconds is: $qzssTow + (fQzssTow * 1e-9)$	
12	I2	qzssWno	-	-	QZSS week number of the navigation epoch	
14	I1	leapS	-	s	QZSS leap seconds (QZSS-UTC)	
15	X1	valid	-	-	Validity Flags	
	bit 0	U:1	qzssTowValid	-	-	1 = Valid QZSS time of week (qzssTow and fQzssTow)
	bit 1	U:1	qzssWnoValid	-	-	1 = Valid QZSS week number
	bit 2	U:1	leapSValid	-	-	1 = Valid QZSS leap seconds
16	U4	tAcc	-	ns	Time Accuracy Estimate	

3.15.25 UBX-NAV-TIMEUTC (0x01 0x21)

3.15.25.1 UTC time solution

Message	UBX-NAV-TIMEUTC						
	UTC time solution						
Type	Periodic/poll						
Comment	Note that during a leap second there may be more or less than 60 seconds in a minute. See the description of leap seconds in the integration manual for details.						
Message structure	Header	Class	ID	Length (Bytes)		Payload	Checksum
	0xb5 0x62	0x01	0x21	20		see below	CK_A CK_B
Payload description:							
Byte offset	Type	Name		Scale	Unit	Description	
0	U4	iTOW		-	ms	GPS time of week of the navigation epoch. See section iTOW timestamps in the integration manual for details.	
4	U4	tAcc		-	ns	Time accuracy estimate (UTC)	
8	I4	nano		-	ns	Fraction of second, range -1e9 .. 1e9 (UTC)	
12	U2	year		-	y	Year, range 1999..2099 (UTC)	
14	U1	month		-	month	Month, range 1..12 (UTC)	
15	U1	day		-	d	Day of month, range 1..31 (UTC)	
16	U1	hour		-	h	Hour of day, range 0..23 (UTC)	
17	U1	min		-	min	Minute of hour, range 0..59 (UTC)	
18	U1	sec		-	s	Seconds of minute, range 0..60 (UTC)	
19	X1	valid		-	-	Validity Flags	
	bit 0	U:1	validTOW	-	-	1 = Valid Time of Week (see section Time validity in the integration manual for details)	
	bit 1	U:1	validWKN	-	-	1 = Valid Week Number (see section Time validity in the integration manual for details)	
	bit 2	U:1	validUTC	-	-	1 = Valid UTC Time	
	bit 3	U:1	authStatus	-	-	Indicates if the parameters used to convert GNSS time into UTC time have been authenticated. 0 = Unknown 1 = Authenticated	

Note that currently the only data authentication function is provided by Galileo Open Service Navigation Message Authentication (OSNMA) protocol for E1 I/NAV message. Systems other than EU UTC can be authenticated indirectly only using the above information.

bits 7...4	U:4	utcStandard	-	-	UTC standard identifier. (Not supported for protocol versions less than 15.00)
0 = Information not available 1 = Communications Research Laboratory (CRL), Tokyo, Japan 2 = National Institute of Standards and Technology (NIST) 3 = U.S. Naval Observatory (USNO) 4 = International Bureau of Weights and Measures (BIPM) 5 = European laboratories 6 = Former Soviet Union (SU) 7 = National Time Service Center (NTSC), China 8 = National Physics Laboratory India (NPLI) 15 = Unknown					

3.15.26 UBX-NAV-VELECEF (0x01 0x11)

3.15.26.1 Velocity solution in ECEF

Message	UBX-NAV-VELECEF				
	Velocity solution in ECEF				
Type	Periodic/poll				
Comment	See important comments concerning validity of position given in section Navigation output filters in the integration manual.				
Message structure	Header	Class	ID	Length (Bytes)	Payload
	0xb5 0x62	0x01	0x11	20	see below
Payload description:					
Byte offset	Type	Name	Scale	Unit	Description
0	U4	iTOW	-	ms	GPS time of week of the navigation epoch. See section iTOW timestamps in the integration manual for details.
4	I4	ecefVX	-	cm/s	ECEF X velocity
8	I4	ecefVY	-	cm/s	ECEF Y velocity
12	I4	ecefVZ	-	cm/s	ECEF Z velocity
16	U4	sAcc	-	cm/s	Speed accuracy estimate

3.15.27 UBX-NAV-VELNED (0x01 0x12)

3.15.27.1 Velocity solution in NED frame

Message	UBX-NAV-VELNED
	Velocity solution in NED frame
Type	Periodic/poll

Comment	See important comments concerning validity of position given in section Navigation output filters in the integration manual.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x01	0x12	36	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U4	iTOW		-	ms	GPS time of week of the navigation epoch. See section iTOW timestamps in the integration manual for details.
4	I4	velN		-	cm/s	North velocity component
8	I4	velE		-	cm/s	East velocity component
12	I4	velD		-	cm/s	Down velocity component
16	U4	speed		-	cm/s	Speed (3-D)
20	U4	gSpeed		-	cm/s	Ground speed (2-D)
24	I4	heading		1e-5	deg	Heading of motion 2-D
28	U4	sAcc		-	cm/s	Speed accuracy Estimate
32	U4	cAcc		1e-5	deg	Course / Heading accuracy estimate

3.16 UBX-RXM (0x02)

The messages in the UBX-RXM class are used to output status and result data from the receiver manager as well as sending commands to the receiver manager.

3.16.1 UBX-RXM-MEASX (0x02 0x14)

3.16.1.1 Satellite measurements for RRLP

Message	UBX-RXM-MEASX Satellite measurements for RRLP					
Type	Periodic/pollled					
Comment	The message payload data is, where possible and appropriate, according to the Radio Resource LCS (Location Services) Protocol (RRLP) [1]. One exception is the satellite and GNSS IDs, which here are given according to the Satellite Numbering scheme. The correct satellites have to be selected and their satellite ID translated accordingly [1, tab. A.10.14] for use in a RRLP Measure Position Response Component. Similarly, the measurement reference time of week has to be forwarded correctly (modulo 14400000 for the 24 LSB GPS measurements variant, modulo 3600000 for the 22 LSB Galileo and Additional Navigation Satellite Systems (GANSS) measurements variant) of the RRLP measure position response to the SMLC. Reference: [1] ETSI TS 144 031 V11.0.0 (2012-10), Digital cellular telecommunications system (Phase 2+), Location Services (LCS), Mobile Station (MS) - Serving Mobile Location Centre (SMLC), Radio Resource LCS Protocol (RRLP), (3GPP TS 44.031 version 11.0.0 Release 11).					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x02	0x14	44 + numSV·24	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U1	version		-	-	Message version, currently 0x01
1	U1[3]	reserved0		-	-	Reserved
4	U4	gpsTOW		-	ms	GPS measurement reference time
8	U4	gloTOW		-	ms	GLONASS measurement reference time
12	U4	bdsTOW		-	ms	BeiDou measurement reference time

16	U1[4]	reserved1	-	-	Reserved
20	U4	qzssTOW	-	ms	QZSS measurement reference time
24	U2	gpsTOWacc	2 ⁻⁴	ms	GPS measurement reference time accuracy (0xffff = > 4s)
26	U2	gloTOWacc	2 ⁻⁴	ms	GLONASS measurement reference time accuracy (0xffff = > 4s)
28	U2	bdsTOWacc	2 ⁻⁴	ms	BeiDou measurement reference time accuracy (0xffff = > 4s)
30	U1[2]	reserved2	-	-	Reserved
32	U2	qzssTOWacc	2 ⁻⁴	ms	QZSS measurement reference time accuracy (0xffff = > 4s)
34	U1	numSV	-	-	Number of satellites in repeated block
35	U1	flags	-	-	Flags
bits 1...0	U ₂	towSet	-	-	TOW set (0 = no, 1 or 2 = yes)
36	U1[8]	reserved3	-	-	Reserved
<i>Start of repeated group (numSV times)</i>					
44 + n·24	U1	gnssId	-	-	GNSS ID (see Satellite Numbering)
45 + n·24	U1	svId	-	-	Satellite ID (see Satellite Numbering)
46 + n·24	U1	cNo	-	-	carrier noise ratio (0..63)
47 + n·24	U1	mpathIndic	-	-	multipath index (according to [1]) (0 = not measured, 1 = low, 2 = medium, 3 = high)
48 + n·24	I4	dopplerMS	0.04	m/s	Doppler measurement
52 + n·24	I4	dopplerHz	0.2	Hz	Doppler measurement
56 + n·24	U2	wholeChips	-	-	whole value of the code phase measurement (0..1022 for GPS)
58 + n·24	U2	fracChips	-	-	fractional value of the code phase measurement (0..1023)
60 + n·24	U4	codePhase	2 ⁻²¹	ms	Code phase
64 + n·24	U1	intCodePhase	-	ms	Integer (part of the) code phase
65 + n·24	U1	pseuRangeRMS Err	-	-	pseudorange RMS error index (according to [1]) (0..63)
66 + n·24	U1[2]	reserved4	-	-	Reserved
<i>End of repeated group (numSV times)</i>					

3.16.2 UBX-RXM-PMREQ (0x02 0x41)

3.16.2.1 Power management request

Message	UBX-RXM-PMREQ Power management request					
Type	Command					
Comment	This message requests a power management related task of the receiver.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x02	0x41	16	see below	CK_A CK_B

Payload description:

Byte offset	Type	Name	Scale	Unit	Description
0	U1	version	-	-	Message version (0x00 for this version)
1	U1[3]	reserved0	-	-	Reserved
4	U4	duration	-	ms	Duration of the requested task. The maximum supported value is 12 days. Set to 0 to wait for a wakeup signal on a pin
8	X4	flags	-	-	task flags
bit 1	U ₁	backup	-	-	
bit 2	U ₁	force	-	-	
12	X4	wakeupSources	-	-	Configure pins to wake up the receiver. The receiver wakes up if there is either a falling or a rising edge on one of the configured pins.
bit 3	U ₁	uartRx	-	-	Wake up the receiver if there is an edge on the UART RX pin
bit 5	U ₁	extint0	-	-	Wake up the receiver if there is an edge on the EXTINT0 pin
bit 6	U ₁	extint1	-	-	Wake up the receiver if there is an edge on the EXTINT1 pin
bit 7	U ₁	spics	-	-	Wake up the receiver if there is an edge on the SPI CS pin

3.16.3 UBX-RXM-RAWX (0x02 0x15)

3.16.3.1 Multi-GNSS raw measurements

Message	UBX-RXM-RAWX					
	Multi-GNSS raw measurements					
Type	Periodic/pollled					
Comment	This message contains the information needed to be able to generate a RINEX 3 multi-GNSS observation file (see ftp://ftp.igs.org/pub/data/format/). This message contains pseudorange, Doppler, carrier phase, phase lock and signal quality information for GNSS satellites once signals have been synchronized. This message supports all active GNSS.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x02	0x15	16 + numMeas·32	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	R8	rcvTow	-	s	Measurement time of week in receiver local time approximately aligned to the GPS time system. The receiver local time of week, week number and leap second information can be used to translate the time to other time systems. More information about the difference in time systems can be found in the RINEX 3 format documentation. For a receiver operating in GLONASS only mode, UTC time can be determined by subtracting the leapS field from GPS time regardless of whether the GPS leap seconds are valid.	
8	U2	week	-	weeks	GPS week number in receiver local time.	
10	I1	leapS	-	s	GPS leap seconds (GPS-UTC). This field represents the receiver's best knowledge of the leap seconds offset. A flag is given in the recStat bitfield to indicate if the leap seconds are known.	

11	U1	numMeas	-	-	Number of measurements to follow
12	X1	recStat	-	-	Receiver tracking status bitfield
	bit 0 U ₁	leapSec	-	-	Leap seconds have been determined
	bit 1 U ₁	clkReset	-	-	Clock reset applied. Typically the receiver clock is changed in increments of integer milliseconds.
13	U1	version	-	-	Message version (0x01 for this version)
14	U1[2]	reserved0	-	-	Reserved
<i>Start of repeated group (numMeas times)</i>					
16 + n·32	R8	prMes	-	m	Pseudorange measurement [m]. GLONASS inter frequency channel delays are compensated with an internal calibration table.
24 + n·32	R8	cpMes	-	cycles	Carrier phase measurement [cycles]. The carrier phase initial ambiguity is initialized using an approximate value to make the magnitude of the phase close to the pseudorange measurement. Clock resets are applied to both phase and code measurements in accordance with the RINEX specification.
32 + n·32	R4	doMes	-	Hz	Doppler measurement (positive sign for approaching satellites) [Hz]
36 + n·32	U1	gnssId	-	-	GNSS identifier (see Satellite Numbering for a list of identifiers)
37 + n·32	U1	svId	-	-	Satellite identifier (see Satellite Numbering)
38 + n·32	U1	sigId	-	-	New style signal identifier (see Signal Identifiers). (not supported for protocol versions less than 27.00)
39 + n·32	U1	freqId	-	-	Only used for GLONASS: This is the frequency slot + 7 (range from 0 to 13)
40 + n·32	U2	locktime	-	ms	Carrier phase locktime counter (maximum 64500ms)
42 + n·32	U1	cno	-	dBHz	Carrier-to-noise density ratio (signal strength) [dB-Hz]
43 + n·32	X1	prStdev	0.01·2 ⁿ	m	Estimated pseudorange measurement standard deviation
	bits 3...0 U ₄	prStd	-	-	Estimated pseudorange standard deviation
44 + n·32	X1	cpStdev	0.004	cycles	Estimated carrier phase measurement standard deviation (note a raw value of 0x0F indicates the value is invalid)
	bits 3...0 U ₄	cpStd	-	-	Estimated carrier phase standard deviation
45 + n·32	X1	doStdev	0.002·2 ⁿ	Hz	Estimated Doppler measurement standard deviation.
	bits 3...0 U ₄	doStd	-	-	Estimated Doppler standard deviation
46 + n·32	X1	trkStat	-	-	Tracking status bitfield
	bit 0 U ₁	prValid	-	-	Pseudorange valid
	bit 1 U ₁	cpValid	-	-	Carrier phase valid
	bit 2 U ₁	halfCyc	-	-	Half cycle valid
	bit 3 U ₁	subHalfCyc	-	-	Half cycle subtracted from phase
47 + n·32	U1	reserved1	-	-	Reserved

End of repeated group (*numMeas* times)

3.16.4 UBX-RXM-RLM (0x02 0x59)

3.16.4.1 Galileo SAR short-RLM report

Message	UBX-RXM-RLM Galileo SAR short-RLM report					
Type	Output					
Comment	This message contains the contents of any Galileo Search and Rescue (SAR) Short Return Link Message detected by the receiver.					
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i>	<i>Checksum</i>
	0xb5 0x62	0x02	0x59	16	see below	CK_A CK_B
<i>Payload description:</i>						
<i>Byte offset</i>	<i>Type</i>	<i>Name</i>	<i>Scale</i>	<i>Unit</i>	<i>Description</i>	
0	U1	version	-	-	Message version (0x00 for this version)	
1	U1	type	-	-	Message type (0x01 for Short-RLM)	
2	U1	svId	-	-	Identifier of transmitting satellite (see Satellite Numbering)	
3	U1	reserved0	-	-	Reserved	
4	U1[8]	beacon	-	-	Beacon identifier (60 bits), with bytes ordered by earliest transmitted (most significant) first. Top four bits of first byte are zero.	
12	U1	message	-	-	Message code (4 bits)	
13	U1[2]	params	-	-	Parameters (16 bits), with bytes ordered by earliest transmitted (most significant) first.	
15	U1	reserved1	-	-	Reserved	

3.16.4.2 Galileo SAR long-RLM report

Message	UBX-RXM-RLM Galileo SAR long-RLM report					
Type	Output					
Comment	This message contains the contents of any Galileo Search and Rescue (SAR) Long Return Link Message detected by the receiver.					
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i>	<i>Checksum</i>
	0xb5 0x62	0x02	0x59	28	see below	CK_A CK_B
<i>Payload description:</i>						
<i>Byte offset</i>	<i>Type</i>	<i>Name</i>	<i>Scale</i>	<i>Unit</i>	<i>Description</i>	
0	U1	version	-	-	Message version (0x00 for this version)	
1	U1	type	-	-	Message type (0x02 for Long-RLM)	
2	U1	svId	-	-	Identifier of transmitting satellite (see Satellite Numbering)	
3	U1	reserved0	-	-	Reserved	
4	U1[8]	beacon	-	-	Beacon identifier (60 bits), with bytes ordered by earliest transmitted (most significant) first. Top four bits of first byte are zero.	
12	U1	message	-	-	Message code (4 bits)	

13	U1[12]	params	-	-	Parameters (96 bits), with bytes ordered by earliest transmitted (most significant) first.
25	U1[3]	reserved1	-	-	Reserved

3.16.5 UBX-RXM-SFRBX (0x02 0x13)

3.16.5.1 Broadcast navigation data subframe

Message	UBX-RXM-SFRBX					
	Broadcast navigation data subframe					
Type	Output					
Comment	This message reports a complete subframe of broadcast navigation data decoded from a single signal. The number of data words reported in each message depends on the nature of the signal.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x02	0x13	8 + numWords·4	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name		Scale	Unit	Description
0	U1	gnssId		-	-	GNSS identifier (see Satellite Numbering)
1	U1	svId		-	-	Satellite identifier (see Satellite Numbering)
2	U1	sigId		-	-	Signal identifier (see Signal Identifiers)
3	U1	freqId		-	-	Only used for GLONASS: This is the frequency slot + 7 (range from 0 to 13)
4	U1	numWords		-	-	The number of data words contained in this message (up to 16, for currently supported signals)
5	U1	chn		-	-	The tracking channel number the message was received on
6	U1	version		-	-	Message version, (0x02 for this version)
7	U1	reserved0		-	-	Reserved
Start of repeated group (numWords times)						
8 + n·4	U4	dwrđ		-	-	The data words
End of repeated group (numWords times)						

3.17 UBX-SEC (0x27)

The messages in the UBX-SEC class are used for security features of the receiver.

3.17.1 UBX-SEC-SIG (0x27 0x09)

3.17.1.1 Signal security information

Message	UBX-SEC-SIG Signal security information					
Type	Periodic/pollēd					
Comment	Information related to the security, i.e. availability and integrity, of the signals.					
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i>	<i>Checksum</i>
	0xb5 0x62	0x27	0x09	4 + jamNumCentFreqs·4	see below	CK_A CK_B
Payload description:						
<i>Byte offset</i>	<i>Type</i>	<i>Name</i>	<i>Scale</i>	<i>Unit</i>	<i>Description</i>	
0	U1	version	-	-	Message version (0x02 for this version)	

1	X1	sigSecFlags	-	-	Signal security flags, providing high-level jamming and spoofing detector information	
	bit 0	U;1	jamDetEnabled	-	-	Flag indicates whether jamming detection is enabled
	bits 2...1	U;2	jamState	-	-	Jamming state 0: Unknown, denotes that the currently available information is not sufficient to judge whether the receiver is jammed or not. This may occur at receiver start up (or more generally when the receiver is in a mode, where jamming detection is hindered) or when the jamming indicator is disabled. 1: No jamming indicated: the jamming indicator is enabled and does not detect any significant jamming. 2: Warning; jamming indicated: the jamming indicator is indicating jamming which has a significant impact on the signal tracking. (The list <i>jamPerCentFreq</i> can be checked to find out which frequency bands are jammed.)
	bit 3	U;1	spfDetEnabled	-	-	Flag indicates whether spoofing detection is enabled
	bits 6...4	U;3	spfState	-	-	Spoofing state 0: Unknown 1: No spoofing indicated 2: Spoofing indicated 3: Spoofing affirmed
2	U1	reserved0	-	-	Reserved	
3	U1	jamNumCentFreqs	-	-	The number of center frequencies we provide jamming information for (subsequent messages)	
Start of repeated group (<i>jamNumCentFreqs times</i>)						
4 + n·4	X4	jamStateCentFreq	-	-	Jamming state of signals sharing a given center frequency Note that jamming information is only provided for center frequencies related to at least one in-use signal, for which a sufficient amount of information is currently available to judge if it is affected by jamming.	
	bits 23...0	U;24	centFreq	-	kHz	Center frequency in [kHz], floored to the nearest kHz multiple
	bit 24	U;1	jammed	-	-	Flag indicates whether signals on the given center frequency are considered jammed
End of repeated group (<i>jamNumCentFreqs times</i>)						

3.17.2 UBX-SEC-SIGLOG (0x27 0x10)

3.17.2.1 Signal security log

Message	UBX-SEC-SIGLOG
	Signal security log
Type	Periodic/pollled

Comment	This message provides a log of past signal security related events, that is, events related to jamming and spoofing. Each event is a combination of a detection type and a event type, where the event type 'indication started' and 'indication stopped' and also the event type 'indication triggered' and 'indication timed-out' form a pair. A maximum of 16 events are logged; after the log is filled, recent events take precedence over past events in the log. Power cycles and restarts of the receiver reset the log, deleting its content. Note: It is advised not to restart the receiver while it's indicating spoofing.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x27	0x10	8 + numEvents*8	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U1	version	-	-	Message version (0x01 for this version)	
1	U1	numEvents	-	-	Number of events	
2	U1[6]	reserved0	-	-	Reserved	
Start of repeated group (numEvents times)						
8 + n*8	U4	timeElapsed	-	s	Seconds elapsed since this event Special value 0xFFFFFFFF: more than 45 days	
12 + n*8	U1	detectionType	-	-	Type of the spoofing or jamming detection: 0 = simulated signal 1 = abnormal signal 2 = INS/GNSS mismatch 3 = abrupt changes in GNSS signal 4 = jamming indicated 5 = authentication failed 6 = replayed signals	
13 + n*8	U1	eventType	-	-	Type of the event: 0 = indication started 1 = indication stopped 2 = indication triggered 3 = indication timed-out Note: Single epoch events, caused by abrupt changes due to switching from the real to the spoofing signal or vice versa are handled as time-out events. This means that the time-out event is reported after a certain cool off period which is not related to any observations in the signal. The other detection types make use of 'start' and 'stop' event types.	
14 + n*8	U1[2]	reserved1	-	-	Reserved	
End of repeated group (numEvents times)						

3.17.3 UBX-SEC-UNIQID (0x27 0x03)

3.17.3.1 Unique chip ID

Message	UBX-SEC-UNIQID Unique chip ID					
Type	Output					
Comment	This message is used to retrieve a unique chip identifier (48 bits, 6 bytes).					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x27	0x03	10	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	

0	U1	version	-	-	Message version (0x02 for this version)
1	U1[3]	reserved0	-	-	Reserved
4	U1[6]	uniqueId	-	-	Unique chip ID

3.18 UBX-TIM (0x0d)

The messages in the UBX-TIM class are used to output timing information from the receiver, such as time pulse and time mark measurements.

3.18.1 UBX-TIM-TM2 (0x0d 0x03)

3.18.1.1 Time mark data

Message	UBX-TIM-TM2						
	Time mark data						
Type	Periodic/pollled						
Comment	This message contains information for high precision time stamping / pulse counting. The delay figures and timebase given in CFG-TP configuration items are also applied to the time results output in this message.						
Message structure	Header	Class	ID	Length (Bytes)		Payload	Checksum
	0xb5 0x62	0x0d	0x03	28		see below	CK_A CK_B
Payload description:							
Byte offset	Type	Name		Scale	Unit	Description	
0	U1	ch		-	-	Channel (i.e. EXTINT) upon which the pulse was measured	
1	X1	flags		-	-	Bitmask	
bit 0	U:1	mode		-	-	0=single 1=running	
bit 1	U:1	run		-	-	0=armed 1=stopped	
bit 2	U:1	newFallingEdge		-	-	New falling edge detected	
bits 4...3	U:2	timeBase		-	-	0=Time base is Receiver time 1=Time base is GNSS time (the system according to the configuration in CFG-TP configuration items for tpIdx=0) 2=Time base is UTC (the variant according to the configuration in CFG-NAVSPG-* configuration items)	
bit 5	U:1	utc		-	-	0=UTC not available 1=UTC available	
bit 6	U:1	time		-	-	0=Time is not valid 1=Time is valid (Valid GNSS fix)	
bit 7	U:1	newRisingEdge		-	-	New rising edge detected	
2	U2	count		-	-	Rising edge counter	
4	U2	wnR		-	-	Week number of last rising edge	
6	U2	wnF		-	-	Week number of last falling edge	

8	U4	towMsR	-	ms	Tow of rising edge
12	U4	towSubMsR	-	ns	Millisecond fraction of tow of rising edge in nanoseconds
16	U4	towMsF	-	ms	Tow of falling edge
20	U4	towSubMsF	-	ns	Millisecond fraction of tow of falling edge in nanoseconds
24	U4	accEst	-	ns	Accuracy estimate

3.18.2 UBX-TIM-TP (0x0d 0x01)

3.18.2.1 Time pulse time data

Message	UBX-TIM-TP					
	Time pulse time data					
Type	Periodic/pollled					
Comment	This message contains information on the timing of the next pulse at the time pulse output. The recommended configuration when using this message is to set both the measurement rate (CFG-RATE) and the timepulse frequency (CFG-TP) to 1 Hz.					
Message structure	Header	Class	ID	Length (Bytes)	Payload	Checksum
	0xb5 0x62	0x0d	0x01	16	see below	CK_A CK_B
Payload description:						
Byte offset	Type	Name	Scale	Unit	Description	
0	U4	towMS	-	ms	Time pulse time of week according to time base	
4	U4	towSubMS	2^32	ms	Submillisecond part of towMS	
8	I4	qErr	-	ps	Quantization error of time pulse	
12	U2	week	-	weeks	Time pulse week number according to time base	
14	X1	flags	-	-	Flags	
bit 0	U:1	timeBase	-	-	0 = Time base is GNSS 1 = Time base is UTC	
bit 1	U:1	utc	-	-	0 = UTC not available 1 = UTC available	
bits 3...2	U:2	raim	-	-	(T)RAIM information 0 = Information not available 1 = Not active 2 = Active	
bit 4	U:1	qErrInvalid	-	-	0 = Quantization error valid 1 = Quantization error invalid	
bit 5	U:1	TpNotLocked	-	-	0 = Next TP is locked to GNSS 1 = Next TP is based on local time and not locked to GNSS - week/tow may be invalid	
15	X1	refInfo	-	-	Time reference information	
bits 3...0	U:4	timeRefGnss	-	-	GNSS reference information. Only valid if time base is GNSS (timeBase=0). 0 = GPS 1 = GLONASS 2 = BeiDou	

					3 = Galileo 4 = NavIC 15 = Unknown
bits 7...4	U:4	utcStandard	-	-	UTC standard identifier. Only valid if time base is UTC (timeBase=1). 0 = Information not available 1 = Communications Research Laboratory (CRL), Tokyo, Japan 2 = National Institute of Standards and Technology (NIST) 3 = U.S. Naval Observatory (USNO) 4 = International Bureau of Weights and Measures (BIPM) 5 = European laboratories 6 = Former Soviet Union (SU) 7 = National Time Service Center (NTSC), China 8 = National Physics Laboratory India (NPLI) 15 = Unknown

3.18.3 UBX-TIM-VRFY (0x0d 0x06)

3.18.3.1 Sourced time verification

Message	UBX-TIM-VRFY Sourced time verification					
Type	Periodic/pollled					
Comment	This message contains verification information about previous time received via assistance data or from RTC.					
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i>	<i>Checksum</i>
	0xb5 0x62	0x0d	0x06	20	see below	CK_A CK_B
Payload description:						
<i>Byte offset</i>	<i>Type</i>	<i>Name</i>	<i>Scale</i>	<i>Unit</i>	<i>Description</i>	
0	I4	itow	-	ms	integer millisecond tow received by source	
4	I4	frac	-	ns	sub-millisecond part of tow	
8	I4	deltaMs	-	ms	integer milliseconds of delta time (current time minus sourced time)	
12	I4	deltaNs	-	ns	Sub-millisecond part of delta time	
16	U2	wno	-	week	Week number	
18	X1	flags	-	-	Flags	
bits 2...0	U:3	src	-	-	Aiding time source 0 = no time aiding done 2 = source was RTC 3 = source was assistance data	
19	U1	reserved0	-	-	Reserved	

3.19 UBX-UPD (0x09)

The messages in the UBX-UPD class are used to download a firmware to the receiver and to update the firmware on the flash.

3.19.1 UBX-UPD-SOS (0x09 0x14)

3.19.1.1 Poll backup restore status

Message	UBX-UPD-SOS Poll backup restore status					
Type	Poll request					
Comment	Sending this (empty) message to the receiver results in the receiver returning a <i>System restored from backup</i> message as defined below.					
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i>	<i>Checksum</i>
	0xb5 0x62	0x09	0x14	0	see below	CK_A CK_B
Payload	This message has no payload.					

3.19.1.2 Create backup in flash

Message	UBX-UPD-SOS Create backup in flash					
Type	Command					
Comment	The host can send this message in order to save part of the battery-backed memory (BBR) in a file in the flash file system. The feature is designed in order to emulate the presence of the backup battery even if it is not present; the host can issue the save on shutdown command before switching off the device supply. It is recommended to issue a GNSS stop command using UBX-CFG-RST before in order to keep the BBR memory content consistent.					
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i>	<i>Checksum</i>
	0xb5 0x62	0x09	0x14	4	see below	CK_A CK_B
Payload description:						
<i>Byte offset</i>	<i>Type</i>	<i>Name</i>		<i>Scale</i>	<i>Unit</i>	<i>Description</i>
0	U1	cmd		-	-	Command (must be 0)
1	U1[3]	reserved0		-	-	Reserved

3.19.1.3 Clear backup in flash

Message	UBX-UPD-SOS Clear backup in flash					
Type	Command					
Comment	The host can send this message in order to erase the backup file present in flash. It is recommended that the clear operation is issued after the host has received the notification that the memory has been restored after a reset. Alternatively the host can parse the startup string <i>Restored data saved on shutdown</i> or poll the UBX-UPD-SOS message for obtaining the status.					
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i>	<i>Checksum</i>
	0xb5 0x62	0x09	0x14	4	see below	CK_A CK_B
Payload description:						
<i>Byte offset</i>	<i>Type</i>	<i>Name</i>		<i>Scale</i>	<i>Unit</i>	<i>Description</i>
0	U1	cmd		-	-	Command (must be 1)
1	U1[3]	reserved0		-	-	Reserved

3.19.1.4 Backup creation acknowledge

Message	UBX-UPD-SOS Backup creation acknowledge					
Type	Output					
Comment	The message is sent from the device as confirmation of creation of a backup file in flash. The host can safely shut down the device after having received this message.					
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i>	<i>Checksum</i>
	0xb5 0x62	0x09	0x14	8	see below	CK_A CK_B
<i>Payload description:</i>						
<i>Byte offset</i>	<i>Type</i>	<i>Name</i>		<i>Scale</i>	<i>Unit</i>	<i>Description</i>
0	U1	cmd		-	-	Command (must be 2)
1	U1[3]	reserved0		-	-	Reserved
4	U1	response		-	-	0 = Not acknowledged 1 = Acknowledged
5	U1[3]	reserved1		-	-	Reserved

3.19.1.5 System restored from backup

Message	UBX-UPD-SOS System restored from backup					
Type	Output					
Comment	The message is sent from the device to notify the host the BBR has been restored from a backup file in the flash file system. The host should clear the backup file after receiving this message. If the UBX-UPD-SOS message is polled, this message is resent.					
Message structure	<i>Header</i>	<i>Class</i>	<i>ID</i>	<i>Length (Bytes)</i>	<i>Payload</i>	<i>Checksum</i>
	0xb5 0x62	0x09	0x14	8	see below	CK_A CK_B
<i>Payload description:</i>						
<i>Byte offset</i>	<i>Type</i>	<i>Name</i>		<i>Scale</i>	<i>Unit</i>	<i>Description</i>
0	U1	cmd		-	-	Command (must be 3)
1	U1[3]	reserved0		-	-	Reserved
4	U1	response		-	-	0 = Unknown 1 = Failed restoring from backup 2 = Restored from backup 3 = Not restored (no backup)
5	U1[3]	reserved1		-	-	Reserved

4 Configuration interface

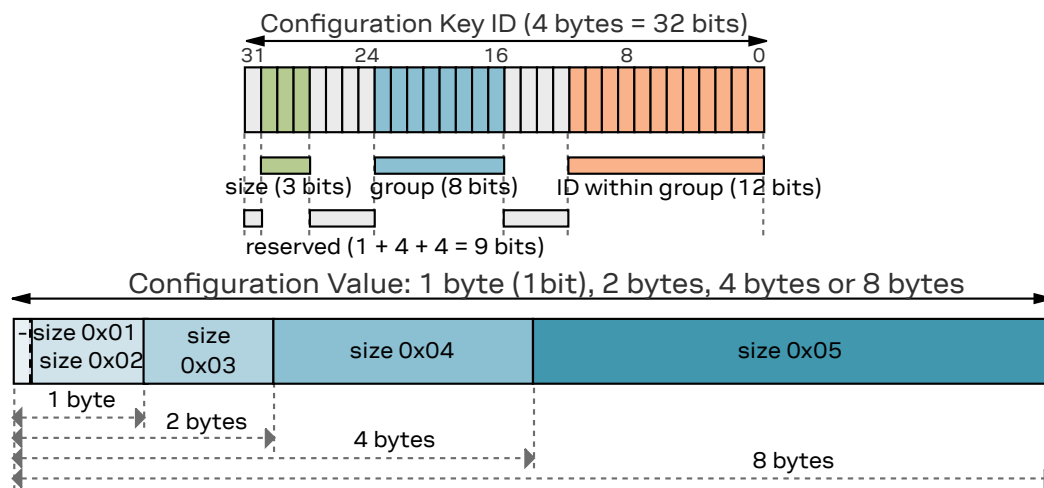
This chapter describes the receiver configuration interface.

4.1 Configuration database

The configuration database in the receiver's RAM stores the current receiver settings used during runtime. This database is constructed from multiple sources known as *configuration layers* when the receiver starts up. The active settings, known as the current configuration, are stored in the *RAM layer*. Each configuration layer is organized into *configuration items*, which are uniquely identified by a *configuration key ID* and hold a single *configuration value*.

4.2 Configuration items

The following figure shows the structure of a *configuration item*, which consists of a (*configuration*) *key ID* and its (*configuration*) *value*:



A configuration key ID is a 32-bit integer value, which is split into the following parts:

- Bit 31: Currently unused. Reserved for future use.
- Bits 30...28: Three bits that indicate the storage size of a configuration value (range 0x01-0x05, see below)
- Bits 27...24: Currently unused. Reserved for future use.
- Bits 23...16: Eight bits that define a unique group ID (range 0x01-0xfe)
- Bits 15...12: Currently unused. Reserved for future use.
- Bits 11...0: Twelve bits that define a unique item ID within a group (range 0x001-0xffe)

The entire 32-bit value is the unique key ID, which uniquely identifies a particular item. The numeric representation of the key ID uses the lower-case hexadecimal format, such as `0x20c400a1`. An easier, more readable text representation uses the form `CFG-GROUP-ITEM`. This is also referred to as the (*configuration*) *key name*.

Supported storage size identifiers (bits 30...28 of the key ID) are:

- 0x01: one bit (the actual storage used is one byte, but only the least significant bit is used)
- 0x02: one byte
- 0x03: two bytes
- 0x04: four bytes
- 0x05: eight bytes

Each configuration item is of a certain type, which defines the interpretation of the raw binary data (see also [UBX data types](#)):

- U1, U2, U4, U8: unsigned little-endian integers of 8-, 16-, 32- and 64-bit widths
- I1, I2, I4, I8: signed little-endian, two's complement integers of 8-, 16-, 32- and 64-bit widths
- R4, R8: IEEE 754 single (32-bit) and double (64-bit) precision floats
- E1, E2, E4: unsigned little-endian enumeration of 8-, 16-, and 32-bit widths
- X1, X2, X4, X8: unsigned little-endian integers of 8-, 16-, 32- and 64-bit widths for bitfields and other binary data, such as strings
- L: single-bit boolean (true = 1, false = 0), stored as U1

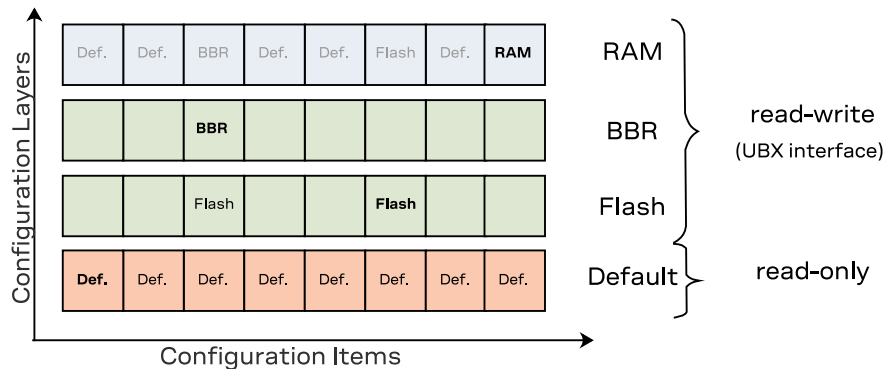
4.3 Configuration layers

The receiver has several *configuration layers*. They are separate sources of configuration items. Some of the layers are read-only and others are modifiable. Layers are organized in terms of priority. Values in a high-priority layer replace values stored in a low-priority layer. At startup, the receiver reads all configuration layers and stacks up the items to create the *current configuration*, which is used by the receiver at run-time.

The following configuration layers are available (in order of priority, highest priority first):

- **RAM:** This layer contains items stored in volatile RAM. This is the current configuration. The configuration items in this layer can be set at run-time and are effective immediately.
- **BBR:** This layer contains items stored in the battery-backed RAM. The contents in this layer are preserved as long as a battery backup supply is provided during off periods. The configuration items in this layer can be set at run-time and they become effective when the receiver is restarted.
- **Flash:** This layer contains configuration items stored permanently in the external flash memory and it is available only if external flash memory is used. The configuration items in this layer can be set at run-time and they become effective when the receiver is restarted.
- **Default:** This layer contains all items known to the running receiver software and the hard-coded default values. Data in this layer cannot be modified during run-time. The default layer includes limited one-time programmable (OTP) memory for setting customized default values during device production.

The stacking of the configuration items from the different layers (sources) in order to construct the current configuration in the RAM layer is depicted in the following figure. For each defined item, i.e. for each item in the default layer, the receiver software goes through the layers above and stacks all the found items on top. Some items may not be present in every layer. The result is the RAM layer filled with all configuration items given configuration values coming from the highest priority layer the corresponding item was present. In the example figure, bold text indicates the source of the value in the current configuration (the RAM layer). Empty boxes mean that the layer can hold the item but that it is not currently stored there. Boxes with text mean that an item is currently stored in the layer.



In the example figure above several items (e.g. the first item) are only set in the default layer and hence, the default value ends up in the current configuration in the RAM layer. The third item is present in the Default, flash and BBR layers. The value from the BBR layer has the highest priority and therefore it ends up in the RAM layer. On the other hand, the default value of the sixth item is changed by the value in the flash layer. The value of the last item is changed in the RAM layer only, i.e. upon startup the value in the RAM layer was the value from the default layer, but the value in the RAM layer was changed at runtime.

4.4 Configuration interface access

The following sections describe the existing interfaces to access the Configuration Database.

4.4.1 UBX protocol interface

The following [UBX protocol](#) messages are available to access the configuration database:

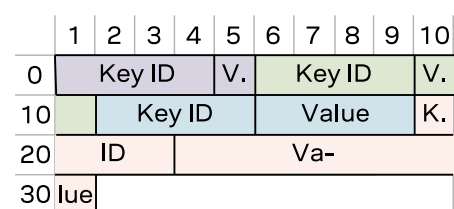
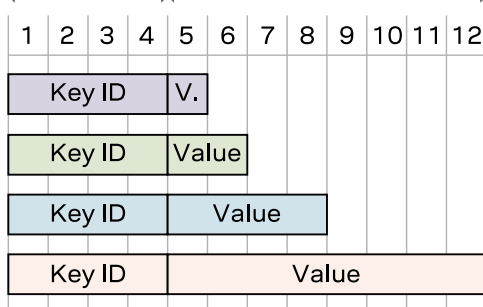
- [UBX-CFG-VALGET](#) to read configuration items from the database
- [UBX-CFG-VALSET](#) to set configuration items in the database
- [UBX-CFG-VALDEL](#) to delete configuration items from the database

4.5 Configuration data

Configuration data is the binary representation of a list of key ID and value pairs. It is formed by concatenating keys (U4 values) and values (variable type) without any padding. This format is used in the [UBX-CFG-VALSET](#) and [UBX-CFG-VALGET](#) messages.

The figure below shows an example. The four items (key ID - value pairs) on the left use the four fundamental storage sizes: one byte (L, U1, I1, E1 and X1 types), 2 bytes (U2, I2, E2 and X2 types), four byte (U4, I4, E4, X4 and R4 types) and eight bytes (U8, I8, X8 and R8 types). When concatenated (right) the key IDs and values are not aligned and there is no padding.

Key ID: 4 bytes Value: 1, 2, 4 or 8 bytes



Total 31 bytes

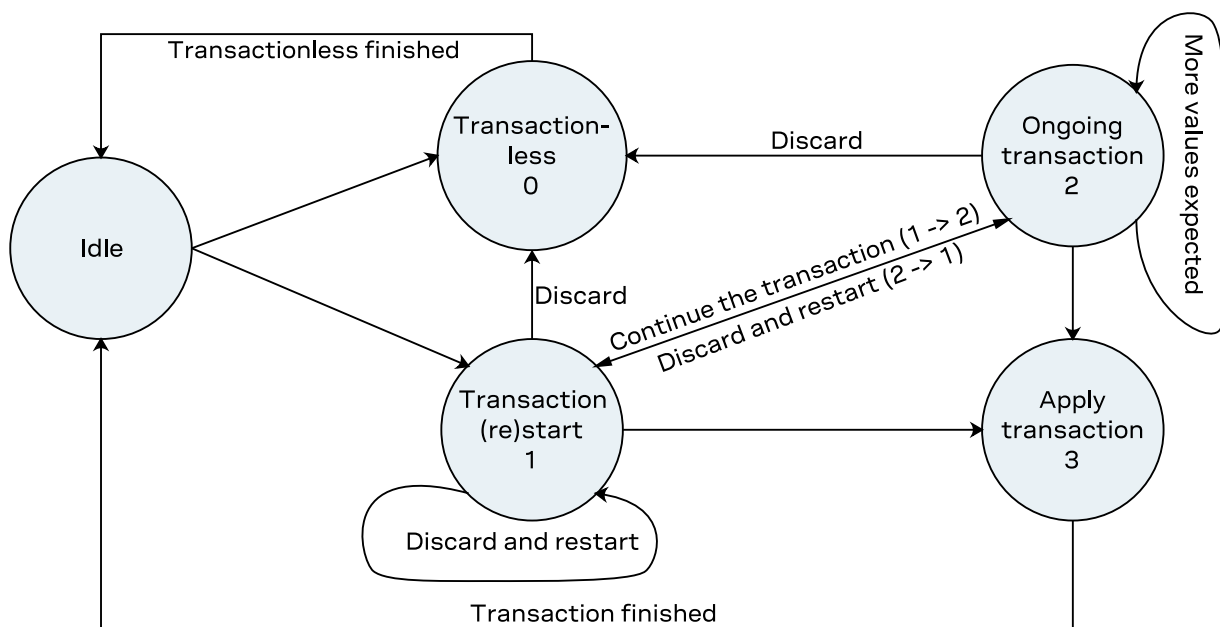
Note that this is an arbitrary example and any number of items of any value storage size can be concatenated the same way.

4.6 Configuration transactions

The configuration interface supports two mechanisms of configuration: the first is a transactionless mechanism where sent configuration changes are applied immediately to the configuration layer(s) requested. The second mechanism is a configuration transaction.

A transaction offers a way of queuing multiple configuration changes. It is particularly useful where different configuration keys depend on each other in such a way that sending one before the other can cause the configuration to be rejected. The queued configuration change requests are stored then checked collectively before being applied to the receiver.

A transaction can have the following states described in the figure below.



When starting a transaction, specify the layer(s) to apply the changes to. This list of configuration layer(s) must be observed throughout the transaction states. Modifying the configuration layer(s) mid-transaction causes the transaction to be aborted and consequently, no queued changes will be applied.

In the start transaction state, the receiver locks the configuration database so that changes from another entity or message cannot be applied. It is possible to send a configuration key-value pairs with the start transaction state. These are queued waiting to be applied.

In the ongoing state, a configuration key and value must be sent. The receiver aborts the transaction and does not apply any changes if this condition is violated. Key-value pairs sent in the ongoing state are queued waiting to be applied.

In the apply state, the receiver collectively checks the queued changes and applied them to the requested configuration layer(s). Note that any additional key-value pairs sent within the apply state are ignored.

Note that a transaction can only come from a single source, a [UBX-CFG-VALSET](#) message or a [UBX-CFG-VALDEL](#) message. This means that in any given transaction it is not possible to mix a delete

and a save request. Starting a transaction from a different source aborts the current transaction and the queued changes are not applied.

Refer to [UBX-CFG-VALSET](#) and [UBX-CFG-VALDEL](#) messages for a detailed description of how to set up a configuration transaction, its limitations and conditions that would cause the transaction to be rejected.

4.7 Configuration reset behavior

The RAM layer is always rebuilt from the layers below when the chip's processor comes out from reset. When using [UBX-CFG-RST](#) the processor goes through a reset cycle with these reset types (`resetMode` field):

- 0x00 hardware reset (watchdog) immediately
- 0x01 controlled software reset
- 0x04 hardware reset (watchdog) after shutdown

See section Forcing a receiver reset in the integration manual.

4.8 Configuration overview

Group	Description
CFG-ANA	AssistNow Autonomous and Offline configuration
CFG-BATCH	Batched output configuration
CFG-BDS	BeiDou system configuration
CFG-GEOFENCE	Geofencing configuration
CFG-HW	Hardware configuration
CFG-I2C	Configuration of the I2C interface
CFG-I2CINPROT	Input protocol configuration of the I2C interface
CFG-I2COUTPROT	Output protocol configuration of the I2C interface
CFG-INFMSG	Information message configuration
CFG-LOGFILTER	Data logger configuration
CFG-MOT	Motion detector configuration
CFG-MSGOUT	Message output configuration
CFG-NAVSPG	Standard precision navigation configuration
CFG-NMEA	NMEA protocol configuration
CFG-ODO	Odometer and low-speed course over ground filter configuration
CFG-PM	Configuration for receiver power management
CFG-QZSS	QZSS system configuration
CFG-RATE	Navigation and measurement rate configuration
CFG-RINV	Remote inventory
CFG-SBAS	SBAS configuration
CFG-SEC	Security configuration
CFG-SIGNAL	Satellite systems (GNSS) signal configuration
CFG-SPI	Configuration of the SPI interface
CFG-SPIINPROT	Input protocol configuration of the SPI interface
CFG-SPIOUTPROT	Output protocol configuration of the SPI interface

Group	Description
CFG-TP	Time pulse configuration
CFG-TXREADY	TX ready configuration
CFG-UART1	Configuration of the UART1 interface
CFG-UART1INPROT	Input protocol configuration of the UART1 interface
CFG-UART1OUTPROT	Output protocol configuration of the UART1 interface

4.9 Configuration reference

4.9.1 CFG-ANA: AssistNow Autonomous and Offline configuration

Configuration for the AssistNow Autonomous feature. See section AssistNow Autonomous in the integration manual for feature details.

Configuration item	Key ID	Type	Scale	Unit	Description
CFG-ANA-USE_ANA	0x10230001	L	-	-	Use AssistNow Autonomous
CFG-ANA-ORBMAXERR	0x30230002	U2	-	m	Maximum acceptable (modeled) orbit error Range is from 5 to 1000.

Table 5: CFG-ANA configuration items

4.9.2 CFG-BATCH: Batched output configuration

Use this group to configure the data batching feature which allows position fixes to be stored in the RAM of the receiver to be retrieved later in one batch.

Configuration item	Key ID	Type	Scale	Unit	Description
CFG-BATCH-ENABLE	0x10260013	L	-	-	Enable data batching Enable the feature. Note that it will do nothing unless a positive value is set for CFG-BATCH-MAXENTRIES.
CFG-BATCH-PIOENABLE	0x10260014	L	-	-	Enable PIO notification Enable PIO notification when the buffer fill level exceeds WARNTHRS.
CFG-BATCH-MAXENTRIES	0x30260015	U2	-	-	Maximum entries in buffer Size of buffer in number of epochs to store. The firmware will reject this configuration if it exceeds the available memory.
CFG-BATCH-WARNTHRS	0x30260016	U2	-	-	Buffer fill level warning threshold Buffer fill level that triggers PIO notification, in number of epochs stored.
CFG-BATCH-PIOACTIVELOW	0x10260018	L	-	-	PIO is active low If this is set the PIO selected with CFG-BATCH-PIOID will be driven low when the buffer fill level reaches WARNTHRS. Otherwise the polarity of the PIO will be the opposite.
CFG-BATCH-PIOID	0x20260019	U1	-	-	PIO ID for buffer level notification PIO that is used for buffer fill level notification. It must not be assigned to a different function.
CFG-BATCH-EXTRAPVT	0x1026001a	L	-	-	Include extra PVT data Include additional PVT information in UBX-LOG-BATCH messages. If not selected only basic information is included. The fields iTOW, tAcc, numSV, hMSL, vAcc, velN, velE, velD, sAcc, headAcc and pDOP in UBX-LOG-BATCH are only valid if this flag is set.
CFG-BATCH-EXTRAODO	0x1026001b	L	-	-	Include odometer data

Configuration item	Key ID	Type	Scale	Unit	Description
The fields <code>distance</code> , <code>totalDistance</code> and <code>distanceStd</code> in UBX-LOG-BATCH are only valid if this flag is set.					

Table 6: CFG-BATCH configuration items

4.9.3 CFG-BDS: BeiDou system configuration

Note that enabling and disabling of individual GNSS is done via the [CFG-SIGNAL](#) configuration group.

Configuration item	Key ID	Type	Scale	Unit	Description
CFG-BDS-D1D2_NAVDATA	0x20340009	E1	-	-	Enable only the given BDS D1/D2 navigation data streams, ignoring the others

See [Table 8](#) below for a list of possible constants for this item.

Table 7: CFG-BDS configuration items

Constant	Value	Description
ALL	0	Enable all BDS D1D2 navigation data streams (default)
B1I	1	Force B1I navigation data, ignoring B2I and B3I

Table 8: Constants for CFG-BDS-D1D2_NAVDATA

4.9.4 CFG-GEOFENCE: Geofencing configuration

Configuration for the geofencing feature. See section Geofencing in the integration manual for feature details.

If the receiver is sent a valid new configuration, it will respond with a [UBX-ACK-ACK](#) message and immediately change to the new configuration. Otherwise the receiver will reject the request, by issuing a [UBX-ACK-NAK](#) and continuing operation with the previous configuration.

Note that the acknowledge message does not indicate whether the PIO configuration has been successfully applied (pin assigned), it only indicates the successful configuration of the feature. The configured PIO must be previously unoccupied for successful assignment.

Configuration item	Key ID	Type	Scale	Unit	Description
CFG-GEOFENCE-CONFLVL	0x20240011	E1	-	-	Required confidence level for state evaluation This value times the position's standard deviation (sigma) defines the confidence band. See Table 10 below for a list of possible constants for this item.
CFG-GEOFENCE-USE_PIO	0x10240012	L	-	-	Use PIO combined fence state output
CFG-GEOFENCE-PINPOL	0x20240013	E1	-	-	PIO pin polarity See Table 11 below for a list of possible constants for this item.
CFG-GEOFENCE-PIN	0x20240014	U1	-	-	PIO pin number
CFG-GEOFENCE-POLYGON1_NUM_VERTICES	0x20240017	U1	-	-	Number of vertices for polygon 1: 0 = disabled, 1..20 = enabled with defined amount of vertices, range for valid polygon = 3..20
CFG-GEOFENCE-POLYGON2_NUM_VERTICES	0x20240018	U1	-	-	Number of vertices for polygon 2: 0 = disabled, 1..20 = enabled with defined amount of vertices, range for valid polygon = 3..20
CFG-GEOFENCE-POLYGON3_NUM_VERTICES	0x20240019	U1	-	-	Number of vertices for polygon 3: 0 = disabled, 1..20 = enabled with defined amount of vertices, range for valid polygon = 3..20
CFG-GEOFENCE-POLYGON4_NUM_VERTICES	0x2024001a	U1	-	-	Number of vertices for polygon 4: 0 = disabled, 1..20 = enabled with defined amount of vertices, range for valid polygon = 3..20

Configuration item	Key ID	Type	Scale	Unit	Description
CFG-GEOFENCE-POLYGON5_NUM_VERTICES	0x2024001b	U1	-	-	Number of vertices for polygon 5: 0 = disabled, 1..20 = enabled with defined amount of vertices, range for valid polygon = 3..20
CFG-GEOFENCE-USE_FENCE1	0x10240020	L	-	-	Use first geofence
CFG-GEOFENCE-FENCE1_LAT	0x40240021	I4	1e-7	deg	Latitude of the first geofence circle center
CFG-GEOFENCE-FENCE1_LON	0x40240022	I4	1e-7	deg	Longitude of the first geofence circle center
CFG-GEOFENCE-FENCE1_RAD	0x40240023	U4	0.01	m	Radius of the first geofence circle
CFG-GEOFENCE-USE_FENCE2	0x10240030	L	-	-	Use second geofence
CFG-GEOFENCE-FENCE2_LAT	0x40240031	I4	1e-7	deg	Latitude of the second geofence circle center
CFG-GEOFENCE-FENCE2_LON	0x40240032	I4	1e-7	deg	Longitude of the second geofence circle center
CFG-GEOFENCE-FENCE2_RAD	0x40240033	U4	0.01	m	Radius of the second geofence circle
CFG-GEOFENCE-USE_FENCE3	0x10240040	L	-	-	Use third geofence
CFG-GEOFENCE-FENCE3_LAT	0x40240041	I4	1e-7	deg	Latitude of the third geofence circle center
CFG-GEOFENCE-FENCE3_LON	0x40240042	I4	1e-7	deg	Longitude of the third geofence circle center
CFG-GEOFENCE-FENCE3_RAD	0x40240043	U4	0.01	m	Radius of the third geofence circle
CFG-GEOFENCE-USE_FENCE4	0x10240050	L	-	-	Use fourth geofence
CFG-GEOFENCE-FENCE4_LAT	0x40240051	I4	1e-7	deg	Latitude of the fourth geofence circle center
CFG-GEOFENCE-FENCE4_LON	0x40240052	I4	1e-7	deg	Longitude of the fourth geofence circle center
CFG-GEOFENCE-FENCE4_RAD	0x40240053	U4	0.01	m	Radius of the fourth geofence circle
CFG-GEOFENCE-VERTEX1_LAT	0x40240061	I4	1e-7	deg	Latitude of the 1st geofence vertex
CFG-GEOFENCE-VERTEX1_LON	0x40240062	I4	1e-7	deg	Longitude of the 1st geofence vertex
CFG-GEOFENCE-VERTEX2_LAT	0x40240063	I4	1e-7	deg	Latitude of the 2nd geofence vertex
CFG-GEOFENCE-VERTEX2_LON	0x40240064	I4	1e-7	deg	Longitude of the 2nd geofence vertex
CFG-GEOFENCE-VERTEX3_LAT	0x40240065	I4	1e-7	deg	Latitude of the 3rd geofence vertex
CFG-GEOFENCE-VERTEX3_LON	0x40240066	I4	1e-7	deg	Longitude of the 3rd geofence vertex
CFG-GEOFENCE-VERTEX4_LAT	0x40240067	I4	1e-7	deg	Latitude of the 4th geofence vertex
CFG-GEOFENCE-VERTEX4_LON	0x40240068	I4	1e-7	deg	Longitude of the 4th geofence vertex
CFG-GEOFENCE-VERTEX5_LAT	0x40240069	I4	1e-7	deg	Latitude of the 5th geofence vertex
CFG-GEOFENCE-VERTEX5_LON	0x4024006a	I4	1e-7	deg	Longitude of the 5th geofence vertex
CFG-GEOFENCE-VERTEX6_LAT	0x4024006b	I4	1e-7	deg	Latitude of the 6th geofence vertex
CFG-GEOFENCE-VERTEX6_LON	0x4024006c	I4	1e-7	deg	Longitude of the 6th geofence vertex
CFG-GEOFENCE-VERTEX7_LAT	0x4024006d	I4	1e-7	deg	Latitude of the 7th geofence vertex
CFG-GEOFENCE-VERTEX7_LON	0x4024006e	I4	1e-7	deg	Longitude of the 7th geofence vertex
CFG-GEOFENCE-VERTEX8_LAT	0x4024006f	I4	1e-7	deg	Latitude of the 8th geofence vertex
CFG-GEOFENCE-VERTEX8_LON	0x40240070	I4	1e-7	deg	Longitude of the 8th geofence vertex
CFG-GEOFENCE-VERTEX9_LAT	0x40240071	I4	1e-7	deg	Latitude of the 9th geofence vertex
CFG-GEOFENCE-VERTEX9_LON	0x40240072	I4	1e-7	deg	Longitude of the 9th geofence vertex
CFG-GEOFENCE-VERTEX10_LAT	0x40240073	I4	1e-7	deg	Latitude of the 10th geofence vertex
CFG-GEOFENCE-VERTEX10_LON	0x40240074	I4	1e-7	deg	Longitude of the 10th geofence vertex
CFG-GEOFENCE-VERTEX11_LAT	0x40240075	I4	1e-7	deg	Latitude of the 11th geofence vertex
CFG-GEOFENCE-VERTEX11_LON	0x40240076	I4	1e-7	deg	Longitude of the 11th geofence vertex
CFG-GEOFENCE-VERTEX12_LAT	0x40240077	I4	1e-7	deg	Latitude of the 12th geofence vertex

Configuration item	Key ID	Type	Scale	Unit	Description
CFG-GEOFENCE-VERTEX12_LON	0x40240078	I4	1e-7	deg	Longitude of the 12th geofence vertex
CFG-GEOFENCE-VERTEX13_LAT	0x40240079	I4	1e-7	deg	Latitude of the 13th geofence vertex
CFG-GEOFENCE-VERTEX13_LON	0x4024007a	I4	1e-7	deg	Longitude of the 13th geofence vertex
CFG-GEOFENCE-VERTEX14_LAT	0x4024007b	I4	1e-7	deg	Latitude of the 14th geofence vertex
CFG-GEOFENCE-VERTEX14_LON	0x4024007c	I4	1e-7	deg	Longitude of the 14th geofence vertex
CFG-GEOFENCE-VERTEX15_LAT	0x4024007d	I4	1e-7	deg	Latitude of the 15th geofence vertex
CFG-GEOFENCE-VERTEX15_LON	0x4024007e	I4	1e-7	deg	Longitude of the 15th geofence vertex
CFG-GEOFENCE-VERTEX16_LAT	0x4024007f	I4	1e-7	deg	Latitude of the 16th geofence vertex
CFG-GEOFENCE-VERTEX16_LON	0x40240080	I4	1e-7	deg	Longitude of the 16th geofence vertex
CFG-GEOFENCE-VERTEX17_LAT	0x40240081	I4	1e-7	deg	Latitude of the 17th geofence vertex
CFG-GEOFENCE-VERTEX17_LON	0x40240082	I4	1e-7	deg	Longitude of the 17th geofence vertex
CFG-GEOFENCE-VERTEX18_LAT	0x40240083	I4	1e-7	deg	Latitude of the 18th geofence vertex
CFG-GEOFENCE-VERTEX18_LON	0x40240084	I4	1e-7	deg	Longitude of the 18th geofence vertex
CFG-GEOFENCE-VERTEX19_LAT	0x40240085	I4	1e-7	deg	Latitude of the 19th geofence vertex
CFG-GEOFENCE-VERTEX19_LON	0x40240086	I4	1e-7	deg	Longitude of the 19th geofence vertex
CFG-GEOFENCE-VERTEX20_LAT	0x40240087	I4	1e-7	deg	Latitude of the 20th geofence vertex
CFG-GEOFENCE-VERTEX20_LON	0x40240088	I4	1e-7	deg	Longitude of the 20th geofence vertex

Table 9: CFG-GEOFENCE configuration items

Constant	Value	Description
L000	0	No confidence
L680	1	68%
L950	2	95%
L997	3	99.7%
L9999	4	99.99%
L999999	5	99.9999%

Table 10: Constants for CFG-GEOFENCE-CONFLVL

Constant	Value	Description
LOW_IN	0	PIO low means inside geofence
LOW_OUT	1	PIO low means outside geofence

Table 11: Constants for CFG-GEOFENCE-PINPOL

4.9.5 CFG-HW: Hardware configuration

Hardware configuration settings.

Note that not all settings are available for all products. See the applicable data sheet for supported features.

Configuration item	Key ID	Type	Scale	Unit	Description
CFG-HW-ANT_CFG_VOLTCTRL	0x10a3002e	L	-	-	Active antenna voltage control flag Enable active antenna voltage control flag. Used by EXT and MADC engines.
CFG-HW-ANT_CFG_SHORTDET	0x10a3002f	L	-	-	Short antenna detection flag Enable short antenna detection flag. Used by EXT and MADC engines.
CFG-HW-ANT_CFG_SHORTDET_POL	0x10a30030	L	-	-	Short antenna detection polarity

Configuration item	Key ID	Type	Scale	Unit	Description
Set to true if polarity of the antenna short detection is active low. Used by EXT engine.					
CFG-HW-ANT_CFG_OPENDET	0x10a30031	L	-	-	Open antenna detection flag
Enable open antenna detection flag. Used by EXT and MADC engines.					
CFG-HW-ANT_CFG_OPENDET_POL	0x10a30032	L	-	-	Open antenna detection polarity
Set to true if polarity of the antenna open detection is active low. Used by EXT engine.					
CFG-HW-ANT_CFG_PWRDOWN	0x10a30033	L	-	-	Power down antenna flag
Enable power down antenna logic in the event of antenna short circuit. CFG-HW-ANT_CFG_SHORTDET must be enabled to use this feature. Used by EXT and MADC engines.					
CFG-HW-ANT_CFG_PWRDOWN_POL	0x10a30034	L	-	-	Power down antenna logic polarity
Set to true if polarity of the antenna power down logic is active high. Used by EXT and MADC engines.					
CFG-HW-ANT_CFG_RECOVER	0x10a30035	L	-	-	Automatic recovery from short state flag
Enable automatic recovery from short state. Used by EXT and MADC engines.					
CFG-HW-ANT_SUP_SWITCH_PIN	0x20a30036	U1	-	-	Antenna switch PIO number
Antenna switch PIO number. Used by EXT and MADC engines.					
CFG-HW-ANT_SUP_SHORT_PIN	0x20a30037	U1	-	-	Antenna short detection PIO number
Antenna short detection PIO number. Used by EXT engine.					
CFG-HW-ANT_SUP_OPEN_PIN	0x20a30038	U1	-	-	Antenna open detection PIO number
Antenna open detection PIO number. Used by EXT engine.					
CFG-HW-ANT_ON_SHORT_US	0x30a3003c	U2	-	-	ANT on->short timeout[μs]
Delay in microseconds between turning the antenna power supply on and enabling the antenna short circuit detection.					
CFG-HW-ANT_SUP_ENGINE	0x20a30054	E1	-	-	Antenna supervisor engine selection
Select the engine used to evaluate antenna state.					
The EXT engine uses an external comparator for current measurement.					
The MADC engine uses a built-in measurement ADC and requires only a shunt resistor for current measurement. The MADC engine is supported only in selected u-blox generation 9 receivers.					
See Table 13 below for a list of possible constants for this item.					
CFG-HW-RF_LNA_MODE	0x20a30057	E1	-	-	Mode for internal LNA
Sets the operating mode for the RF LNA. Lowgain or bypass options can be used if there is already a external LNA in front of the chip with sufficient gain.					
See Table 14 below for a list of possible constants for this item.					

Table 12: CFG-HW configuration items

Constant	Value	Description
EXT	0	Use the EXT engine (not available in all products)
MADC	1	Use the MADC engine (not available in all products)

Table 13: Constants for CFG-HW-ANT_SUP_ENGINE

Constant	Value	Description
NORMAL	0	All RFs. Normal operation, internal LNA enabled at full gain
LOWGAIN	1	All RFs. LNA enabled in low gain mode
BYPASS	2	All RFs. Bypass LNA

Table 14: Constants for CFG-HW-RF_LNA_MODE

4.9.6 CFG-I2C: Configuration of the I2C interface

Settings needed to configure the I2C communication interface.

Configuration item	Key ID	Type	Scale	Unit	Description
<i>CFG-I2C-ADDRESS</i>	0x20510001	U1	-	-	I2C address of the receiver (7 bits)
<i>CFG-I2C-EXTENDEDTIMEOUT</i>	0x10510002	L	-	-	Flag to disable timeouting the interface after 1.5 s
<i>CFG-I2C-ENABLED</i>	0x10510003	L	-	-	Flag to indicate if the I2C interface should be enabled

Table 15: CFG-I2C configuration items

4.9.7 CFG-I2CINPROT: Input protocol configuration of the I2C interface

Input protocol enable flags of the I2C interface.

Configuration item	Key ID	Type	Scale	Unit	Description
<i>CFG-I2CINPROT-UBX</i>	0x10710001	L	-	-	Flag to indicate if UBX should be an input protocol on I2C
<i>CFG-I2CINPROT-NMEA</i>	0x10710002	L	-	-	Flag to indicate if NMEA should be an input protocol on I2C

Table 16: CFG-I2CINPROT configuration items

4.9.8 CFG-I2COUTPROT: Output protocol configuration of the I2C interface

Output protocol enable flags of the I2C interface.

Configuration item	Key ID	Type	Scale	Unit	Description
<i>CFG-I2COUTPROT-UBX</i>	0x10720001	L	-	-	Flag to indicate if UBX should be an output protocol on I2C
<i>CFG-I2COUTPROT-NMEA</i>	0x10720002	L	-	-	Flag to indicate if NMEA should be an output protocol on I2C

Table 17: CFG-I2COUTPROT configuration items

4.9.9 CFG-INFMSG: Information message configuration

Information message configuration for the NMEA and UBX protocols.

Configuration item	Key ID	Type	Scale	Unit	Description
<i>CFG-INFMSG-UBX_I2C</i>	0x20920001	X1	-	-	Information message enable flags for the UBX protocol on the I2C interface
See Table 19 below for a list of possible constants for this item.					
<i>CFG-INFMSG-UBX_UART1</i>	0x20920002	X1	-	-	Information message enable flags for the UBX protocol on the UART1 interface
See Table 19 below for a list of possible constants for this item.					
<i>CFG-INFMSG-UBX_SPI</i>	0x20920005	X1	-	-	Information message enable flags for the UBX protocol on the SPI interface
See Table 19 below for a list of possible constants for this item.					
<i>CFG-INFMSG-NMEA_I2C</i>	0x20920006	X1	-	-	Information message enable flags for the NMEA protocol on the I2C interface
See Table 19 below for a list of possible constants for this item.					
<i>CFG-INFMSG-NMEA_UART1</i>	0x20920007	X1	-	-	Information message enable flags for the NMEA protocol on the UART1 interface
See Table 19 below for a list of possible constants for this item.					
<i>CFG-INFMSG-NMEA_SPI</i>	0x2092000a	X1	-	-	Information message enable flags for the NMEA protocol on the SPI interface
See Table 19 below for a list of possible constants for this item.					

Table 18: CFG-INFMSG configuration items

Constant	Value	Description
<i>ERROR</i>	0x01	Enable ERROR information messages
<i>WARNING</i>	0x02	Enable WARNING information messages
<i>NOTICE</i>	0x04	Enable NOTICE information messages
<i>TEST</i>	0x08	Enable TEST information messages
<i>DEBUG</i>	0x10	Enable DEBUG information messages

Table 19: Constants for CFG-INFMSG-UBX_I2C, CFG-INFMSG-UBX_UART1, CFG-INFMSG-UBX_SPI, CFG-INFMSG-NMEA_I2C, CFG-INFMSG-NMEA_UART1, CFG-INFMSG-NMEA_SPI

4.9.10 CFG-LOGFILTER: Data logger configuration

This group can be used to configure the data logger, i.e. to enable/disable the log recording and to get/set the position entry filter settings.

Position entries can be filtered based on time difference, position difference or current speed thresholds. Position and speed filtering also have a minimum time interval. A position is logged if any of the thresholds are exceeded. If a threshold is set to zero it is ignored. The maximum rate of position logging is 1 Hz.

The filter settings will be configured to the provided values only if the APPLY_ALL_FILTERS flag is set. This allows the recording to be enabled/disabled independently of configuring the filter settings.

It is possible to configure the data logger in the absence of a logging file. By doing so, once the logging file is created, the data logger configuration will take effect immediately and logging recording and filtering will activate according to the configuration.

Configuration item	Key ID	Type	Scale	Unit	Description
<i>CFG-LOGFILTER-RECORD_ENA</i> Set to true when recording enabled.	0x10de0002	L	-	-	Recording enabled
<i>CFG-LOGFILTER-ONCE_PER_WAKE_UP_ENA</i> Set to true recording only one single position per Long Interval Tracking mode wake-up period is enabled. Note: the value set here does not take effect unless CFG-LOGFILTER-APPLY_ALL_FILTERS is enabled.	0x10de0003	L	-	-	Once per wake up
<i>CFG-LOGFILTER-APPLY_ALL_FILTERS</i> Set to true when all filter settings are to be applied, not just recording enabling/disabling.	0x10de0004	L	-	-	Apply all filter settings
<i>CFG-LOGFILTER-MIN_INTERVAL</i> Minimum time interval between logged positions (0 = not set). This is only applied in combination with the speed and/or position thresholds. If both MIN_INTERVAL and TIME_THRS are set, MIN_INTERVAL must be less than or equal to TIME_THRS. Note: the value set here does not take effect unless CFG-LOGFILTER-APPLY_ALL_FILTERS is enabled.	0x30de0005	U2	-	s	Minimum time interval between logged positions
<i>CFG-LOGFILTER-TIME_THRS</i> If the time difference is greater than the threshold then the position is logged (0 = not set). Note: the value set here does not take effect unless CFG-LOGFILTER-APPLY_ALL_FILTERS is enabled.	0x30de0006	U2	-	s	Time threshold
<i>CFG-LOGFILTER-SPEED_THRS</i> If the current speed is greater than the threshold then the position is logged (0 = not set). MIN_INTERVAL also applies. Note: value set here does not take effect unless CFG-LOGFILTER-APPLY_ALL_FILTERS is enabled.	0x30de0007	U2	-	m/s	Speed threshold
<i>CFG-LOGFILTER-POSITION_THRS</i>	0x40de0008	U4	-	m	Position threshold

Configuration item	Key ID	Type	Scale	Unit	Description
If the 3D position difference is greater than the threshold then the position is logged (0 = not set). MIN_INTERVAL also applies.					
Note: the value set here does not take effect unless CFG-LOGFILTER-APPLY_ALL_FILTERS is enabled.					

Table 20: CFG-LOGFILTER configuration items

4.9.11 CFG-MOT: Motion detector configuration

The items in this group specify the parameters used for the internal receiver motion detector. The platform motion is assessed by combining the detected motion of different detectors looking at specific data types (i.e. GNSS, gyroscopes, accelerometers, wheel ticks). The decision thresholds of the internal detectors can be specified using the configuration items in this group.

Configuration item	Key ID	Type	Scale	Unit	Description
CFG-MOT-GNSSSPEED_THRS	0x20250038	U1	0.01	m/s	Static hold speed threshold, below which the receiver is considered to be stationary
Set this parameter to 0 to enable the default firmware value or behavior.					
CFG-MOT-GNSSDIST_THRS	0x3025003b	U2	1.0	m	Static hold distance threshold, within which the receiver is considered to be stationary
Set this parameter to 0 to enable the default firmware value or behavior.					

Table 21: CFG-MOT configuration items

4.9.12 CFG-MSGOUT: Message output configuration

For each message and port a separate output rate (per second, per epoch) can be configured.

Configuration item	Key ID	Type	Scale	Unit	Description
CFG-MSGOUT-NMEA_ID_DTM_I2C	0x209100a6	U1	-	-	Output rate of the NMEA-GX-DTM message on the I2C port
CFG-MSGOUT-NMEA_ID_DTM_SPI	0x209100aa	U1	-	-	Output rate of the NMEA-GX-DTM message on the SPI port
CFG-MSGOUT-NMEA_ID_DTM_UART1	0x209100a7	U1	-	-	Output rate of the NMEA-GX-DTM message on the UART1 port
CFG-MSGOUT-NMEA_ID_GBS_I2C	0x209100dd	U1	-	-	Output rate of the NMEA-GX-GBS message on the I2C port
CFG-MSGOUT-NMEA_ID_GBS_SPI	0x209100e1	U1	-	-	Output rate of the NMEA-GX-GBS message on the SPI port
CFG-MSGOUT-NMEA_ID_GBS_UART1	0x209100de	U1	-	-	Output rate of the NMEA-GX-GBS message on the UART1 port
CFG-MSGOUT-NMEA_ID_GGA_I2C	0x209100ba	U1	-	-	Output rate of the NMEA-GX-GGA message on the I2C port
CFG-MSGOUT-NMEA_ID_GGA_SPI	0x209100be	U1	-	-	Output rate of the NMEA-GX-GGA message on the SPI port
CFG-MSGOUT-NMEA_ID_GGA_UART1	0x209100bb	U1	-	-	Output rate of the NMEA-GX-GGA message on the UART1 port
CFG-MSGOUT-NMEA_ID_GLL_I2C	0x209100c9	U1	-	-	Output rate of the NMEA-GX-GLL message on the I2C port
CFG-MSGOUT-NMEA_ID_GLL_SPI	0x209100cd	U1	-	-	Output rate of the NMEA-GX-GLL message on the SPI port
CFG-MSGOUT-NMEA_ID_GLL_UART1	0x209100ca	U1	-	-	Output rate of the NMEA-GX-GLL message on the UART1 port
CFG-MSGOUT-NMEA_ID_GNS_I2C	0x209100b5	U1	-	-	Output rate of the NMEA-GX-GNS message on the I2C port
CFG-MSGOUT-NMEA_ID_GNS_SPI	0x209100b9	U1	-	-	Output rate of the NMEA-GX-GNS message on the SPI port

Configuration item	Key ID	Type	Scale	Unit	Description
<i>CFG-MSGOUT-NMEA_ID_GNS_UART1</i>	0x209100b6	U1	-	-	Output rate of the NMEA-GX-GNS message on the UART1 port
<i>CFG-MSGOUT-NMEA_ID_GRS_I2C</i>	0x209100ce	U1	-	-	Output rate of the NMEA-GX-GRS message on the I2C port
<i>CFG-MSGOUT-NMEA_ID_GRS_SPI</i>	0x209100d2	U1	-	-	Output rate of the NMEA-GX-GRS message on the SPI port
<i>CFG-MSGOUT-NMEA_ID_GRS_UART1</i>	0x209100cf	U1	-	-	Output rate of the NMEA-GX-GRS message on the UART1 port
<i>CFG-MSGOUT-NMEA_ID_GSA_I2C</i>	0x209100bf	U1	-	-	Output rate of the NMEA-GX-GSA message on the I2C port
<i>CFG-MSGOUT-NMEA_ID_GSA_SPI</i>	0x209100c3	U1	-	-	Output rate of the NMEA-GX-GSA message on the SPI port
<i>CFG-MSGOUT-NMEA_ID_GSA_UART1</i>	0x209100c0	U1	-	-	Output rate of the NMEA-GX-GSA message on the UART1 port
<i>CFG-MSGOUT-NMEA_ID_GST_I2C</i>	0x209100d3	U1	-	-	Output rate of the NMEA-GX-GST message on the I2C port
<i>CFG-MSGOUT-NMEA_ID_GST_SPI</i>	0x209100d7	U1	-	-	Output rate of the NMEA-GX-GST message on the SPI port
<i>CFG-MSGOUT-NMEA_ID_GST_UART1</i>	0x209100d4	U1	-	-	Output rate of the NMEA-GX-GST message on the UART1 port
<i>CFG-MSGOUT-NMEA_ID_GSV_I2C</i>	0x209100c4	U1	-	-	Output rate of the NMEA-GX-GSV message on the I2C port
<i>CFG-MSGOUT-NMEA_ID_GSV_SPI</i>	0x209100c8	U1	-	-	Output rate of the NMEA-GX-GSV message on the SPI port
<i>CFG-MSGOUT-NMEA_ID_GSV_UART1</i>	0x209100c5	U1	-	-	Output rate of the NMEA-GX-GSV message on the UART1 port
<i>CFG-MSGOUT-NMEA_ID_RLM_I2C</i>	0x20910400	U1	-	-	Output rate of the NMEA-GX-RLM message on the I2C port
<i>CFG-MSGOUT-NMEA_ID_RLM_SPI</i>	0x20910404	U1	-	-	Output rate of the NMEA-GX-RLM message on the SPI port
<i>CFG-MSGOUT-NMEA_ID_RLM_UART1</i>	0x20910401	U1	-	-	Output rate of the NMEA-GX-RLM message on the UART1 port
<i>CFG-MSGOUT-NMEA_ID_RMC_I2C</i>	0x209100ab	U1	-	-	Output rate of the NMEA-GX-RMC message on the I2C port
<i>CFG-MSGOUT-NMEA_ID_RMC_SPI</i>	0x209100af	U1	-	-	Output rate of the NMEA-GX-RMC message on the SPI port
<i>CFG-MSGOUT-NMEA_ID_RMC_UART1</i>	0x209100ac	U1	-	-	Output rate of the NMEA-GX-RMC message on the UART1 port
<i>CFG-MSGOUT-NMEA_ID_VLW_I2C</i>	0x209100e7	U1	-	-	Output rate of the NMEA-GX-VLW message on the I2C port
<i>CFG-MSGOUT-NMEA_ID_VLW_SPI</i>	0x209100eb	U1	-	-	Output rate of the NMEA-GX-VLW message on the SPI port
<i>CFG-MSGOUT-NMEA_ID_VLW_UART1</i>	0x209100e8	U1	-	-	Output rate of the NMEA-GX-VLW message on the UART1 port
<i>CFG-MSGOUT-NMEA_ID_VTG_I2C</i>	0x209100b0	U1	-	-	Output rate of the NMEA-GX-VTG message on the I2C port
<i>CFG-MSGOUT-NMEA_ID_VTG_SPI</i>	0x209100b4	U1	-	-	Output rate of the NMEA-GX-VTG message on the SPI port
<i>CFG-MSGOUT-NMEA_ID_VTG_UART1</i>	0x209100b1	U1	-	-	Output rate of the NMEA-GX-VTG message on the UART1 port
<i>CFG-MSGOUT-NMEA_ID_ZDA_I2C</i>	0x209100d8	U1	-	-	Output rate of the NMEA-GX-ZDA message on the I2C port

Configuration item	Key ID	Type	Scale	Unit	Description
CFG-MSGOUT-NMEA_ID_ZDA_SPI	0x209100dc	U1	-	-	Output rate of the NMEA-GX-ZDA message on the SPI port
CFG-MSGOUT-NMEA_ID_ZDA_UART1	0x209100d9	U1	-	-	Output rate of the NMEA-GX-ZDA message on the UART1 port
CFG-MSGOUT-PUBX_ID_POLYP_I2C	0x209100ec	U1	-	-	Output rate of the NMEA-GX-PUBX00 message on the I2C port
CFG-MSGOUT-PUBX_ID_POLYP_SPI	0x209100f0	U1	-	-	Output rate of the NMEA-GX-PUBX00 message on the SPI port
CFG-MSGOUT-PUBX_ID_POLYP_UART1	0x209100ed	U1	-	-	Output rate of the NMEA-GX-PUBX00 message on the UART1 port
CFG-MSGOUT-PUBX_ID_POLYS_I2C	0x209100f1	U1	-	-	Output rate of the NMEA-GX-PUBX03 message on the I2C port
CFG-MSGOUT-PUBX_ID_POLYS_SPI	0x209100f5	U1	-	-	Output rate of the NMEA-GX-PUBX03 message on the SPI port
CFG-MSGOUT-PUBX_ID_POLYS_UART1	0x209100f2	U1	-	-	Output rate of the NMEA-GX-PUBX03 message on the UART1 port
CFG-MSGOUT-PUBX_ID_POLYT_I2C	0x209100f6	U1	-	-	Output rate of the NMEA-GX-PUBX04 message on the I2C port
CFG-MSGOUT-PUBX_ID_POLYT_SPI	0x209100fa	U1	-	-	Output rate of the NMEA-GX-PUBX04 message on the SPI port
CFG-MSGOUT-PUBX_ID_POLYT_UART1	0x209100f7	U1	-	-	Output rate of the NMEA-GX-PUBX04 message on the UART1 port
CFG-MSGOUT-UBX_LOG_INFO_I2C	0x20910259	U1	-	-	Output rate of the UBX-LOG-INFO message on the I2C port
CFG-MSGOUT-UBX_LOG_INFO_SPI	0x2091025d	U1	-	-	Output rate of the UBX-LOG-INFO message on the SPI port
CFG-MSGOUT-UBX_LOG_INFO_UART1	0x2091025a	U1	-	-	Output rate of the UBX-LOG-INFO message on the UART1 port
CFG-MSGOUT-UBX_MON_COMMS_I2C	0x2091034f	U1	-	-	Output rate of the UBX-MON-COMMS message on the I2C port
CFG-MSGOUT-UBX_MON_COMMS_SPI	0x20910353	U1	-	-	Output rate of the UBX-MON-COMMS message on the SPI port
CFG-MSGOUT-UBX_MON_COMMS_UART1	0x20910350	U1	-	-	Output rate of the UBX-MON-COMMS message on the UART1 port
CFG-MSGOUT-UBX_MON_HW3_I2C	0x20910354	U1	-	-	Output rate of the UBX-MON-HW3 message on the I2C port
CFG-MSGOUT-UBX_MON_HW3_SPI	0x20910358	U1	-	-	Output rate of the UBX-MON-HW3 message on the SPI port
CFG-MSGOUT-UBX_MON_HW3_UART1	0x20910355	U1	-	-	Output rate of the UBX-MON-HW3 message on the UART1 port
CFG-MSGOUT-UBX_MON_IO_I2C	0x209101a5	U1	-	-	Output rate of the UBX-MON-IO message on the I2C port
CFG-MSGOUT-UBX_MON_IO_SPI	0x209101a9	U1	-	-	Output rate of the UBX-MON-IO message on the SPI port
CFG-MSGOUT-UBX_MON_IO_UART1	0x209101a6	U1	-	-	Output rate of the UBX-MON-IO message on the UART1 port
CFG-MSGOUT-UBX_MON_MSGPP_I2C	0x20910196	U1	-	-	Output rate of the UBX-MON-MSGPP message on the I2C port
CFG-MSGOUT-UBX_MON_MSGPP_SPI	0x2091019a	U1	-	-	Output rate of the UBX-MON-MSGPP message on the SPI port
CFG-MSGOUT-UBX_MON_MSGPP_UART1	0x20910197	U1	-	-	Output rate of the UBX-MON-MSGPP message on the UART1 port

Configuration item	Key ID	Type	Scale	Unit	Description
CFG-MSGOUT-UBX_MON_RF_I2C	0x20910359	U1	-	-	Output rate of the UBX-MON-RF message on the I2C port
CFG-MSGOUT-UBX_MON_RF_SPI	0x2091035d	U1	-	-	Output rate of the UBX-MON-RF message on the SPI port
CFG-MSGOUT-UBX_MON_RF_UART1	0x2091035a	U1	-	-	Output rate of the UBX-MON-RF message on the UART1 port
CFG-MSGOUT-UBX_MON_RXBUF_I2C	0x209101a0	U1	-	-	Output rate of the UBX-MON-RXBUF message on the I2C port
CFG-MSGOUT-UBX_MON_RXBUF_SPI	0x209101a4	U1	-	-	Output rate of the UBX-MON-RXBUF message on the SPI port
CFG-MSGOUT-UBX_MON_RXBUF_UART1	0x209101a1	U1	-	-	Output rate of the UBX-MON-RXBUF message on the UART1 port
CFG-MSGOUT-UBX_MON_RXR_I2C	0x20910187	U1	-	-	Output rate of the UBX-MON-RXR message on the I2C port
CFG-MSGOUT-UBX_MON_RXR_SPI	0x2091018b	U1	-	-	Output rate of the UBX-MON-RXR message on the SPI port
CFG-MSGOUT-UBX_MON_RXR_UART1	0x20910188	U1	-	-	Output rate of the UBX-MON-RXR message on the UART1 port
CFG-MSGOUT-UBX_MON_SPAN_I2C	0x2091038b	U1	-	-	Output rate of the UBX-MON-SPAN message on the I2C port
CFG-MSGOUT-UBX_MON_SPAN_SPI	0x2091038f	U1	-	-	Output rate of the UBX-MON-SPAN message on the SPI port
CFG-MSGOUT-UBX_MON_SPAN_UART1	0x2091038c	U1	-	-	Output rate of the UBX-MON-SPAN message on the UART1 port
CFG-MSGOUT-UBX_MON_SYS_I2C	0x2091069d	U1	-	-	Output rate of the UBX-MON-SYS message on the I2C port
CFG-MSGOUT-UBX_MON_SYS_SPI	0x209106a1	U1	-	-	Output rate of the UBX-MON-SYS message on the SPI port
CFG-MSGOUT-UBX_MON_SYS_UART1	0x2091069e	U1	-	-	Output rate of the UBX-MON-SYS message on the UART1 port
CFG-MSGOUT-UBX_MON_TXBUF_I2C	0x2091019b	U1	-	-	Output rate of the UBX-MON-TXBUF message on the I2C port
CFG-MSGOUT-UBX_MON_TXBUF_SPI	0x2091019f	U1	-	-	Output rate of the UBX-MON-TXBUF message on the SPI port
CFG-MSGOUT-UBX_MON_TXBUF_UART1	0x2091019c	U1	-	-	Output rate of the UBX-MON-TXBUF message on the UART1 port
CFG-MSGOUT-UBX_NAV_AOPSTATUS_I2C	0x20910079	U1	-	-	Output rate of the UBX-NAV-AOPSTATUS message on the I2C port
CFG-MSGOUT-UBX_NAV_AOPSTATUS_SPI	0x2091007d	U1	-	-	Output rate of the UBX-NAV-AOPSTATUS message on the SPI port
CFG-MSGOUT-UBX_NAV_AOPSTATUS_UART1	0x2091007a	U1	-	-	Output rate of the UBX-NAV-AOPSTATUS message on the UART1 port
CFG-MSGOUT-UBX_NAV_CLOCK_I2C	0x20910065	U1	-	-	Output rate of the UBX-NAV-CLOCK message on the I2C port
CFG-MSGOUT-UBX_NAV_CLOCK_SPI	0x20910069	U1	-	-	Output rate of the UBX-NAV-CLOCK message on the SPI port
CFG-MSGOUT-UBX_NAV_CLOCK_UART1	0x20910066	U1	-	-	Output rate of the UBX-NAV-CLOCK message on the UART1 port
CFG-MSGOUT-UBX_NAV_COV_I2C	0x20910083	U1	-	-	Output rate of the UBX-NAV-COV message on the I2C port
CFG-MSGOUT-UBX_NAV_COV_SPI	0x20910087	U1	-	-	Output rate of the UBX-NAV-COV message on the SPI port

Configuration item	Key ID	Type	Scale	Unit	Description
CFG-MSGOUT-UBX_NAV_COV_UART1	0x20910084	U1	-	-	Output rate of the UBX-NAV-COV message on the UART1 port
CFG-MSGOUT-UBX_NAV_DOP_I2C	0x20910038	U1	-	-	Output rate of the UBX-NAV-DOP message on the I2C port
CFG-MSGOUT-UBX_NAV_DOP_SPI	0x2091003c	U1	-	-	Output rate of the UBX-NAV-DOP message on the SPI port
CFG-MSGOUT-UBX_NAV_DOP_UART1	0x20910039	U1	-	-	Output rate of the UBX-NAV-DOP message on the UART1 port
CFG-MSGOUT-UBX_NAV_EOE_I2C	0x2091015f	U1	-	-	Output rate of the UBX-NAV-EOE message on the I2C port
CFG-MSGOUT-UBX_NAV_EOE_SPI	0x20910163	U1	-	-	Output rate of the UBX-NAV-EOE message on the SPI port
CFG-MSGOUT-UBX_NAV_EOE_UART1	0x20910160	U1	-	-	Output rate of the UBX-NAV-EOE message on the UART1 port
CFG-MSGOUT-UBX_NAV_GEOFENCE_I2C	0x209100a1	U1	-	-	Output rate of the UBX-NAV-GEOFENCE message on the I2C port
CFG-MSGOUT-UBX_NAV_GEOFENCE_SPI	0x209100a5	U1	-	-	Output rate of the UBX-NAV-GEOFENCE message on the SPI port
CFG-MSGOUT-UBX_NAV_GEOFENCE_UART1	0x209100a2	U1	-	-	Output rate of the UBX-NAV-GEOFENCE message on the UART1 port
CFG-MSGOUT-UBX_NAV_ODO_I2C	0x2091007e	U1	-	-	Output rate of the UBX-NAV-ODO message on the I2C port
CFG-MSGOUT-UBX_NAV_ODO_SPI	0x20910082	U1	-	-	Output rate of the UBX-NAV-ODO message on the SPI port
CFG-MSGOUT-UBX_NAV_ODO_UART1	0x2091007f	U1	-	-	Output rate of the UBX-NAV-ODO message on the UART1 port
CFG-MSGOUT-UBX_NAV_ORB_I2C	0x20910010	U1	-	-	Output rate of the UBX-NAV-ORB message on the I2C port
CFG-MSGOUT-UBX_NAV_ORB_SPI	0x20910014	U1	-	-	Output rate of the UBX-NAV-ORB message on the SPI port
CFG-MSGOUT-UBX_NAV_ORB_UART1	0x20910011	U1	-	-	Output rate of the UBX-NAV-ORB message on the UART1 port
CFG-MSGOUT-UBX_NAV_POSECEF_I2C	0x20910024	U1	-	-	Output rate of the UBX-NAV-POSECEF message on the I2C port
CFG-MSGOUT-UBX_NAV_POSECEF_SPI	0x20910028	U1	-	-	Output rate of the UBX-NAV-POSECEF message on the SPI port
CFG-MSGOUT-UBX_NAV_POSECEF_UART1	0x20910025	U1	-	-	Output rate of the UBX-NAV-POSECEF message on the UART1 port
CFG-MSGOUT-UBX_NAV_POSLLH_I2C	0x20910029	U1	-	-	Output rate of the UBX-NAV-POSLLH message on the I2C port
CFG-MSGOUT-UBX_NAV_POSLLH_SPI	0x2091002d	U1	-	-	Output rate of the UBX-NAV-POSLLH message on the SPI port
CFG-MSGOUT-UBX_NAV_POSLLH_UART1	0x2091002a	U1	-	-	Output rate of the UBX-NAV-POSLLH message on the UART1 port
CFG-MSGOUT-UBX_NAV_PVT_I2C	0x20910006	U1	-	-	Output rate of the UBX-NAV-PVT message on the I2C port
CFG-MSGOUT-UBX_NAV_PVT_SPI	0x2091000a	U1	-	-	Output rate of the UBX-NAV-PVT message on the SPI port
CFG-MSGOUT-UBX_NAV_PVT_UART1	0x20910007	U1	-	-	Output rate of the UBX-NAV-PVT message on the UART1 port
CFG-MSGOUT-UBX_NAV_SAT_I2C	0x20910015	U1	-	-	Output rate of the UBX-NAV-SAT message on the I2C port

Configuration item	Key ID	Type	Scale	Unit	Description
CFG-MSGOUT-UBX_NAV_SAT_SPI	0x20910019	U1	-	-	Output rate of the UBX-NAV-SAT message on the SPI port
CFG-MSGOUT-UBX_NAV_SAT_UART1	0x20910016	U1	-	-	Output rate of the UBX-NAV-SAT message on the UART1 port
CFG-MSGOUT-UBX_NAV_SBAS_I2C	0x2091006a	U1	-	-	Output rate of the UBX-NAV-SBAS message on the I2C port
CFG-MSGOUT-UBX_NAV_SBAS_SPI	0x2091006e	U1	-	-	Output rate of the UBX-NAV-SBAS message on the SPI port
CFG-MSGOUT-UBX_NAV_SBAS_UART1	0x2091006b	U1	-	-	Output rate of the UBX-NAV-SBAS message on the UART1 port
CFG-MSGOUT-UBX_NAV_SIG_I2C	0x20910345	U1	-	-	Output rate of the UBX-NAV-SIG message on the I2C port
CFG-MSGOUT-UBX_NAV_SIG_SPI	0x20910349	U1	-	-	Output rate of the UBX-NAV-SIG message on the SPI port
CFG-MSGOUT-UBX_NAV_SIG_UART1	0x20910346	U1	-	-	Output rate of the UBX-NAV-SIG message on the UART1 port
CFG-MSGOUT-UBX_NAV_SLAS_I2C	0x20910336	U1	-	-	Output rate of the UBX-NAV-SLAS message on the I2C port
CFG-MSGOUT-UBX_NAV_SLAS_SPI	0x2091033a	U1	-	-	Output rate of the UBX-NAV-SLAS message on the SPI port
CFG-MSGOUT-UBX_NAV_SLAS_UART1	0x20910337	U1	-	-	Output rate of the UBX-NAV-SLAS message on the UART1 port
CFG-MSGOUT-UBX_NAV_STATUS_I2C	0x2091001a	U1	-	-	Output rate of the UBX-NAV-STATUS message on the I2C port
CFG-MSGOUT-UBX_NAV_STATUS_SPI	0x2091001e	U1	-	-	Output rate of the UBX-NAV-STATUS message on the SPI port
CFG-MSGOUT-UBX_NAV_STATUS_UART1	0x2091001b	U1	-	-	Output rate of the UBX-NAV-STATUS message on the UART1 port
CFG-MSGOUT-UBX_NAV_TIMEBDS_I2C	0x20910051	U1	-	-	Output rate of the UBX-NAV-TIMEBDS message on the I2C port
CFG-MSGOUT-UBX_NAV_TIMEBDS_SPI	0x20910055	U1	-	-	Output rate of the UBX-NAV-TIMEBDS message on the SPI port
CFG-MSGOUT-UBX_NAV_TIMEBDS_UART1	0x20910052	U1	-	-	Output rate of the UBX-NAV-TIMEBDS message on the UART1 port
CFG-MSGOUT-UBX_NAV_TIMEGAL_I2C	0x20910056	U1	-	-	Output rate of the UBX-NAV-TIMEGAL message on the I2C port
CFG-MSGOUT-UBX_NAV_TIMEGAL_SPI	0x2091005a	U1	-	-	Output rate of the UBX-NAV-TIMEGAL message on the SPI port
CFG-MSGOUT-UBX_NAV_TIMEGAL_UART1	0x20910057	U1	-	-	Output rate of the UBX-NAV-TIMEGAL message on the UART1 port
CFG-MSGOUT-UBX_NAV_TIMEGPS_I2C	0x20910047	U1	-	-	Output rate of the UBX-NAV-TIMEGPS message on the I2C port
CFG-MSGOUT-UBX_NAV_TIMEGPS_SPI	0x2091004b	U1	-	-	Output rate of the UBX-NAV-TIMEGPS message on the SPI port
CFG-MSGOUT-UBX_NAV_TIMEGPS_UART1	0x20910048	U1	-	-	Output rate of the UBX-NAV-TIMEGPS message on the UART1 port
CFG-MSGOUT-UBX_NAV_TIMELS_I2C	0x20910060	U1	-	-	Output rate of the UBX-NAV-TIMELS message on the I2C port
CFG-MSGOUT-UBX_NAV_TIMELS_SPI	0x20910064	U1	-	-	Output rate of the UBX-NAV-TIMELS message on the SPI port
CFG-MSGOUT-UBX_NAV_TIMELS_UART1	0x20910061	U1	-	-	Output rate of the UBX-NAV-TIMELS message on the UART1 port

Configuration item	Key ID	Type	Scale	Unit	Description
CFG-MSGGOUT-UBX_NAV_TIMENAVIC_I2C	0x209106a2	U1	-	-	Output rate of the UBX-NAV-TIMENAVIC message on the I2C port
CFG-MSGGOUT-UBX_NAV_TIMENAVIC_SPI	0x209106a6	U1	-	-	Output rate of the UBX-NAV-TIMENAVIC message on the SPI port
CFG-MSGGOUT-UBX_NAV_TIMENAVIC_UART1	0x209106a3	U1	-	-	Output rate of the UBX-NAV-TIMENAVIC message on the UART1 port
CFG-MSGGOUT-UBX_NAV_TIMEQZSS_I2C	0x20910386	U1	-	-	Output rate of the UBX-NAV-TIMEQZSS message on the I2C port
CFG-MSGGOUT-UBX_NAV_TIMEQZSS_SPI	0x2091038a	U1	-	-	Output rate of the UBX-NAV-TIMEQZSS message on the SPI port
CFG-MSGGOUT-UBX_NAV_TIMEQZSS_UART1	0x20910387	U1	-	-	Output rate of the UBX-NAV-TIMEQZSS message on the UART1 port
CFG-MSGGOUT-UBX_NAV_TIMEUTC_I2C	0x2091005b	U1	-	-	Output rate of the UBX-NAV-TIMEUTC message on the I2C port
CFG-MSGGOUT-UBX_NAV_TIMEUTC_SPI	0x2091005f	U1	-	-	Output rate of the UBX-NAV-TIMEUTC message on the SPI port
CFG-MSGGOUT-UBX_NAV_TIMEUTC_UART1	0x2091005c	U1	-	-	Output rate of the UBX-NAV-TIMEUTC message on the UART1 port
CFG-MSGGOUT-UBX_NAV_VELECEF_I2C	0x2091003d	U1	-	-	Output rate of the UBX-NAV-VELECEF message on the I2C port
CFG-MSGGOUT-UBX_NAV_VELECEF_SPI	0x20910041	U1	-	-	Output rate of the UBX-NAV-VELECEF message on the SPI port
CFG-MSGGOUT-UBX_NAV_VELECEF_UART1	0x2091003e	U1	-	-	Output rate of the UBX-NAV-VELECEF message on the UART1 port
CFG-MSGGOUT-UBX_NAV_VELNED_I2C	0x20910042	U1	-	-	Output rate of the UBX-NAV-VELNED message on the I2C port
CFG-MSGGOUT-UBX_NAV_VELNED_SPI	0x20910046	U1	-	-	Output rate of the UBX-NAV-VELNED message on the SPI port
CFG-MSGGOUT-UBX_NAV_VELNED_UART1	0x20910043	U1	-	-	Output rate of the UBX-NAV-VELNED message on the UART1 port
CFG-MSGGOUT-UBX_RXM_MEASX_I2C	0x20910204	U1	-	-	Output rate of the UBX-RXM-MEASX message on the I2C port
CFG-MSGGOUT-UBX_RXM_MEASX_SPI	0x20910208	U1	-	-	Output rate of the UBX-RXM-MEASX message on the SPI port
CFG-MSGGOUT-UBX_RXM_MEASX_UART1	0x20910205	U1	-	-	Output rate of the UBX-RXM-MEASX message on the UART1 port
CFG-MSGGOUT-UBX_RXM_RAWX_I2C	0x209102a4	U1	-	-	Output rate of the UBX-RXM-RAWX message on the I2C port
CFG-MSGGOUT-UBX_RXM_RAWX_SPI	0x209102a8	U1	-	-	Output rate of the UBX-RXM-RAWX message on the SPI port
CFG-MSGGOUT-UBX_RXM_RAWX_UART1	0x209102a5	U1	-	-	Output rate of the UBX-RXM-RAWX message on the UART1 port
CFG-MSGGOUT-UBX_RXM_RLM_I2C	0x2091025e	U1	-	-	Output rate of the UBX-RXM-RLM message on the I2C port
CFG-MSGGOUT-UBX_RXM_RLM_SPI	0x20910262	U1	-	-	Output rate of the UBX-RXM-RLM message on the SPI port
CFG-MSGGOUT-UBX_RXM_RLM_UART1	0x2091025f	U1	-	-	Output rate of the UBX-RXM-RLM message on the UART1 port
CFG-MSGGOUT-UBX_RXM_SFRBX_I2C	0x20910231	U1	-	-	Output rate of the UBX-RXM-SFRBX message on the I2C port
CFG-MSGGOUT-UBX_RXM_SFRBX_SPI	0x20910235	U1	-	-	Output rate of the UBX-RXM-SFRBX message on the SPI port

Configuration item	Key ID	Type	Scale	Unit	Description
CFG-MSGOUT-UBX_RXM_SFRBX_UART1	0x20910232	U1	-	-	Output rate of the UBX-RXM-SFRBX message on the UART1 port
CFG-MSGOUT-UBX_SEC_SIGLOG_I2C	0x20910689	U1	-	-	Output rate of the UBX-SEC-SIGLOG message on the I2C port
CFG-MSGOUT-UBX_SEC_SIGLOG_SPI	0x2091068d	U1	-	-	Output rate of the UBX-SEC-SIGLOG message on the SPI port
CFG-MSGOUT-UBX_SEC_SIGLOG_UART1	0x2091068a	U1	-	-	Output rate of the UBX-SEC-SIGLOG message on the UART1 port
CFG-MSGOUT-UBX_SEC_SIG_I2C	0x20910634	U1	-	-	Output rate of the UBX-SEC-SIG message on the I2C port
CFG-MSGOUT-UBX_SEC_SIG_SPI	0x20910638	U1	-	-	Output rate of the UBX-SEC-SIG message on the SPI port
CFG-MSGOUT-UBX_SEC_SIG_UART1	0x20910635	U1	-	-	Output rate of the UBX-SEC-SIG message on the UART1 port
CFG-MSGOUT-UBX_TIM_TM2_I2C	0x20910178	U1	-	-	Output rate of the UBX-TIM-TM2 message on the I2C port
CFG-MSGOUT-UBX_TIM_TM2_SPI	0x2091017c	U1	-	-	Output rate of the UBX-TIM-TM2 message on the SPI port
CFG-MSGOUT-UBX_TIM_TM2_UART1	0x20910179	U1	-	-	Output rate of the UBX-TIM-TM2 message on the UART1 port
CFG-MSGOUT-UBX_TIM_TP_I2C	0x2091017d	U1	-	-	Output rate of the UBX-TIM-TP message on the I2C port
CFG-MSGOUT-UBX_TIM_TP_SPI	0x20910181	U1	-	-	Output rate of the UBX-TIM-TP message on the SPI port
CFG-MSGOUT-UBX_TIM_TP_UART1	0x2091017e	U1	-	-	Output rate of the UBX-TIM-TP message on the UART1 port
CFG-MSGOUT-UBX_TIM_VRFY_I2C	0x20910092	U1	-	-	Output rate of the UBX-TIM-VRFY message on the I2C port
CFG-MSGOUT-UBX_TIM_VRFY_SPI	0x20910096	U1	-	-	Output rate of the UBX-TIM-VRFY message on the SPI port
CFG-MSGOUT-UBX_TIM_VRFY_UART1	0x20910093	U1	-	-	Output rate of the UBX-TIM-VRFY message on the UART1 port

Table 22: CFG-MSGOUT configuration items

4.9.13 CFG-NAVSPG: Standard precision navigation configuration

This group contains configuration items related to the operation of the receiver at standard precision, including configuring position fix mode, ionospheric model selection and other related items.

Configuration item	Key ID	Type	Scale	Unit	Description
CFG-NAVSPG-FIXMODE	0x20110011	E1	-	-	Position fix mode
See Table 24 below for a list of possible constants for this item.					
CFG-NAVSPG-INIFIX3D	0x10110013	L	-	-	Initial fix must be a 3D fix
CFG-NAVSPG-WKNROLLOVER	0x30110017	U2	-	-	GPS week rollover number
GPS week numbers are set correctly from this week up to 1024 weeks after this week. The range is from 1 to 4096.					
CFG-NAVSPG-UTCSTANDARD	0x2011001c	E1	-	-	UTC standard to be used
See section GNSS time base in the integration manual. See Table 25 below for a list of possible constants for this item.					
CFG-NAVSPG-DYNMODEL	0x20110021	E1	-	-	Dynamic platform model

Configuration item	Key ID	Type	Scale	Unit	Description
See Table 26 below for a list of possible constants for this item.					
CFG-NAVSPG-ACKAIDING	0x10110025	L	-	-	Acknowledge assistance input messages
CFG-NAVSPG-USE_USRDAT	0x10110061	L	-	-	Use user geodetic datum parameters
User specified geodetic system can be used instead of the receiver's default WGS84 ellipsoid. All of the CFG-NAVSPG-USERDAT_* user geodetic datum parameters listed here must be configured before enabling the user specified geodetic system.					
CFG-NAVSPG-USRDAT_MAJA	0x50110062	R8	-	m	Geodetic datum semi-major axis
Accepted range is from 6,300,000.0 to 6,500,000.0 meters					
CFG-NAVSPG-USRDAT_FLAT	0x50110063	R8	-	-	Geodetic datum 1.0 / flattening
Accepted range is 0.0 to 500.0.					
CFG-NAVSPG-USRDAT_DX	0x40110064	R4	-	m	Geodetic datum X axis shift at the origin
Accepted range is +/- 5000.0 meters.					
CFG-NAVSPG-USRDAT_DY	0x40110065	R4	-	m	Geodetic datum Y axis shift at the origin
Accepted range is +/- 5000.0 meters.					
CFG-NAVSPG-USRDAT_DZ	0x40110066	R4	-	m	Geodetic datum Z axis shift at the origin
Accepted range is +/- 5000.0 meters.					
CFG-NAVSPG-USRDAT_ROT_X	0x40110067	R4	-	arcsec	Geodetic datum rotation about the X axis
Accepted range is +/- 20.0 milli arc seconds.					
CFG-NAVSPG-USRDAT_ROT_Y	0x40110068	R4	-	arcsec	Geodetic datum rotation about the Y axis ()
Accepted range is +/- 20.0 milli-arc seconds.					
CFG-NAVSPG-USRDAT_ROT_Z	0x40110069	R4	-	arcsec	Geodetic datum rotation about the Z axis
Accepted range is +/- 20.0 milli-arc seconds.					
CFG-NAVSPG-USRDAT_SCALE	0x4011006a	R4	-	ppm	Geodetic datum scale factor
Accepted range is 0.0 to 50.0 parts per million.					
CFG-NAVSPG-INFIL_MINSVS	0x201100a1	U1	-	-	Minimum number of satellites for navigation
CFG-NAVSPG-INFIL_MAXSVS	0x201100a2	U1	-	-	Maximum number of satellites for navigation
CFG-NAVSPG-INFIL_MINCNO	0x201100a3	U1	-	dBHz	Minimum satellite signal level for navigation
CFG-NAVSPG-INFIL_MINELEV	0x201100a4	I1	-	deg	Minimum elevation for a GNSS satellite to be used in navigation
CFG-NAVSPG-INFIL_NCNOTHRS	0x201100aa	U1	-	-	Number of satellites required to have C/N0 above CFG-NAVSPG-INFIL_CNOTHRS for a fix to be attempted
CFG-NAVSPG-INFIL_CNOTHRS	0x201100ab	U1	-	-	C/N0 threshold for deciding whether to attempt a fix
CFG-NAVSPG-OUTFIL_PDOP	0x301100b1	U2	0.1	-	Output filter position DOP mask (threshold)
CFG-NAVSPG-OUTFIL_TDOP	0x301100b2	U2	0.1	-	Output filter time DOP mask (threshold)
CFG-NAVSPG-OUTFIL_PACC	0x301100b3	U2	-	m	Output filter position accuracy mask (threshold)
CFG-NAVSPG-OUTFIL_TACC	0x301100b4	U2	-	m	Output filter time accuracy mask (threshold)
CFG-NAVSPG-OUTFIL_FACC	0x301100b5	U2	0.01	m/s	Output filter frequency accuracy mask (threshold)
CFG-NAVSPG-CONSTR_ALT	0x401100c1	I4	0.01	m	Fixed altitude (mean sea level) for 2D fix mode
CFG-NAVSPG-CONSTR_ALTVAR	0x401100c2	U4	0.0001	m^2	Fixed altitude variance for 2D mode
CFG-NAVSPG-CONSTR_DGNSSSTO	0x201100c4	U1	-	s	DGNSS timeout. Maximum value is 255.
CFG-NAVSPG-CONSTR_DGNSSSTO_SCALE	0x201100c5	U1	-	-	DGNSS timeout value scale for CFG-NAVSPG-CONSTR_DGNSSSTO

Configuration item	Key ID	Type	Scale	Unit	Description
DGNSS timeout value is calculated as "CFG-NAVSPG-CONSTR_DGNSSTO * CFG-NAVSPG-CONSTR_DGNSSTO_SCALE". The default scaling is 1, meaning the CFG-NAVSPG-CONSTR_DGNSSTO value is used as-is. When set, the value must be 1 or greater.					

CFG-NAVSPG-SIGATTCOMP	0x201100d6	E1	-	-	Permanently attenuated signal compensation mode
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See [Table 27](#) below for a list of possible constants for this item.

Table 23: CFG-NAVSPG configuration items

Constant	Value	Description
2DONLY	1	2D only
3DONLY	2	3D only
AUTO	3	Auto 2D/3D

Table 24: Constants for CFG-NAVSPG-FIXMODE

Constant	Value	Description
AUTO	0	Automatic; receiver selects based on GNSS configuration
USNO	3	UTC as operated by the U.S. Naval Observatory (USNO); derived from GPS time
EU	5	UTC as combined from multiple European laboratories; derived from Galileo time
SU	6	UTC as operated by the former Soviet Union (SU); derived from GLONASS time
NTSC	7	UTC as operated by the National Time Service Center (NTSC), China; derived from BeiDou time
NPLI	8	UTC as operated by the National Physics Laboratory, India (NPLI); derived from NavIC time
NICT	9	UTC as operated by the National Institute of Information and Communications Technology, Japan (NICT); derived from QZSS time

Table 25: Constants for CFG-NAVSPG-UTCSTANDARD

Constant	Value	Description
PORT	0	Portable
STAT	2	Stationary
PED	3	Pedestrian
AUTOMOT	4	Automotive
SEA	5	Sea
AIR1	6	Airborne with <1g acceleration
AIR2	7	Airborne with <2g acceleration
AIR4	8	Airborne with <4g acceleration
WRIST	9	Wrist-worn watch (not available in all products)
BIKE	10	Motorbike (not available in all products)
MOWER	11	Robotic lawn mower (not available in all products)
ESCOOTER	12	E-scooter (not available in all products)
RAIL	13	Rail vehicles (trains, trams) (not available in all products)
SWIM	14	Wrist-worn watch for swimming activity (not available in all products)
WRIST_LP	15	Wrist-worn watch, optimized for low power (not available in all products)

Table 26: Constants for CFG-NAVSPG-DYNMODEL

Constant	Value	Description
<i>DIS</i>	0	Disable signal attenuation compensation
<i>AUTO</i>	255	Automatic signal attenuation compensation
<i>01DBHZ</i>	1	Maximum expected C/NO level is 1 dBHz
<i>02DBHZ</i>	2	Maximum expected C/NO level is 2 dBHz
<i>03DBHZ</i>	3	Maximum expected C/NO level is 3 dBHz
<i>04DBHZ</i>	4	Maximum expected C/NO level is 4 dBHz
<i>05DBHZ</i>	5	Maximum expected C/NO level is 5 dBHz
<i>06DBHZ</i>	6	Maximum expected C/NO level is 6 dBHz
<i>07DBHZ</i>	7	Maximum expected C/NO level is 7 dBHz
<i>08DBHZ</i>	8	Maximum expected C/NO level is 8 dBHz
<i>09DBHZ</i>	9	Maximum expected C/NO level is 9 dBHz
<i>10DBHZ</i>	10	Maximum expected C/NO level is 10 dBHz
<i>11DBHZ</i>	11	Maximum expected C/NO level is 11 dBHz
<i>12DBHZ</i>	12	Maximum expected C/NO level is 12 dBHz
<i>13DBHZ</i>	13	Maximum expected C/NO level is 13 dBHz
<i>14DBHZ</i>	14	Maximum expected C/NO level is 14 dBHz
<i>15DBHZ</i>	15	Maximum expected C/NO level is 15 dBHz
<i>16DBHZ</i>	16	Maximum expected C/NO level is 16 dBHz
<i>17DBHZ</i>	17	Maximum expected C/NO level is 17 dBHz
<i>18DBHZ</i>	18	Maximum expected C/NO level is 18 dBHz
<i>19DBHZ</i>	19	Maximum expected C/NO level is 19 dBHz
<i>20DBHZ</i>	20	Maximum expected C/NO level is 20 dBHz
<i>21DBHZ</i>	21	Maximum expected C/NO level is 21 dBHz
<i>22DBHZ</i>	22	Maximum expected C/NO level is 22 dBHz
<i>23DBHZ</i>	23	Maximum expected C/NO level is 23 dBHz
<i>24DBHZ</i>	24	Maximum expected C/NO level is 24 dBHz
<i>25DBHZ</i>	25	Maximum expected C/NO level is 25 dBHz
<i>26DBHZ</i>	26	Maximum expected C/NO level is 26 dBHz
<i>27DBHZ</i>	27	Maximum expected C/NO level is 27 dBHz
<i>28DBHZ</i>	28	Maximum expected C/NO level is 28 dBHz
<i>29DBHZ</i>	29	Maximum expected C/NO level is 29 dBHz
<i>30DBHZ</i>	30	Maximum expected C/NO level is 30 dBHz
<i>31DBHZ</i>	31	Maximum expected C/NO level is 31 dBHz
<i>32DBHZ</i>	32	Maximum expected C/NO level is 32 dBHz
<i>33DBHZ</i>	33	Maximum expected C/NO level is 33 dBHz
<i>34DBHZ</i>	34	Maximum expected C/NO level is 34 dBHz
<i>35DBHZ</i>	35	Maximum expected C/NO level is 35 dBHz
<i>36DBHZ</i>	36	Maximum expected C/NO level is 36 dBHz
<i>37DBHZ</i>	37	Maximum expected C/NO level is 37 dBHz
<i>38DBHZ</i>	38	Maximum expected C/NO level is 38 dBHz
<i>39DBHZ</i>	39	Maximum expected C/NO level is 39 dBHz

Constant	Value	Description
40DBHZ	40	Maximum expected C/NO level is 40 dBHz
41DBHZ	41	Maximum expected C/NO level is 41 dBHz
42DBHZ	42	Maximum expected C/NO level is 42 dBHz
43DBHZ	43	Maximum expected C/NO level is 43 dBHz
44DBHZ	44	Maximum expected C/NO level is 44 dBHz
45DBHZ	45	Maximum expected C/NO level is 45 dBHz
46DBHZ	46	Maximum expected C/NO level is 46 dBHz
47DBHZ	47	Maximum expected C/NO level is 47 dBHz
48DBHZ	48	Maximum expected C/NO level is 48 dBHz
49DBHZ	49	Maximum expected C/NO level is 49 dBHz
50DBHZ	50	Maximum expected C/NO level is 50 dBHz
51DBHZ	51	Maximum expected C/NO level is 51 dBHz
52DBHZ	52	Maximum expected C/NO level is 52 dBHz
53DBHZ	53	Maximum expected C/NO level is 53 dBHz
54DBHZ	54	Maximum expected C/NO level is 54 dBHz
55DBHZ	55	Maximum expected C/NO level is 55 dBHz
56DBHZ	56	Maximum expected C/NO level is 56 dBHz
57DBHZ	57	Maximum expected C/NO level is 57 dBHz
58DBHZ	58	Maximum expected C/NO level is 58 dBHz
59DBHZ	59	Maximum expected C/NO level is 59 dBHz
60DBHZ	60	Maximum expected C/NO level is 60 dBHz
61DBHZ	61	Maximum expected C/NO level is 61 dBHz
62DBHZ	62	Maximum expected C/NO level is 62 dBHz
63DBHZ	63	Maximum expected C/NO level is 63 dBHz

Table 27: Constants for CFG-NAVSPG-SIGATTCOMP

4.9.14 CFG-NMEA: NMEA protocol configuration

This group configures the [NMEA protocol](#). See section [NMEA protocol configuration](#) for a detailed description of the configuration effects on NMEA output.

Configuration item	Key ID	Type	Scale	Unit	Description
CFG-NMEA-PROTVR	0x20930001	E1	-	-	NMEA protocol version
See Table 29 below for a list of possible constants for this item.					
CFG-NMEA-MAXSVS	0x20930002	E1	-	-	Maximum number of SVs to report per Talker ID
See Table 30 below for a list of possible constants for this item.					
CFG-NMEA-COMPAT	0x10930003	L	-	-	Enable compatibility mode
This might be needed for certain applications, e.g. for an NMEA parser that expects a fixed number of digits in position coordinates.					
CFG-NMEA-CONSIDER	0x10930004	L	-	-	Enable considering mode
This affects the way the used satellite count in NMEA output is calculated. If set, also considered but rejected satellites (e.g. RAIMED) are counted as used satellites as well.					
CFG-NMEA-LIMIT82	0x10930005	L	-	-	Enable strict limit to 82 characters maximum NMEA message length
CFG-NMEA-HIGHPREC	0x10930006	L	-	-	Enable high precision mode

Configuration item	Key ID	Type	Scale	Unit	Description
This flag cannot be set in conjunction with either CFG-NMEA-COMPAT or CFG-NMEA-LIMIT82 mode.					
CFG-NMEA-SVNUMBERING	0x20930007	E1	-	-	Display configuration for SVs that do not have value defined in NMEA
Configures the display of satellites that do not have an NMEA-defined value. Note: this does not apply to satellites with an unknown ID. See also Satellite Numbering. See Table 31 below for a list of possible constants for this item.					
CFG-NMEA-FILT_GPS	0x10930011	L	-	-	Disable reporting of GPS satellites
CFG-NMEA-FILT_SBAS	0x10930012	L	-	-	Disable reporting of SBAS satellites
CFG-NMEA-FILT_GAL	0x10930013	L	-	-	Disable reporting of Galileo satellites
CFG-NMEA-FILT_QZSS	0x10930015	L	-	-	Disable reporting of QZSS satellites
CFG-NMEA-FILT_BDS	0x10930017	L	-	-	Disable reporting of BeiDou satellites
CFG-NMEA-FILT_NAVIC	0x10930018	L	-	-	Disable reporting of NavIC satellites
CFG-NMEA-OUT_INVFIX	0x10930021	L	-	-	Enable position output for failed or invalid fixes
CFG-NMEA-OUT_MSKFIX	0x10930022	L	-	-	Enable position output for invalid fixes
CFG-NMEA-OUT_INVTIME	0x10930023	L	-	-	Enable time output for invalid times
CFG-NMEA-OUT_INVDATE	0x10930024	L	-	-	Enable date output for invalid dates
CFG-NMEA-OUT_ONLYGPS	0x10930025	L	-	-	Restrict output to GPS satellites only
CFG-NMEA-OUT_FROZENCOG	0x10930026	L	-	-	Enable course over ground output even if it is frozen
CFG-NMEA-MAINTALKERID	0x20930031	E1	-	-	Main Talker ID
By default the main Talker ID (i.e. the Talker ID used for all messages other than GSV) is determined by the GNSS assignment of the receiver's channels (see CFG-SIGNAL). This field enables the main Talker ID to be overridden. See Table 32 below for a list of possible constants for this item.					
CFG-NMEA-GSVTALKERID	0x20930032	E1	-	-	Talker ID for GSV NMEA messages
By default the Talker ID for GSV messages is GNSS-specific (as defined by NMEA). This field enables the GSV Talker ID to be overridden. See Table 33 below for a list of possible constants for this item.					
CFG-NMEA-BDSTALKERID	0x30930033	U2	-	-	BeiDou Talker ID
Sets the two ASCII characters that should be used for the BeiDou Talker ID. If these are set to zero, the receiver uses the default BeiDou Talker ID.					

Table 28: CFG-NMEA configuration items

Constant	Value	Description
V21	21	NMEA protocol version 2.1
V23	23	NMEA protocol version 2.3
V40	40	NMEA protocol version 4.0 (not available in all products)
V41	41	NMEA protocol version 4.10 (not available in all products)
V411	42	NMEA protocol version 4.11 (not available in all products)

Table 29: Constants for CFG-NMEA-PROTVER

Constant	Value	Description
UNLIM	0	Unlimited
8SVS	8	8 SVs

Constant	Value	Description
<i>12SVS</i>	12	12 SVs
<i>16SVS</i>	16	16 SVs

Table 30: Constants for CFG-NMEA-MAXSVS

Constant	Value	Description
<i>STRICT</i>	0	Strict - satellites are not output
<i>EXTENDED</i>	1	Extended - use proprietary numbering

Table 31: Constants for CFG-NMEA-SVNUMBERING

Constant	Value	Description
<i>AUTO</i>	0	Main Talker ID is not overridden
<i>GP</i>	1	Set main Talker ID to 'GP'
<i>GL</i>	2	Set main Talker ID to 'GL'
<i>GN</i>	3	Set main Talker ID to 'GN'
<i>GA</i>	4	Set main Talker ID to 'GA' (not available in all products)
<i>GB</i>	5	Set main Talker ID to 'GB' (not available in all products)
<i>GQ</i>	7	Set main Talker ID to 'GQ' (not available in all products)

Table 32: Constants for CFG-NMEA-MAINTALKERID

Constant	Value	Description
<i>GNSS</i>	0	Use GNSS-specific Talker ID (as defined by NMEA)
<i>MAIN</i>	1	Use the main Talker ID

Table 33: Constants for CFG-NMEA-GSVTALKERID

4.9.15 CFG-ODO: Odometer and low-speed course over ground filter configuration

The items in this group allow the user to configure the Odometer feature and Low-Speed Course Over Ground Filter.

Configuration item	Key ID	Type	Scale	Unit	Description
<i>CFG-ODO-USE_ODO</i>	0x10220001	L	-	-	Use odometer
<i>CFG-ODO-USE_COG</i>	0x10220002	L	-	-	Use low-speed course over ground filter
<i>CFG-ODO-OUTLPVEL</i>	0x10220003	L	-	-	Output low-pass filtered velocity
<i>CFG-ODO-OUTLPCOG</i>	0x10220004	L	-	-	Output low-pass filtered course over ground (heading)
<i>CFG-ODO-PROFILE</i>	0x20220005	E1	-	-	Odometer profile configuration See Table 35 below for a list of possible constants for this item.
<i>CFG-ODO-COGMAXSPEED</i>	0x20220021	U1	1e-1	m/s	Upper speed limit for low-speed course over ground filter
<i>CFG-ODO-COGMAXPOSACC</i>	0x20220022	U1	-	-	Maximum acceptable position accuracy for computing low-speed filtered course over ground
<i>CFG-ODO-VELLPGAIN</i>	0x20220031	U1	-	-	Velocity low-pass filter level Range is from 0 to 255.
<i>CFG-ODO-COGLPGAIN</i>	0x20220032	U1	-	-	Course over ground low-pass filter level (at speed < 8 m/s)

Configuration item	Key ID	Type	Scale	Unit	Description
Range is from 0 to 255.					

Table 34: CFG-ODO configuration items

Constant	Value	Description
<i>RUN</i>	0	Running
<i>CYCL</i>	1	Cycling
<i>SWIM</i>	2	Swimming
<i>CAR</i>	3	Car
<i>CUSTOM</i>	4	Custom

Table 35: Constants for CFG-ODO-PROFILE

4.9.16 CFG-PM: Configuration for receiver power management

Use this configuration group to manage the two main receiver power save modes (Long Interval Tracking or LEAP).

Configuration item	Key ID	Type	Scale	Unit	Description
<i>CFG-PM-OPERATEMODE</i>	0x20d00001	E1	-	-	General mode of operation. Setting this to either Long Interval Tracking or LEAP will turn the corresponding mode on. Setting this to FULL will turn any PSM off. See the integration manual for details. See Table 37 below for a list of possible constants for this item.
<i>CFG-PM-POSUPDATEPERIOD</i>	0x40d00002	U4	-	s	Position update period for Long Interval Tracking. Allowed range: >= 5 s and smaller than the number of seconds in a week. If set to 0, the receiver will never retry a fix and will wait for external events.
<i>CFG-PM-MAXACQTIME</i>	0x20d00007	U1	-	s	Maximum time to spend in Acquisition state If 0: bound disabled (see the Maximum startup state duration section in the integration manual for details).
<i>CFG-PM-UPDATEEPH</i>	0x10d0000a	L	-	-	Update ephemeris regularly. Disable to not wake up to update ephemeris data, enable adds extra wakeup cycles to update the ephemeris data.
<i>CFG-PM-EXTINTWAKE</i>	0x10d0000c	L	-	-	EXTINT pin control (Wake) Enable to keep receiver awake as long as selected EXTINT pin is "high".
<i>CFG-PM-EXTINTBACKUP</i>	0x10d0000d	L	-	-	EXTINT pin control (Backup) Enable to force receiver into BACKUP mode when selected EXTINT pin is "low".
<i>CFG-PM-EXTINTINACTIVE</i>	0x10d0000e	L	-	-	EXTINT pin control (Inactive) Enable to force backup in case EXTINT pin is inactive for time longer than CFG-PM-EXTINTINACTIVITY.
<i>CFG-PM-EXTINTINACTIVITY</i>	0x40d0000f	U4	0.001	s	Inactivity time out on EXTINT pin if enabled
<i>CFG-PM-LIMITPEAKCURR</i>	0x10d00010	L	-	-	Limit peak current
<i>CFG-PM-AUTOSAVEPERIOD</i>	0x20d0001d	U1	-	min	Time between automatic SOS BACKUP operations Defines the maximum time between automatic SOS (save-on-shutdown) BACKUP operations in minutes. If set to 0, automatic SOS BACKUP feature is disabled.
<i>CFG-PM-SELECTIVE_L5</i>	0x10d0001e	L	-	-	Enable the selective L5 mode Automatically toggle L5 tracking based on environmental and signal conditions to achieve the best balance of accuracy and power consumption.
<i>CFG-PM-POSRETRYPERIOD</i>	0x40d00021	U4	-	s	Acquisition period used if the receiver previously failed to achieve a position fix. If set to 0, the retry interval is the same as the POSUPDATEPERIOD.

Table 36: CFG-PM configuration items

Constant	Value	Description
<i>FULL</i>	0	normal operation, no power save mode active
<i>PSMOO</i>	1	Long Interval Tracking
<i>PSMCT</i>	2	LEAP

Table 37: Constants for CFG-PM-OPERATEMODE

4.9.17 CFG-QZSS: QZSS system configuration

Note that enabling and disabling of individual GNSS is done via the [CFG-SIGNAL](#) configuration group.

Configuration item	Key ID	Type	Scale	Unit	Description
<i>CFG-QZSS-USE_SLAS_DGNSS</i>	0x10370005	L	-	-	Apply QZSS SLAS DGNSS corrections
<i>CFG-QZSS-USE_SLAS_TESTMODE</i>	0x10370006	L	-	-	Use QZSS SLAS data when it is in test mode (SLAS msg 0)
<i>CFG-QZSS-USE_SLAS_RAIM_UNCORR</i>	0x10370007	L	-	-	Raim out measurements that are not corrected by QZSS SLAS, if at least 5 measurements are corrected
<i>CFG-QZSS-SLAS_MAX_BASELINE</i>	0x30370008	U2	-	km	Maximum baseline distance to closest GMS

SLAS corrections are only applied if the receiver is at most this far away from the closest ground monitoring station (GMS). Note that due to the nature of the service, the usefulness of corrections degrades with distance. When far away from GMS, SBAS may be a better correction source.

Table 38: CFG-QZSS configuration items

4.9.18 CFG-RATE: Navigation and measurement rate configuration

The configuration items in this group allow the user to alter the rate at which navigation solutions (and the measurements that they depend on) are generated by the receiver. The calculation of the navigation solution is aligned to the top of a second zero (first second of the week) of the configured reference time system. The navigation period is an integer multiple of the measurement period.

Configuration item	Key ID	Type	Scale	Unit	Description
<i>CFG-RATE-MEAS</i>	0x30210001	U2	0.001	s	Nominal time between GNSS measurements E.g. 100 ms results in 10 Hz measurement rate, 1000 ms = 1 Hz measurement rate.
<i>CFG-RATE-NAV</i>	0x30210002	U2	-	-	Ratio of number of measurements to number of navigation solutions E.g. 5 means five measurements for every navigation solution. The minimum value is 1. The maximum value is 127.
<i>CFG-RATE-TIMEREF</i>	0x20210003	E1	-	-	Time system to which measurements are aligned

See [Table 40](#) below for a list of possible constants for this item.

Table 39: CFG-RATE configuration items

Constant	Value	Description
<i>UTC</i>	0	Align measurements to UTC time
<i>GPS</i>	1	Align measurements to GPS time
<i>GLO</i>	2	Align measurements to GLONASS time
<i>BDS</i>	3	Align measurements to BeiDou time
<i>GAL</i>	4	Align measurements to Galileo time

Constant	Value	Description
NAVIC	5	Align measurements to NavIC time

Table 40: Constants for CFG-RATE-TIMEREF

4.9.19 CFG-RINV: Remote inventory

The remote inventory enables storing user-defined data in the receiver's non-volatile memory. The data can be either binary or a string of ASCII characters. In the latter case, it can optionally be output at startup after the boot screen.

Configuration item	Key ID	Type	Scale	Unit	Description
CFG-RINV-DUMP	0x10c70001	L	-	-	Dump data at startup When true, data is dumped to the interface at startup, unless CFG-RINV-BINARY is set.
CFG-RINV-BINARY	0x10c70002	L	-	-	Data is binary When true, the data is treated as binary data.
CFG-RINV-DATA_SIZE	0x20c70003	U1	-	-	Size of data Size of data to store/stored in the remote inventory (maximum 30 bytes).
CFG-RINV-CHUNK0	0x50c70004	X8	-	-	Data bytes 1-8 (LSB) Data to store/stored in remote inventory - max 8 bytes, left-most in LSB, e.g. string ABCD will appear as 0x44434241.
CFG-RINV-CHUNK1	0x50c70005	X8	-	-	Data bytes 9-16 Data to store/stored in remote inventory - max 8 bytes, left-most in LSB, e.g. string ABCD will appear as 0x44434241.
CFG-RINV-CHUNK2	0x50c70006	X8	-	-	Data bytes 17-24 Data to store/stored in remote inventory - max 8 bytes, left-most in LSB, e.g. string ABCD will appear as 0x44434241.
CFG-RINV-CHUNK3	0x50c70007	X8	-	-	Data bytes 25-30 (MSB) Data to store/stored in remote inventory - max 6 bytes, left-most in LSB, e.g. string ABCD will appear as 0x44434241.

Table 41: CFG-RINV configuration items

4.9.20 CFG-SBAS: SBAS configuration

This group configures the SBAS receiver subsystem (i.e. WAAS, EGNOS, MSAS). See SBAS configuration settings description in the integration manual for a detailed description of how these settings affect receiver operation.

Configuration item	Key ID	Type	Scale	Unit	Description
CFG-SBAS-USE_TESTMODE	0x10360002	L	-	-	Use SBAS data when it is in test mode (SBAS msg 0)
CFG-SBAS-USE_RANGING	0x10360003	L	-	-	Use SBAS GEOs as a ranging source (for navigation)
CFG-SBAS-USE_DIFFCORR	0x10360004	L	-	-	Use SBAS differential corrections
CFG-SBAS-USE_INTEGRITY	0x10360005	L	-	-	Use SBAS integrity information If enabled, the receiver uses only GPS satellites for which integrity information is available
CFG-SBAS-ACCEPT_NOT_IN_PRN_MASK	0x30360008	X2	-	-	Accept corrections from SBAS SV, even if not self included in PRN MASK (Message Type 1) If enabled, the receiver will still use the SBAS data, even when the SBAS SV itself is not included in its PRN MASK. This is only useful for BDSBAS and not compatible with current EGNOS implementation. See Table 43 below for a list of possible constants for this item.
CFG-SBAS-USE_IONOONLY	0x10360007	L	-	-	Use SBAS ionosphere correction only
CFG-SBAS-PRNSCANMASK	0x50360006	X8	-	-	SBAS PRN search configuration

Configuration item	Key ID	Type	Scale	Unit	Description
This configuration item determines which SBAS PRNs should be searched. Setting it to 0 indicates auto-scanning all SBAS PRNs. For non-zero values the bits correspond to the allocated SBAS PRNs ranging from PRN120 (bit 0) to PRN158 (bit 38), where a bit set enables searching for the corresponding PRN.					
See Table 44 below for a list of possible constants for this item.					

Table 42: CFG-SBAS configuration items

Constant	Value	Description
WAAS	0x01	WAAS bit
1 = Use WAAS provider Id.		
EGNOS	0x02	EGNOS bit
1 = Use EGNOS provider Id.		
MSAS	0x04	MSAS bit
1 = Use MSAS provider Id.		
GAGAN	0x08	GAGAN bit
1 = Use GAGAN provider Id.		
SDCM	0x10	SDCM bit
1 = Use SDCM provider Id.		
BDSBAS	0x20	BDSBAS bit
1 = Use BDSBAS provider Id.		
KASS	0x40	KASS bit
1 = Use KASS provider Id.		

Table 43: Constants for CFG-SBAS-ACCEPT_NOT_IN_PRNMASK

Constant	Value	Description
ALL	0x0000000000000000	Enable search for all SBAS PRNs
PRN120	0x0000000000000001	Enable search for SBAS PRN120
PRN121	0x0000000000000002	Enable search for SBAS PRN121
PRN122	0x0000000000000004	Enable search for SBAS PRN122
PRN123	0x0000000000000008	Enable search for SBAS PRN123
PRN124	0x0000000000000010	Enable search for SBAS PRN124
PRN125	0x0000000000000020	Enable search for SBAS PRN125
PRN126	0x0000000000000040	Enable search for SBAS PRN126
PRN127	0x0000000000000080	Enable search for SBAS PRN127
PRN128	0x0000000000000100	Enable search for SBAS PRN128
PRN129	0x0000000000000200	Enable search for SBAS PRN129
PRN130	0x0000000000000400	Enable search for SBAS PRN130
PRN131	0x0000000000000800	Enable search for SBAS PRN131
PRN132	0x0000000000001000	Enable search for SBAS PRN132
PRN133	0x0000000000002000	Enable search for SBAS PRN133
PRN134	0x0000000000004000	Enable search for SBAS PRN134
PRN135	0x0000000000008000	Enable search for SBAS PRN135
PRN136	0x0000000000010000	Enable search for SBAS PRN136
PRN137	0x0000000000020000	Enable search for SBAS PRN137
PRN138	0x0000000000040000	Enable search for SBAS PRN138

Constant	Value	Description
PRN139	0x00000000000080000	Enable search for SBAS PRN139
PRN140	0x00000000000100000	Enable search for SBAS PRN140
PRN141	0x00000000000200000	Enable search for SBAS PRN141
PRN142	0x00000000000400000	Enable search for SBAS PRN142
PRN143	0x00000000000800000	Enable search for SBAS PRN143
PRN144	0x00000000001000000	Enable search for SBAS PRN144
PRN145	0x00000000002000000	Enable search for SBAS PRN145
PRN146	0x00000000004000000	Enable search for SBAS PRN146
PRN147	0x00000000008000000	Enable search for SBAS PRN147
PRN148	0x00000000010000000	Enable search for SBAS PRN148
PRN149	0x00000000020000000	Enable search for SBAS PRN149
PRN150	0x00000000040000000	Enable search for SBAS PRN150
PRN151	0x00000000080000000	Enable search for SBAS PRN151
PRN152	0x00000001000000000	Enable search for SBAS PRN152
PRN153	0x00000002000000000	Enable search for SBAS PRN153
PRN154	0x00000004000000000	Enable search for SBAS PRN154
PRN155	0x00000008000000000	Enable search for SBAS PRN155
PRN156	0x00000010000000000	Enable search for SBAS PRN156
PRN157	0x00000020000000000	Enable search for SBAS PRN157
PRN158	0x00000040000000000	Enable search for SBAS PRN158

Table 44: Constants for CFG-SBAS-PRNSCANMASK

4.9.21 CFG-SEC: Security configuration

Security configuration.

Configuration item	Key ID	Type	Scale	Unit	Description
CFG-SEC-CFG_LOCK	0x10f60009	L	-	-	Configuration lockdown When set, the receiver configuration is locked and cannot be changed any more.
CFG-SEC-CFG_LOCK_UNLOCKGRP1	0x30f6000a	U2	-	-	Configuration lockdown exempted group 1 This item can be set before enabling the configuration lockdown. It enables writing to the specified group even after the configuration lockdown has been enabled.
CFG-SEC-CFG_LOCK_UNLOCKGRP2	0x30f6000b	U2	-	-	Configuration lockdown exempted group 2 This item can be set before enabling the configuration lockdown. It enables writing to the specified group even after the configuration lockdown has been enabled.
CFG-SEC-SPOOFDET_SIM_SIG_DIS	0x10f6005d	L	-	-	Disabling the simulated signal spoofing detection.
CFG-SEC-JAMDET_SENSITIVITY_HI	0x10f60051	L	-	-	When set, go for a more sensitive jamming detection (at the cost of increased false alarm rate).

Table 45: CFG-SEC configuration items

4.9.22 CFG-SIGNAL: Satellite systems (GNSS) signal configuration

The enable items for individual signals are governed by their corresponding constellation enable item. It is necessary that at least one signal from a major GNSS constellation is enabled. See GNSS signal configuration in the integration manual for more details.

Configuration specific to a GNSS system is available in other groups (e.g. CFG-SBAS).

Note that some changes to items within this group trigger a reset to the GNSS subsystem. The reset takes some time, so wait first for the acknowledgement from the receiver and then 0.5 seconds before sending the next command.

Configuration item	Key ID	Type	Scale	Unit	Description
CFG-SIGNAL-GPS_ENA	0x1031001f	L	-	-	GPS enable
CFG-SIGNAL-GPS_L1CA_ENA	0x10310001	L	-	-	GPS L1C/A
CFG-SIGNAL-GPS_L5_ENA	0x10310004	L	-	-	GPS L5
CFG-SIGNAL-SBAS_ENA	0x10310020	L	-	-	SBAS enable
CFG-SIGNAL-SBAS_L1CA_ENA	0x10310005	L	-	-	SBAS L1C/A
CFG-SIGNAL-GAL_ENA	0x10310021	L	-	-	Galileo enable
CFG-SIGNAL-GAL_E1_ENA	0x10310007	L	-	-	Galileo E1
CFG-SIGNAL-GAL_E5A_ENA	0x10310009	L	-	-	Galileo E5a
CFG-SIGNAL-BDS_ENA	0x10310022	L	-	-	BeiDou Enable
CFG-SIGNAL-BDS_B1_ENA	0x1031000d	L	-	-	BeiDou B1I
CFG-SIGNAL-BDS_B1C_ENA	0x1031000f	L	-	-	BeiDou B1C
CFG-SIGNAL-BDS_B2A_ENA	0x10310028	L	-	-	BeiDou B2a
CFG-SIGNAL-QZSS_ENA	0x10310024	L	-	-	QZSS enable
CFG-SIGNAL-QZSS_L1CA_ENA	0x10310012	L	-	-	QZSS L1C/A
CFG-SIGNAL-QZSS_L1S_ENA	0x10310014	L	-	-	QZSS L1S
CFG-SIGNAL-QZSS_L5_ENA	0x10310017	L	-	-	QZSS L5
CFG-SIGNAL-NAVIC_ENA	0x10310026	L	-	-	NavIC enable
CFG-SIGNAL-NAVIC_L5_ENA	0x1031001d	L	-	-	NavIC L5
CFG-SIGNAL-PLAN	0x2031003a	E1	-	-	Active signal plan

Select the active signal plan of the receiver. Multiple options exist. Not all signal plans are available in this product. See [UBX-MON-GNSS](#) for the signal plans available in this product.

See [Table 47](#) below for a list of possible constants for this item.

Table 46: CFG-SIGNAL configuration items

Constant	Value	Description
SP1	0x01	Signal plan 1
SP2	0x02	Signal plan 2
SP3	0x03	Signal plan 3
SP4	0x04	Signal plan 4
SP5	0x05	Signal plan 5
SP6	0x06	Signal plan 6
SP7	0x07	Signal plan 7
SP8	0x08	Signal plan 8
SP9	0x09	Signal plan 9

Constant	Value	Description
<i>SP10</i>	0x0a	Signal plan 10

Table 47: Constants for CFG-SIGNAL-PLAN

4.9.23 CFG-SPI: Configuration of the SPI interface

Settings needed to configure the SPI communication interface.

Configuration item	Key ID	Type	Scale	Unit	Description
<i>CFG-SPI-MAXFF</i>	0x20640001	U1	-	-	Number of bytes containing 0xFF to receive before switching off reception. Range: 0 (mechanism off) - 63
<i>CFG-SPI-CPOLARITY</i>	0x10640002	L	-	-	Clock polarity select: 0: Active Hight Clock, SCLK idles low, 1: Active Low Clock, SCLK idles high
<i>CFG-SPI-CPHASE</i>	0x10640003	L	-	-	Clock phase select: 0: Data captured on first edge of SCLK, 1: Data captured on second edge of SCLK
<i>CFG-SPI-EXTENDEDTIMEOUT</i>	0x10640005	L	-	-	Flag to disable timeouting the interface after 1.5s
<i>CFG-SPI-ENABLED</i>	0x10640006	L	-	-	Flag to indicate if the SPI interface should be enabled

Table 48: CFG-SPI configuration items

4.9.24 CFG-SPIINPROT: Input protocol configuration of the SPI interface

Input protocol enable flags of the SPI interface.

Configuration item	Key ID	Type	Scale	Unit	Description
<i>CFG-SPIINPROT-UBX</i>	0x10790001	L	-	-	Flag to indicate if UBX should be an input protocol on SPI
<i>CFG-SPIINPROT-NMEA</i>	0x10790002	L	-	-	Flag to indicate if NMEA should be an input protocol on SPI

Table 49: CFG-SPIINPROT configuration items

4.9.25 CFG-SPIOUTPROT: Output protocol configuration of the SPI interface

Output protocol enable flags of the SPI interface.

Configuration item	Key ID	Type	Scale	Unit	Description
<i>CFG-SPIOUTPROT-UBX</i>	0x107a0001	L	-	-	Flag to indicate if UBX should be an output protocol on SPI
<i>CFG-SPIOUTPROT-NMEA</i>	0x107a0002	L	-	-	Flag to indicate if NMEA should be an output protocol on SPI

Table 50: CFG-SPIOUTPROT configuration items

4.9.26 CFG-TP: Time pulse configuration

Use this group to configure the generation of time pulses.

Configuration item	Key ID	Type	Scale	Unit	Description
<i>CFG-TP-PULSE_DEF</i>	0x20050023	E1	-	-	Determines whether the time pulse is interpreted as frequency or period
See Table 52 below for a list of possible constants for this item.					
<i>CFG-TP-PULSE_LENGTH_DEF</i>	0x20050030	E1	-	-	Determines whether the time pulse length is interpreted as length[μs] or pulse ratio[%]
See Table 53 below for a list of possible constants for this item.					

Configuration item	Key ID	Type	Scale	Unit	Description
<i>CFG-TP-ANT_CABLEDELAY</i>	0x30050001	I2	1e-9	s	Antenna cable delay in [ns]
<i>CFG-TP-PERIOD_TP1</i>	0x40050002	U4	1e-6	s	Time pulse period (TP1) in [μs] This is used only if CFG-TP-PULSE_DEF=PERIOD.
<i>CFG-TP-PERIOD_LOCK_TP1</i>	0x40050003	U4	1e-6	s	Time pulse period when locked to GNSS time (TP1) in [μs] Only used if CFG-TP-PULSE_DEF=PERIOD and CFG-TP-USE_LOCKED_TP1 is set.
<i>CFG-TP-FREQ_TP1</i>	0x40050024	U4	-	Hz	Time pulse frequency (TP1) in [Hz] This is used only if CFG-TP-PULSE_DEF=FREQ.
<i>CFG-TP-FREQ_LOCK_TP1</i>	0x40050025	U4	-	Hz	Time pulse frequency when locked to GNSS time (TP1) in [Hz] Only used if CFG-TP-PULSE_DEF=FREQ and CFG-TP-USE_LOCKED_TP1 is set.
<i>CFG-TP-LEN_TP1</i>	0x40050004	U4	1e-6	s	Time pulse length (TP1) in [μs] Only used if CFG-TP-PULSE_LENGTH_DEF=LENGTH is set.
<i>CFG-TP-LEN_LOCK_TP1</i>	0x40050005	U4	1e-6	s	Time pulse length when locked to GNSS time (TP1) in [μs] Only used if CFG-TP-PULSE_LENGTH_DEF=LENGTH and CFG-TP-USE_LOCKED_TP1 is set.
<i>CFG-TP-DUTY_TP1</i>	0x5005002a	R8	-	%	Time pulse duty cycle (TP1) in [%] Only used if CFG-TP-PULSE_LENGTH_DEF=RATIO is set.
<i>CFG-TP-DUTY_LOCK_TP1</i>	0x5005002b	R8	-	%	Time pulse duty cycle when locked to GNSS time (TP1) in [%] Only used if CFG-TP-PULSE_LENGTH_DEF=RATIO and CFG-TP-USE_LOCKED_TP1 are set.
<i>CFG-TP-USER_DELAY_TP1</i>	0x40050006	I4	1e-9	s	User-configurable time pulse delay (TP1) in [ns]
<i>CFG-TP-TP1_ENA</i>	0x10050007	L	-	-	Enable the time pulse (TP1) if pin associated with time pulse is assigned for another function, the other function takes precedence. Must be set for frequency-time products.
<i>CFG-TP-SYNC_GNSS_TP1</i>	0x10050008	L	-	-	Sync time pulse to GNSS time or local clock (TP1) If set, sync to GNSS if GNSS time is valid. Otherwise, use local clock. This flag can be unset only in Timing product variants.
<i>CFG-TP-USE_LOCKED_TP1</i>	0x10050009	L	-	-	Use locked parameters when possible (TP1) If set, use CFG-TP-PERIOD_LOCK_TP1 and CFG-TP-LEN_LOCK_TP1 as soon as GNSS time is valid. Otherwise, use CFG-TP-PERIOD_TP1 and CFG-TP-LEN_TP1.
<i>CFG-TP-ALIGN_TO_TOW_TP1</i>	0x1005000a	L	-	-	Align time pulse to top of second (TP1) To use this feature, CFG-TP-SYNC_GNSS_TP1 must be set. Time pulse period must be an integer fraction of 1 second.
<i>CFG-TP-POL_TP1</i>	0x1005000b	L	-	-	Set time pulse polarity (TP1) false (0) : falling edge at top of second. true (1) : rising edge at top of second.
<i>CFG-TP-TIMEGRID_TP1</i>	0x2005000c	E1	-	-	Time grid to use (TP1) Only relevant if CFG-TP-SYNC_GNSS_TP1 is set. Note that configured GNSS time is estimated by the receiver if locked to any GNSS system. If the receiver has a valid GNSS fix it attempts to steer the TP to the specified time grid even if the specified time is not based on information from the constellation's satellites. To ensure timing based purely on a given GNSS, restrict the supported constellations in CFG-SIGNAL-*. See Table 54 below for a list of possible constants for this item.
<i>CFG-TP-DRSTR_TP1</i>	0x20050035	E1	-	-	Set drive strength of TP1

Configuration item	Key ID	Type	Scale	Unit	Description
Time Pulse pin 1 (TP1) can support 4 possible drive strength cases: 2, 4, 8 and 12 mA					
See Table 55 below for a list of possible constants for this item.					

Table 51: CFG-TP configuration items

Constant	Value	Description
<i>PERIOD</i>	0	Time pulse period [μ s]
<i>FREQ</i>	1	Time pulse frequency [Hz]

Table 52: Constants for CFG-TP-PULSE_DEF

Constant	Value	Description
<i>RATIO</i>	0	Time pulse ratio
<i>LENGTH</i>	1	Time pulse length

Table 53: Constants for CFG-TP-PULSE_LENGTH_DEF

Constant	Value	Description
<i>UTC</i>	0	UTC time reference
<i>GPS</i>	1	GPS time reference
<i>GLO</i>	2	GLONASS time reference
<i>BDS</i>	3	BeiDou time reference
<i>GAL</i>	4	Galileo time reference
<i>NAVIC</i>	5	NavIC time reference

Table 54: Constants for CFG-TP-TIMEGRID_TP1

Constant	Value	Description
<i>DRIVE_STRENGTH_2MA</i>	0	2 mA drive strength
<i>DRIVE_STRENGTH_4MA</i>	1	4 mA drive strength
<i>DRIVE_STRENGTH_8MA</i>	2	8 mA drive strength
<i>DRIVE_STRENGTH_12MA</i>	3	12 mA drive strength

Table 55: Constants for CFG-TP-DRSTR_TP1

4.9.27 CFG-TXREADY: TX ready configuration

Configuration of the TX ready pin.

Configuration item	Key ID	Type	Scale	Unit	Description
<i>CFG-TXREADY-ENABLED</i>	0x10a20001	L	-	-	Flag to indicate if TX ready pin mechanism should be enabled
<i>CFG-TXREADY-POLARITY</i>	0x10a20002	L	-	-	The polarity of the TX ready pin: false:high-active, true:low-active
<i>CFG-TXREADY-PIN</i>	0x20a20003	U1	-	-	Pin number to use for the TX ready functionality
<i>CFG-TXREADY-THRESHOLD</i>	0x30a20004	U2	-	-	Amount of data that should be ready on the interface before triggering the TX ready pin
The value is amount of 8-byte chunks. For example, value of 250 sets the trigger to 2000 bytes.					
<i>CFG-TXREADY-INTERFACE</i>	0x20a20005	E1	-	-	Interface where the TX ready feature should be linked to

See [Table 57](#) below for a list of possible constants for this item.

Table 56: CFG-TXREADY configuration items

Constant	Value	Description
<i>I2C</i>	0	I2C interface
<i>SPI</i>	1	SPI interface

Table 57: Constants for CFG-TXREADY-INTERFACE

4.9.28 CFG-UART1: Configuration of the UART1 interface

Settings needed to configure the UART1 communication interface.

Configuration item	Key ID	Type	Scale	Unit	Description
<i>CFG-UART1-BAUDRATE</i>	0x40520001	U4	-	-	The baud rate that should be configured on the UART1
<i>CFG-UART1-STOPBITS</i>	0x20520002	E1	-	-	Number of stopbits that should be used on UART1
See Table 59 below for a list of possible constants for this item.					
<i>CFG-UART1-DATABITS</i>	0x20520003	E1	-	-	Number of databits that should be used on UART1
See Table 60 below for a list of possible constants for this item.					
<i>CFG-UART1-PARITY</i>	0x20520004	E1	-	-	Parity mode that should be used on UART1
See Table 61 below for a list of possible constants for this item.					
<i>CFG-UART1-ENABLED</i>	0x10520005	L	-	-	Flag to indicate if the UART1 should be enabled

Table 58: CFG-UART1 configuration items

Constant	Value	Description
<i>HALF</i>	0	0.5 stopbits
<i>ONE</i>	1	1.0 stopbits
<i>ONEHALF</i>	2	1.5 stopbits
<i>TWO</i>	3	2.0 stopbits

Table 59: Constants for CFG-UART1-STOPBITS

Constant	Value	Description
<i>EIGHT</i>	0	8 databits
<i>SEVEN</i>	1	7 databits

Table 60: Constants for CFG-UART1-DATABITS

Constant	Value	Description
<i>NONE</i>	0	No parity bit
<i>ODD</i>	1	Add an odd parity bit
<i>EVEN</i>	2	Add an even parity bit

Table 61: Constants for CFG-UART1-PARITY

4.9.29 CFG-UART1INPROT: Input protocol configuration of the UART1 interface

Input protocol enable flags of the UART1 interface.

Configuration item	Key ID	Type	Scale	Unit	Description
<i>CFG-UART1INPROT-UBX</i>	0x10730001	L	-	-	Flag to indicate if UBX should be an input protocol on UART1

Configuration item	Key ID	Type	Scale	Unit	Description
<i>CFG-UART1INPROT-NMEA</i>	0x10730002	L	-	-	Flag to indicate if NMEA should be an input protocol on UART1

Table 62: CFG-UART1INPROT configuration items

4.9.30 CFG-UART1OUTPROT: Output protocol configuration of the UART1 interface

Output protocol enable flags of the UART1 interface.

Configuration item	Key ID	Type	Scale	Unit	Description
<i>CFG-UART1OUTPROT-UBX</i>	0x10740001	L	-	-	Flag to indicate if UBX should be an output protocol on UART1
<i>CFG-UART1OUTPROT-NMEA</i>	0x10740002	L	-	-	Flag to indicate if NMEA should be an output protocol on UART1

Table 63: CFG-UART1OUTPROT configuration items

Configuration defaults

The following tables contain the configuration defaults for the firmware. Some of these values may be changed in production. Refer to the integration manual for product-specific details.

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-ANA-USE_ANA	0x10230001	L	-	-	0 (false)
CFG-ANA-ORBMAXERR	0x30230002	U2	-	m	100

Table 64: CFG-ANA configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-BATCH-ENABLE	0x10260013	L	-	-	0 (false)
CFG-BATCH-PIOENABLE	0x10260014	L	-	-	0 (false)
CFG-BATCH-MAXENTRIES	0x30260015	U2	-	-	0
CFG-BATCH-WARNTHRS	0x30260016	U2	-	-	0
CFG-BATCH-PIOACTIVELOW	0x10260018	L	-	-	0 (false)
CFG-BATCH-PIOID	0x20260019	U1	-	-	0
CFG-BATCH-EXTRAPVT	0x1026001a	L	-	-	0 (false)
CFG-BATCH-EXTRAODO	0x1026001b	L	-	-	0 (false)

Table 65: CFG-BATCH configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-BDS-D1D2_NAVDATA	0x20340009	E1	-	-	0 (ALL)

Table 66: CFG-BDS configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-GEOFENCE-CONFLVL	0x20240011	E1	-	-	0 (L000)
CFG-GEOFENCE-USE_PIO	0x10240012	L	-	-	0 (false)
CFG-GEOFENCE-PINPOL	0x20240013	E1	-	-	0 (LOW_IN)
CFG-GEOFENCE-PIN	0x20240014	U1	-	-	0
CFG-GEOFENCE-POLYGON1_NUM_VERTICES	0x20240017	U1	-	-	0
CFG-GEOFENCE-POLYGON2_NUM_VERTICES	0x20240018	U1	-	-	0
CFG-GEOFENCE-POLYGON3_NUM_VERTICES	0x20240019	U1	-	-	0
CFG-GEOFENCE-POLYGON4_NUM_VERTICES	0x2024001a	U1	-	-	0
CFG-GEOFENCE-POLYGON5_NUM_VERTICES	0x2024001b	U1	-	-	0
CFG-GEOFENCE-USE_FENCE1	0x10240020	L	-	-	0 (false)
CFG-GEOFENCE-FENCE1_LAT	0x40240021	I4	1e-7	deg	0
CFG-GEOFENCE-FENCE1_LON	0x40240022	I4	1e-7	deg	0
CFG-GEOFENCE-FENCE1_RAD	0x40240023	U4	0.01	m	0
CFG-GEOFENCE-USE_FENCE2	0x10240030	L	-	-	0 (false)
CFG-GEOFENCE-FENCE2_LAT	0x40240031	I4	1e-7	deg	0
CFG-GEOFENCE-FENCE2_LON	0x40240032	I4	1e-7	deg	0
CFG-GEOFENCE-FENCE2_RAD	0x40240033	U4	0.01	m	0
CFG-GEOFENCE-USE_FENCE3	0x10240040	L	-	-	0 (false)
CFG-GEOFENCE-FENCE3_LAT	0x40240041	I4	1e-7	deg	0

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-GEOFENCE-FENCE3_LON	0x40240042	I4	1e-7	deg	0
CFG-GEOFENCE-FENCE3_RAD	0x40240043	U4	0.01	m	0
CFG-GEOFENCE-USE_FENCE4	0x10240050	L	-	-	0 (false)
CFG-GEOFENCE-FENCE4_LAT	0x40240051	I4	1e-7	deg	0
CFG-GEOFENCE-FENCE4_LON	0x40240052	I4	1e-7	deg	0
CFG-GEOFENCE-FENCE4_RAD	0x40240053	U4	0.01	m	0
CFG-GEOFENCE-VERTEX1_LAT	0x40240061	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX1_LON	0x40240062	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX2_LAT	0x40240063	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX2_LON	0x40240064	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX3_LAT	0x40240065	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX3_LON	0x40240066	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX4_LAT	0x40240067	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX4_LON	0x40240068	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX5_LAT	0x40240069	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX5_LON	0x4024006a	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX6_LAT	0x4024006b	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX6_LON	0x4024006c	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX7_LAT	0x4024006d	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX7_LON	0x4024006e	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX8_LAT	0x4024006f	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX8_LON	0x40240070	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX9_LAT	0x40240071	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX9_LON	0x40240072	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX10_LAT	0x40240073	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX10_LON	0x40240074	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX11_LAT	0x40240075	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX11_LON	0x40240076	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX12_LAT	0x40240077	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX12_LON	0x40240078	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX13_LAT	0x40240079	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX13_LON	0x4024007a	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX14_LAT	0x4024007b	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX14_LON	0x4024007c	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX15_LAT	0x4024007d	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX15_LON	0x4024007e	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX16_LAT	0x4024007f	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX16_LON	0x40240080	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX17_LAT	0x40240081	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX17_LON	0x40240082	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX18_LAT	0x40240083	I4	1e-7	deg	0

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-GEOFENCE-VERTEX18_LON	0x40240084	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX19_LAT	0x40240085	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX19_LON	0x40240086	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX20_LAT	0x40240087	I4	1e-7	deg	0
CFG-GEOFENCE-VERTEX20_LON	0x40240088	I4	1e-7	deg	0

Table 67: CFG-GEOFENCE configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-HW-ANT_CFG_VOLTCTRL	0x10a3002e	L	-	-	0 (false)
CFG-HW-ANT_CFG_SHORTDET	0x10a3002f	L	-	-	0 (false)
CFG-HW-ANT_CFG_SHORTDET_POL	0x10a30030	L	-	-	1 (true)
CFG-HW-ANT_CFG_OPENDET	0x10a30031	L	-	-	0 (false)
CFG-HW-ANT_CFG_OPENDET_POL	0x10a30032	L	-	-	1 (true)
CFG-HW-ANT_CFG_PWRDOWN	0x10a30033	L	-	-	0 (false)
CFG-HW-ANT_CFG_PWRDOWN_POL	0x10a30034	L	-	-	1 (true)
CFG-HW-ANT_CFG_RECOVER	0x10a30035	L	-	-	0 (false)
CFG-HW-ANT_SUP_SWITCH_PIN	0x20a30036	U1	-	-	7
CFG-HW-ANT_SUP_SHORT_PIN	0x20a30037	U1	-	-	6
CFG-HW-ANT_SUP_OPEN_PIN	0x20a30038	U1	-	-	5
CFG-HW-ANT_ON_SHORT_US	0x30a3003c	U2	-	-	500
CFG-HW-ANT_SUP_ENGINE	0x20a30054	E1	-	-	0 (EXT)
CFG-HW-RF_LNA_MODE	0x20a30057	E1	-	-	0 (NORMAL)

Table 68: CFG-HW configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-I2C-ADDRESS	0x20510001	U1	-	-	132
CFG-I2C-EXTENDEDTIMEOUT	0x10510002	L	-	-	0 (false)
CFG-I2C-ENABLED	0x10510003	L	-	-	0 (false)

Table 69: CFG-I2C configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-I2CINPROT-UBX	0x10710001	L	-	-	1 (true)
CFG-I2CINPROT-NMEA	0x10710002	L	-	-	1 (true)

Table 70: CFG-I2CINPROT configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-I2COUTPROT-UBX	0x10720001	L	-	-	1 (true)
CFG-I2COUTPROT-NMEA	0x10720002	L	-	-	1 (true)

Table 71: CFG-I2COUTPROT configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-INFMSG-UBX_I2C	0x20920001	X1	-	-	0x00
CFG-INFMSG-UBX_UART1	0x20920002	X1	-	-	0x00
CFG-INFMSG-UBX_SPI	0x20920005	X1	-	-	0x00

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-INFMSG-NMEA_I2C	0x20920006	X1	-	-	0x07 (ERROR WARNING NOTICE)
CFG-INFMSG-NMEA_UART1	0x20920007	X1	-	-	0x07 (ERROR WARNING NOTICE)
CFG-INFMSG-NMEA_SPI	0x2092000a	X1	-	-	0x07 (ERROR WARNING NOTICE)

Table 72: CFG-INFMSG configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-LOGFILTER-RECORD_ENA	0x10de0002	L	-	-	0 (false)
CFG-LOGFILTER-ONCE_PER_WAKE_UP_ENA	0x10de0003	L	-	-	0 (false)
CFG-LOGFILTER-APPLY_ALL_FILTERS	0x10de0004	L	-	-	0 (false)
CFG-LOGFILTER-MIN_INTERVAL	0x30de0005	U2	-	s	0
CFG-LOGFILTER-TIME_THRS	0x30de0006	U2	-	s	0
CFG-LOGFILTER-SPEED_THRS	0x30de0007	U2	-	m/s	0
CFG-LOGFILTER-POSITION_THRS	0x40de0008	U4	-	m	0

Table 73: CFG-LOGFILTER configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-MOT-GNSSSPEED_THRS	0x20250038	U1	0.01	m/s	0
CFG-MOT-GNSSDIST_THRS	0x3025003b	U2	1.0	m	0

Table 74: CFG-MOT configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-MSGOUT-NMEA_ID_DTM_I2C	0x209100a6	U1	-	-	0
CFG-MSGOUT-NMEA_ID_DTM_SPI	0x209100aa	U1	-	-	0
CFG-MSGOUT-NMEA_ID_DTM_UART1	0x209100a7	U1	-	-	0
CFG-MSGOUT-NMEA_ID_GBS_I2C	0x209100dd	U1	-	-	0
CFG-MSGOUT-NMEA_ID_GBS_SPI	0x209100e1	U1	-	-	0
CFG-MSGOUT-NMEA_ID_GBS_UART1	0x209100de	U1	-	-	0
CFG-MSGOUT-NMEA_ID_GGA_I2C	0x209100ba	U1	-	-	1
CFG-MSGOUT-NMEA_ID_GGA_SPI	0x209100be	U1	-	-	1
CFG-MSGOUT-NMEA_ID_GGA_UART1	0x209100bb	U1	-	-	1
CFG-MSGOUT-NMEA_ID_GLL_I2C	0x209100c9	U1	-	-	1
CFG-MSGOUT-NMEA_ID_GLL_SPI	0x209100cd	U1	-	-	1
CFG-MSGOUT-NMEA_ID_GLL_UART1	0x209100ca	U1	-	-	1
CFG-MSGOUT-NMEA_ID_GNS_I2C	0x209100b5	U1	-	-	0
CFG-MSGOUT-NMEA_ID_GNS_SPI	0x209100b9	U1	-	-	0
CFG-MSGOUT-NMEA_ID_GNS_UART1	0x209100b6	U1	-	-	0
CFG-MSGOUT-NMEA_ID_GRS_I2C	0x209100ce	U1	-	-	0
CFG-MSGOUT-NMEA_ID_GRS_SPI	0x209100d2	U1	-	-	0
CFG-MSGOUT-NMEA_ID_GRS_UART1	0x209100cf	U1	-	-	0
CFG-MSGOUT-NMEA_ID_GSA_I2C	0x209100bf	U1	-	-	1
CFG-MSGOUT-NMEA_ID_GSA_SPI	0x209100c3	U1	-	-	1

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-MSGOUT-NMEA_ID_GSA_UART1	0x209100c0	U1	-	-	1
CFG-MSGOUT-NMEA_ID_GST_I2C	0x209100d3	U1	-	-	0
CFG-MSGOUT-NMEA_ID_GST_SPI	0x209100d7	U1	-	-	0
CFG-MSGOUT-NMEA_ID_GST_UART1	0x209100d4	U1	-	-	0
CFG-MSGOUT-NMEA_ID_GSV_I2C	0x209100c4	U1	-	-	1
CFG-MSGOUT-NMEA_ID_GSV_SPI	0x209100c8	U1	-	-	1
CFG-MSGOUT-NMEA_ID_GSV_UART1	0x209100c5	U1	-	-	1
CFG-MSGOUT-NMEA_ID_RLM_I2C	0x20910400	U1	-	-	0
CFG-MSGOUT-NMEA_ID_RLM_SPI	0x20910404	U1	-	-	0
CFG-MSGOUT-NMEA_ID_RLM_UART1	0x20910401	U1	-	-	0
CFG-MSGOUT-NMEA_ID_RMC_I2C	0x209100ab	U1	-	-	1
CFG-MSGOUT-NMEA_ID_RMC_SPI	0x209100af	U1	-	-	1
CFG-MSGOUT-NMEA_ID_RMC_UART1	0x209100ac	U1	-	-	1
CFG-MSGOUT-NMEA_ID_VLW_I2C	0x209100e7	U1	-	-	0
CFG-MSGOUT-NMEA_ID_VLW_SPI	0x209100eb	U1	-	-	0
CFG-MSGOUT-NMEA_ID_VLW_UART1	0x209100e8	U1	-	-	0
CFG-MSGOUT-NMEA_ID_VTG_I2C	0x209100b0	U1	-	-	1
CFG-MSGOUT-NMEA_ID_VTG_SPI	0x209100b4	U1	-	-	1
CFG-MSGOUT-NMEA_ID_VTG_UART1	0x209100b1	U1	-	-	1
CFG-MSGOUT-NMEA_ID_ZDA_I2C	0x209100d8	U1	-	-	0
CFG-MSGOUT-NMEA_ID_ZDA_SPI	0x209100dc	U1	-	-	0
CFG-MSGOUT-NMEA_ID_ZDA_UART1	0x209100d9	U1	-	-	0
CFG-MSGOUT-PUBX_ID_POLYP_I2C	0x209100ec	U1	-	-	0
CFG-MSGOUT-PUBX_ID_POLYP_SPI	0x209100f0	U1	-	-	0
CFG-MSGOUT-PUBX_ID_POLYP_UART1	0x209100ed	U1	-	-	0
CFG-MSGOUT-PUBX_ID_POLYS_I2C	0x209100f1	U1	-	-	0
CFG-MSGOUT-PUBX_ID_POLYS_SPI	0x209100f5	U1	-	-	0
CFG-MSGOUT-PUBX_ID_POLYS_UART1	0x209100f2	U1	-	-	0
CFG-MSGOUT-PUBX_ID_POLYT_I2C	0x209100f6	U1	-	-	0
CFG-MSGOUT-PUBX_ID_POLYT_SPI	0x209100fa	U1	-	-	0
CFG-MSGOUT-PUBX_ID_POLYT_UART1	0x209100f7	U1	-	-	0
CFG-MSGOUT-UBX_LOG_INFO_I2C	0x20910259	U1	-	-	0
CFG-MSGOUT-UBX_LOG_INFO_SPI	0x2091025d	U1	-	-	0
CFG-MSGOUT-UBX_LOG_INFO_UART1	0x2091025a	U1	-	-	0
CFG-MSGOUT-UBX_MON_COMMS_I2C	0x2091034f	U1	-	-	0
CFG-MSGOUT-UBX_MON_COMMS_SPI	0x20910353	U1	-	-	0
CFG-MSGOUT-UBX_MON_COMMS_UART1	0x20910350	U1	-	-	0
CFG-MSGOUT-UBX_MON_HW3_I2C	0x20910354	U1	-	-	0
CFG-MSGOUT-UBX_MON_HW3_SPI	0x20910358	U1	-	-	0
CFG-MSGOUT-UBX_MON_HW3_UART1	0x20910355	U1	-	-	0
CFG-MSGOUT-UBX_MON_IO_I2C	0x209101a5	U1	-	-	0

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-MSGOUT-UBX_MON_IO_SPI	0x209101a9	U1	-	-	0
CFG-MSGOUT-UBX_MON_IO_UART1	0x209101a6	U1	-	-	0
CFG-MSGOUT-UBX_MON_MSGPP_I2C	0x20910196	U1	-	-	0
CFG-MSGOUT-UBX_MON_MSGPP_SPI	0x2091019a	U1	-	-	0
CFG-MSGOUT-UBX_MON_MSGPP_UART1	0x20910197	U1	-	-	0
CFG-MSGOUT-UBX_MON_RF_I2C	0x20910359	U1	-	-	0
CFG-MSGOUT-UBX_MON_RF_SPI	0x2091035d	U1	-	-	0
CFG-MSGOUT-UBX_MON_RF_UART1	0x2091035a	U1	-	-	0
CFG-MSGOUT-UBX_MON_RXBUF_I2C	0x209101a0	U1	-	-	0
CFG-MSGOUT-UBX_MON_RXBUF_SPI	0x209101a4	U1	-	-	0
CFG-MSGOUT-UBX_MON_RXBUF_UART1	0x209101a1	U1	-	-	0
CFG-MSGOUT-UBX_MON_RXR_I2C	0x20910187	U1	-	-	0
CFG-MSGOUT-UBX_MON_RXR_SPI	0x2091018b	U1	-	-	0
CFG-MSGOUT-UBX_MON_RXR_UART1	0x20910188	U1	-	-	0
CFG-MSGOUT-UBX_MON_SPAN_I2C	0x2091038b	U1	-	-	0
CFG-MSGOUT-UBX_MON_SPAN_SPI	0x2091038f	U1	-	-	0
CFG-MSGOUT-UBX_MON_SPAN_UART1	0x2091038c	U1	-	-	0
CFG-MSGOUT-UBX_MON_SYS_I2C	0x2091069d	U1	-	-	0
CFG-MSGOUT-UBX_MON_SYS_SPI	0x209106a1	U1	-	-	0
CFG-MSGOUT-UBX_MON_SYS_UART1	0x2091069e	U1	-	-	0
CFG-MSGOUT-UBX_MON_TXBUF_I2C	0x2091019b	U1	-	-	0
CFG-MSGOUT-UBX_MON_TXBUF_SPI	0x2091019f	U1	-	-	0
CFG-MSGOUT-UBX_MON_TXBUF_UART1	0x2091019c	U1	-	-	0
CFG-MSGOUT-UBX_NAV_AOPSTATUS_I2C	0x20910079	U1	-	-	0
CFG-MSGOUT-UBX_NAV_AOPSTATUS_SPI	0x2091007d	U1	-	-	0
CFG-MSGOUT-UBX_NAV_AOPSTATUS_UART1	0x2091007a	U1	-	-	0
CFG-MSGOUT-UBX_NAV_CLOCK_I2C	0x20910065	U1	-	-	0
CFG-MSGOUT-UBX_NAV_CLOCK_SPI	0x20910069	U1	-	-	0
CFG-MSGOUT-UBX_NAV_CLOCK_UART1	0x20910066	U1	-	-	0
CFG-MSGOUT-UBX_NAV_COV_I2C	0x20910083	U1	-	-	0
CFG-MSGOUT-UBX_NAV_COV_SPI	0x20910087	U1	-	-	0
CFG-MSGOUT-UBX_NAV_COV_UART1	0x20910084	U1	-	-	0
CFG-MSGOUT-UBX_NAV_DOP_I2C	0x20910038	U1	-	-	0
CFG-MSGOUT-UBX_NAV_DOP_SPI	0x2091003c	U1	-	-	0
CFG-MSGOUT-UBX_NAV_DOP_UART1	0x20910039	U1	-	-	0
CFG-MSGOUT-UBX_NAV_EOE_I2C	0x2091015f	U1	-	-	0
CFG-MSGOUT-UBX_NAV_EOE_SPI	0x20910163	U1	-	-	0
CFG-MSGOUT-UBX_NAV_EOE_UART1	0x20910160	U1	-	-	0
CFG-MSGOUT-UBX_NAV_GEOFENCE_I2C	0x209100a1	U1	-	-	0
CFG-MSGOUT-UBX_NAV_GEOFENCE_SPI	0x209100a5	U1	-	-	0
CFG-MSGOUT-UBX_NAV_GEOFENCE_UART1	0x209100a2	U1	-	-	0

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-MSGOUT-UBX_NAV_ODO_I2C	0x2091007e	U1	-	-	0
CFG-MSGOUT-UBX_NAV_ODO_SPI	0x20910082	U1	-	-	0
CFG-MSGOUT-UBX_NAV_ODO_UART1	0x2091007f	U1	-	-	0
CFG-MSGOUT-UBX_NAV_ORB_I2C	0x20910010	U1	-	-	0
CFG-MSGOUT-UBX_NAV_ORB_SPI	0x20910014	U1	-	-	0
CFG-MSGOUT-UBX_NAV_ORB_UART1	0x20910011	U1	-	-	0
CFG-MSGOUT-UBX_NAV_POSECEF_I2C	0x20910024	U1	-	-	0
CFG-MSGOUT-UBX_NAV_POSECEF_SPI	0x20910028	U1	-	-	0
CFG-MSGOUT-UBX_NAV_POSECEF_UART1	0x20910025	U1	-	-	0
CFG-MSGOUT-UBX_NAV_POSLLH_I2C	0x20910029	U1	-	-	0
CFG-MSGOUT-UBX_NAV_POSLLH_SPI	0x2091002d	U1	-	-	0
CFG-MSGOUT-UBX_NAV_POSLLH_UART1	0x2091002a	U1	-	-	0
CFG-MSGOUT-UBX_NAV_PVT_I2C	0x20910006	U1	-	-	0
CFG-MSGOUT-UBX_NAV_PVT_SPI	0x2091000a	U1	-	-	0
CFG-MSGOUT-UBX_NAV_PVT_UART1	0x20910007	U1	-	-	0
CFG-MSGOUT-UBX_NAV_SAT_I2C	0x20910015	U1	-	-	0
CFG-MSGOUT-UBX_NAV_SAT_SPI	0x20910019	U1	-	-	0
CFG-MSGOUT-UBX_NAV_SAT_UART1	0x20910016	U1	-	-	0
CFG-MSGOUT-UBX_NAV_SBAS_I2C	0x2091006a	U1	-	-	0
CFG-MSGOUT-UBX_NAV_SBAS_SPI	0x2091006e	U1	-	-	0
CFG-MSGOUT-UBX_NAV_SBAS_UART1	0x2091006b	U1	-	-	0
CFG-MSGOUT-UBX_NAV_SIG_I2C	0x20910345	U1	-	-	0
CFG-MSGOUT-UBX_NAV_SIG_SPI	0x20910349	U1	-	-	0
CFG-MSGOUT-UBX_NAV_SIG_UART1	0x20910346	U1	-	-	0
CFG-MSGOUT-UBX_NAV_SLAS_I2C	0x20910336	U1	-	-	0
CFG-MSGOUT-UBX_NAV_SLAS_SPI	0x2091033a	U1	-	-	0
CFG-MSGOUT-UBX_NAV_SLAS_UART1	0x20910337	U1	-	-	0
CFG-MSGOUT-UBX_NAV_STATUS_I2C	0x2091001a	U1	-	-	0
CFG-MSGOUT-UBX_NAV_STATUS_SPI	0x2091001e	U1	-	-	0
CFG-MSGOUT-UBX_NAV_STATUS_UART1	0x2091001b	U1	-	-	0
CFG-MSGOUT-UBX_NAV_TIMEBDS_I2C	0x20910051	U1	-	-	0
CFG-MSGOUT-UBX_NAV_TIMEBDS_SPI	0x20910055	U1	-	-	0
CFG-MSGOUT-UBX_NAV_TIMEBDS_UART1	0x20910052	U1	-	-	0
CFG-MSGOUT-UBX_NAV_TIMEGAL_I2C	0x20910056	U1	-	-	0
CFG-MSGOUT-UBX_NAV_TIMEGAL_SPI	0x2091005a	U1	-	-	0
CFG-MSGOUT-UBX_NAV_TIMEGAL_UART1	0x20910057	U1	-	-	0
CFG-MSGOUT-UBX_NAV_TIMEGPS_I2C	0x20910047	U1	-	-	0
CFG-MSGOUT-UBX_NAV_TIMEGPS_SPI	0x2091004b	U1	-	-	0
CFG-MSGOUT-UBX_NAV_TIMEGPS_UART1	0x20910048	U1	-	-	0
CFG-MSGOUT-UBX_NAV_TIMELS_I2C	0x20910060	U1	-	-	0
CFG-MSGOUT-UBX_NAV_TIMELS_SPI	0x20910064	U1	-	-	0

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-MSGOUT-UBX_NAV_TIMELS_UART1	0x20910061	U1	-	-	0
CFG-MSGOUT-UBX_NAV_TIMENAVIC_I2C	0x209106a2	U1	-	-	0
CFG-MSGOUT-UBX_NAV_TIMENAVIC_SPI	0x209106a6	U1	-	-	0
CFG-MSGOUT-UBX_NAV_TIMENAVIC_UART1	0x209106a3	U1	-	-	0
CFG-MSGOUT-UBX_NAV_TIMEQZSS_I2C	0x20910386	U1	-	-	0
CFG-MSGOUT-UBX_NAV_TIMEQZSS_SPI	0x2091038a	U1	-	-	0
CFG-MSGOUT-UBX_NAV_TIMEQZSS_UART1	0x20910387	U1	-	-	0
CFG-MSGOUT-UBX_NAV_TIMEUTC_I2C	0x2091005b	U1	-	-	0
CFG-MSGOUT-UBX_NAV_TIMEUTC_SPI	0x2091005f	U1	-	-	0
CFG-MSGOUT-UBX_NAV_TIMEUTC_UART1	0x2091005c	U1	-	-	0
CFG-MSGOUT-UBX_NAV_VELECEF_I2C	0x2091003d	U1	-	-	0
CFG-MSGOUT-UBX_NAV_VELECEF_SPI	0x20910041	U1	-	-	0
CFG-MSGOUT-UBX_NAV_VELECEF_UART1	0x2091003e	U1	-	-	0
CFG-MSGOUT-UBX_NAV_VELNED_I2C	0x20910042	U1	-	-	0
CFG-MSGOUT-UBX_NAV_VELNED_SPI	0x20910046	U1	-	-	0
CFG-MSGOUT-UBX_NAV_VELNED_UART1	0x20910043	U1	-	-	0
CFG-MSGOUT-UBX_RXM_MEASX_I2C	0x20910204	U1	-	-	0
CFG-MSGOUT-UBX_RXM_MEASX_SPI	0x20910208	U1	-	-	0
CFG-MSGOUT-UBX_RXM_MEASX_UART1	0x20910205	U1	-	-	0
CFG-MSGOUT-UBX_RXM_RAWX_I2C	0x209102a4	U1	-	-	0
CFG-MSGOUT-UBX_RXM_RAWX_SPI	0x209102a8	U1	-	-	0
CFG-MSGOUT-UBX_RXM_RAWX_UART1	0x209102a5	U1	-	-	0
CFG-MSGOUT-UBX_RXM_RLM_I2C	0x2091025e	U1	-	-	0
CFG-MSGOUT-UBX_RXM_RLM_SPI	0x20910262	U1	-	-	0
CFG-MSGOUT-UBX_RXM_RLM_UART1	0x2091025f	U1	-	-	0
CFG-MSGOUT-UBX_RXM_SFRBX_I2C	0x20910231	U1	-	-	0
CFG-MSGOUT-UBX_RXM_SFRBX_SPI	0x20910235	U1	-	-	0
CFG-MSGOUT-UBX_RXM_SFRBX_UART1	0x20910232	U1	-	-	0
CFG-MSGOUT-UBX_SEC_SIGLOG_I2C	0x20910689	U1	-	-	0
CFG-MSGOUT-UBX_SEC_SIGLOG_SPI	0x2091068d	U1	-	-	0
CFG-MSGOUT-UBX_SEC_SIGLOG_UART1	0x2091068a	U1	-	-	0
CFG-MSGOUT-UBX_SEC_SIG_I2C	0x20910634	U1	-	-	0
CFG-MSGOUT-UBX_SEC_SIG_SPI	0x20910638	U1	-	-	0
CFG-MSGOUT-UBX_SEC_SIG_UART1	0x20910635	U1	-	-	0
CFG-MSGOUT-UBX_TIM_TM2_I2C	0x20910178	U1	-	-	0
CFG-MSGOUT-UBX_TIM_TM2_SPI	0x2091017c	U1	-	-	0
CFG-MSGOUT-UBX_TIM_TM2_UART1	0x20910179	U1	-	-	0
CFG-MSGOUT-UBX_TIM_TP_I2C	0x2091017d	U1	-	-	0
CFG-MSGOUT-UBX_TIM_TP_SPI	0x20910181	U1	-	-	0
CFG-MSGOUT-UBX_TIM_TP_UART1	0x2091017e	U1	-	-	0
CFG-MSGOUT-UBX_TIM_VRFY_I2C	0x20910092	U1	-	-	0

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-MSGOUT-UBX_TIM_VRFY_SPI	0x20910096	U1	-	-	0
CFG-MSGOUT-UBX_TIM_VRFY_UART1	0x20910093	U1	-	-	0

Table 75: CFG-MSGOUT configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-NAVSPG-FIXMODE	0x20110011	E1	-	-	3 (AUTO)
CFG-NAVSPG-INIFIX3D	0x10110013	L	-	-	0 (false)
CFG-NAVSPG-WKNROLLOVER	0x30110017	U2	-	-	2425
CFG-NAVSPG-UTCSTANDARD	0x2011001c	E1	-	-	0 (AUTO)
CFG-NAVSPG-DYNMODEL	0x20110021	E1	-	-	0 (PORT)
CFG-NAVSPG-ACKAIDING	0x10110025	L	-	-	1 (true)
CFG-NAVSPG-USE_USRDAT	0x10110061	L	-	-	0 (false)
CFG-NAVSPG-USRDAT_MAJA	0x50110062	R8	-	m	6378137
CFG-NAVSPG-USRDAT_FLAT	0x50110063	R8	-	-	298.25722356300002502
CFG-NAVSPG-USRDAT_DX	0x40110064	R4	-	m	0
CFG-NAVSPG-USRDAT_DY	0x40110065	R4	-	m	0
CFG-NAVSPG-USRDAT_DZ	0x40110066	R4	-	m	0
CFG-NAVSPG-USRDAT_ROT_X	0x40110067	R4	-	arcsec	0
CFG-NAVSPG-USRDAT_ROT_Y	0x40110068	R4	-	arcsec	0
CFG-NAVSPG-USRDAT_ROT_Z	0x40110069	R4	-	arcsec	0
CFG-NAVSPG-USRDAT_SCALE	0x4011006a	R4	-	ppm	0
CFG-NAVSPG-INFIL_MINSVS	0x201100a1	U1	-	-	4
CFG-NAVSPG-INFIL_MAXSVS	0x201100a2	U1	-	-	32
CFG-NAVSPG-INFIL_MINCNO	0x201100a3	U1	-	dBHz	6
CFG-NAVSPG-INFIL_MINELEV	0x201100a4	I1	-	deg	5
CFG-NAVSPG-INFIL_NCNOTHS	0x201100aa	U1	-	-	0
CFG-NAVSPG-INFIL_CNOTHS	0x201100ab	U1	-	-	0
CFG-NAVSPG-OUTFIL_PDOP	0x301100b1	U2	0.1	-	250
CFG-NAVSPG-OUTFIL_TDOP	0x301100b2	U2	0.1	-	250
CFG-NAVSPG-OUTFIL_PACC	0x301100b3	U2	-	m	30
CFG-NAVSPG-OUTFIL_TACC	0x301100b4	U2	-	m	350
CFG-NAVSPG-OUTFIL_FACC	0x301100b5	U2	0.01	m/s	150
CFG-NAVSPG-CONSTR_ALT	0x401100c1	I4	0.01	m	0
CFG-NAVSPG-CONSTR_ALTVAR	0x401100c2	U4	0.0001	m^2	10000
CFG-NAVSPG-CONSTR_DGNSSTO	0x201100c4	U1	-	s	60
CFG-NAVSPG-CONSTR_DGNSSTO_SCALE	0x201100c5	U1	-	-	1
CFG-NAVSPG-SIGATTCOMP	0x201100d6	E1	-	-	0 (DIS)

Table 76: CFG-NAVSPG configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-NMEA-PROTVER	0x20930001	E1	-	-	42 (V411)
CFG-NMEA-MAXSVS	0x20930002	E1	-	-	0 (UNLIM)

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-NMEA-COMPAT	0x10930003	L	-	-	0 (false)
CFG-NMEA-CONSIDER	0x10930004	L	-	-	1 (true)
CFG-NMEA-LIMIT82	0x10930005	L	-	-	0 (false)
CFG-NMEA-HIGHPREC	0x10930006	L	-	-	0 (false)
CFG-NMEA-SVNUMBERING	0x20930007	E1	-	-	0 (STRICT)
CFG-NMEA-FILT_GPS	0x10930011	L	-	-	0 (false)
CFG-NMEA-FILT_SBAS	0x10930012	L	-	-	0 (false)
CFG-NMEA-FILT_GAL	0x10930013	L	-	-	0 (false)
CFG-NMEA-FILT_QZSS	0x10930015	L	-	-	0 (false)
CFG-NMEA-FILT_BDS	0x10930017	L	-	-	0 (false)
CFG-NMEA-FILT_NAVIC	0x10930018	L	-	-	0 (false)
CFG-NMEA-OUT_INVFIX	0x10930021	L	-	-	0 (false)
CFG-NMEA-OUT_MSKFIX	0x10930022	L	-	-	0 (false)
CFG-NMEA-OUT_INVTIME	0x10930023	L	-	-	0 (false)
CFG-NMEA-OUT_INVDATE	0x10930024	L	-	-	0 (false)
CFG-NMEA-OUT_ONLYGPS	0x10930025	L	-	-	0 (false)
CFG-NMEA-OUT_FROZENCOG	0x10930026	L	-	-	0 (false)
CFG-NMEA-MAINTALKERID	0x20930031	E1	-	-	0 (AUTO)
CFG-NMEA-GSVTALKERID	0x20930032	E1	-	-	0 (GNSS)
CFG-NMEA-BDSTALKERID	0x30930033	U2	-	-	0

Table 77: CFG-NMEA configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-ODO-USE_ODO	0x10220001	L	-	-	1 (true)
CFG-ODO-USE_COG	0x10220002	L	-	-	0 (false)
CFG-ODO-OUTLPVEL	0x10220003	L	-	-	0 (false)
CFG-ODO-OUTLPCOG	0x10220004	L	-	-	0 (false)
CFG-ODO-PROFILE	0x20220005	E1	-	-	0 (RUN)
CFG-ODO-COGMAXSPEED	0x20220021	U1	1e-1	m/s	10
CFG-ODO-COGMAXPOSACC	0x20220022	U1	-	-	50
CFG-ODO-VELLPGAIN	0x20220031	U1	-	-	153
CFG-ODO-COGLPGAIN	0x20220032	U1	-	-	76

Table 78: CFG-ODO configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-PM-OPERATEMODE	0x20d00001	E1	-	-	0 (FULL)
CFG-PM-POSUPDATEPERIOD	0x40d00002	U4	-	s	300
CFG-PM-MAXACQTIME	0x20d00007	U1	-	s	0
CFG-PM-UPDATEEPH	0x10d0000a	L	-	-	1 (true)
CFG-PM-EXTINTWAKE	0x10d0000c	L	-	-	0 (false)
CFG-PM-EXTINTBACKUP	0x10d0000d	L	-	-	0 (false)
CFG-PM-EXTINTINACTIVE	0x10d0000e	L	-	-	0 (false)

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-PM-EXTINTINACTIVITY	0x40d0000f	U4	0.001	s	0
CFG-PM-LIMITPEAKCURR	0x10d00010	L	-	-	0 (false)
CFG-PM-AUTOSAVEPERIOD	0x20d0001d	U1	-	min	0
CFG-PM-SELECTIVE_L5	0x10d0001e	L	-	-	0 (false)
CFG-PM-POSRETRYPERIOD	0x40d00021	U4	-	s	0

Table 79: CFG-PM configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-QZSS-USE_SLAS_DGNSS	0x10370005	L	-	-	1 (true)
CFG-QZSS-USE_SLAS_TESTMODE	0x10370006	L	-	-	0 (false)
CFG-QZSS-USE_SLAS_RAIM_UNCORR	0x10370007	L	-	-	0 (false)
CFG-QZSS-SLAS_MAX_BASELINE	0x30370008	U2	-	km	350

Table 80: CFG-QZSS configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-RATE-MEAS	0x30210001	U2	0.001	s	1000
CFG-RATE-NAV	0x30210002	U2	-	-	1
CFG-RATE-TIMEREF	0x20210003	E1	-	-	1 (GPS)

Table 81: CFG-RATE configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-RINV-DUMP	0x10c70001	L	-	-	0 (false)
CFG-RINV-BINARY	0x10c70002	L	-	-	0 (false)
CFG-RINV-DATA_SIZE	0x20c70003	U1	-	-	22
CFG-RINV-CHUNK0	0x50c70004	X8	-	-	0x203a656369746f4e ("Notice: ")
CFG-RINV-CHUNK1	0x50c70005	X8	-	-	0x2061746164206f6e ("no data ")
CFG-RINV-CHUNK2	0x50c70006	X8	-	-	0x0000216465766173 ("saved!\0\0")
CFG-RINV-CHUNK3	0x50c70007	X8	-	-	0x0000000000000000

Table 82: CFG-RINV configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-SBAS-USE_TESTMODE	0x10360002	L	-	-	0 (false)
CFG-SBAS-USE_RANGING	0x10360003	L	-	-	1 (true)
CFG-SBAS-USE_DIFFCORR	0x10360004	L	-	-	1 (true)
CFG-SBAS-USE_INTEGRITY	0x10360005	L	-	-	0 (false)
CFG-SBAS-ACCEPT_NOT_IN_PRNMASK	0x30360008	X2	-	-	0x0000
CFG-SBAS-USE_IONOONLY	0x10360007	L	-	-	0 (false)
CFG-SBAS-PRNSCANMASK	0x50360006	X8	-	-	0x000000000083eb8e (ALL PRN121 PRN122 PRN123 PRN127 PRN128 PRN129 PRN131 PRN133 PRN134 PRN135 PRN136 PRN137 PRN143)

Table 83: CFG-SBAS configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-SEC-CFG_LOCK	0x10f60009	L	-	-	0 (false)
CFG-SEC-CFG_LOCK_UNLOCKGRP1	0x30f6000a	U2	-	-	0
CFG-SEC-CFG_LOCK_UNLOCKGRP2	0x30f6000b	U2	-	-	0
CFG-SEC-SPOOFDET_SIM_SIG_DIS	0x10f6005d	L	-	-	0 (false)
CFG-SEC-JAMDET_SENSITIVITY_HI	0x10f60051	L	-	-	1 (true)

Table 84: CFG-SEC configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-SIGNAL-GPS_ENA	0x1031001f	L	-	-	1 (true)
CFG-SIGNAL-GPS_L1CA_ENA	0x10310001	L	-	-	1 (true)
CFG-SIGNAL-GPS_L5_ENA	0x10310004	L	-	-	1 (true)
CFG-SIGNAL-SBAS_ENA	0x10310020	L	-	-	1 (true)
CFG-SIGNAL-SBAS_L1CA_ENA	0x10310005	L	-	-	1 (true)
CFG-SIGNAL-GAL_ENA	0x10310021	L	-	-	1 (true)
CFG-SIGNAL-GAL_E1_ENA	0x10310007	L	-	-	1 (true)
CFG-SIGNAL-GAL_E5A_ENA	0x10310009	L	-	-	1 (true)
CFG-SIGNAL-BDS_ENA	0x10310022	L	-	-	1 (true)
CFG-SIGNAL-BDS_B1_ENA	0x1031000d	L	-	-	1 (true)
CFG-SIGNAL-BDS_B1C_ENA	0x1031000f	L	-	-	1 (true)
CFG-SIGNAL-BDS_B2A_ENA	0x10310028	L	-	-	1 (true)
CFG-SIGNAL-QZSS_ENA	0x10310024	L	-	-	0 (false)
CFG-SIGNAL-QZSS_L1CA_ENA	0x10310012	L	-	-	1 (true)
CFG-SIGNAL-QZSS_L1S_ENA	0x10310014	L	-	-	1 (true)
CFG-SIGNAL-QZSS_L5_ENA	0x10310017	L	-	-	1 (true)
CFG-SIGNAL-NAVIC_ENA	0x10310026	L	-	-	0 (false)
CFG-SIGNAL-NAVIC_L5_ENA	0x1031001d	L	-	-	1 (true)
CFG-SIGNAL-PLAN	0x2031003a	E1	-	-	2 (SP2)

Table 85: CFG-SIGNAL configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-SPI-MAXFF	0x20640001	U1	-	-	50
CFG-SPI-CPOLARITY	0x10640002	L	-	-	0 (false)
CFG-SPI-CPHASE	0x10640003	L	-	-	0 (false)
CFG-SPI-EXTENDEDTIMEOUT	0x10640005	L	-	-	0 (false)
CFG-SPI-ENABLED	0x10640006	L	-	-	0 (false)

Table 86: CFG-SPI configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-SPIINPROT-UBX	0x10790001	L	-	-	1 (true)
CFG-SPIINPROT-NMEA	0x10790002	L	-	-	1 (true)

Table 87: CFG-SPIINPROT configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-SPIOUTPROT-UBX	0x107a0001	L	-	-	1 (true)

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-SPIOUTPROT-NMEA	0x107a0002	L	-	-	1 (true)

Table 88: CFG-SPIOUTPROT configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-TP-PULSE_DEF	0x20050023	E1	-	-	0 (PERIOD)
CFG-TP-PULSE_LENGTH_DEF	0x20050030	E1	-	-	1 (LENGTH)
CFG-TP-ANT_CABLEDELAY	0x30050001	I2	1e-9	s	50
CFG-TP-PERIOD_TP1	0x40050002	U4	1e-6	s	1000000
CFG-TP-PERIOD_LOCK_TP1	0x40050003	U4	1e-6	s	1000000
CFG-TP-FREQ_TP1	0x40050024	U4	-	Hz	1
CFG-TP-FREQ_LOCK_TP1	0x40050025	U4	-	Hz	1
CFG-TP-LEN_TP1	0x40050004	U4	1e-6	s	0
CFG-TP-LEN_LOCK_TP1	0x40050005	U4	1e-6	s	100000
CFG-TP-DUTY_TP1	0x5005002a	R8	-	%	0
CFG-TP-DUTY_LOCK_TP1	0x5005002b	R8	-	%	10
CFG-TP-USER_DELAY_TP1	0x40050006	I4	1e-9	s	0
CFG-TP-TP1_ENA	0x10050007	L	-	-	0 (false)
CFG-TP-SYNC_GNSS_TP1	0x10050008	L	-	-	1 (true)
CFG-TP-USE_LOCKED_TP1	0x10050009	L	-	-	1 (true)
CFG-TP-ALIGN_TO_TOW_TP1	0x1005000a	L	-	-	1 (true)
CFG-TP-POL_TP1	0x1005000b	L	-	-	1 (true)
CFG-TP-TIMEGRID_TP1	0x2005000c	E1	-	-	0 (UTC)
CFG-TP-DRSTR_TP1	0x20050035	E1	-	-	1 (DRIVE_STRENGTH_4MA)

Table 89: CFG-TP configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-TXREADY-ENABLED	0x10a20001	L	-	-	0 (false)
CFG-TXREADY-POLARITY	0x10a20002	L	-	-	0 (false)
CFG-TXREADY-PIN	0x20a20003	U1	-	-	0
CFG-TXREADY-THRESHOLD	0x30a20004	U2	-	-	0
CFG-TXREADY-INTERFACE	0x20a20005	E1	-	-	0 (I2C)

Table 90: CFG-TXREADY configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-UART1-BAUDRATE	0x40520001	U4	-	-	38400
CFG-UART1-STOPBITS	0x20520002	E1	-	-	1 (ONE)
CFG-UART1-DATABITS	0x20520003	E1	-	-	0 (EIGHT)
CFG-UART1-PARITY	0x20520004	E1	-	-	0 (NONE)
CFG-UART1-ENABLED	0x10520005	L	-	-	1 (true)

Table 91: CFG-UART1 configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-UART1INPROT-UBX	0x10730001	L	-	-	1 (true)

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-UART1INPROT-NMEA	0x10730002	L	-	-	1 (true)

Table 92: CFG-UART1INPROT configuration defaults

Configuration item	Key ID	Type	Scale	Unit	Default value
CFG-UART1OUTPROT-UBX	0x10740001	L	-	-	1 (true)
CFG-UART1OUTPROT-NMEA	0x10740002	L	-	-	1 (true)

Table 93: CFG-UART1OUTPROT configuration defaults

Related documents

- [1] UBX-M10150-KB Datasheet (NDA required), UBXDOC-304424225-20108
- [2] UBX-M10150-KB Integration manual (NDA required), UBXDOC-304424225-20458



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Revision history

Revision	Date	Status / Comments
R01	09-Jul-2026	Initial release
R02	01-Sept-2026	Firmware version SPG 7.01 included.

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